



US 20250277238A1

(19) **United States**

(12) **Patent Application Publication**
Perkins et al.

(10) **Pub. No.: US 2025/0277238 A1**

(43) **Pub. Date: Sep. 4, 2025**

(54) **LYMPHOCYTE TARGETED LENTIVIRAL VECTORS**

(71) Applicant: **Kelonia Therapeutics, Inc.**, Boston, MA (US)

(72) Inventors: **Molly R. Perkins**, Milton, MA (US);
Kevin M. Friedman, Boston, MA (US)

(21) Appl. No.: **19/215,495**

(22) Filed: **May 22, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/375,868, filed on Oct. 2, 2023, now Pat. No. 12,312,593, which is a continuation of application No. 18/279,201, filed on Aug. 28, 2023, filed as application No. PCT/US2022/018027 on Feb. 25, 2022.

(60) Provisional application No. 63/154,639, filed on Feb. 26, 2021.

Publication Classification

(51) **Int. Cl.**
C12N 15/86 (2006.01)
A61K 40/10 (2025.01)
A61K 40/11 (2025.01)
A61K 40/31 (2025.01)

A61K 40/42 (2025.01)
C07K 14/005 (2006.01)
C07K 14/705 (2006.01)
C07K 16/28 (2006.01)

(52) **U.S. Cl.**

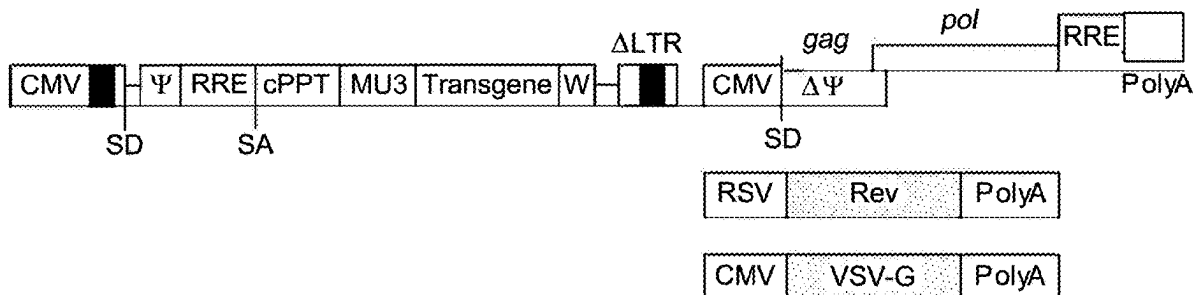
CPC **C12N 15/86** (2013.01); **A61K 40/10** (2025.01); **A61K 40/11** (2025.01); **A61K 40/31** (2025.01); **A61K 40/42II** (2025.01); **A61K 40/4215** (2025.01); **C07K 14/005** (2013.01); **C07K 14/70532** (2013.01); **C07K 16/2803** (2013.01); **C07K 16/2809** (2013.01); **C07K 16/2878** (2013.01); **A61K 2239/31** (2023.05); **A61K 2239/48** (2023.05); **C07K 2317/622** (2013.01); **C07K 2319/03** (2013.01); **C12N 2740/15022** (2013.01); **C12N 2740/15043** (2013.01); **C12N 2740/15051** (2013.01); **C12N 2760/20222** (2013.01)

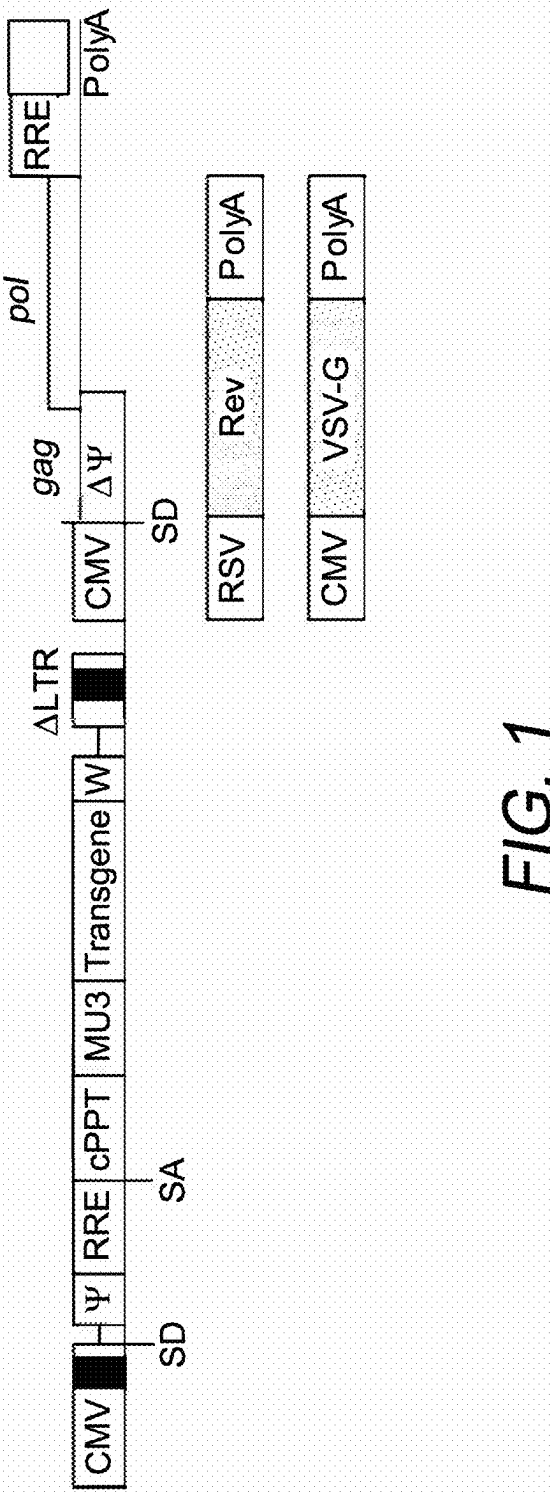
(57)

ABSTRACT

Provided herein are lentiviral vectors comprising a mutated, heterologous envelope protein, a targeting protein, and at least one transgene for delivery to and expression by a cell characterized by the targeting protein. Also provided are methods and materials for producing the lentiviral vectors described herein, methods for transducing target cells, and cells transduced by lentiviral vectors according to the present disclosure.

Specification includes a Sequence Listing.





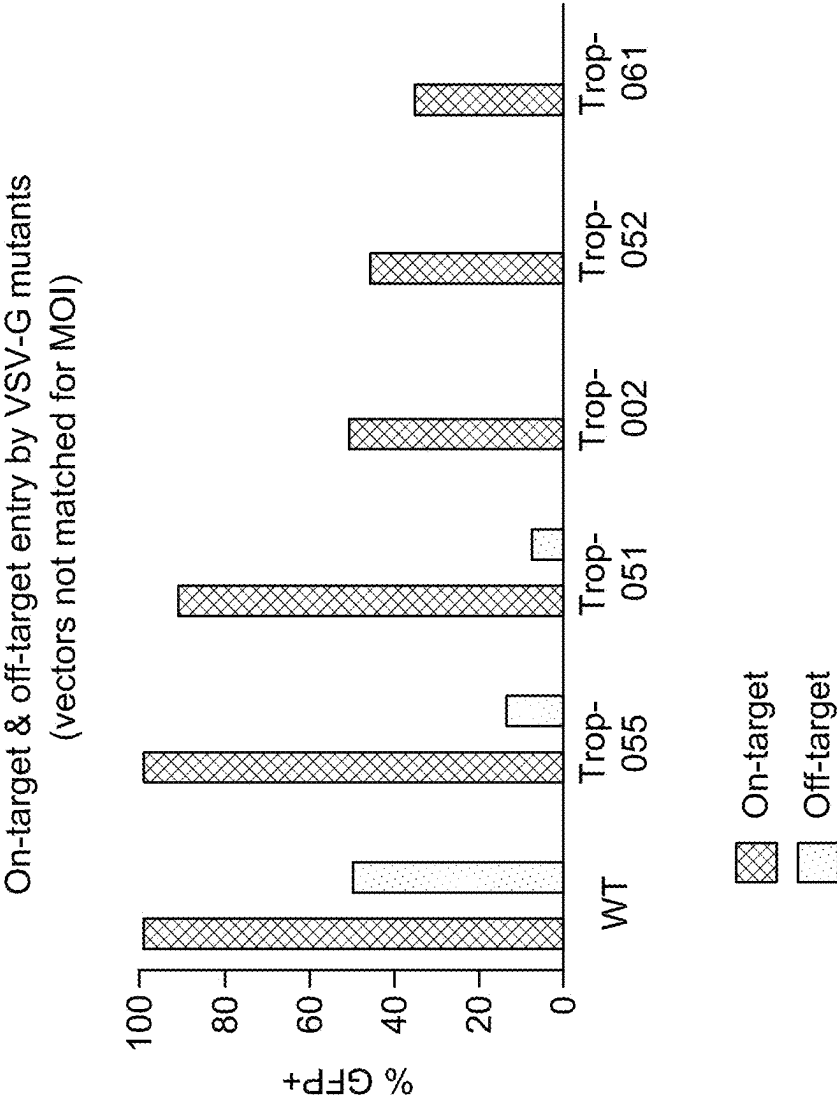


FIG. 2A

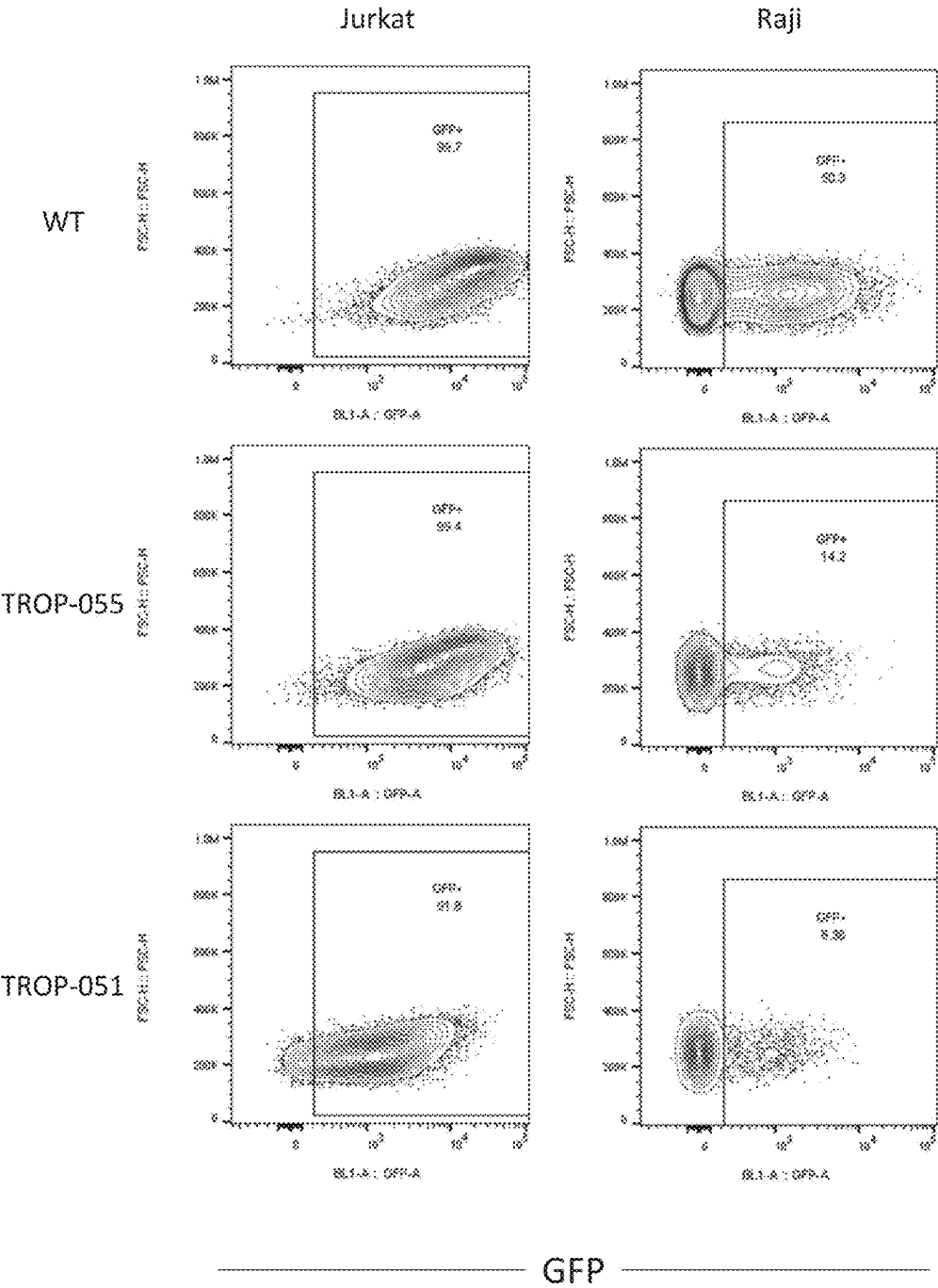


FIG. 2B

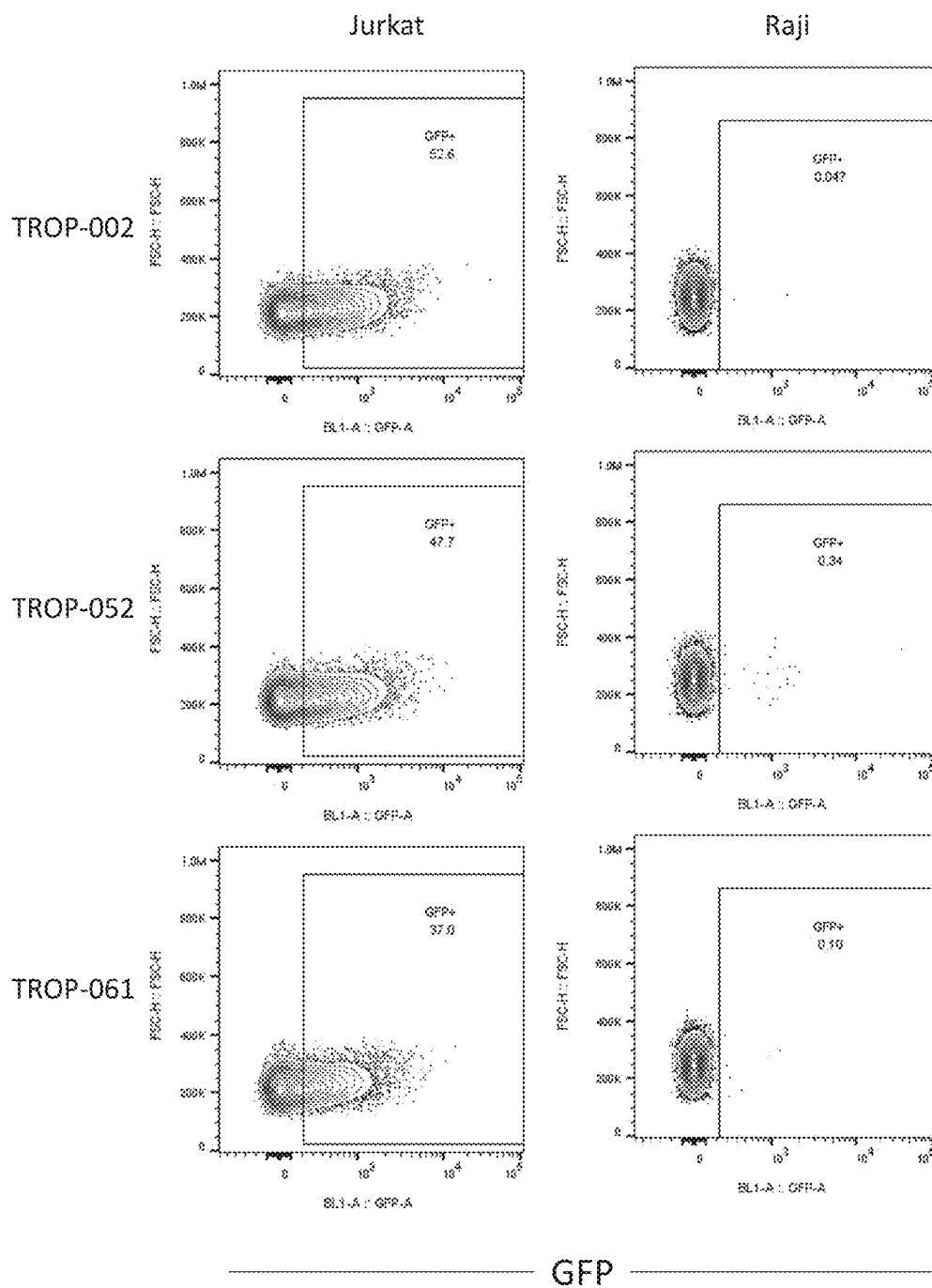


FIG. 2B CONT.

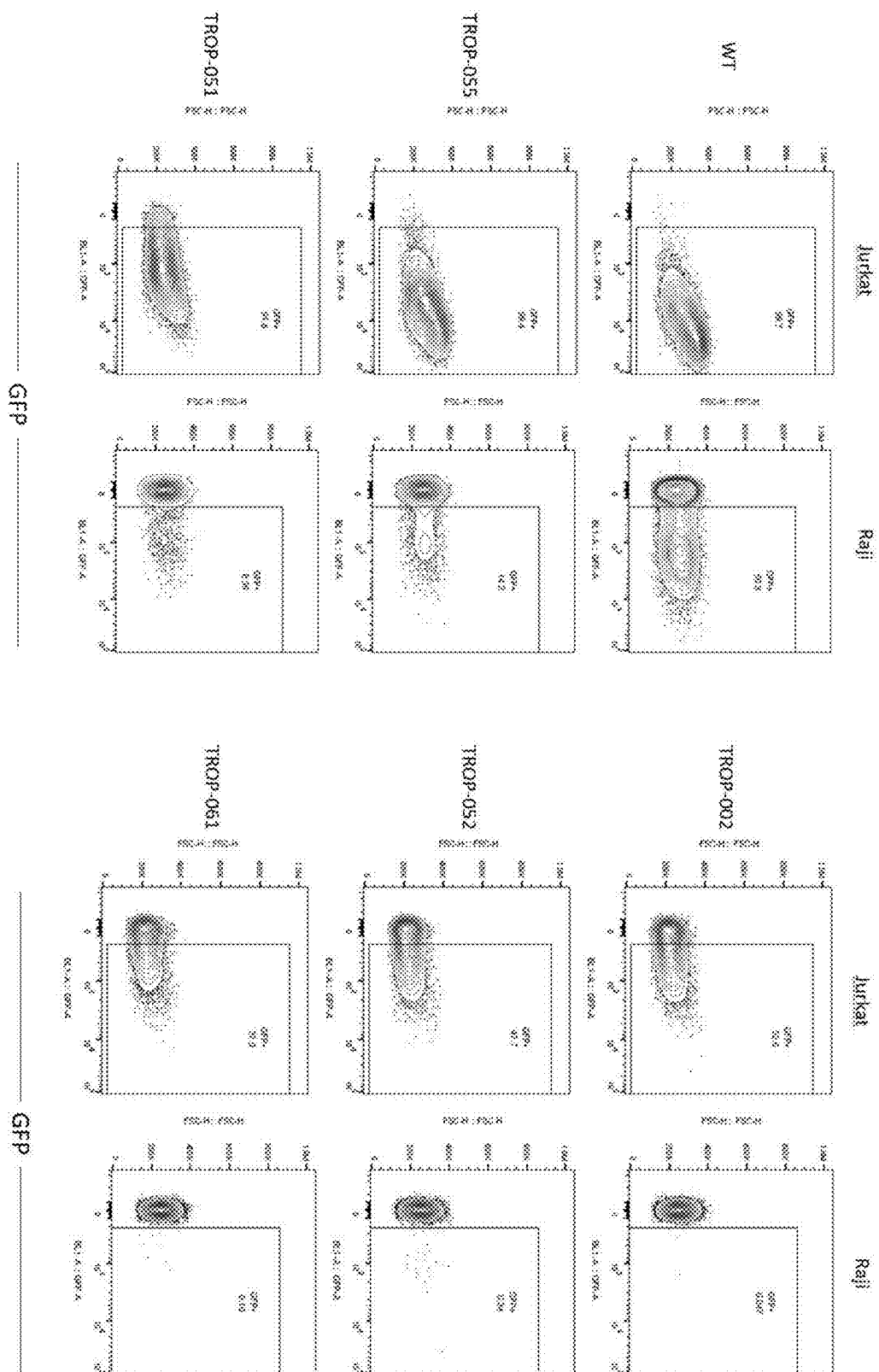


FIG. 2B CONT.

FIG. 3A

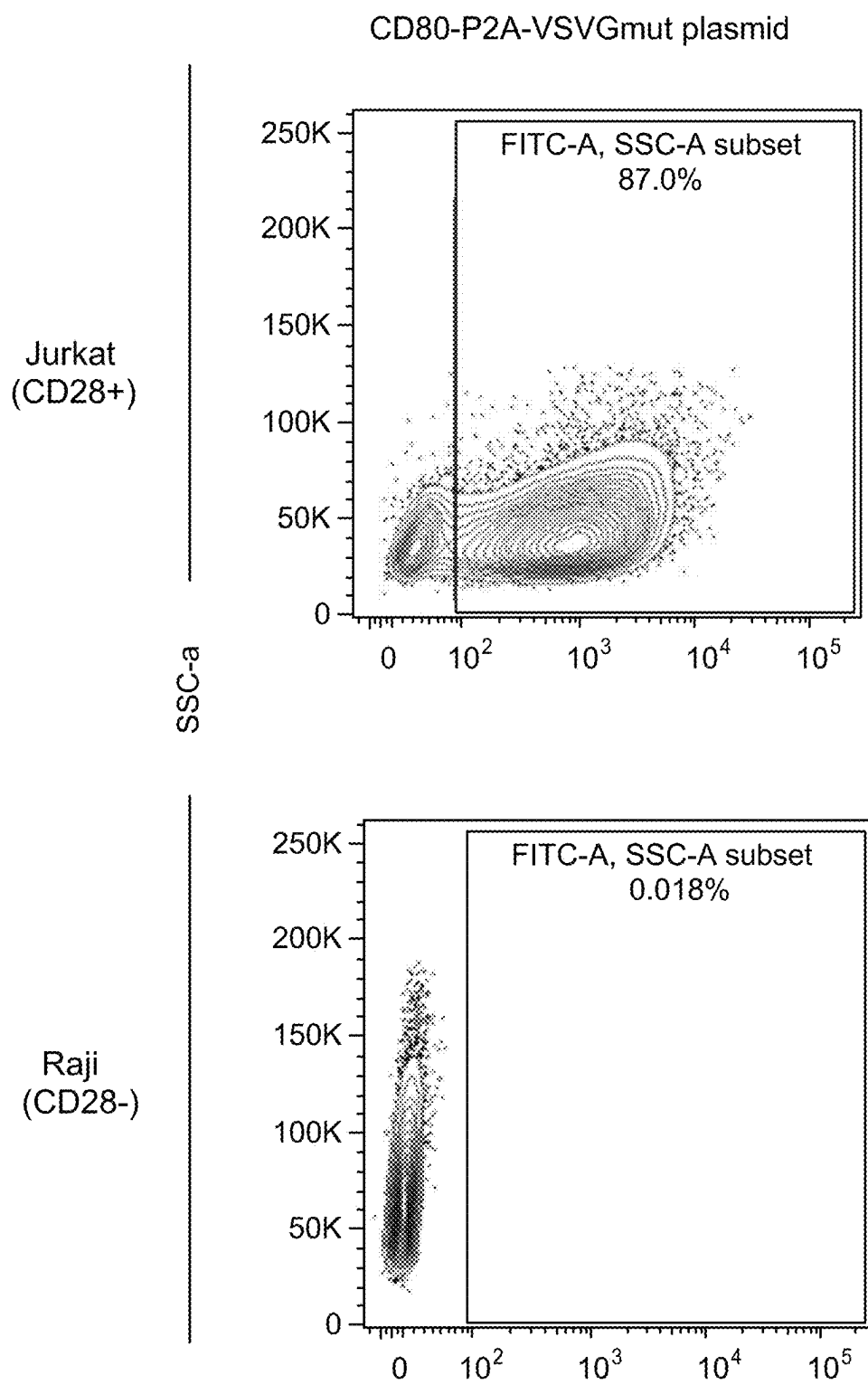


FIG. 3B

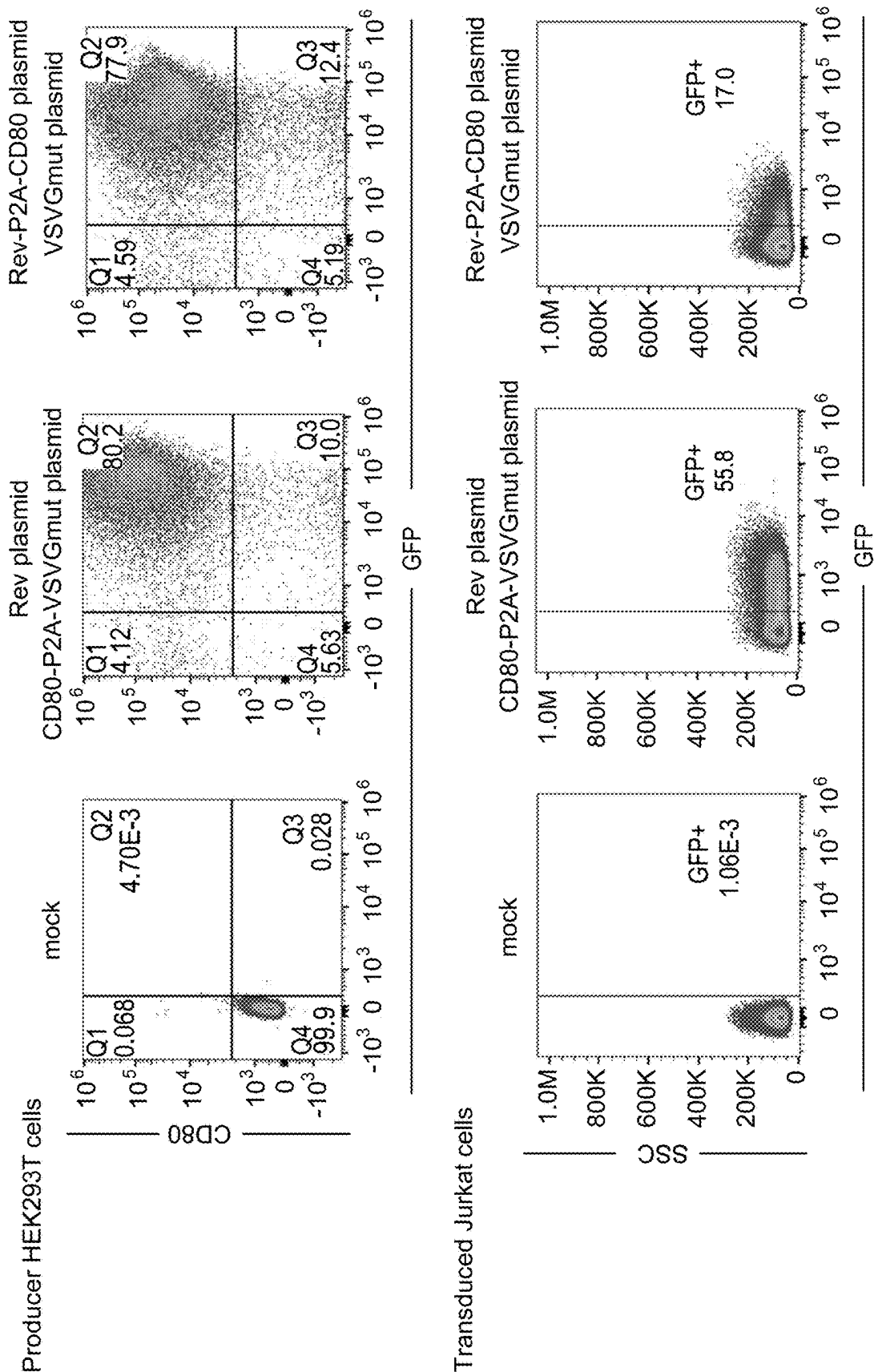


FIG. 4

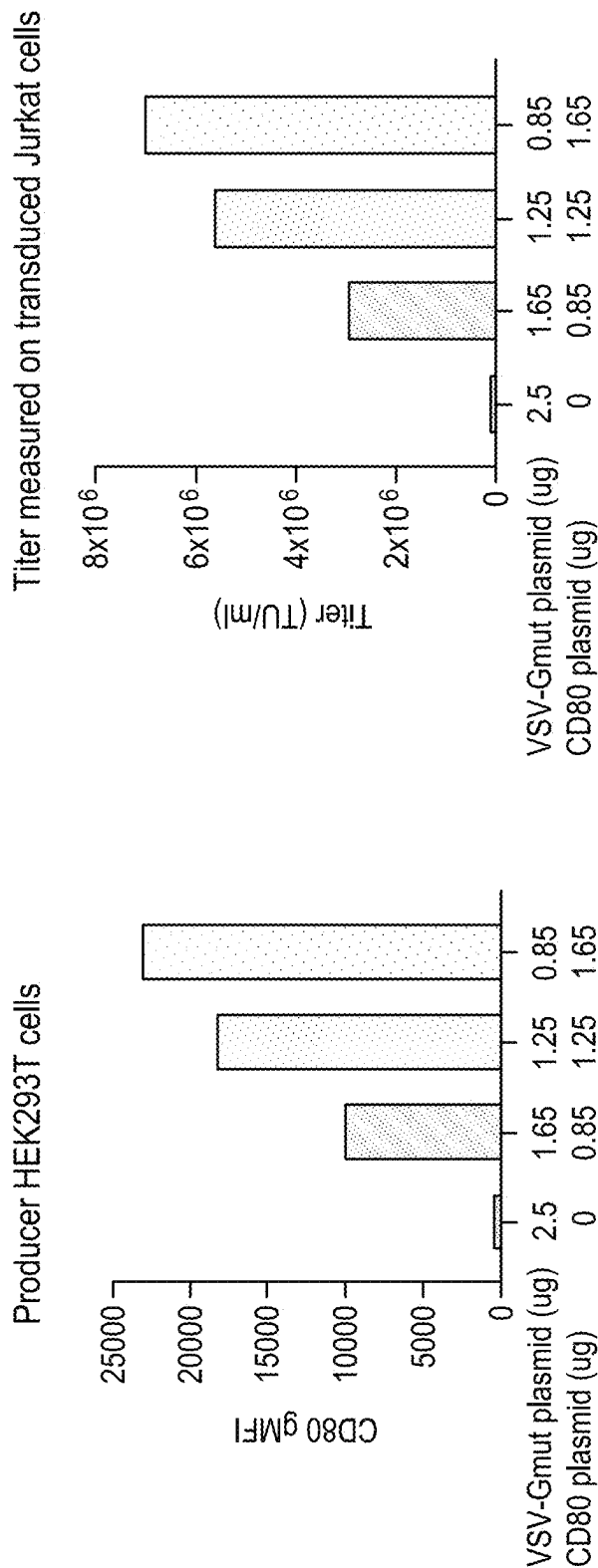


FIG. 5

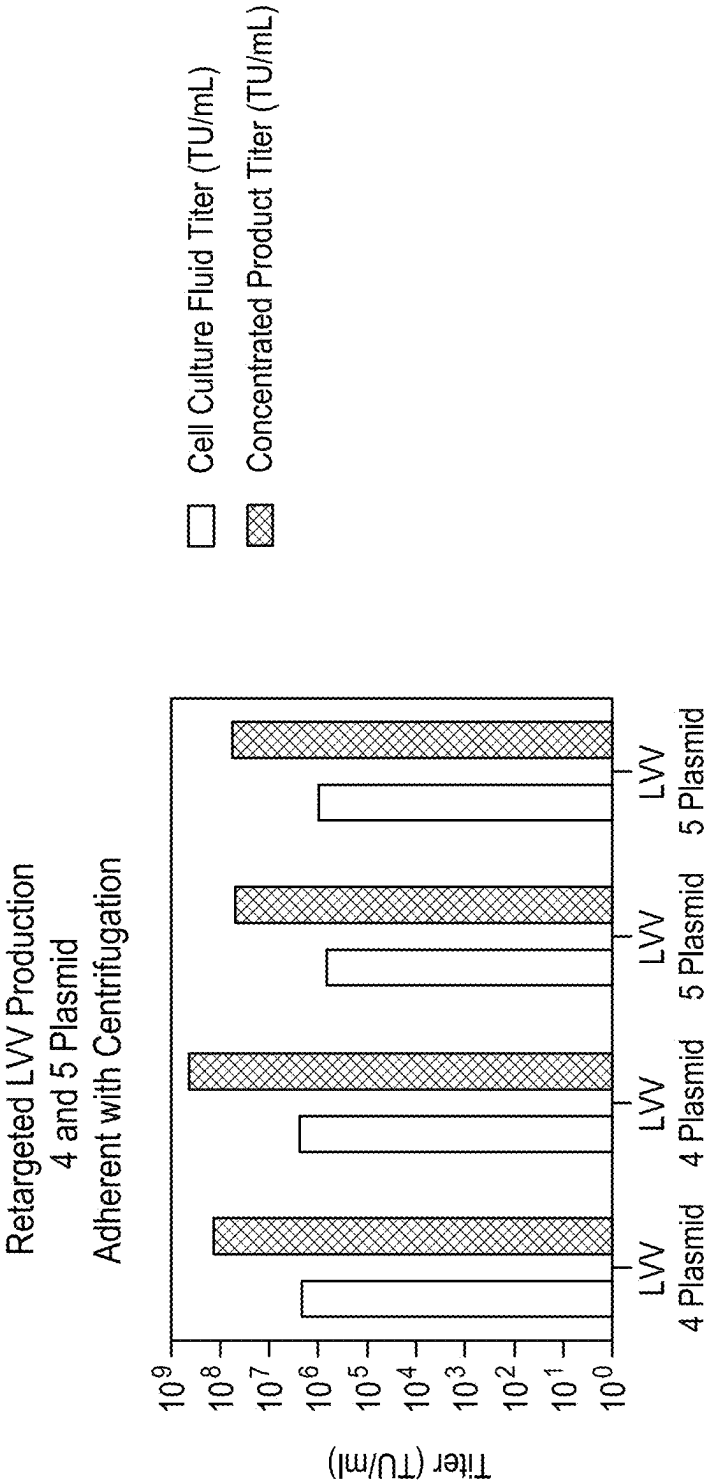


FIG. 6A

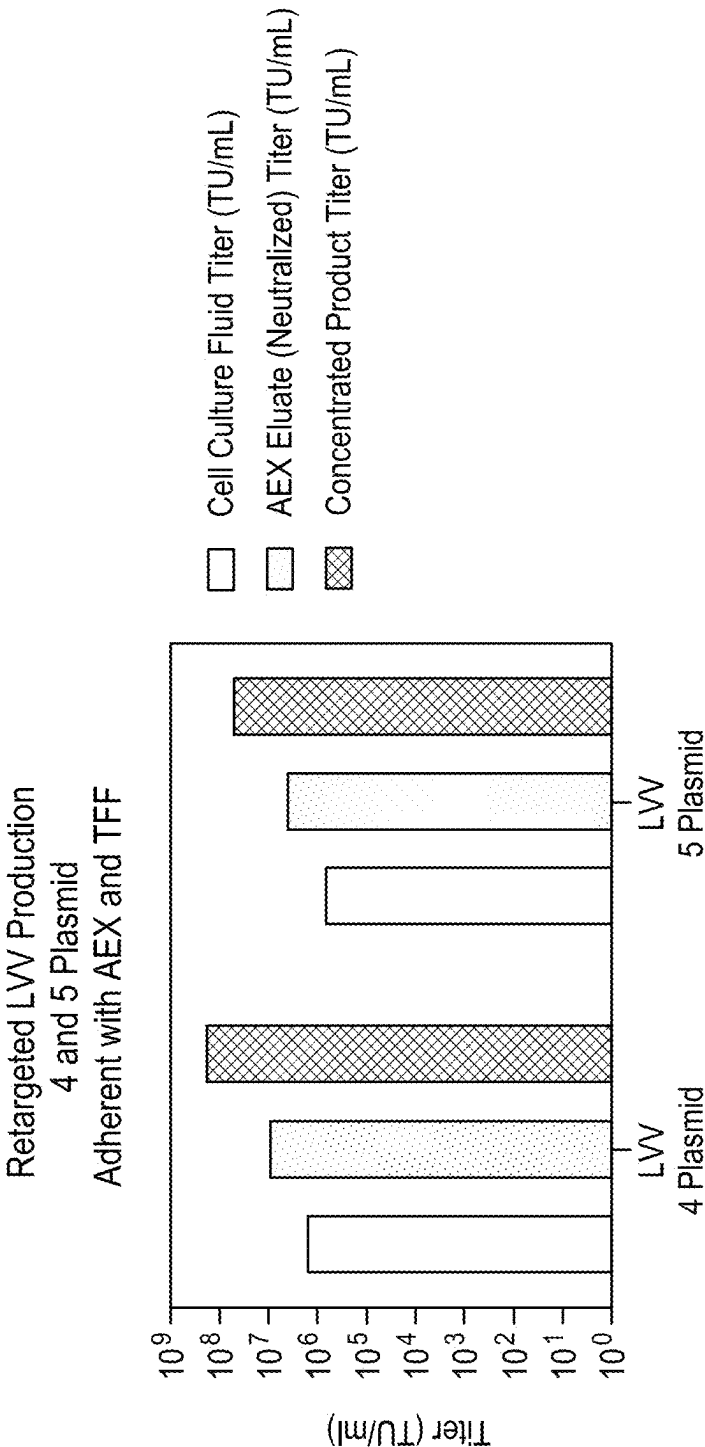


FIG. 6B

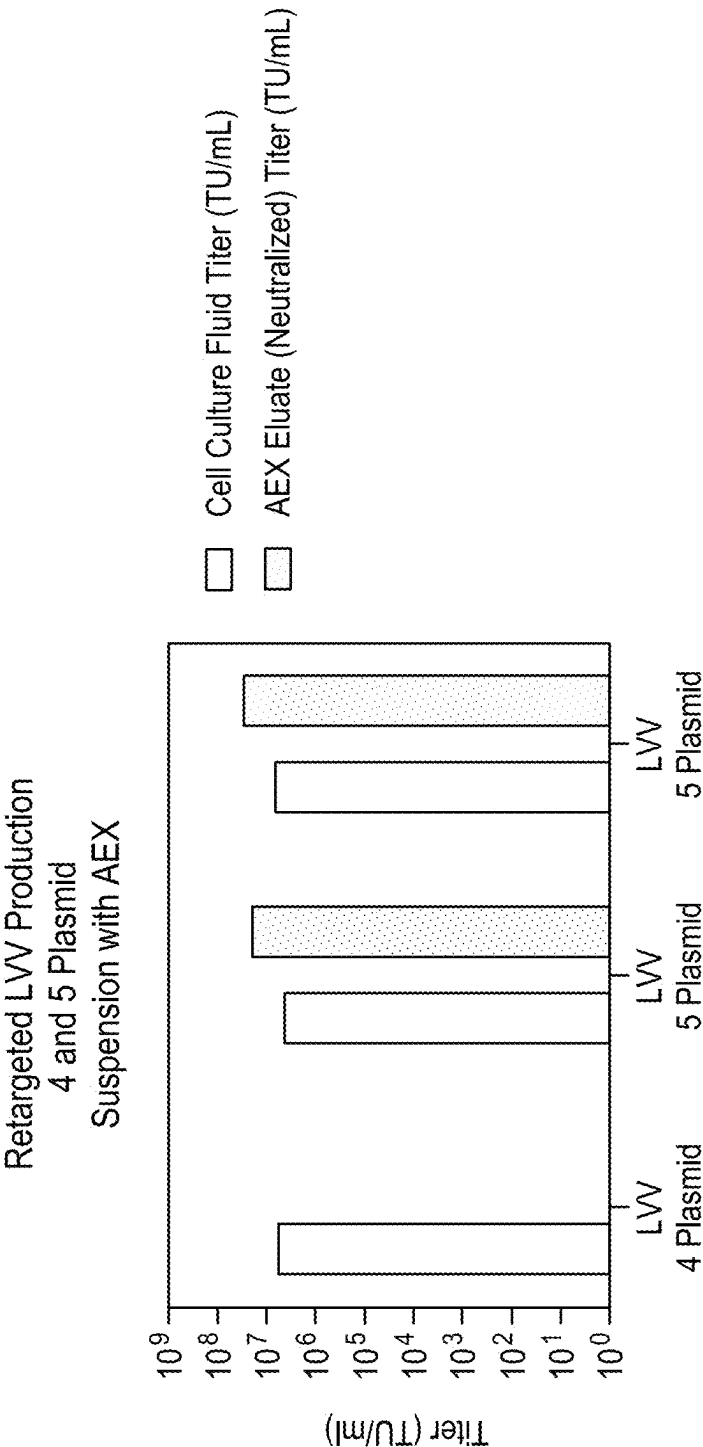


FIG. 6C

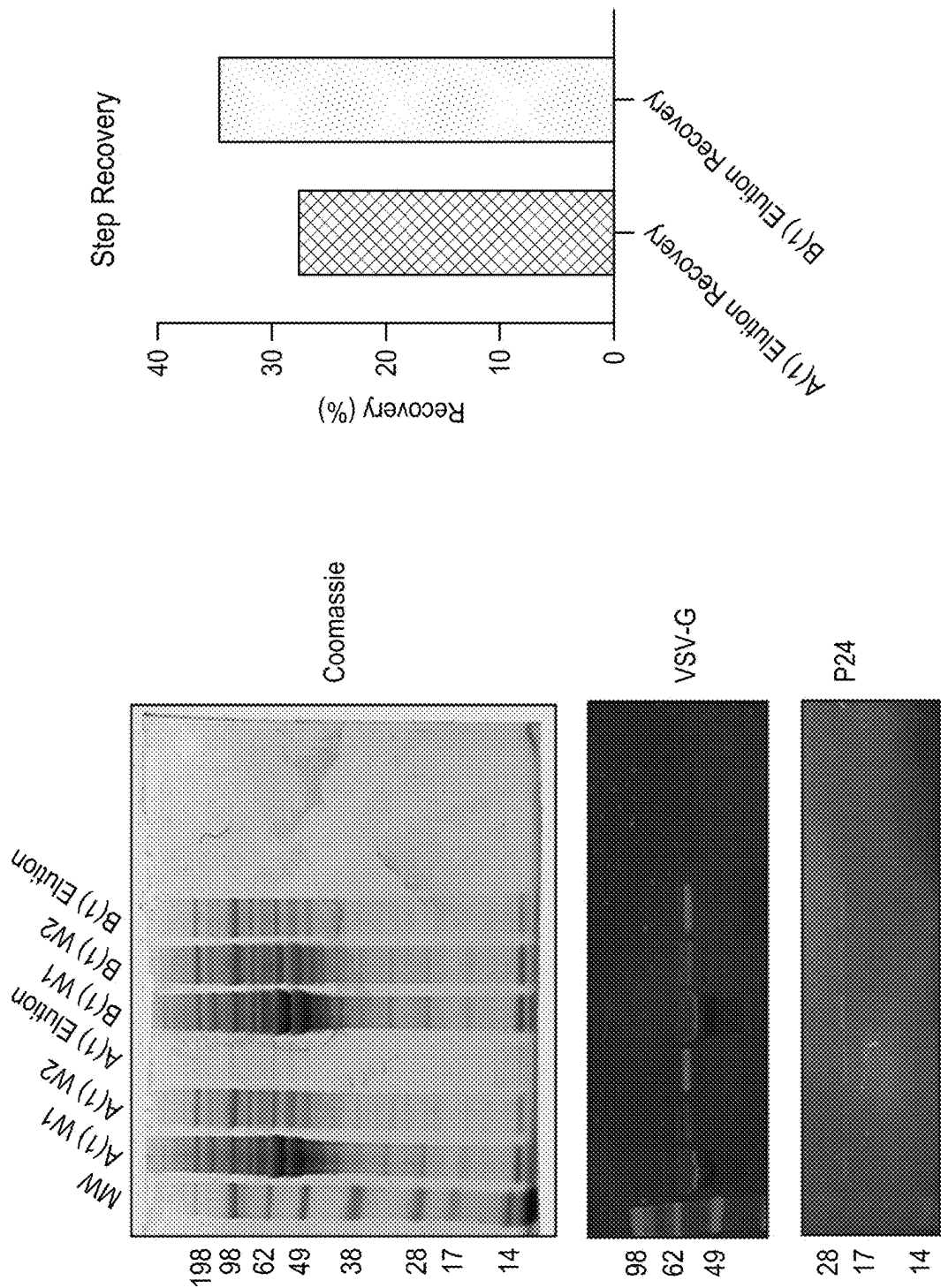


FIG. 6D

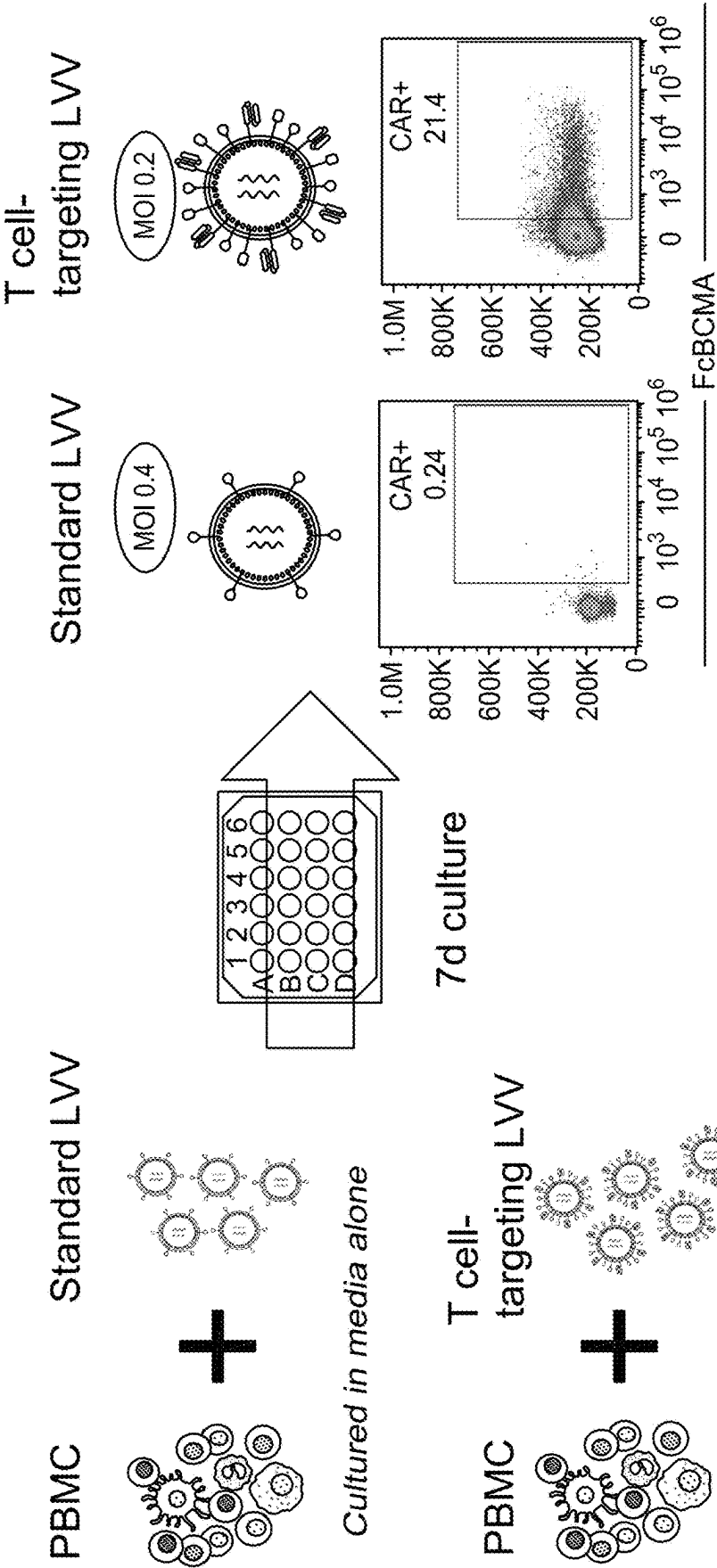


FIG. 7

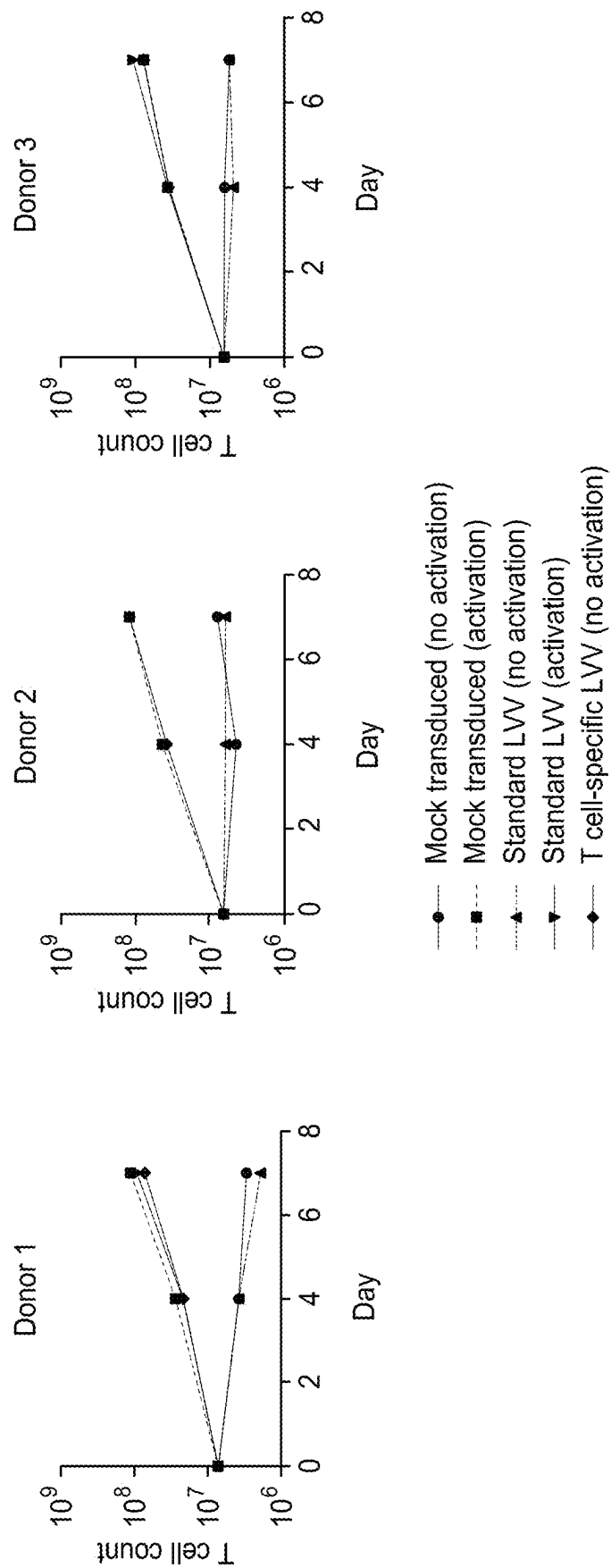


FIG. 8

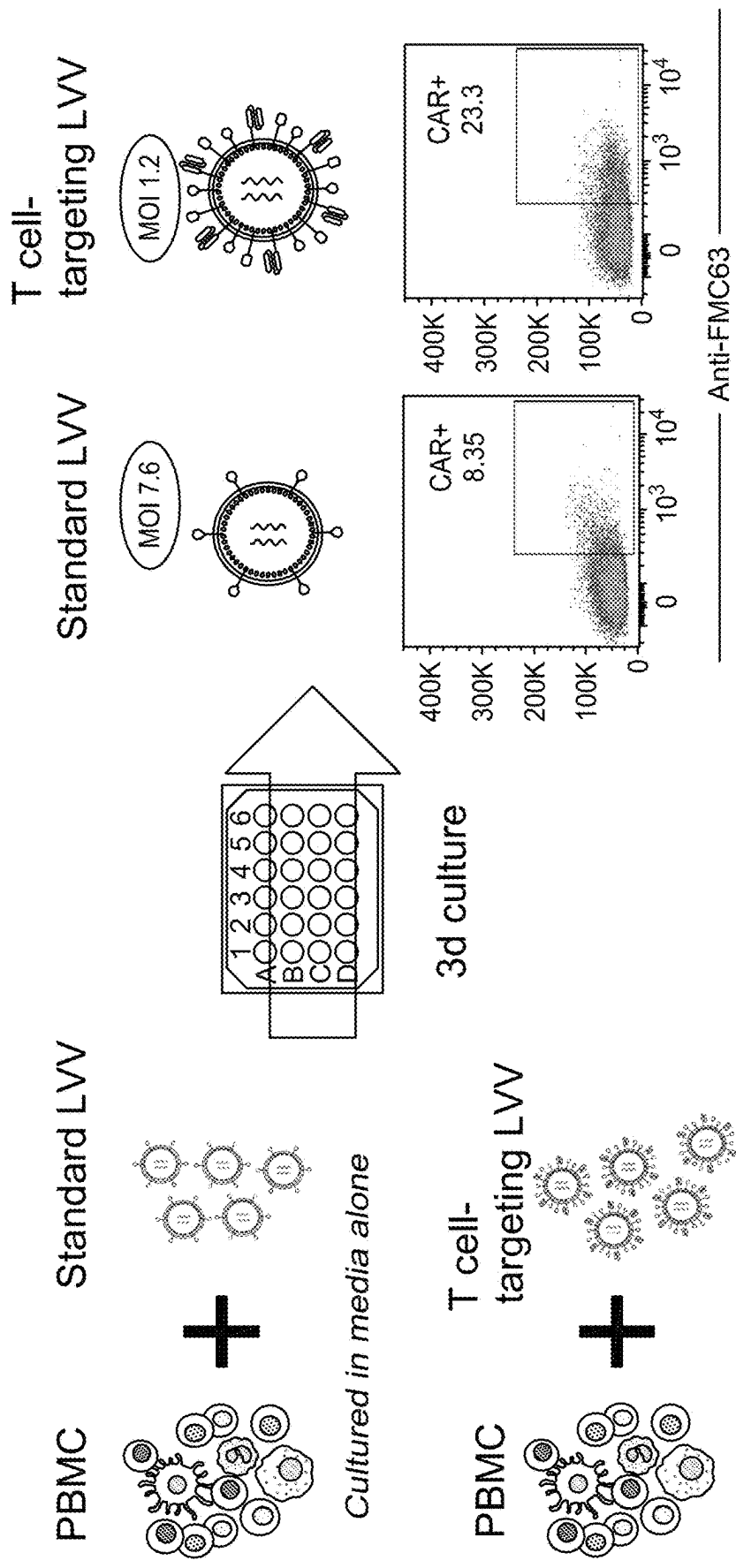


FIG. 9

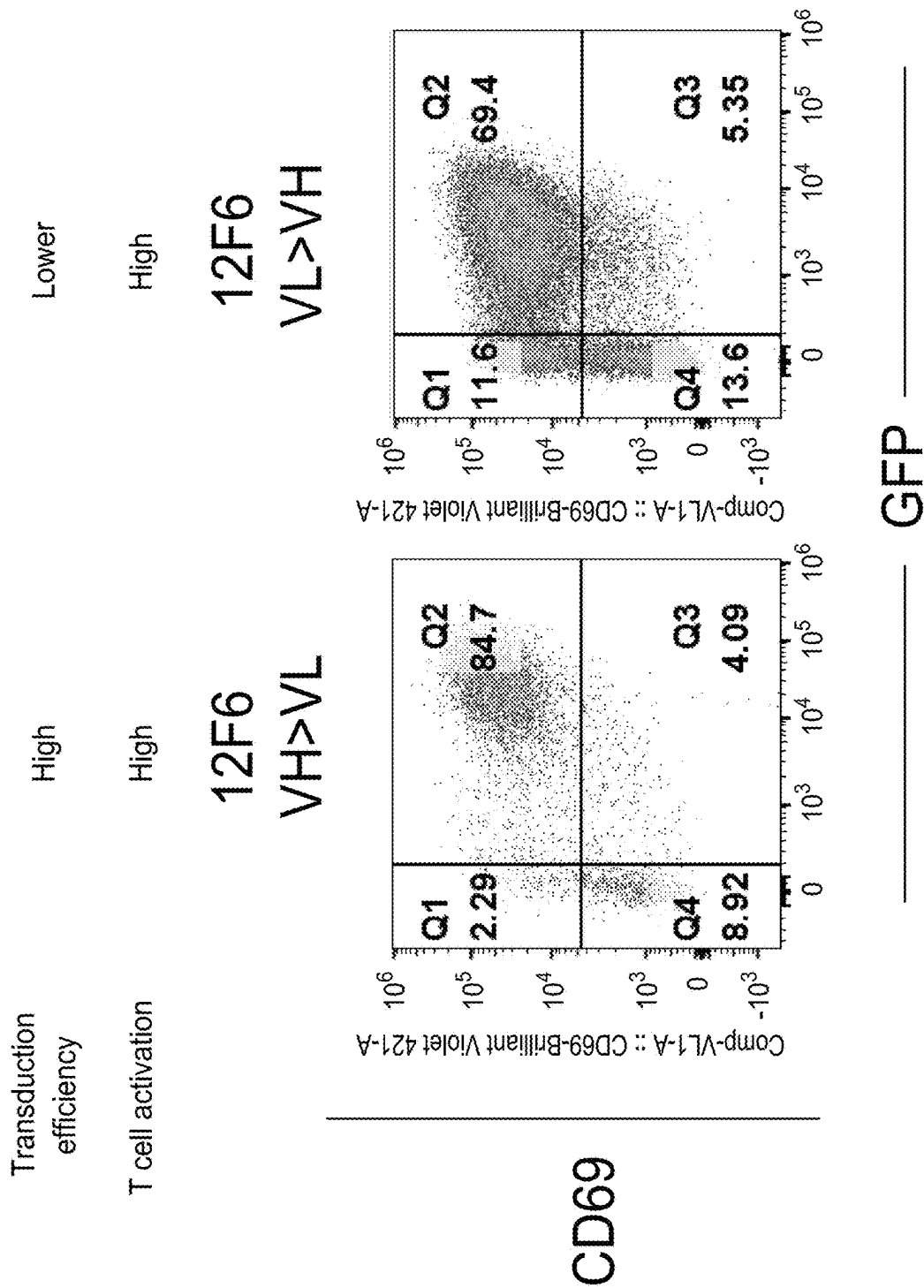


FIG. 10

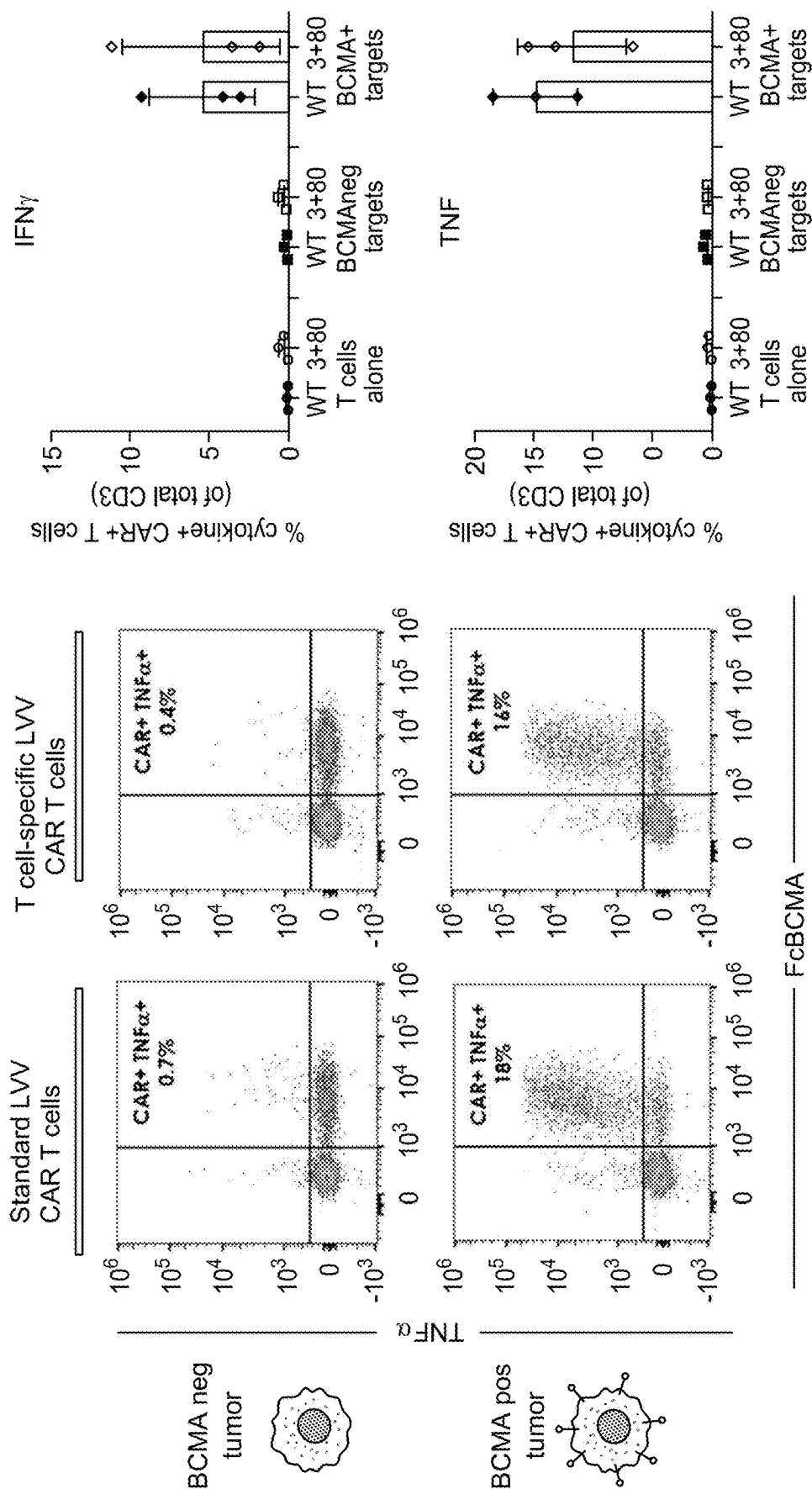


FIG. 11

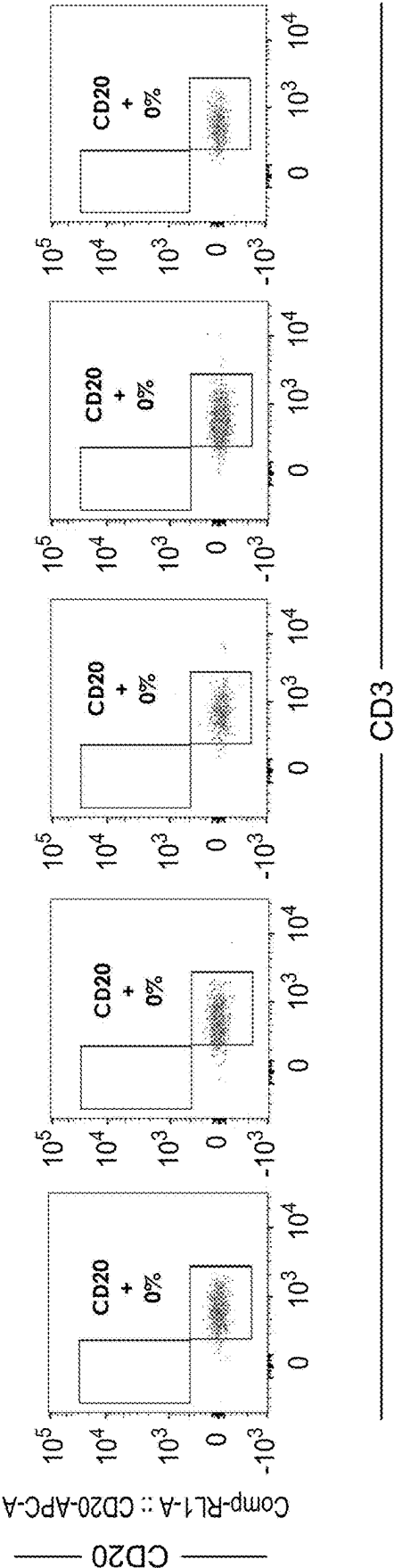


FIG. 12A

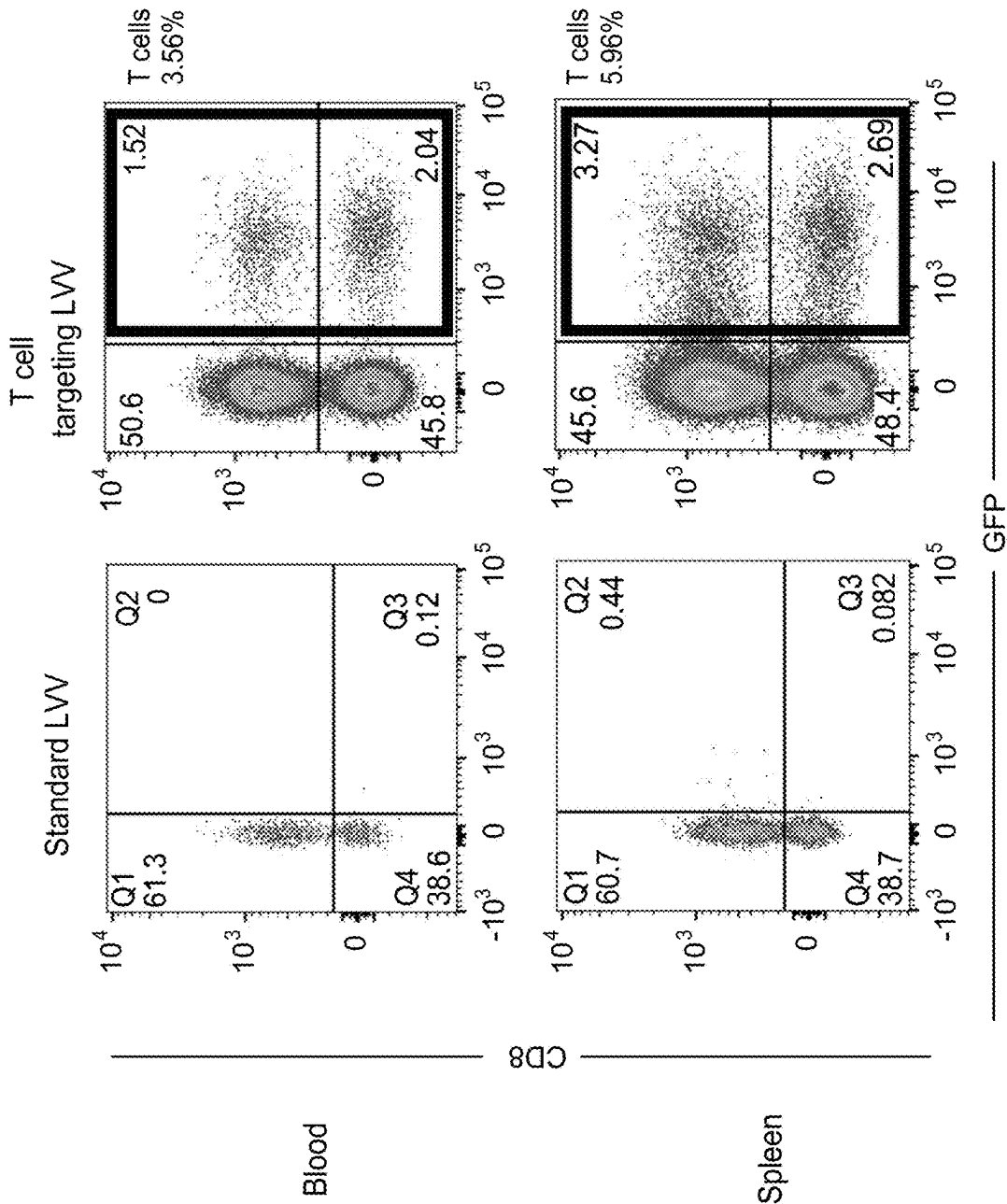
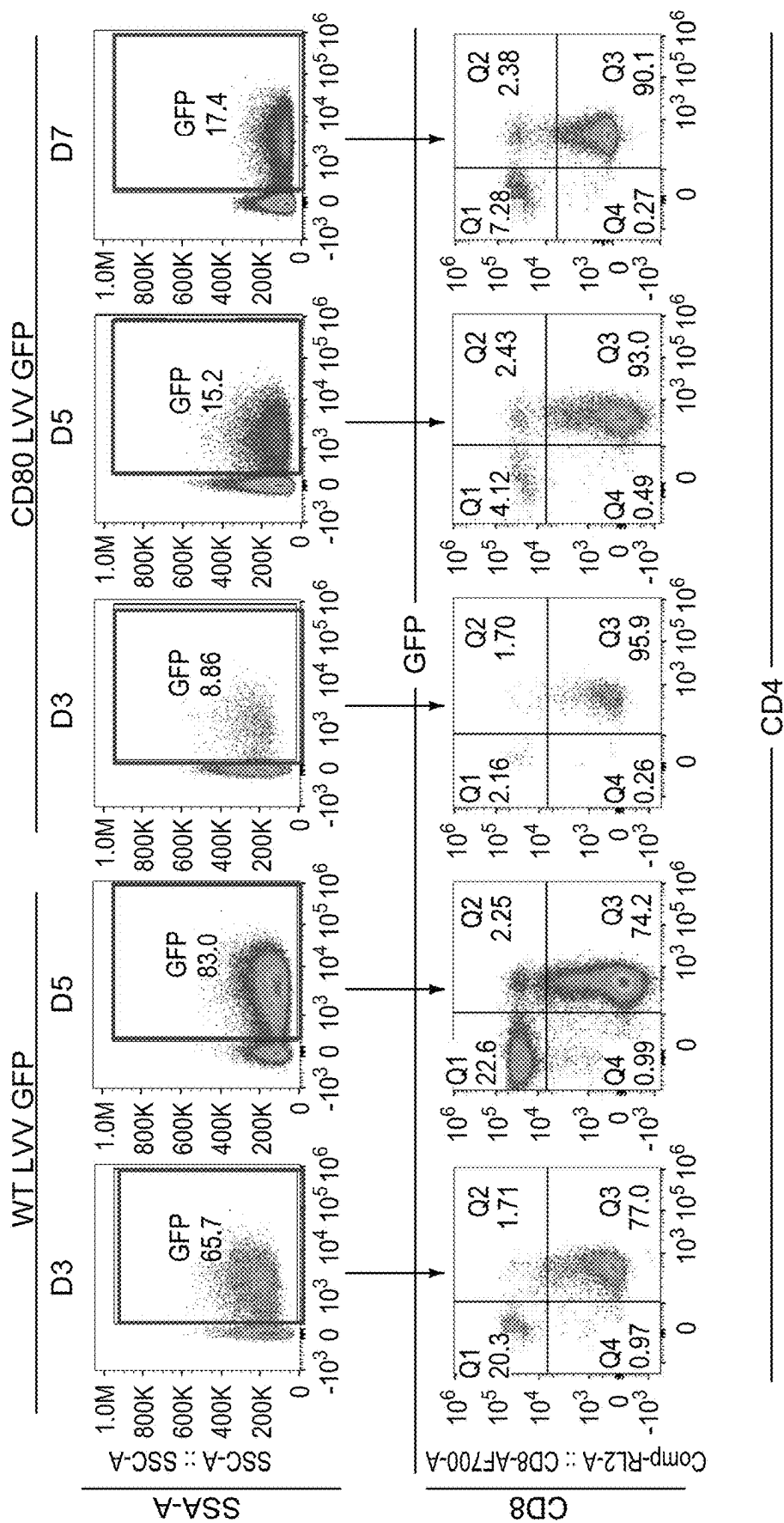


FIG. 12B



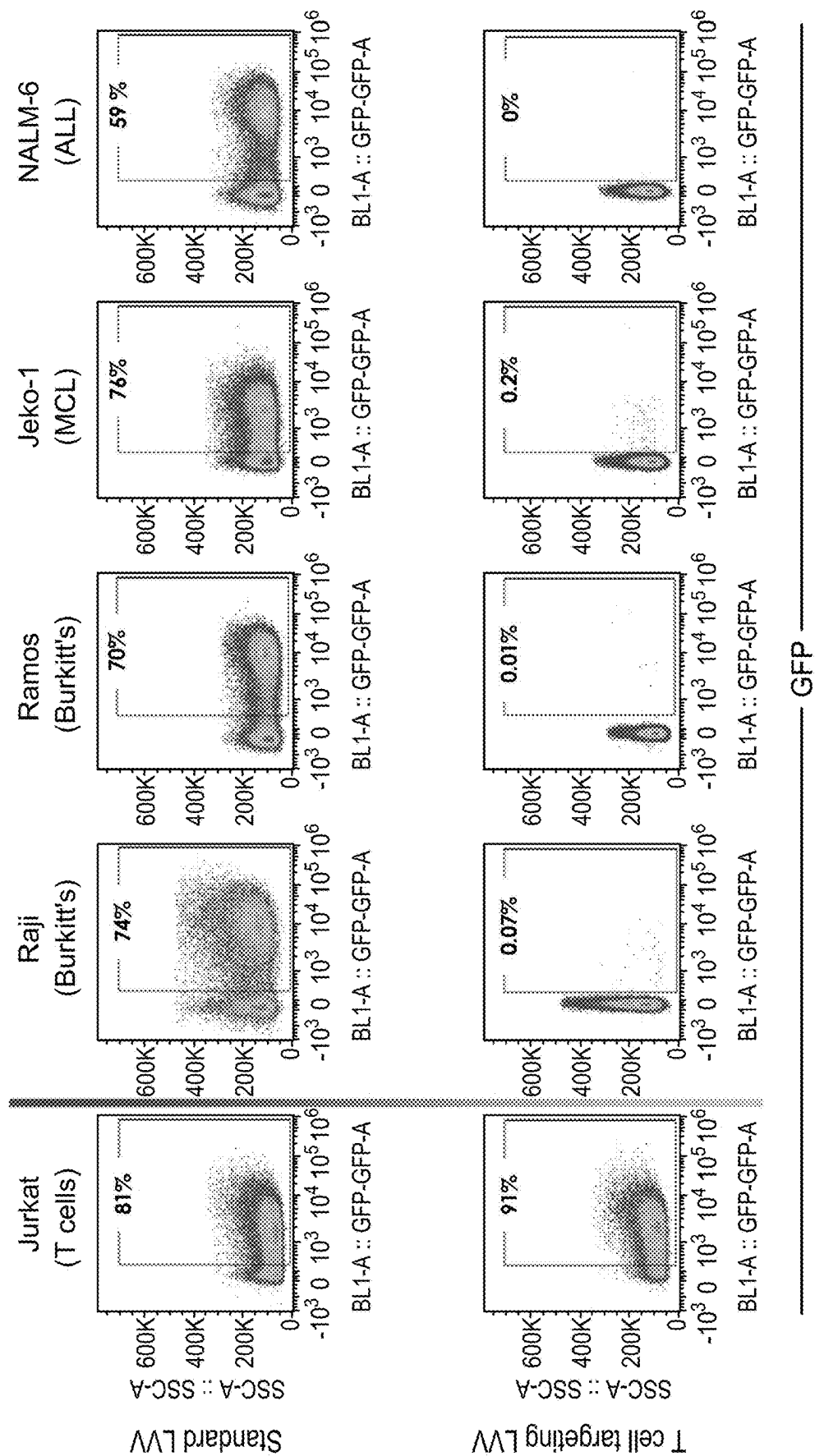


FIG. 14

LYMPHOCYTE TARGETED LENTIVIRAL VECTORS

RELATED APPLICATIONS

[0001] This application is a continuation of U.S. application Ser. No. 18/375,868, filed Oct. 2, 2023, which is a continuation of U.S. application Ser. No. 18/279,201, filed Aug. 28, 2023, which is a national stage filing under 35 U.S.C. 371 of International Application No. PCT/US2022/018027, filed Feb. 25, 2022, which claims the benefit of U.S. Provisional Application No. 63/154,639, filed Feb. 26, 2021. The entire teachings of said applications are incorporated herein by reference. International Application No. PCT/US2022/018027 was published under PCT Article 21(2) in English.

STATEMENT REGARDING SEQUENCE LISTING

[0002] The Sequence Listing associated with this application is provided in text format in lieu of a paper copy, and is hereby incorporated by reference into the specification. The name of the text file containing the Sequence Listing is KELO-401-102X_ST26.xml. The text file is 200 KB, was created on Dec. 20, 2023, and is being submitted electronically via Patent Center.

BACKGROUND

[0003] Lentiviral vectors play a critical role in gene-modified cell therapies, particularly T cell therapies. Recently approved T cell therapies rely on retroviral vectors to transduce the therapeutic molecule (e.g., chimeric antigen receptor (CAR)) into T lymphocytes. An associated risk to CAR T cell production is the transduction of other cell types with the transgene. The use of integrating vectors with broad cell tropism, e.g., lentiviral vectors pseudotyped with a VSV-G envelope protein, can represent a serious, though rare, safety concern.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0004] The patent or application file contains at least one drawing executed in color. Copies of this patent or patent application publication with color drawings will be provided by the Office upon request and payment of the necessary fee.

[0005] FIG. 1 is a schematic representation of helper plasmids suitable for use in a third generation LVV production system.

[0006] FIGS. 2A-2B depict graphs (FIG. 2A) and FACS plots (FIG. 2B) showing on-target and off-target entry of Jurkat T cells and Raji B cells by lentiviral vectors bearing mutated VSV-G envelope to abolish LDL receptor binding (Trop-002, Trop-051, Trop-052, Trop-055, and Trop-061) and T cell targeting protein CD80. In FIG. 2A, on-target entry is the left bar and off-target is the right bar of each sample. Binding of the T cell targeting protein to its cognate ligand on T cells leads to entry of the lentiviral vector, and subsequent expression of reporter green fluorescent protein (GFP) is measured.

[0007] FIGS. 3A-3B depicts graphs showing: (FIG. 3A) T cell targeting protein CD80 expressed from the VSV-G packaging plasmid is expressed at relatively equivalent levels as the mutated VSV-G on the surface of HEK293

producer cells; and (FIG. 3B) LVV generated with this approach can transduce targeted Jurkat T cells but do not transduce Raji B cells.

[0008] FIG. 4 depicts graphs showing: (top row) expression levels of CD80 and mutated VSV-G on the surface of HEK293T producer cells using LVV generated by cloning CD80 targeting protein into the Rev packaging plasmid or into the mutated VSV-G packaging plasmid; and (bottom) LVV generated by cloning CD80 targeting protein into the Rev packaging plasmid or into the mutated VSV-G packaging plasmid transduce targeted Jurkat T cells.

[0009] FIG. 5 depicts (left) CD80 targeting protein expression on surface of HEK293T producer cells using a five plasmid packaging system as a function of CD80 plasmid concentration; and (right) LVV generated with the five plasmid packaging system transduce targeted Jurkat T cells and transduction efficiency was associated with CD80 packaging plasmid concentration.

[0010] FIGS. 6A-6D depict titers of lymphocyte targeting LVV produced in adherent HEK293 or suspension HEK293 producer cells. LVV harvested from HEK293 adherent cell culture medium by centrifugation (FIG. 6A) or by anion exchange chromatography followed by tangential flow filtration (FIG. 6B). LVV harvested from HEK293 suspension cell culture medium by anion exchange chromatography (FIG. 6C). Concentration by AEX/TFF resulted in LVV preparations with a high level of purity and recovery (FIG. 6D).

[0011] FIG. 7 depicts a schematic for testing T cell transduction in PBMCs from healthy human donors with LVV comprising a BCMA CAR transgene or T cell targeting LVV (anti-CD3 and CD80) comprising a BCMA CAR transgene. Graphs shown on lower right indicate that even at low MOI, the T cell targeting LVV transduced T cells at a higher level than standard LVV and that T cell targeting LVV is capable of transducing T cells without the presence of IL-2 and exogenous activating antibodies (anti-CD3 and anti-CD28) in contrast to standard LVV.

[0012] FIG. 8 depicts graphs showing T cell expansion from PBMCs obtained from three different donors and transduced using standard LVV or T cell redirected LVV (anti-CD3 and CD80) in the presence or absence of exogenous activating anti-CD3 and anti-CD28 antibodies

[0013] FIG. 9 depicts a schematic for testing T cell transduction in PBMCs from healthy human donors with LVV comprising a CD19 CAR transgene or T cell targeting LVV (anti-CD3 and CD80) comprising a CD19 CAR transgene. Graphs shown on lower right show that the T cell targeting LVV transduced T cells at a higher level than standard LVV and that T cell targeting LVV is capable of transducing T cells without the presence of exogenous activating antibodies (anti-CD3 and anti-CD28) in contrast to standard LVV.

[0014] FIG. 10 depicts graphs showing levels of T cell transduction efficiency and T cell activation with anti-CD3 targeting proteins (12F6 in VH-VL orientation and VL-VH orientation) used to generate T cell targeting LVV.

[0015] FIG. 11 depicts graphs showing that BCMA CAR T cells exhibited increased expression of T cell effector cytokines (TNF α —left; TNF α and IFN γ —right) after culture with BCMA-positive cell lines that is not observed with BCMA-negative cell lines whether generated by standard LVV or T cell-redirection LVV (anti-CD3 and CD80).

[0016] FIGS. 12A-12B depict graphs showing in vivo delivery of transgene using T cell targeting LVV: (FIG. 12A)

T cell targeting LVV (anti-CD3 and CD80) specifically transduce human T cells (CD3+) and not human B cells (CD20+) in humanized mouse model (n=5); and (FIG. 12B) T cell targeting LVV transduce both CD8+ and CD8-(CD4) T cells compared to standard LVV which did not.

[0017] FIG. 13 depicts graphs showing that CD80 targeting LVVs enhance transduction of CD4 T cells compared to standard LVVs.

[0018] FIG. 14 depicts graphs showing that T cell targeting (anti-CD3 and CD80) LVVs transduce target Jurkat T cells but do not transduce off-target tumor cells (Raji, Ramos, Jeko-1, and NALM-6) compared to standard LVV.

DETAILED DESCRIPTION

[0019] Engineered lentiviral vectors are described herein. The lentiviral vectors include a mutated, heterologous envelope protein, a targeting protein, and at least one transgene for delivery to and expression by a cell characterized by the targeting protein. In some embodiments, the targeting protein is selected to target an immune cell, including, for example a lymphocyte or a T cell. In certain such embodiments, the lentiviral vectors described herein are capable of selectively targeting and efficiently transducing resting lymphocytes, e.g., T cells. In some embodiments, lentiviral vectors described herein are capable of transducing and/or activating T cells in the absence of an exogenous T cell stimulating agent. In some embodiments, lentiviral vectors described herein enhance transduction of CD4 T cells compared to standard lentiviral vectors.

[0020] In some embodiments, the lentiviral vectors incorporating a mutated env and a targeting protein as described herein are capable of producing a high titer LVV product, as compared to standard LVV incorporating another fusogenic env protein (e.g., cocl env, paramyxovirus env, truncated VSV-G env).

[0021] Also provided are methods and materials for producing the lentiviral vectors described herein, methods for transducing target cells, and cells transduced by lentiviral vectors according to the present disclosure. In some embodiments, a lentiviral vector as described herein and/or cells transduced by such a vector may be used in treating a disease or disorder responsive to the presence of cells expressing the transgene delivered by the vector.

Definitions

[0022] Prior to setting forth this disclosure in more detail, it may be helpful to an understanding thereof to provide definitions of certain terms to be used herein.

[0023] In the present description, any concentration range, percentage range, ratio range, or integer range is to be understood to include the value of any integer within the recited range and, when appropriate, fractions thereof (such as one tenth and one hundredth of an integer), unless otherwise indicated. Also, any number range recited herein relating to any physical feature, such as polymer subunits, size or thickness, are to be understood to include any integer within the recited range, unless otherwise indicated. As used herein, the term “about” means $\pm 20\%$ of the indicated range, value, or structure, unless otherwise indicated. It should be understood that the terms “a” and “an” as used herein refer to “one or more” of the enumerated components. The use of the alternative (e.g., “or”) should be understood to mean either one, both, or any combination thereof of the alterna-

tives. As used herein, the terms “include,” “have” and “comprise” are used synonymously, which terms and variants thereof are intended to be construed as non-limiting.

[0024] Terms understood by those in the art of antibody technology are each given the meaning acquired in the art, unless expressly defined differently herein. The term “antibody” is used in the broadest sense and includes polyclonal and monoclonal antibodies. An “antibody” may refer to an intact antibody comprising at least two heavy (H) chains and two light (L) chains inter-connected by disulfide bonds, as well as an antigen-binding portion (or antigen-binding domain) of an intact antibody that has or retains the capacity to bind a target molecule. An antibody may be naturally occurring, recombinantly produced, genetically engineered, or modified forms of immunoglobulins, for example intrabodies, peptibodies, nanobodies, single domain antibodies, SMIPs, multispecific antibodies (e.g., bispecific antibodies, diabodies, triabodies, tetrabodies, tandem di-scFv, tandem tri-scFv, ADAPTIR). A monoclonal antibody or antigen-binding portion thereof may be non-human, chimeric, humanized, or human, preferably humanized or human. Immunoglobulin structure and function are reviewed, for example, in Harlow et al., Eds., *Antibodies: A Laboratory Manual*, Chapter 14 (Cold Spring Harbor Laboratory, Cold Spring Harbor, 1988). “Antigen-binding portion” or “antigen-binding domain” of an intact antibody is meant to encompass an “antibody fragment,” which indicates a portion of an intact antibody and refers to the antigenic determining variable regions or complementary determining regions of an intact antibody. Examples of antibody fragments include, but are not limited to, Fab, Fab', F(ab')₂, and Fv fragments, Fab'-SH, F(ab')₂, diabodies, linear antibodies, scFv antibodies, VH, and multispecific antibodies formed from antibody fragments. A “Fab” (fragment antigen binding) is a portion of an antibody that binds to antigens and includes the variable region and CH1 of the heavy chain linked to the light chain via an inter-chain disulfide bond. An antibody may be of any class or subclass, including IgG and subclasses thereof (IgG₁, IgG₂, IgG₃, IgG₄), IgM, IgE, IgA, and IgD.

[0025] The term “variable region” or “variable domain” in the context of an antibody refers to the domain of an antibody heavy or light chain that is involved in binding of the antibody to antigen. The variable domains (or regions) of the heavy chain and light chain (VH and VL, respectively) of a native antibody generally have similar structures, with each domain comprising four conserved framework regions (FRs) and three complementary determining regions (CDRs). (See, e.g., Kindt et al. *Kuby Immunology*, 6th ed., W.H. Freeman and Co., page 91 (2007)). A single VH or VL domain may be sufficient to confer antigen-binding specificity. Furthermore, antibodies that bind a particular antigen may be isolated using a VH or VL domain from an antibody that binds the antigen to screen a library of complementary VL or VH domains, respectively. See, e.g., Portolano et al., *J. Immunol.* 150:880-887 (1993); Clarkson et al., *Nature* 352:624-628 (1991).

[0026] The terms “complementarity determining region” and “CDR,” which are synonymous with “hypervariable region” or “HVR,” are known in the art to refer to non-contiguous sequences of amino acids within antibody variable regions, which confer antigen specificity and/or binding affinity. In general, there are three CDRs in each heavy chain

variable region (HCDR1, HCDR2, HCDR3) and three CDRs in each light chain variable region (LCDR1, LCDR2, LCDR3).

[0027] As used herein, the terms “binding domain”, “binding region”, and “binding moiety” refer to a molecule, such as a peptide, oligopeptide, polypeptide, or protein that possesses the ability to specifically and non-covalently bind, associate, unite, recognize, or combine with a target molecule (e.g., tumor antigen). A binding domain includes any naturally occurring, synthetic, semi-synthetic, or recombinantly produced binding partner for a biological molecule or other target of interest. In some embodiments, the binding domain is an antigen-binding domain, such as an antibody or functional binding domain or antigen-binding portion thereof. Exemplary binding domains include single chain antibody variable regions (e.g., domain antibodies, scFv, scFv, Fab), receptor ectodomains (e.g., TNF- α), ligands (e.g., cytokines, chemokines), or synthetic polypeptides selected for the specific ability to bind to a biological molecule.

[0028] “Major histocompatibility complex molecule” (MHC molecule) refers to a glycoprotein that delivers a peptide antigen to a cell surface. MHC class I molecules are heterodimers composed of a membrane spanning α chain (with three α domains) and a non-covalently associated J2 microglobulin. MHC class II molecules are composed of two transmembrane glycoproteins, α and β , both of which span the membrane. Each chain has two domains. MHC class I molecules deliver peptides originating in the cytosol to the cell surface, where peptide:MHC complex is recognized by CD8⁺ T cells. MHC class II molecules deliver peptides originating in the vesicular system to the cell surface, where they are recognized by CD4⁺ T cells. An MHC molecule may be from various animal species, including human, mouse, rat, or other mammals.

[0029] “Chimeric antigen receptor” (CAR) refers to a chimeric fusion protein comprising two or more distinct domains linked together in a way that does not occur naturally in a host cell and can function as a receptor when expressed on the surface of a cell. CARs are generally composed of an extracellular domain comprising a binding domain that binds a target antigen, an optional extracellular spacer domain, a transmembrane domain, and an intracellular signaling domain (e.g., comprising an immunoreceptor tyrosine-based activation motif (ITAM)), and optionally an intracellular costimulatory domain. In certain embodiments, an intracellular signaling domain of a CAR has an ITAM (e.g., CD3 ζ) containing intracellular signaling domain and an intracellular costimulatory domain (e.g., 4-1BB). In certain embodiments, a CAR is synthesized as a single polypeptide chain or is encoded by a nucleic acid molecule as a single chain polypeptide.

[0030] A variety of assays are known for identifying binding domains of the present disclosure that specifically bind a particular target, as well as determining binding domain affinities, such as Western blot, ELISA, analytical ultracentrifugation, spectroscopy, surface plasmon resonance (BIACORE®) analysis, and MHC tetramer analysis (see also, e.g., Scatchard et al., *Ann. N.Y. Acad. Sci.* 51:660, 1949; Wilson, *Science* 295:2103, 2002; Wolff et al., *Cancer Res.* 53:2560, 1993; Altman et al., *Science* 274:94-96, 1996; and U.S. Pat. Nos. 5,283,173, 5,468,614, or the equivalent). As used herein, “specifically binds” refers to an association or union of a binding domain, or a fusion protein thereof, to

a target molecule with an affinity or K_d (i.e., an equilibrium association constant of a particular binding interaction with units of 1/M) equal to or greater than 10^5 M^{-1} , while not significantly associating or uniting with any other molecules or components in a sample.

[0031] The terms “antigen” and “Ag” refer to a molecule that is capable of inducing an immune response. The immune response that is induced may involve antibody production, the activation of specific immunologically-competent cells, or both.

[0032] Macromolecules, including proteins, glycoproteins, and glycolipids, can serve as an antigen. Antigens can be derived from recombinant or genomic DNA. As contemplated herein, an antigen need not be encoded (i) solely by a full-length nucleotide sequence of a gene or (ii) by a “gene” at all. An antigen can be generated or synthesized, or an antigen can be derived from a biological sample. Such a biological sample can include, but is not limited to, a tissue sample, a tumor sample, a cell, or a biological fluid.

[0033] The term “epitope” or “antigenic epitope” includes any molecule, structure, amino acid sequence or protein determinant within an antigen that is specifically bound by a cognate immune binding molecule, such as an antibody or fragment thereof (e.g., scFv), T cell receptor (TCR), CAR, or other binding molecule, domain or protein. Epitopic determinants generally contain chemically active surface groupings of molecules, such as amino acids or sugar side chains, and can have specific three-dimensional structural characteristics, as well as specific charge characteristics. An epitope may be a linear epitope or a conformational epitope.

[0034] As used herein, an “effector domain” is an intracellular portion of a fusion protein or chimeric receptor that can directly or indirectly promote a biological or physiological response in a cell expressing the effector domain when receiving the appropriate signal. In certain embodiments, an effector domain is part of a protein or protein complex that receives a signal when bound. In other embodiments, the effector domain is part of a protein or protein complex that binds directly to a target molecule, which triggers a signal from the effector domain. For example, in response to binding of a CAR to a target molecule, the effector domain may transduce a signal to the interior of the host cell, eliciting an effector function. An effector domain may directly promote a cellular response when it contains one or more signaling domains or motifs. In other embodiments, an effector domain will indirectly promote a cellular response by associating with one or more other proteins that directly promote a cellular response.

[0035] “Junction amino acids” or “junction amino acid residues” refer to one or more (e.g., about 2-20) amino acid residues between two adjacent motifs, regions or domains of a polypeptide. Junction amino acids may result from the construct design of a chimeric protein (e.g., amino acid residues resulting from the use of a restriction enzyme site during the construction of a nucleic acid molecule encoding a fusion protein).

[0036] A “disease” is a state of health of a subject wherein the subject cannot maintain homeostasis, and wherein, if the disease is not ameliorated, then the subject’s health continues to deteriorate. In contrast, a “disorder” or “undesirable condition” in a subject is a state of health in which the subject is able to maintain homeostasis, but in which the subject’s state of health is less favorable than it would be in the absence of the disorder or undesirable condition. Left

untreated, a disorder or undesirable condition does not necessarily result in a further decrease in the subject's state of health.

[0037] “Nucleic acid molecule” and “polynucleotide” can be in the form of RNA or DNA, which includes cDNA, genomic DNA, and synthetic DNA. A nucleic acid molecule may be composed of naturally occurring nucleotides (such as deoxyribonucleotides and ribonucleotides), analogs of naturally occurring nucleotides (e.g., α -enantiomeric forms of naturally occurring nucleotides), or a combination of both. Modified nucleotides can have modifications in or replacement of sugar moieties, or pyrimidine or purine base moieties. Nucleic acid monomers can be linked by phosphodiester bonds or analogs of such linkages. Analogs of phosphodiester linkages include phosphorothioate, phosphorodithioate, phosphoroselenoate, phosphorodiselenoate, phosphorothioanilothioate, phosphoranilidate, phosphoramidate, and the like. A nucleic acid molecule may be double stranded or single stranded, and if single stranded, may be the coding strand or non-coding (anti-sense strand). A coding molecule may have a coding sequence identical to a coding sequence known in the art or may have a different coding sequence, which, as the result of the redundancy or degeneracy of the genetic code, or by splicing, can encode the same polypeptide.

[0038] “Encoding” refers to the inherent property of specific polynucleotide sequences, such as DNA, cDNA, and mRNA sequences, to serve as templates for synthesis of other polymers and macromolecules in biological processes having either a defined sequence of nucleotides (i.e., rRNA, tRNA and mRNA) or a defined sequence of amino acids and the biological properties resulting therefrom. Thus, a polynucleotide encodes a protein if transcription and translation of mRNA corresponding to that polynucleotide produces the protein in a cell or other biological system. Both a coding strand and a non-coding strand can be referred to as encoding a protein or other product of the polynucleotide. Unless otherwise specified, a “nucleotide sequence encoding an amino acid sequence” includes all nucleotide sequences that are degenerate versions of each other and that encode the same amino acid sequence.

[0039] As used herein, the term “endogenous” or “native” refers to a gene, protein, compound, molecule or activity that is normally present in a host or host cell, including naturally occurring variants of the gene, protein, compound, molecule, or activity.

[0040] As used herein, “homologous” or “homolog” refers to a molecule or activity from a host cell that is related by ancestry to a second gene or activity, e.g., from the same host cell, from a different host cell, from a different organism, from a different strain, from a different species. For example, a heterologous molecule or heterologous gene encoding the molecule may be homologous to a native host cell molecule or gene that encodes the molecule, respectively, and may optionally have an altered structure, sequence, expression level or any combination thereof.

[0041] As used herein, “heterologous” nucleic acid molecule, construct or sequence refers to a nucleic acid molecule or portion of a nucleic acid molecule that is not native to a host cell, but can be homologous to a nucleic acid molecule or portion of a nucleic acid molecule from the host cell. The source of the heterologous nucleic acid molecule, construct or sequence can be from a different genus or species. In some embodiments, the heterologous nucleic

acid molecules are not naturally occurring. In certain embodiments, a heterologous nucleic acid molecule is added (i.e., not endogenous or native) into a host cell or host genome by, for example, conjugation, transformation, transfection, transduction, electroporation, or the like, wherein the added molecule can integrate into the host cell genome or exist as extra-chromosomal genetic material (e.g., as a plasmid or other form of self-replicating vector), and can be present in multiple copies. In addition, “heterologous” refers to a non-native enzyme, protein or other activity encoded by a non-endogenous nucleic acid molecule introduced into the host cell, even if the host cell encodes a homologous protein or activity.

[0042] As used herein, the term “engineered,” “recombinant,” “mutant,” “modified” or “non-natural” refers to an organism, microorganism, cell, nucleic acid molecule, or vector that has been modified by introduction of a heterologous nucleic acid molecule, or refers to a cell or microorganism that has been genetically engineered by human intervention—that is, modified by introduction of a heterologous nucleic acid molecule, or refers to a cell or microorganism that has been altered such that expression of an endogenous nucleic acid molecule or gene is controlled, deregulated or constitutive, where such alterations or modifications can be introduced by genetic engineering. Human-generated genetic alterations can include, for example, modifications introducing nucleic acid molecules (which may include an expression control element, such as a promoter) encoding one or more proteins, chimeric receptors, or enzymes, or other nucleic acid molecule additions, deletions, substitutions, or other functional disruption of or addition to a cell's genetic material. Exemplary modifications include those in coding regions or functional fragments thereof heterologous or homologous polypeptides from a reference or parent molecule. Additional exemplary modifications include, for example, modifications in non-coding regulatory regions in which the modifications alter expression of a gene or operon.

[0043] As used here, the term “transgene” refers to a gene or polynucleotide encoding a protein of interest (e.g., a CAR) whose expression is desired in a host cell and that has been transferred by genetic engineering techniques into a cell. A transgene may encode proteins of therapeutic interest as well as proteins that are reporters, tags, markers, suicide proteins, etc. A transgene may be from a natural source, modification of a natural gene, or a recombinant or synthetic molecule. In certain embodiments, a transgene is a component of a vector.

[0044] The term “overexpressed” or “overexpression” of an antigen refers to an abnormally high level of antigen expression in a cell. Overexpressed antigen or overexpression of antigen is often associated with a disease state, such as in hematological malignancies and cells forming a solid tumor within a specific tissue or organ of a subject. Solid tumors or hematological malignancies characterized by overexpression of a tumor antigen can be determined by standard assays known in the art.

[0045] As used herein, the terms “peptide,” “polypeptide,” and “protein” are used interchangeably, and refer to a compound comprised of amino acid residues covalently linked by peptide bonds. A protein or peptide must contain at least two amino acids, and no limitation is placed on the maximum number of amino acids that can comprise a protein's or peptide's sequence. Polypeptides include any

peptide or protein comprising two or more amino acids joined to each other by peptide bonds. As used herein, the term refers to both short chains, which also commonly are referred to in the art as peptides, oligopeptides and oligomers, for example, and to longer chains, which generally are referred to in the art as proteins, of which there are many types. "Polypeptides" include, for example, biologically active fragments, substantially homologous polypeptides, oligopeptides, homodimers, heterodimers, variants of polypeptides, modified polypeptides, derivatives, analogs, fusion proteins, among others. The polypeptides include natural peptides, recombinant peptides, synthetic peptides, or a combination thereof.

[0046] As used herein, the term "mature polypeptide" or "mature protein" refers to a protein or polypeptide that is secreted or localized in the cell membrane or inside certain cell organelles (e.g., the endoplasmic reticulum, golgi, or endosome) and includes a partially cleaved N-terminal signal sequence (e.g., one or more amino acids of the signal sequence remaining but less than the whole signal sequence) or does not include an N-terminal signal sequence (i.e., the N-terminal signal sequence has been entirely removed, such as by an endogenous cleavage process, from the protein or polypeptide).

[0047] A "signal sequence", also referred to as "signal peptide" "leader sequence", "leader peptide", "localization signal" or "localization sequence", is a short peptide (usually 13-36 amino acids in length) present at the N-terminus of newly synthesized proteins that are destined for the plasma membrane or a secretory pathway. A signal sequence typically comprises a short stretch of hydrophilic, positively charged amino acids at the N-terminus, a central hydrophobic domain of 5-15 residues, and a C-terminal region with a cleavage site for a signal sequence. In eukaryotes, a signal sequence prompts translocation of the newly synthesized protein to the endoplasmic reticulum where it is cleaved by the signal peptidase, creating a mature protein that then proceeds to its appropriate destination. The diversity of signal sequence length and amino acid composition makes it difficult to precisely predict the cleavage site. For polypeptide sequences disclosures herein, where a signal sequence is noted, the polypeptide sequence absent the signal sequence or having a partial signal sequence is also contemplated.

[0048] The "percent identity" between two or more nucleic acid or amino acid sequences is a function of the number of identical positions shared by the sequences (i.e., % identity=number of identical positions/total number of positions×100), taking into account the number of gaps, and the length of each gap that needs to be introduced to optimize alignment of two or more sequences. The comparison of sequences and determination of percent identity between two or more sequences can be accomplished using a mathematical algorithm, such as BLAST and Gapped BLAST programs at their default parameters (e.g., Altschul et al., *J. Mol. Biol.* 215:403, 1990; see also BLASTN at www.ncbi.nlm.nih.gov/BLAST).

[0049] A "conservative substitution" is recognized in the art as a substitution of one amino acid for another amino acid that has similar properties. Exemplary conservative substitutions are well known in the art (see, e.g., WO 97/09433, page 10, published Mar. 13, 1997; Lehninger, *Biochemistry*, Second Edition; Worth Publishers, Inc. NY:NY (1975), pp.

71-77; Lewin, *Genes IV*, Oxford University Press, NY and Cell Press, Cambridge, MA (1990), p. 8).

[0050] The term "chimeric" refers to any nucleic acid molecule or protein that is not endogenous and comprises a combination of sequences joined or linked together that are not naturally found joined or linked together in nature. For example, a chimeric nucleic acid molecule may comprise nucleic acids encoding various domains from multiple different genes. In another example, a chimeric nucleic acid molecule may comprise regulatory sequences and coding sequences that are derived from different sources, or regulatory sequences and coding sequences that are derived from the same source but arranged in a manner different than that found in nature.

[0051] The term "promoter" as used herein is defined as a DNA sequence recognized by the synthetic machinery of the cell, or introduced synthetic machinery, required to initiate the specific transcription of a polynucleotide sequence.

[0052] As used herein, the term "promoter/regulatory sequence" means a nucleic acid sequence which is required for expression of a gene product operably linked to the promoter/regulatory sequence. In some instances, this sequence may be the core promoter sequence and in other instances, this sequence may include an enhancer sequence and other regulatory elements that are required for expression of the gene product. The promoter/regulatory sequence may, for example, be one that expresses the gene product in a tissue specific manner.

[0053] A "constitutive" promoter is a nucleotide sequence which, when operably linked with a polynucleotide which encodes or specifies a gene product, causes the gene product to be produced in a cell under most or all physiological conditions of the cell.

[0054] An "inducible" promoter is a nucleotide sequence which, when operably linked with a polynucleotide which encodes or specifies a gene product, causes the gene product to be produced in a cell substantially only when an inducer which corresponds to the promoter is present in the cell.

[0055] A "tissue-specific" promoter is a nucleotide sequence which, when operably linked with a polynucleotide encodes or specified by a gene, causes the gene product to be produced in a cell substantially only if the cell is a cell of the tissue type corresponding to the promoter.

[0056] The phrase "under transcriptional control" or "operatively linked" as used herein means that a promoter is in the correct location and orientation in relation to a polynucleotide to control the initiation of transcription by RNA polymerase and expression of the polynucleotide.

[0057] A "vector" is a nucleic acid molecule that is capable of transporting another nucleic acid. Vectors may be, for example, plasmids, cosmids, viruses, or phage. The term should also be construed to include non-plasmid and non-viral compounds that facilitate transfer of nucleic acid into cells. An "expression vector" is a vector that is capable of directing the expression of a protein encoded by one or more transgenes carried by the vector when it is present in the appropriate environment. The one or more transgenes to be expressed by a vector as described herein are encoded in an expression cassette.

[0058] A "lentiviral vector" is a vector derived from a lentivirus and includes one or more lentiviral packaging proteins and/or one or more lentiviral proteins necessary for expression of the one or more genes carried by the vector.

The abbreviation “LVV” is used herein to refer to a lentiviral vector (singular), as well as multiple lentiviral vectors (plural).

[0059] “Lentivirus” refers to a genus of retroviruses that are capable of infecting dividing and non-dividing cells. Examples of lentiviruses include, but are not limited to HIV (human immunodeficiency virus, including HIV type 1 and HIV type 2, equine infectious anemia virus, feline immunodeficiency virus (FIV), bovine immune deficiency virus (BIV), and simian immunodeficiency virus (SIV).

[0060] “Retrovirus” refers to are RNA viruses with a single strand positive-sense RNA molecule. Retroviruses comprise a reverse transcriptase enzyme and an integrase enzyme. Upon entry into a target cell, retroviruses utilize their reverse transcriptase to transcribe their RNA molecule into a DNA molecule. Subsequently, the integrase enzyme is used to integrate the DNA molecule into the host cell genome. Upon integration into the host cell genome, the sequence from the retrovirus is referred to as a provirus (e.g., proviral sequence or provims sequence).

[0061] As used herein, the term “expression cassette” refers to a distinct component of a vector nucleic acid comprising at least one transgene and regulatory sequences controlling its expression (e.g., promoter, 3'UTR) in a host cell. A tandem expression cassette refers to a component of a vector nucleic acid comprising at least two transgenes under the control of the same set of regulatory sequences for tandem expression of the at least two transgenes. In certain embodiments, the tandem expression cassette comprises at least two transgenes under the control of the same promoter. In certain embodiments, the first transgene and second transgene are separated by an internal ribosome entry site (IRES), furin cleavage site, or self-cleaving viral 2A peptide to allow for co-expression of two proteins from a single mRNA.

[0062] The term “immune system cell” or “immune cell” means any cell of the immune system that originates from a hematopoietic stem cell in the bone marrow. Hematopoietic stem cells give rise to two major lineages: myeloid progenitor cells (which give rise to myeloid cells such as monocytes, macrophages, dendritic cells, megakaryocytes and granulocytes) and lymphoid progenitor cells (which give rise to lymphoid cells such as T cells, B cells and natural killer (NK) cells). Exemplary immune system cells include a CD4⁺ T cell, a CD8⁺ T cell, a CD4⁺ CD8⁺ double negative T cell, a $\gamma\delta$ T cell, a regulatory T cell, a natural killer cell, and a dendritic cell. Macrophages and dendritic cells may also be referred to as “antigen presenting cells” or “APCs,” which are specialized cells that can activate T cells when a major histocompatibility complex (MHC) receptor on the surface of the APC complexed with a peptide interacts with a TCR on the surface of a T cell.

[0063] The term “lymphocyte” refers to immune cells of lymphoid origin, which are cells that show at least one phenotype characteristic of a lymphocyte or a precursor or progenitor thereof that distinguishes the cells from cells of erythroid or myeloid lineages. The term “lymphocytes” encompasses T cells, B cells, and natural killer (NK) cells.

[0064] The term “T cells” refers to cells of T cell lineage. “Cells of T cell lineage” refer to cells that show at least one phenotypic characteristic of a T cell or a precursor or progenitor thereof that distinguishes the cells from other lymphoid cells, and cells of the erythroid or myeloid lineages. Such phenotypic characteristics can include expres-

sion of one or more proteins specific for T cells (e.g., CD3⁺, CD4⁺, CD8⁺), or a physiological, morphological, functional, or immunological feature specific for a T cell. For example, cells of the T cell lineage may be progenitor or precursor cells committed to the T cell lineage; CD25⁺ immature and inactivated T cells; cells that have undergone CD4 or CD8 lineage commitment; thymocyte progenitor cells that are CD4⁺CD8⁺ double positive; single positive CD4⁺ or CD8⁺; TCR $\alpha\beta$ or TCR $\gamma\delta$; or mature and functional or activated T cells. The term “T cells” encompasses naïve T cells (CD45RA⁺, CCR7⁺, CD62L⁺, CD27⁺, CD45RO⁻), central memory T cells (CD45RA⁻, CD45RO⁺, CD62L⁺, CCR7⁺, CD27⁺), effector memory T cells (CD45RA⁻, CD45RO⁺, CCR7⁻, CD62L⁻, CD27⁻), mucosal-associated invariant T (MAIT) cells, $\gamma\delta$ T cells, Tregs, natural killer T cells, and tissue resident T cells.

[0065] The term “natural killer cells” or “NK cells” refers to large granular lymphocytes (LGL) and constitute the third kind of cells differentiated from the common lymphoid progenitor-generating B and T lymphocytes. NK cells are known to differentiate and mature in the bone marrow, lymph nodes, spleen, tonsils, and thymus, where they then enter into the circulation. NK cells differ from natural killer T cells (NKTs) phenotypically, by origin and by respective effector functions; often, NKT cell activity promotes NK cell activity by secreting IFN γ . In contrast to NKT cells, NK cells do not express T-cell antigen receptors (TCR) or pan T marker CD3 or surface immunoglobulins (Ig) B cell receptors, but they usually express the surface markers CD16 (Fc γ RIII) and CD56 in humans, NK1.1 or NK1.2 in C57BL/6 mice. Up to 80% of human NK cells also express CD8.

[0066] The term “B cells” refers to cells of the B cell lineage. “Cells of B cell lineage” refers to cells that show at least one phenotypic characteristic of a B cell or a precursor or progenitor thereof that distinguishes the cells from other lymphoid cells, and cells of the erythroid or myeloid lineages. Such phenotypic characteristics can include expression of one or more proteins specific for B cells (e.g., cells positive for one or more of CD19⁺, CD72⁺, CD24⁺, CD20⁺, CD21⁺, CD22⁺, CD38⁺, CD40⁺, CD72⁺, CD32b⁺, CD268⁺, CD269⁺, CD267⁺, CD86⁺, CD80⁺, CD40⁺, CD52⁺, CD138⁺, CD27⁺, CD28⁺, CD21⁺, CD23⁺, CD84⁺, CD257⁺, CD270⁺, CD37⁺, and CD74⁺), or a physiological, morphological, functional, or immunological feature specific for a B cell. For example, cells of the B cell lineage may be progenitor or precursor cells committed to the B cell lineage (e.g., pre-pro-B cells, pro-B cells, and pre-B cells); immature and inactivated B cells or mature and functional or activated B cells. Thus, “B cells” encompass naïve B cells, plasma cells, regulatory B cells, marginal zone B cells, follicular B cells, lymphoplasmacytoid cells, plasmablast cells, and memory B cells (e.g., CD27⁺, IgD⁻).

[0067] The term “cytotoxic activity,” also referred to as “cytolytic activity,” with respect to a cell (e.g., T cell) that expresses an immune receptor (e.g., a CAR) on its surface, means that upon antigen-specific signaling (e.g., via the CAR) the cell induces a target cell to undergo apoptosis. In some embodiments, a cytotoxic cell may induce apoptosis in a target cell via the release of cytotoxins, such as perforin, granzyme, and granulysin, from granules. Perforins insert into the target cell membrane and form pores that allow water and salts to rapidly enter the target cell. Granzymes are

serine proteases that induce apoptosis in the target cell. Granulysin is also capable of forming pores in the target cell membrane and is a proinflammatory molecule. In some embodiments, a cytotoxic cell may induce apoptosis in a target cell via interaction of Fas ligand, which is upregulated on T cell following antigen-specific signaling, with Fas molecules expressed on the target cell. Fas is an apoptosis-signaling receptor molecule on the surface of a number of different cells.

[0068] A “disease” is a state of health of a subject wherein the subject cannot maintain homeostasis, and wherein, if the disease is not ameliorated, then the subject’s health continues to deteriorate. In contrast, a “disorder” or “undesirable condition” in a subject is a state of health in which the subject is able to maintain homeostasis, but in which the subject’s state of health is less favorable than it would be in the absence of the disorder or undesirable condition. Left untreated, a disorder or undesirable condition does not necessarily result in a further decrease in the subject’s state of health.

[0069] The term “cancer” as used herein is defined as disease characterized by the rapid and uncontrolled growth of aberrant cells. The aberrant cells may form solid tumors or constitute a hematological malignancy. Cancer cells can spread locally or through the bloodstream and lymphatic system to other parts of the body. Examples of various cancers include, but are not limited to, breast cancer, prostate cancer, ovarian cancer, cervical cancer, skin cancer, pancreatic cancer, colorectal cancer, renal cancer, liver cancer, brain cancer, lymphoma, leukemia, lung cancer and the like.

[0070] The term “subject,” “patient” and “individual” are used interchangeably herein and are intended to include living organisms in which an immune response can be elicited (e.g., mammals). Examples of subjects include humans, primates, cows, horses, goats, sheep, dogs, cats, mice, rats, rabbits, guinea pigs, pigs, and transgenic species thereof.

[0071] “Adoptive cellular immunotherapy” or “adoptive immunotherapy” refers to the administration of naturally occurring or genetically engineered, disease antigen-specific immune cells (e.g., T cells). Adoptive cellular immunotherapy may be autologous (immune cells are from the recipient), allogeneic (immune cells are from a donor of the same species) or syngeneic (immune cells are from a donor genetically identical to the recipient).

[0072] “Autologous” refers to a graft (e.g. organ, tissue, cells) derived from the same subject to which it is later to be re-introduced.

[0073] “Allogeneic” refers to a graft derived from a different subject of the same species.

[0074] A “therapeutically effective amount” or “effective amount” of a lentiviral vector or a cell transduced by a lentiviral vector as described herein (e.g., a T cell expressing a CAR as encoded by the transgene of the lentiviral vector) refers to that amount of lentiviral particles or cells sufficient to result in amelioration of one or more symptoms of the disease, disorder, or undesired condition being treated.

[0075] “Treat” or “treatment” or “ameliorate” refers to medical management of a disease, disorder, or undesired condition of a subject. In general, an appropriate dose or treatment regimen comprising a lentiviral vector or a cell expressing a CAR of this disclosure is administered in an amount sufficient to elicit a therapeutic or prophylactic benefit. Therapeutic or prophylactic/preventive benefit

includes improved clinical outcome; lessening or alleviation of symptoms associated with a disease, disorder, or undesired condition; decreased occurrence of symptoms; improved quality of life; longer disease-free status; diminishment of extent of disease, disorder, or undesired condition; stabilization of disease state; delay of disease progression; remission; survival; prolonged survival; or any combination thereof.

[0076] The term “anti-tumor effect” refers to a biological effect which can be manifested by a decrease in tumor volume, a decrease in the number of tumor cells, a decrease in the number of metastases, an increase in life expectancy, or amelioration of various physiological symptoms associated with a cancerous condition. An “anti-tumor effect” can also be manifested by prevention of a hematological malignancy or tumor formation.

[0077] Additional definitions are provided throughout the present disclosure.

Lentiviral Vector

[0078] Lentivirus is a genus of retroviruses that typically gives rise to slowly developing diseases due to their ability to incorporate into a host genome. Modified lentiviral genomes are useful as viral vectors for the delivery of a nucleic acids to a host cell.

[0079] The present disclosure provides self-inactivating lentiviral vectors (“LVV”) that include a viral envelope comprising a mutated, heterologous envelope protein and a tropism defining molecule (also referred to as a “targeting protein”). The LVV described herein further includes a transgene, with the LVV being capable of specifically binding to a target immune cell and transducing the target immune cell such that the transgene is expressed by the immune cell. In some embodiments, the LVV may carry more than one transgene. In further specific embodiments, the transgene encodes a chimeric antigen receptor (CAR).

Mutated Viral Envelope Protein

[0080] Lentiviral vectors as described herein are pseudotyped with a mutated heterologous viral envelope protein that, in the absence of mutation, mediates both cellular attachment and membrane fusion. In particular embodiments, the mutated envelope protein includes at least one mutation that inhibits the envelope protein’s ability to bind its native target, while preserving the envelope protein’s fusogenic properties. In specific embodiments, the heterologous envelope protein is a vesicular stomatitis virus G protein (“VSV-G”, “VSV-G envelope protein” or “VSV-G protein”) that includes one or more mutations that inhibit binding of VSV-G to the low-density lipoprotein receptor (“LDL-R”), while preserving the VSV-G protein’s fusogenic function.

[0081] When referring to the envelope protein’s ability to bind its native target, the terms “inhibit” and “inhibits” encompass both complete elimination of binding by the envelope protein to its native target, as well as a significant reduction in binding by the envelope protein to its native target. In particular embodiments, “significant reduction” refers to a reduction selected from a reduction of at least 95%, at least 90%, at least 85%, at least 80%, at least 75%, at least 70%, at least 65%, at least 60%, at least 55%, at least

50%, at least 45%, at least 40%, at least 35%, at least 30%, at least 25%, at least 20%, at least 15%, and at least 10% in binding to the native target.

[0082] The nucleic acid and amino acid sequences set forth in Table 10 provide references sequences for a wild type VSV-G envelope protein and examples of mutated VSV-G envelope proteins according to the present disclosure. SEQ ID NO: 77 is a reference nucleic acid sequence encoding the VSV-G envelope protein, and SEQ ID NO: 78 is an amino acid sequence of a reference VSV-G envelope protein. SEQ ID NO: 87 is a nucleic acid sequence encoding a cleavable signal peptide of the reference VSV-G envelope protein. SEQ ID NO: 88 is an amino acid sequence of the cleavable signal peptide of the reference VSV-G envelope protein. SEQ ID NO: 89 is a reference nucleic acid sequence encoding the VSV-G envelope protein, absent the signal sequence. SEQ ID NO: 90 is an amino acid sequence of a reference VSV-G envelope protein absent the sequence. The position and nature of VSV-G mutations disclosed herein are described in reference to the nucleic acid sequence and the amino acid sequence provided by SEQ ID NO: 77 absent the N-terminal signal sequence of SEQ ID NO: 87 and SEQ ID NO: 78 absent the N-terminal signal sequence of SEQ ID NO: 88, respectively. Thus, reference to a position 1 of the VSV-G env amino acid sequence for the purpose of identifying mutation position refers to position 17 of the wildtype VSV-G amino acid sequence of SEQ ID NO: 78, which is a lysine (K), or position 1 of the wildtype VSV-G amino acid sequence of SEQ ID NO: 90.

[0083] In some embodiments, the mutated VSV-G envelope protein includes a mutation at amino acid position H8, N9, Q10, K47, K50, A51, S183, S179, N180, I182, M184, Y209, I347, T350, T352, E353, R354, an insertion of TT between N9 and Q10, an insertion of GGS between H8 and N9, an insertion of GGS between N9 and Q10, an insertion of TT between N208 and Y209, an insertion of GGS between P46 and K47, an insertion of GGS between N208 and Y209, and/or a deletion of residues 1-8. In certain embodiments, the VSV-G envelope protein includes two or more mutations at amino acid positions selected from H8, N9, Q10, K47, K50, A51, S183, S179, N180, I182, M184, Y209, I347, T350, T352, E353, R354, an insertion of TT between N9 and Q10, an insertion of GGS between H8 and N9, an insertion of GGS between N9 and Q10, an insertion of TT between N208 and Y209, an insertion of GGS between P46 and K47, an insertion of GGS between N208 and Y209, and a deletion of residues 1-8. In other embodiments, the VSV-G envelope protein includes three or more mutations at amino acid positions selected from H8, N9, Q10, K47, K50, A51, S183, S179, N180, I182, M184, Y209, I347, T350, T352, E353, R354, an insertion of TT between N9 and Q10, an insertion of GGS between H8 and N9, an insertion of GGS between N9 and Q10, an insertion of TT between N208 and Y209, an insertion of GGS between P46 and K47, an insertion of GGS between N208 and Y209, and a deletion of residues 1-8. In other embodiments, the mutated VSV-G envelope protein comprises a H8A, K47A, K47Q, Y209A, R354A, and/or R354Q mutation. In still other embodiments, the VSV-G envelope protein includes one or more mutations selected from N9, Q10, K50, A51, S183, S179, N180, I182, M184, I347, T350, T352, and E353, an insertion of TT between N9 and Q10, an insertion of GGS between H8 and N9, an insertion of GGS between N9 and Q10, an insertion of TT between N208 and Y209, an insertion of GGS between P46 and K47, an insertion of GGS between N208 and Y209, and a deletion of residues 1-8. In yet other embodiments, the VSV-G envelope protein includes two or more mutations selected from a mutation at one or more of N9, Q10, K50, A51, S183, S179, N180, I182, M184, I347, T350, T352, and E353, an insertion of TT between N9 and Q10, an insertion of GGS between H8 and N9, an insertion of GGS between N9 and Q10, an insertion of TT between N208 and Y209, an insertion of GGS between P46 and K47, an insertion of GGS between N208 and Y209, and a deletion of residues 1-8. In some embodiments, a mutated VSV-G envelope protein is selected from one or more of an H8A, N9A, Q10A, K47A, K47Q, N180A, I182A, Y209A, T352A, T352W, E353A, R354A, and R354Q mutation. In some embodiments, a mutated VSV-G envelope protein is selected from one or more of an H8A, K47A, K47Q, Y209A, R354A, and R354Q mutation. In some embodiments, the mutated VSV-G envelope protein comprises a K47 mutation and a R354 mutation. In some embodiments, the mutated VSV-G envelope protein comprises a K47Q mutation and a R354A mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N180 mutation, a I182 mutation, a T352 mutation, and a E353 mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N180A mutation, a I182A mutation, a T352A mutation, and a E353A mutation. In some embodiments, the mutated VSV-G envelope protein comprises a T352 mutation and E353 mutation. In further embodiments, the mutated VSV-G envelope protein comprises a T352W mutation and E353A mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N9 mutation, a Q10 mutation, and a N180 mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N9A mutation, a Q10A mutation, and a N180A mutation. In some embodiments, the mutated VSV-G envelope protein comprises an insertion of GGS between H8 and N9 and an insertion of GGS between N9 and Q10. In some embodiments, the mutated VSV-G envelope protein comprises an insertion of TT between N9 and Q10. In some embodiments, the mutated VSV-G envelope protein comprises an insertion of GGS between P46 and K47. In other embodiments, the mutated VSV-G envelope protein is as described in Nikolic et al., "Structural basis for the recognition of LDL-receptor family members by VSV glycoprotein." *Nature Comm.*, 2018, 9:1029, the relevant disclosures of which are incorporated by reference herein.

insertion of GGS between P46 and K47, an insertion of GGS between N208 and Y209, and a deletion of residues 1-8. In yet other embodiments, the VSV-G envelope protein includes two or more mutations selected from a mutation at one or more of N9, Q10, K50, A51, S183, S179, N180, I182, M184, I347, T350, T352, and E353, an insertion of TT between N9 and Q10, an insertion of GGS between H8 and N9, an insertion of GGS between N9 and Q10, an insertion of TT between N208 and Y209, an insertion of GGS between P46 and K47, an insertion of GGS between N208 and Y209, and a deletion of residues 1-8. In some embodiments, a mutated VSV-G envelope protein is selected from one or more of an H8A, N9A, Q10A, K47A, K47Q, N180A, I182A, Y209A, T352A, T352W, E353A, R354A, and R354Q mutation. In some embodiments, a mutated VSV-G envelope protein is selected from one or more of an H8A, K47A, K47Q, Y209A, R354A, and R354Q mutation. In some embodiments, the mutated VSV-G envelope protein comprises a K47 mutation and a R354 mutation. In some embodiments, the mutated VSV-G envelope protein comprises a K47Q mutation and a R354A mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N180 mutation, a I182 mutation, a T352 mutation, and a E353 mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N180A mutation, a I182A mutation, a T352A mutation, and a E353A mutation. In some embodiments, the mutated VSV-G envelope protein comprises a T352 mutation and E353 mutation. In further embodiments, the mutated VSV-G envelope protein comprises a T352W mutation and E353A mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N9 mutation, a Q10 mutation, and a N180 mutation. In some embodiments, the mutated VSV-G envelope protein comprises a N9A mutation, a Q10A mutation, and a N180A mutation. In some embodiments, the mutated VSV-G envelope protein comprises an insertion of GGS between H8 and N9 and an insertion of GGS between N9 and Q10. In some embodiments, the mutated VSV-G envelope protein comprises an insertion of TT between N9 and Q10. In some embodiments, the mutated VSV-G envelope protein comprises an insertion of GGS between P46 and K47. In other embodiments, the mutated VSV-G envelope protein is as described in Nikolic et al., "Structural basis for the recognition of LDL-receptor family members by VSV glycoprotein." *Nature Comm.*, 2018, 9:1029, the relevant disclosures of which are incorporated by reference herein.

[0084] In specific embodiments, the mutant VSV envelope protein comprises a K47Q mutation and R354A mutation. In specific embodiments, the mutant VSV-G envelope protein has an amino acid sequence according to SEQ ID NO: 74.

[0085] In specific embodiments, the mutant VSV envelope protein comprises a N180A mutation, an I182A mutation, a T352A mutation, and an E353A mutation. In specific embodiments, the mutant VSV-G envelope protein has an amino acid sequence according to SEQ ID NO: 93.

[0086] In specific embodiments, the mutant VSV envelope protein comprises a T352W mutation and an E353A mutation. In specific embodiments, the mutant VSV-G envelope protein has an amino acid sequence according to SEQ ID NO: 95.

[0087] In specific embodiments, the mutant VSV envelope protein comprises a N9A mutation, a Q10A mutation, and a N180A mutation. In specific embodiments, the mutant

VSV-G envelope protein has an amino acid sequence according to SEQ ID NO: 97.

[0088] In specific embodiments, the mutant VSV envelope protein comprises an insertion of TT between N9 and Q10 mutation. In specific embodiments, the mutant VSV-G envelope protein has an amino acid sequence according to SEQ ID NO: 99.

[0089] In specific embodiments, the mutant VSV envelope protein comprises an insertion of GGS between P46 and K47 mutation. In specific embodiments, the mutant VSV-G envelope protein has an amino acid sequence according to SEQ ID NO: 101.

[0090] In specific embodiments, the mutant VSV envelope protein comprises an insertion of GGS between H8 and N9 and an insertion of GGS between N9 and Q10. In specific embodiments, the mutant VSV-G envelope protein has an amino acid sequence according to SEQ ID NO: 103.

Targeting Protein

[0091] The targeting protein is a membrane-bound protein displayed on the viral envelope and includes an extracellular domain that includes a lymphocyte targeting domain and a transmembrane domain. A lymphocyte targeting domain is any protein or peptide that has an amino acid sequence and is a binding partner for a target molecule or ligand (e.g., a cognate protein or ligand) on a target lymphocyte cell surface. In some embodiments, a lymphocyte targeting domain is a T cell targeting domain. In some embodiments, a lymphocyte targeting domain is a NK cell targeting domain. In some embodiments, a lymphocyte targeting domain is a B cell targeting domain. For example, in certain embodiments, the extracellular domain of the targeting protein includes a lymphocyte targeting domain that specifically binds a protein or ligand on the surface of a defined population of cells, such as a population of lymphocytes characterized by the presence of the targeted protein or ligand on the cell surface. In some embodiments, a lymphocyte targeting domain targets a specific type of lymphocyte, such as for example, T cells, B cells, or natural killer (NK) cells. In some embodiments, the targeting protein additionally includes an extracellular linker or hinge domain positioned between the transmembrane domain and the extracellular domain.

[0092] In some embodiments, the targeting protein is a full-length cell surface protein or receptor, wherein each component (e.g., each of the extracellular domain, transmembrane domain, and intracellular domain) derived from a single protein or receptor. In other embodiments, the targeting protein is a chimeric protein, having at least two components derived from a different protein or receptor. In some embodiments, the targeting protein is obtained from a mammalian species, including humans, primates, cows, horses, goats, sheep, dogs, cats, mice, rats, rabbits, guinea pigs, and pigs. In some embodiments, the targeting protein is a fully human sequence. In some embodiments, the targeting protein is a chimeric sequence comprising human sequence component(s) and component(s) from another species or synthetic component(s). In some embodiments, a chimeric targeting protein may be a fully synthetic protein. In still other embodiments, a chimeric targeting protein may include one or more of an extracellular domain, an extracellular binding domain, a transmembrane domain, an intracellular domain, a hinge domain, or a linker that is a fully synthetic protein.

[0093] In some embodiments, the lymphocyte targeting domain includes an antibody. In such embodiments, the extracellular targeting domain may be a full-length antibody, an antibody fragment, a nanobody, or a single chain Fv fragment (scFv).

[0094] In particular embodiments, the targeting molecule includes a T cell targeting domain that specifically binds to a T cell surface marker, including, for example, a T cell antigen, a T cell surface receptor, or any other protein present on the surface of a targeted T cell. In some embodiments, the targeting protein specifically binds to a T cell marker selected from CD3, CD28, CD80, 4-1BB, AhR, CD3, CD2, CD7, CD4, CD8, CD25, CD44, CD45RA, CD47, CD62L, CD69, CD94, CD95, CD127, CD161, CD183 (CXCR3), CD184 (CXCR4), CD185 (CXCR5), CD193 (CCR3), CD194 (CCR4), CD195 (CCR5), CD196 (CCR6), CD197 (CCR7), CCR10, PD-1, TCRA/b, CD5, CD27, CD45RO, CD45RB, CD57, CD103, CD122, P2RX7, TIGIT, LAG-3, TIM-3, IL6ST, and any combination thereof.

[0095] In further embodiments, the targeting protein specifically binds to a $\gamma\delta$ T cell marker selected from $\gamma\delta$ TCR, Vdelta1, Vdelta2, NKG2D (KLRK1, CD314), and any combination thereof.

[0096] In other embodiments, the targeting protein specifically binds to an NK T cell marker selected from Invariant TCR (Va24-Ja18), CD185 (CXCR5), CXCR6, IL-21R, and any combination thereof.

[0097] In other embodiments, the targeting protein specifically binds to a MAIT cell marker selected from Va7.2, Ja33, CXCR6, IL-18R, KLRB1 (CD161), VLA4 (alpha4beta1 integrin), and any combination thereof.

[0098] In other embodiments, the targeting protein specifically binds to an NK cell marker selected from CD56, NKp46, CD16, KIR(s), NKG2 proteins (e.g., NKG2D (KLRK1, CD314)), KLRB1 (CD161), KLRD1 (cd94), IL2Rb (CD122), IL-21R, SLAMF6 (CD352), SLAMF7 (CD319), IL-18R, and any combination thereof.

[0099] In other embodiments, the targeting protein specifically binds to a B cell marker selected from CD19, CD20, CD21, CD22, CD24, CD38, CD40, CD72, CD32b, CD268, CD269, CD267, CD86, CD80, CD52, CD138, CD27, CD28, CD23, CD84, CD257, CD270, CD37, CD74, and CD269, and any combination thereof.

[0100] In embodiments, the targeting protein specifically binds to a lymphocyte marker (e.g., B cell and T cells) selected from CD80, CD27, CD28, and any combination thereof.

[0101] In some embodiments of the targeting protein, a linker is positioned between the transmembrane domain and the lymphocyte targeting domain. A linker is an amino acid linker and may be a rigid linker, a flexible linker, or an oligomerized linker. A rigid linker is an amino acid sequence that lacks flexibility (e.g., may comprise at least one proline). In some embodiments, a rigid linker comprises a platelet-derived growth factor receptor (PDGFR) stalk or a CD8a stalk. In some embodiments, a PDGFR stalk comprises an amino acid sequence comprising AVGQDTQEIVVPHSLPFK (SEQ ID NO: 104). In some embodiments, a PDGFR stalk comprises an amino acid sequence comprising

(SEQ ID NO: 105)

ASAKPTTTPAPRPPTPAPTIASQPLSLRPEARPAAGGAVHTRGLDFAK.

[0102] A flexible linker is an amino acid sequence that has many degrees of freedom (e.g., may comprise a plurality of amino acids with small side chains, e.g., glycine or alanine). In some embodiments, a flexible linker comprises an amino acid sequence comprising GAPGAS. In some embodiments, a flexible linker comprises an amino acid sequence consisting of GAPGSGGGGSGGGGSAS (SEQ ID NO: 106). In some embodiments, a flexible linker comprises an amino acid sequence comprising GGGGS. In some embodiments, a flexible linker comprises an amino acid sequence comprising GGGG. In some embodiments, a flexible linker comprises an amino acid sequence comprising (GAPGAS)_N, (G₃S)_N, or (G₄S)_N, wherein N is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, or more.

[0103] An oligomerized linker is an amino acid that can oligomerize to another related amino acid. In some embodiments, an oligomerized linker is an amino acid sequence that can form a dimer, trimer, or tetramer. In some embodiments, an oligomerized linker comprises an IgG4 hinge domain (e.g., ESKYGPPCPPCPAVGQDTQEVIVVPHSLPFK (SEQ ID NO: 107)). In some embodiments, an oligomerized linker comprises an amino acid sequence that can form a tetrameric coiled coil (e.g., ASGGGSGELAAIKQELAAIKKELAAIKWELAAIKQGAG (SEQ ID NO: 108)). In some embodiments, an oligomerized linker comprises an amino acid sequence that can form a dimeric coiled coil (e.g., ASESKYGPCCPPCP (SEQ ID NO: 109)).

[0104] The targeting protein will typically include a signal sequence (also referred to as a signal peptide of localization sequence). The signal sequence can be located at the N- or C-terminal ends of the targeting protein. A signal sequence functions to translocate the targeting protein to the membrane that serves as the envelope of the lentiviral vector. Non-limiting examples of signal sequences that might be included in a targeting protein as described herein include an Ig Kappa leader sequence (e.g., a murine Ig Kappa leader sequence comprising: METDTLLLVLLLVPGSTG (SEQ ID NO: 110)), CD8a signal peptide sequence (e.g., a CD8a signal peptide comprising: MALPVTALLLPLALLLHAARP (SEQ ID NO: 12)), and a B2M signal peptide sequence (e.g., a B2M signal peptide sequence comprising: MSRSVALAVLALLSLSGLEA (SEQ ID NO: 123)). In certain embodiments, a mature targeting protein has a partially cleaved signal sequence that retains one or more amino acids of the full-length signal sequence. In other embodiments, cleavage by a signal peptidase results in complete removal of the signal sequence, resulting in a mature targeting protein that lacks a signal sequence entirely.

[0105] In some embodiments, the targeting protein is a CD80 protein. Where the targeting protein is a CD80 protein, it may include an extracellular domain according to SEQ ID NO: 6. In such embodiments, a CD80 protein may include a signal peptide according to SEQ ID NO: 4 and a transmembrane and intracellular domain according to SEQ ID NO: 8. In a specific embodiment, a CD80 targeting protein included in a lentiviral vector as described herein includes an amino acid sequence according to SEQ ID NO: 2 or SEQ ID NO: 2 absent the signal peptide of SEQ ID NO: 4.

[0106] In some embodiments, where the targeting protein is a CD80 protein, it may include an extracellular domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 6. In such embodiments, a CD80

protein may include a signal peptide that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 4, and/or a transmembrane and intracellular domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 8. In a specific embodiment, a CD80 targeting protein included in a lentiviral vector as described herein includes an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 2 or SEQ ID NO: 2 absent the signal peptide of SEQ ID NO: 4.

TABLE 1	
CD80 T cell targeting molecule	
Full molecule	
Nucleotide	Amino acid (signal peptide underlined)
ATGGGTCATACACGC	<u>MGHTRRQGTSPSKCP</u>
CGCCAAGGAACCTCA	<u>YLNFFQLVLVLAGLSH</u>
CCATCTAAGTGCCCA	<u>FCSGVIHVTKEVKEV</u>
TATCTGAATTTCTTT	<u>ATLSCGHNVSVVEELA</u>
CAACTTCTCGTGCTG	<u>QTRIFYWQKEKMKVLT</u>
GCGGGGCTCAGTCAT	<u>MMSGDMNIWPEYKNR</u>
TTCTGCAGTGGGGTC	<u>TIFDITNNLSIVILA</u>
ATTCACGTTACTAAA	<u>LRPSDEGTYECVVLK</u>
GAGGTCAAGGAGGTC	<u>YEKDAFKREHLAEVT</u>
GCAACATTGAGTTGT	<u>LSVKADFPTPSISDF</u>
GGCCATAACGTATCA	<u>EIPTSNIIRRIICSTS</u>
GTTGAAGAACTCGCG	<u>GGFPPEPHLSWLENGE</u>
CAGACACGGATTTAC	<u>ELNAINTTVSQDPET</u>
TGGCAAAAGGAAAAG	<u>ELYAVSSKLDNFNMTT</u>
AAGATGGTGTGACA	<u>NHSPMCLIKYGHLRV</u>
ATGATGAGCGGTGAC	<u>NQTFNWNNTKQEHFP</u>
ATGAACATTTGGCCA	<u>DNLLPSWAITLISVN</u>
GAGTACAAAAATCGA	<u>GIFVICCLTYCFAPR</u>
ACGATATTCGATATA	<u>CRE</u>
ACCAATAACTTGTC	(SEQ ID NO: 2)
ATAGTAATACTTGCC	
TTGCGACCTTCTGAC	
GAGGGAACGATGAA	
TGTTGAGTGCTTAAG	
TATGAAAAGATGCC	
TTTAAGCGGGAACAC	
TTGGCTGAGGTTACA	
CTCTCCGTTAAGGCG	
GACTTTCCTACGCCG	
TCTATATCCGACTTC	
GAGATACCCACTTCT	
AACATTCGACGCATC	
ATTTGCTCAACCTCA	
GGTGGTTTCCAGAG	
CCTCACTTGAGCTGG	
CTGGAGAATGGCGAA	
GAACCTTAACGCAATC	
AATACCACGGTGTCC	
CAAGACCCGGAGACA	
GAGCTGTACGCCGTG	
TCATCCAAACTGGAT	
TTTAACATGACGACA	
AATCATAGTTTCATG	
TGTTCTGATCAATAT	
GGGCATCTCAGGGTG	
AATCAGACTTTTAAT	
TGGAACACTACCAAA	
CAAGAGCACTCCCA	
GATAATCTGTTGCCA	

TABLE 1-continued

CD80 T cell targeting molecule	
AGCTGGGCGATAACT CTTATCTCCGTCAAC GGTATCTTCGTAATT TGCTGCCTCACCTAT TGTTTCGCGCCTCGA TGCCGAGAA (SEQ ID NO: 1)	
CD80 signal peptide	
Nucleotide	Amino acid
ATGGGTCATACACGC CGCCAAGGAACCTCA CCATCTAAGTGCCCA TATCTGAATTTCTTT CAACTTCTCGTGCTG GCGGGGCTCAGTCAT TTCTGCAGTGGGGTC (SEQ ID NO: 3)	MGHTRRQGTSPSKCP YLNFFQLVLVLAGLSH FCSGV (SEQ ID NO: 4)
CD80 extracellular domain (ECD)	
Nucleotide	Amino acid
ATTACGCTTAATAAA GAGGTCAAGGAGGTC GCAACATTGAGTTGT GGCCATAACGTATCA GTTGAAGAACTCGCG CAGACACGGATTTAC TGGCAAAAAGGAAAAG AAGATGGTGTGAC ATGATGAGCGGTGAC ATGAACATTTGGCCA GAGTACAAAAATCGA ACGATATTCGATATA ACCAATAACTTGTCC ATAGTAATACTTGCC TTGCGACCTCTGAC GAGGGAACGTATGAA TGTGTAGTGCTTAAG TATGAAAAGATGCC TTTAAGCGGGAACAC TTGGCTGAGGTTACA CTCTCCGTTAAGGCG GACTTTCCTACGCCG TCTATATCCGACTTC GAGATACCCACTTCT AACATTCGACGCATC ATTTGCTCAACCTCA GGTGGTTTCCAGAG CCTCACTTGAGCTGG CTGGAGAATGGCGAA GAACTTAACGCAATC AATACCACGGTGTCC CAAGACCCGGAGACA GAGCTGTACGCCGTG TCATCCAACTGGAT TTTAACATGACGACA AATCATAGTTTCATG TGTCTGATCAAATAT GGGCATCTCAGGGTG AATCAGACTTTTAAT TGGAACTACTACAAA CAAGAGCACTTCCCA GATAATCTGTTGCCA AGC (SEQ ID NO: 5)	IHVTKEVKEVATLSC GHNVSVEELAQTRIY WQKEKMLVLTMSGD MNIWPEYKNRTIFDI TNNLSIVILALRPSD EGTYECVVLKYEKDA FKREHLAEVTLVSKA DFPTPSISDFEIPTS NIRRIICSTSGGFPE PHLSWLENGEELNAI NTTVSQDPETELYAV SSKLDFNMTTNHSFM CLIKYGHLRVNQTFN WNTTKQEHFPDNLPS (SEQ ID NO: 6)

TABLE 1-continued

CD80 transmembrane and intracellular domain	
Nucleotide	Amino acid
TGGGCGATAACTCTT ATCTCCGTCAACGGT ATCTTCGTAATTTGC TGCCTCACCTATTGT TTCGCGCCTCGATGC CGAGAA (SEQ ID NO: 7)	WAITLISVNGIFVIC CLTYCFAPRCRE (SEQ ID NO: 8)

[0107] In some embodiments, lentiviral vectors comprising a CD80 targeting protein exhibit enhanced transduction CD4 T cells over CD8 T cells as a relative percentage of total CD4 and CD8 T cells compared to the transduction by standard LVV. As used herein, “standard lentiviral vector” or “standard LVV” refers to a lentiviral vector which does not contain a mutated VSV-G envelope protein and a lymphocyte targeting protein of the present disclosure. In some embodiments, a standard LVV is generated using a 3rd generation LVV packaging system. In some embodiments, lentiviral vectors comprising a CD80 targeting protein enhance transduction of CD4 T cells over CD8 T cells as a relative percentage of total CD4 and CD8 T cells at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, 99% or more compared to standard LVV.

[0108] In some embodiments, the targeting protein is an anti-CD3 targeting protein. In some embodiments, an anti-CD3 targeting protein is an antibody or binding fragment thereof, including for example, a scFv. Where the targeting protein is an anti-CD3 targeting protein, it may include an anti-CD3 scFv having a light chain variable region according to SEQ ID NO: 14, a G₃S linker according to SEQ ID NO: 16, and a heavy chain variable region according to SEQ ID NO: 18. In some embodiments, an anti-CD3 targeting protein may comprise an anti-CD3 scFv having a light chain variable region according to SEQ ID NO: 115, a G₃S linker according to SEQ ID NO: 16, and a heavy chain variable region according to SEQ ID NO: 113. An anti-CD3 targeting protein as described may include a signal peptide according to SEQ ID NO: 12 and a hinge and transmembrane domain according to SEQ ID NO: 20. In a specific embodiment, an anti-CD3 targeting protein included in a lentiviral vector as described herein includes an amino acid sequence according to SEQ ID NO: 10 or SEQ ID NO: 10 absent the signal peptide of SEQ ID NO: 12. In another specific embodiment, an anti-CD3 targeting protein included in a lentiviral vector as described herein includes an amino acid sequence according to SEQ ID NO: 117 or SEQ ID NO: 117 absent the signal peptide of SEQ ID NO: 12.

[0109] In some embodiments, where the targeting protein is an anti-CD3 targeting protein, it may include an anti-CD3 scFv having a light chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 14, and a heavy chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 18. An anti-CD3 targeting protein as described may include a signal peptide that has at least about 75%, 80%,

85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 12 and a hinge and transmembrane domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 20. In a specific embodiment, an anti-CD3 targeting protein included in a lentiviral vector as described herein includes an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 10 or SEQ ID NO: 10 absent the signal peptide of SEQ ID NO: 12.

[0110] In some embodiments, where the targeting protein is an anti-CD3 targeting protein, it may include an anti-CD3 scFv having a light chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 115, and a heavy chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 113. An anti-CD3 targeting protein as described may include a signal peptide that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 20. In a specific embodiment, an anti-CD3 targeting protein included in a lentiviral vector as described herein includes an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 117 or SEQ ID NO: 117 absent the signal peptide of SEQ ID NO: 12.

TABLE 2

Anti-CD3 T cell targeting molecules	
Anti-CD3 (UCHL1) Full molecule	
Nucleotide	Amino acid (signal peptide underlined)
ATGGCCTTACCACTG	MALPVTALLLLPLALL
ACCGCCTTGCTCCTG	<u>LHAARPD</u> IQMTQTTS
CCGCTGGCCTTGCTG	SLASLGDRTVTSR
CTCCACGCCGCCAGG	ASQDIRNYLNWYQK
CCGACATCCAGATG	PDGTVKLLIYYSRL
ACCCAGACCACCTCC	HSGVPSKFSGSGSGT
TCCCTGTCTGCCTCT	DYSLTISNLEQEDIA
CTGGGAGACAGAGTC	TYFCQQGNLTPWTF
ACCATCAGTTGCAGG	GGTKLEIKRAGGSG
GCAAGTCAGGACATT	GGSGGSGGSEVQL
AGAAATTATTAAAC	QQSGPELVKPGASMK
TGGTATCAACAGAAA	ISCKASGYSTGYTM
CCAGATGGAAGCTGT	NWVKQSHGKNLEWMG
AAACTCCTGATCTAC	LINPYKGVSTYNQKF
TACACATCAAGATTA	KDKATLTVDKSSSTA
CACCTAGGAGTCCCA	YMLLSLTSEDSAVY
TCAAAGTTCAGTGGC	YCARSGYYGDSWYF
AGTGGGTCTGGAACA	DVWAGGTTVTVSST
GATTATTCTCTCACC	TPAPRPPTPAPTIAS
ATTAGCAACCTGGAG	QPLSLRPEACRPAAG
CAAGAGGATATTGCC	GAVHTRGLDFACDIY
ACTTACTTTTGCCAA	IWAPLAGTCGVLLLS
CAGGGTAATACGCTT	LVITLYC
CCGTGGACGTTTCGCT	(SEQ ID NO: 10)
GGAGGCACCAAGCTG	
GAAATCAAACGGGCT	
GGAGGCGGTAGTGGC	
GGTGGATCAGGTGGA	
GGCAGCGGTGGCGGA	

TABLE 2-continued

Anti-CD3 T cell targeting molecules	
TCTGAGGTGCAGCTC	
CAGCAGTCTGGACCT	
GAGCTGGTGAAGCCT	
GGAGCTTCAATGAAG	
ATATCCTGCAAGGCT	
TCTGGTTACTCATTC	
ACTGGCTACACCATG	
AACCTGGTGAAGCAG	
AGTCATGGAAAGAAC	
CTTGAGTGGATGGGA	
CTTATTAACTCTTAC	
AAAGGTGTAGTACC	
TACAACCAGAAGTTC	
AAGGACAAGGCCACA	
TTAAGTGTAGACAAG	
TCATCCAGCACAGCC	
TACATGGAACCTCTC	
AGTCTGACATCTGAG	
GACTCTGCAGTCTAT	
TACTGTGCAAGATCG	
GGGTACTACGGTGAT	
AGTGACTGGTACTTC	
GATGTCTGGGGCGCA	
GGGACCACGGTCACC	
GTCTCCTCAACCACT	
ACaCCAGCaCCTaGA	
CCACCAACACCTGCG	
CCaACCATCGCaTCG	
CAGCCaCTGTCTCTG	
CGCCACAGGGCaTGC	
CGGCCAGCaGCTGGG	
GGCGCAGTGCAACa	
AGGGGCTGGACTTC	
GCaTGTGATATCTAC	
ATCTGGGCaCCaTTG	
GCaGGGACTGTGGG	
GTCTTCTCTGTCTCA	
CTGGTTATCACCCCTT	
TACTGC	
(SEQ ID NO: 9)	
CD8a signal peptide	
Nucleotide	Amino acid
ATGGCCTTACCACTG	MALPVTALLLLPLALL
ACCGCCTTGCTCCTG	LHAARP
CCGCTGGCCTTGCTG	(SEQ ID NO: 12)
CTCCACGCCGCCAGG	
CCG	
(SEQ ID NO: 11)	
Anti-CD3 (UCHL1) Light chain variable region	
Nucleotide	Amino acid
GACATCCAGATGACC	DIQMTQTSSLSASL
CAGACCACCTCCTCC	GDRVITISCRASQDIR
CTGTCTGCCTCTCTG	NYLNWYQQKPDGTVK
GGAGACAGAGTCACC	LLIYYTSRLHSGVPS
ATCAGTTGCAGGGCA	KFSGSGSGTDYSLTI
AGTCAGGACATTAGA	SNLEQEDIATYFCQQ
AATTATTTAACTGG	GNLTPWTFAGGTKLE
TATCAACAGAAACCA	IKRA
GATGGAACGTGTTAAA	(SEQ ID NO: 14)
CTCCTGATCTACTAC	
ACATCAAGATTACAC	
TCAGGAGTCCCATCA	
AAGTTCAGTGGCAGT	
GGGTCTGGAACAGAT	
TATTCTCTCACCATT	
AGCAACCTGGAGCAA	
GAGGATATTGCCACT	
TACTTTTGCCAACAG	

TABLE 2-continued

Anti-CD3 T cell targeting molecules	
GGTAATACGCTTCCG TGGACGTTTCGCTGGA GGCACCAAGCTGGAA ATCAAACGGGGCT (SEQ ID NO: 13)	
G ₃ S linker	
Nucleotide	Amino acid
GGAGGCGGTAGTGGC GGTGGATCAGGTGGA GGCAGCGGTGGCGGA TCT (SEQ ID NO: 15)	GGSGGGSGGGSGGG S (SEQ ID NO: 16)
Anti-CD3 (UCHL1) heavy chain variable region	
Nucleotide	Amino acid
GAGGTGCAGCTCCAG CAGTCTGGACCTGAG CTGGTGAAGCCTGGA GCTTCAATGAAGATA TCCTGCAAGGCTTCT GGTTACTCATTCACT GGCTACACCATGAAC TGGGTGAAGCAGAGT CATGAAAGAACCTT GAGTGGATGGGACTT ATTAATCCTTACAAA GGTGTAGTACCTAC AACCAGAAGTTCAAG GACAAGGCCACATTA ACTGTAGACAAGTCA TCCAGCACAGCCTAC ATGGAACCTCCTCAGT CTGACATCTGAGGAC TCTGCAGTCTATTAC TGTGCAAGATCGGGG TACTACGGTGATAGT GACTGGTACTTCGAT GTCTGGGGCGCAGGG ACCACGGTCACCGTC TCCTCA (SEQ ID NO: 17)	EVQLQQSGPELVKPG ASMKISCKASGYST GYTMNWVKQSHGNL EWMGLINPYKGVSTY NQKFKDKATLTVDKS SSTAYMELLSLTSED SAVYYCARSQYVYGD DWYFDVWGAGTTVT SS (SEQ ID NO: 18)
CD8a hinge and transmembrane	
Nucleotide	Amino acid
ACCACTACACCAGCA CCTAGACCACCAACA CCTGCGCAACCATC GCATCGCAGCCACTG TCTCTGCGCCAGAG GCATGCCGGCCAGCA GCTGGGGGCGCAGTG CACACAAGGGGGCTG GACTTCGATGTGAT ATCTACATCTGGGCA CCATTGGCAGGGACT TGTGGGGTCTCTCTC CTGTCACTGGTTATC ACCCTTTACTGC (SEQ ID NO: 19)	TTTPAPRPPTPPTI ASQPLSLRPEACRPA AGGAVHTRGLDFACD IYIWAPLAGTCGVLL LSLVITLYC (SEQ ID NO: 20)
Anti-CD3 (12F6) heavy chain variable region	
Nucleotide	Amino Acid
CAAGTGCAGCTCCAG CAGAGCGGCGCTGAG	QVQLQQSGAELARPG ASVKMSCKASGYTFT

TABLE 2-continued

Anti-CD3 T cell targeting molecules	
CTGGCCCGGCCCGGC GCCAGCGTGAAGATG AGCTGTAAAGCCAGC GGCTATACATTTACC AGCTACACCATGCAC TGGGTCAAGCAGCGG CCTGGCCAGGGCCTG GAATGGATTGGATAT ATCAACCCAGCAGC GGCTACACCAAGTAC AACCAGAAATTCAG GACAAGGCCACCCCTG ACCGCCGACAAGAGC TCCTCAACAGCCTAC ATGCAACTGAGCAGC CTGACCAGCGAGGAT AGCGCCGTGTACTAC TGCGCCAGATGGCAG GACTACGACGTGTAC TTCGACTACTGGGGC CAAGGCACACACTG ACCGTGTCCAGC (SEQ ID NO: 112)	
Anti-CD3 (12F6) light chain variable region	
Nucleotide	Amino Acid
CAGATCGTGTGAGC CAGTCCCCAGCCATC CTGTCTGCCAGCCCT GGCGAGAAGGTGACC ATGACCTGCAGAGCC TCTTCTCTGTTTCC TACATGCACTGGTAT CAGCAAAAGCCCGGC AGCTCTCCTAAGCCT TGGATCTACGCCACA AGCAACCTGGCTAGC GGCGTGCCCTGCTCG TTCAGCGGCAGCGGC AGCGGCACCAGCTAC AGCCTGACCATCAGC AGAGTGAAGCCGAG GACGCCGCCACCTAC TACTGCCAGCAGTGG TCCTCTAATCCTCCA ACATTGCGCGCGCGC ACCAAGCTGGAACCC AAAAGA (SEQ ID NO: 114)	QIVLSQSPAILSASP GEKVTMTCRASSSVS YMHWYQQKPGSSPKP WIYATSNLAGVPPAR FSGSGSGTSYSLTIS RVEAEDAATYYCQQW SSNPPTFGGGTKLET KR (SEQ ID NO: 115)
Anti-CD3 (12F6) Full molecule	
Nucleotide	Amino Acid (signal peptide underlined)
ATGGCCTTACCAGTG ACCGCCTTGCTCCTG CCGCTGGCCTTGCTG CTCCACGCCGCCAGG CCGCAAGTGCAGCTC CAGCAGAGCGGCGCT GAGCTGGCCCGGCC GGCGCCAGCGTGAAG ATGAGCTGTAAAGCC AGCGGCTATACATTT ACCAGCTACACCATG CACTGGGTCAAGCAG CGGCTTGGCCAGGGC CTGGAATGGATTGGA TATATCAACCCAGC AGCGGCTACACCAAG	MALPVTALLPLALL <u>LHAARPQVQLQQSGA</u> ELARPGASVKMSCKA SGYTFTSYTMHWVKQ RPGQGLEWIGYINPS SGYTKYNQKFKDKAT LTADKSSSTAYMQLS SLTSEDSAVYYCARW QDYDVYFDYWGQGT LTVSSGGSGSGSGSG GSGGGSQIVLSQSPA ILSASPGKVTMTCR ASSSVSYMHWYQQKP GSSPKPWIYATSNLA SGVPARFSGSGSGTS YSLTISRVEAEDAAT

TABLE 2-continued

Anti-CD3 T cell targeting molecules	
TACAACCAGAAATTC	YYCQWSSNPPTFGG
AAGGACAAGGCCACC	GTKLETKRTTTPAPR
CTGACCGCCGACAAG	PPTPAPTIASQPLSL
AGCTCCTCAACAGCC	RPEACRPAAGGAVHT
TACATGCAACTGAGC	RGLDFACDIYIWAPL
AGCCTGACCAGCGAG	AGTCGVLLLSLVITL
GATAGCGCCGTGTAC	YC
TACTGCGCCAGATGG	(SEQ ID NO: 117)
CAGGACTACGACGTG	
TACTTCGACTACTGG	
GGCCAAGGCACAACA	
CTGACCGTGTCCAGC	
GGAGGCGGTAGTGGC	
GGTGGATCAGGTGGA	
GGCAGCGGTGGCGGA	
TCTCAGATCGTGCTG	
AGCCAGTCCCCAGCC	
ATCCTGTCTGCCAGC	
CCTGGCAGAAAGGTG	
ACCATGACCTGCAGA	
GCCTCTTCTTCTGTT	
TCCTACATGCACTGG	
TATCAGCAAAAGCCC	
GGCAGCTCTCCTAAG	
CCTTGGATCTACGCC	
ACAAGCAACCTGGCT	
AGCGGCGTGCCTGCT	
CGCTTCAGCGGCAGC	
GGCAGCGGCACACGC	
TACAGCCTGACCATC	
AGCAGAGTGAAGCC	
GAGGACGCCGCCACC	
TACTACTGCCAGCAG	
TGGTCTCTAATCCT	
CCAACATTGCGCGGC	
GGCACCAAGCTGGAA	
ACCAAAAGAACCACT	
ACACCAGCACCTAGA	
CCACCAACACTGCG	
CCAACCATCGCATCG	
CAGCCACTGTCTCTG	
CGCCCAGAGGCATGC	
CGGCCAGCAGCTGGG	
GGCGCAGTGACACA	
AGGGGGCTGGACTTC	
GCATGTGATATCTAC	
ATCTGGGCACCATTG	
GCAGGGACTTGTGGG	
GTCCTTCTCCTGTCA	
CTGTTTATCACCCCTT	
TACTGC	
(SEQ ID NO: 116)	

[0111] In still further embodiments, a lentiviral vector according to the present description may include multiple targeting proteins or a single targeting protein having multiple different lymphocyte targeting domains. For example, the lentiviral vectors described herein can include at least two targeting proteins, with each of the targeting proteins having a NK cell targeting domain that specifically binds a different target. In another example, the lentiviral vectors described herein can include at least two targeting proteins, with each of the targeting proteins having a T cell targeting domain that specifically binds a different target. In another example, the lentiviral vectors described herein can include at least two targeting proteins, with each of the targeting proteins having a B cell targeting domain that specifically binds a different target. In yet another example, the lentiviral vectors described herein can include at least two targeting proteins, with a first targeting protein having a T cell targeting domain and a second targeting protein having a NK

cell targeting domain. In yet another example, the lentiviral vectors described herein can include at least two targeting proteins, with a first targeting protein having a T cell targeting domain and a second targeting protein having a B cell targeting domain. In yet another example, the lentiviral vectors described herein can include at least two targeting proteins, with a first targeting protein having a B cell targeting domain and a second targeting protein having a NK cell targeting domain. In alternative embodiments, the lentiviral vectors described herein include a single targeting protein having two or more different lymphocyte targeting domains, two or more different T cell targeting domains, two or more different NK cell targeting domains, or two or more different B cell targeting domains (e.g., a bi-specific or multi-specific targeting protein). In any such embodiments, the T lymphocyte targeting domains included in the targeting proteins can be selected from any of those described herein. In yet another example, the lentiviral vectors described herein can include at least two targeting proteins, with a first targeting protein having a lymphocyte targeting domain and a second targeting protein having a T cell targeting domain. In yet another example, the lentiviral vectors described herein can include at least two targeting proteins, with a first targeting protein having a lymphocyte targeting domain and a second targeting protein having a B cell targeting domain.

[0112] In an exemplary embodiment, a lentiviral vector having multiple targeting proteins includes a CD80 targeting protein and an anti-CD3 targeting protein as detailed in Table 1 and Table 2, respectively. In some embodiments, a CD80 targeting protein includes an amino acid sequence according to SEQ ID NO: 2 or SEQ ID NO: 2 absent the signal peptide of SEQ ID NO: 4. In some embodiments, an anti-CD3 targeting protein includes an amino acid sequence according to SEQ ID NO: 10 or SEQ ID NO: 10 absent the signal peptide of SEQ ID NO: 12. In some embodiments, an anti-CD3 targeting protein includes an amino acid sequence according to SEQ ID NO: 117 or SEQ ID NO: 117 absent the signal peptide of SEQ ID NO: 12. Where the lentiviral vector includes a CD80 targeting protein and an anti-CD3 targeting protein, the targeting proteins may be encoded on separate expression vectors used in producing the lentiviral vector. Alternatively, both a CD80 targeting protein and an anti-CD3 targeting protein may be encoded on a tandem expression cassette, providing a single expression cassette in a single expression vector that provides expression of both targeting proteins for production of the lentiviral vector. Exemplary embodiments of nucleic acid sequences and the corresponding amino acid sequences for a CD80 targeting protein and an anti-CD3 targeting protein expressed from the same expression cassette are provided in Table 3. In such an embodiment, the two targeting proteins may be linked by a P2A self-cleaving peptide. Table 3 provides an exemplary nucleic acid that may be used in a tandem expression cassette that includes a CD80 binding protein linked to an anti-CD3 binding protein via a P2A sequence. In a particular embodiment, the P2A sequence is encoded by a nucleic acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 29 and the amino acid sequence of the P2A sequence has an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 30. In some embodiments, the

tandem expression cassette that includes a CD80 binding protein linked to an anti-CD3 binding protein comprises the nucleic acid sequence of SEQ ID NO: 21 or has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID: 21. In some embodiments, the tandem expression cassette that includes a CD80 binding protein linked to an anti-CD3 binding protein encodes an amino acid sequence of SEQ ID

NO: 22 or SEQ ID NO: 22 absent the signal peptide of SEQ ID NO: 24. In some embodiments, the tandem expression cassette that includes a CD80 binding protein linked to an anti-CD3 binding protein encodes an amino acid sequence having at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 22 or SEQ ID NO: 22 absent the signal peptide of SEQ ID NO: 24.

TABLE 3

Co-expressed CD80 and anti-CD3 T cell targeting molecules	
Full molecule	
Nucleotide	Amino acid (signal peptide underlined)
ATGGGTCATACACGCCGCCAAGGA	MGHTRRQGTSPSKCPYLNFPQLLVL
ACCTCACCATCTAAGTGCCCATAT	AGLSHFCSGVIHVTKEVEVATLSC
CTGAATTTCTTCAACTTCTCGTGC	GHNVSVEELAQTRIYWQKEKKMVL
TGGCGGGGCTCAGTCATTTCTGCA	TMMSGDMNIWPEYKNRTIFDI TNNL
GTGGGGTCATTCACGTTACTAAAG	SIVILALRPSDEGTYECVVLKYEKDA
AGGTCAAGGAGGTCGCAACATTGA	FKREHLAEVTL SVKADFP TPSISDFEI
GTGTGGCCATAACGTATCAGTTG	PTSNIRRI ICSTSGGFPEPHLSWLENG
AAGAACTCGCGCAGACACGGATT	EELNAINTTVSQD PETELYAVSSKLD
ACTGGCAAAAGGAAAAGAAGATG	FNMTTNHSMCLIKYGH LRVNQTFN
GTGTGACAATGATGAGCGGTGAC	WNTTKQEHFPDNL LPSWAI TLISVN
ATGAACATTTGGCCAGAGTACAAA	GIFVICCLTYCFAPRCREGSGATNFS
AATCGAACGATATTCGATATAACC	LLKQAGDVEENPGPMALFVTALLLP
AATAACTTGTCCATAGTAATACTT	LALLLHAARPD IQMTQTSSLSASLG
GCCTTGGCAGCCTTCTGACGAGGGA	DRV TISCRASQDIRNYLNWYQKPD
ACGTATGAATGTGTAGTGCTTAAG	GTVKLLIYYTSRLHSGVPSKFSGSGS
TATGAAAAGATGCTTTAAGCGG	GTDYSLTI SNLEQEDIATYFCQQGNT
GAACACTTGGCTGAGGTTACACTC	LPWTFAGGTKLEIKRAGGSGGSGS
TCCGTTAAGGCGGACTTCTCTACG	GGSGGSEVQLQQSGPELVKPGAS
CCGTCTATATCCGACTTCGAGATA	MKISCKASGYSFTGYTMNWVKQSH
CCCACTTCTAACATTTCGACGCATC	GKNLEWMGLINPYKGVSTYNQKFK
ATTTGCTCAACCTCAGGTGGTTTCC	DKATLTVDKSSSTAYMELLSLTSED
CAGAGCCTCACTTGAGCTGGCTGG	SAVYYCARSGYGDSDWYFDVWG
AGAATGGCGAAGAACTTAACGCA	AGTTVTVSSTTTPAPRPPTPAPTIASQ
ATCAATACCACGGTGTCCCAAGAC	PLSLRPEACRPAAGVAHVTRGLDFA
CCGGAGACAGAGCTGTACGCCGTG	CDIYIWAPLAGTCGVLLLSLVITLYC
TCATCCAACTGGATTTTAACATG	(SEQ ID NO: 22)
ACGACAAATCATAGTTTCATGTGT	
CTGATCAAAATATGGGCATCTCAGG	
GTGAATCAGACTTTTAATTGGAAC	
ACTACCAACAAGAGCACTTCCCA	
GATAATCTGTTGCCAAGCTGGGCG	
ATAACTCTTATCTCCGTCAACGGT	
ATCTTCGTAATTTGCTGCCTCACCT	
ATTGTTTTCGGCCTCGATGCCGAG	
AAGGCAGCGCGCCACCAACTTCT	
CCCTGCTGAAGCAGGCCGCGACG	
TGGAAGAAAACCTGGCCCCATGG	
CCTTACCAGTGACCGCCTTGCTCCT	
GCCGCTGGCCTTGCTGCTCCACGC	
CGCCAGGCCGACATCCAGATGAC	
CCAGACCACCTCCTCCCTGTCTGCC	
TCTCTGGGAGACAGAGTCACCATC	
AGTTGCAGGGCAAGTCAGGACATT	
AGAAATTATTTAAACTGGTATCAA	
CAGAAACCAGATGGAAGTGTTAAA	
CTCCTGATCTACTACACATCAAGA	
TTCACTCAGGAGTCCCATCAAAG	
TTCAAGTGGCAGTGGGTCTGGAACA	
GATTATTTCTCTACCATTAGCAACC	
TGGAGCAAGAGGATATTGCCACTT	
ACTTTTGCCAACAGGGTAATACGC	
TTCCGTGGAGCTTCGCTGGAGGCA	
CCAAGCTGGAAATCAAACGGCTG	
GAGGCGGTAGTGGCGGTGGATCAG	
GTGGAGGCAGCGGTGGCGGATCTG	
AGGTGCAGCTCCAGCAGTCTGGAC	
CTGAGCTGGTGAAGCCTGGAGCTT	
CAATGAAGATATCTGCAAGGCTT	

TABLE 3-continued

Co-expressed CD80 and anti-CD3 T cell targeting molecules	
CTGGTTACTCATTCACTGGCTACAC CATGAACTGGGTGAAGCAGAGTCA TGGAAAGAACCTTGAGTGGATGGG ACTTATTAATCCTTACAAAGGTGTT AGTACCTACAACCAGAAGTTCAAG GACAAGGCCACATTAAGTGTAGAC AAGTCATCCAGCACAGCCTACATG GAACTCCTCAGTCTGACATCTGAG GACTCTGCAGTCTATTACTGTGCA AGATCGGGGTACTACGGTGATAGT GACTGGTACTTCGATGTCTGGGGC GCAGGGACCACGGTCACCGTCTCC TCAACCACtACaCCAGCaCctAGACC ACCAACACCTGCGCCaACCATCGCa TCGCAGCCaCTGTCTcTGCGCCAG AGGCaTGCCGGCCAGCaGctGGGGG CGCAGTGCAcACaAGGGGGCTGGA CTTCGCaTGTGATATCTACATCTGG GCaCCaTTGGCaGGGACTTGTGGGG TCCTTCTCCTGTCACTGTTATCAC CCTTTACTGC (SEQ ID NO: 21)	
CD80 signal peptide	
Nucleotide	Amino acid
ATGGGTCATACACGCCCAAGGA ACCTCACCATCTAAGTGCCCATAT CTGAATTTCTTTCAACTTCTCGTGC TGGCGGGGCTCAGTCATTCTGCA GTGGGGTC (SEQ ID NO: 23)	MGHTRRQGTSPSKCPYLNFFQLLVL AGLSHFCSGV (SEQ ID NO: 24)
CD80 extracellular domain (ECD)	
Nucleotide	Amino acid
ATTACGTTACTAAAGAGGTCAAG GAGGTCGCAACATTGAGTTGTGGC CATAACGTATCAGTTGAAGAACTC GGCGAGACACGGATTTACTGGCAA AAGGAAAAGAAAGATGGTGTGAC AATGATGAGCGGTGACATGAACAT TTGGCCAGAGTACAAAATCGAAC GATATTCGATATAACCAATAACTT GTCCATAGTAATACTTGCTTGCG ACCTTCTGACGAGGGAACGTATGA ATGTGTAGTGCTTAAGTATGAAA AGATGCCTTTAAGCGGAACACTT GGCTGAGGTTACACTCTCCGTAA GGCGGACTTTCCTACGCCGTCTAT ATCCGACTTCGAGATACCACTTC TAACATTGACGCATCATTGCTC AACCTCAGGTGGTTTCCAGAGCC TCACTTGAGCTGGCTGGAGAAATGG CGAAGAACTTAACGCAATCAATAC CACGGTGTCCCAAGACCCGGAGAC AGAGCTGTACGCCGTGTATCCAA ACTGGATTTTAACATGACGACAAA TCATAGTTTCTATGTGTCTGATCAA TATGGGCATCTCAGGTGAATCAG ACTTTTAATTGGAACACTACCAA CAAGAGCACTTCCAGATAATCTG TTGCCAAGC (SEQ ID NO: 25)	IHVTKEVKEVATLSCGHNVSEELA QTRIYWQKEKKMVLTMMSGDMNI WPEYKNRTIFDITNNLSIVILALRPSD EGTYECVVLKYEKDAFKREHLAEV TLSVKADFPTPSISDFEIPSTNIRRIICS TSGGFPEPHLSWLENGEELNAINTTV SQDPETELYAVSSKLDNMTTNHSF MCLIKYGHLRVNQTFNWNNTKQEH FPDNLLPS (SEQ ID NO: 26)
CD80 transmembrane and intracellular domain	
Nucleotide	Amino acid
TGGGCGATAACTCTTATCTCCGTC AACGGTATCTTCGTAATTGTGCGC TCACCTATTGTTTCGCGCCTCGATG CCGAGAA (SEQ ID NO: 27)	WAITLISVNGIFVICCLTYCFAPRCRE (SEQ ID NO: 28)

TABLE 3-continued

Co-expressed CD80 and anti-CD3 T cell targeting molecules	
P2A self-cleaving peptide	
Nucleotide	Amino acid
GGCAGCGGCGCCACCAACTTCTCC CTGCTGAAGCAGGCCGGCGACGTG GAAGAAAACCCCTGGCCCC (SEQ ID NO: 29)	GSGATNFSLLKQAGDVEENPGP (SEQ ID NO: 30)
CD8a signal peptide	
Nucleotide	Amino acid
ATGGCCTTACCAGTGACCGCCTTG CTCTGCGCGCTGGCCTTGCTGCTCC ACGCCGCCAGGCCG (SEQ ID NO: 31)	MALPVTALLPLALLLHAARP (SEQ ID NO: 32)
Anti-CD3 Light chain variable region	
Nucleotide	Amino acid
GACATCCAGATGACCCAGACCACC TCCTCCCTGTCTGCCTCTCTGGGAG ACAGAGTCACCATCAGTTGCAGGG CAAGTCAGGACATTAGAAATTATT TAAACTGGTATCAACAGAAACCAG ATGGAACCTGTTAAACTCCTGATCT ACTACACATCAAGATTCACTCAG GAGTCCCATCAAAGTTCAGTGGCA GTGGGTCTGGAACAGATTATTCTC TCACCATTAGCAACCTGGAGCAAG AGGATATTGCCACTTACTTTTGCCA ACAGGGTAATACGCTTCCGTGGAC GTTGCTGGAGGCACCAAGCTGGA AATCAAACGGGCT (SEQ ID NO: 33)	DIQMTQTSSLSASLGDRVTISCRAS QDIRNYLNWYQQKPDGTVKLLIYYT SRLHSGVPSKFSGSGSGTDYSLTISN LEQEDIATYFCQQGNLTPWTFAGGT KLEIKRA (SEQ ID NO: 34)
G ₃ S linker	
Nucleotide	Amino acid
GGAGGCGGTAGTGGCGGTGGATCA GGTGGAGGCAGCGGTGGCGGATCT (SEQ ID NO: 35)	GGSGGGSGGGSGGGS (SEQ ID NO: 36)
Anti-CD3 heavy chain variable region	
Nucleotide	Amino acid
GAGGTGCAGCTCCAGCAGTCTGGA CCTGAGCTGGTGAAGCCTGGAGCT TCAATGAAGATATCTGCAAGGCT TCTGGTTACTCATTCACTGGCTACA CCATGAAC TGGGTGAAGCAGAGTC ATGGAAAGAACCTTGAGTGGATGG GACTTATTAATCCTTACAAAGGTG TTAGTACCTACAACCAGAAGTTCA AGGACAAGGCCACATTAAGTGTAG ACAAGTCATCCAGCACAGCCTACA TGGAACCTCCTCAGTCTGACATCTG AGGACTCTGCAGTCTATTACTGTG CAAGATCGGGTACTACGGTGATA GTGACTGGTACTTCGATGTCTGGG GCGCAGGGACCACGGTCACCGTCT CCTCA (SEQ ID NO: 37)	EVQLQQSGPELVKPGASMKISCKAS GYSFTGYTMNWVKQSHGNLEWM GLINPYKGVSTYNOKEFKDKATLTVD KSSSTAYMELLSLTSEDSAVYYCAR SGYYGDSWYFDVWGAGTTVTVSS (SEQ ID NO: 38)
CD8a hinge and transmembrane	
Nucleotide	Amino acid
ACCACACACCCAGCACCTAGACCA CCAACACCTGCCCAACCATCGCA TCGCAGCCACTGCTCTGCGCCCA GAGGCATGCCGCCAGCAGCTGGG GGCGCAGTGCACACAGGGGGCT GGACTTCGATGTGATATCTACAT	TTTPAPRPPTPAPTIASQPLSLRPEAC RPAAGGAVHTRGLDFACDIYIWAPL AGTCGVLLLSLVITLYC (SEQ ID NO: 40)

TABLE 3-continued

Co-expressed CD80 and anti-CD3 T cell targeting molecules
CTGGGCACCATTTGGCAGGGACTTG TGGGGTCCTTCTCCTGTCACTGGTT ATCACCCCTTACTGC (SEQ ID NO: 39)

Transgene

[0113] The lentiviral vectors described herein include a transgene encoding one or more proteins that is delivered to and expressed by a lymphocyte (e.g., T cell, B cell, or NK cell) targeted by the vector. In particular embodiments, the transgene encodes a chimeric antigen receptor (CAR) that includes: an extracellular domain comprising a binding domain that specifically binds to a target molecule; an intracellular signaling domain, wherein the intracellular signaling domain comprises an immunoreceptor tyrosine-based activation motif (ITAM); and a transmembrane domain connecting the extracellular domain and intracellular signaling domain.

[0114] Binding domains suitable for use in CARs of the present disclosure include any antigen-binding polypeptide. A binding domain may comprise an antibody or antigen binding fragment thereof, including for example, a full length heavy chain, Fab fragment, Fab', F(ab')₂, sFv, VH domain, VL domain, dAb, VHH, CDR, and scFv, specific for a target disease antigen. In certain embodiments, a CAR binding domain is murine, chimeric, human, or humanized. In further embodiments, CAR binding domain is an scFv having V_H and V_L regions that are human or humanized. In further embodiments, the CAR binding domain is an scFv having a (GGGS)_N linker or (GGGGGS)_N linker, wherein N=1-10, joining the V_H and V_L regions. In other embodiments, the CAR binding domain is an scFv having a linker comprising SEQ ID NO: 64 joining the V_H and V_L regions.

[0115] A target molecule that is bound by an extracellular domain of a CAR of the present disclosure may be found on or in association with a target cell. Exemplary target cells include a cancer cell, a cell associated with an autoimmune disease or disorder, a neurodegenerative disease, or with an inflammatory disease or disorder, an infectious microbe (e.g., bacteria, virus, or fungi), and an infected cell (e.g., virus-infected cell). A cell of an infectious organism, such as a mammalian parasite, is also contemplated as a target cell.

[0116] The extracellular domain included in a CAR as described herein can include a binding domain that targets one or more of a variety of target molecules. In certain embodiments, the CAR includes a binding domain that specifically binds a tumor antigen. In some such embodiments, the CAR binding domain specifically binds one or more of CD19, BCMA, alpha folate receptor, 5T4, Ab integrin, B7-H3, B7-H6, CAIX, CD20, CD22, CD23, CD30, CD33, CD38, CD44, CD44v6, CD44v7/8, CD52, CD70, CD79a, CD79b, CD80, CD123, CD138, CD171, CEA, CSPG4, EGFR, ErbB2 (HER2), EGFRvIII, EGP2, EGP40, EpCAM, FAP, fetal AchR, FLT3, Fra, GD2, GD3, Glypican-3 (GPC3), HLA-A1+MAGE1, HLA-A2+MAGE1, HLA-A3+MAGE1, HLA-A1+NY-ESO-1, HLA-A2+NY-ESO-1, HLA-A3+NY-ESO-1, HLADR, IL-11Ralpha, IL-13 Ralpha2, Lambda, Lewis-Y, Kappa, mesothelin, Muc1, Muc16, NCAM, NKG2d ligands, NY-ESO-1, PRAME,

PSCA, PSMA, ROR1, SSX, Survivin, TAG72, TEMs, VEGFR2, BAFF-R, Claudin18.2, CD86, FcRL5, GPRC5, and TACI.

[0117] In certain embodiments, the extracellular domain of a CAR encoded by the transgene of the lentiviral vectors described the present disclosure optionally comprises an extracellular, non-signaling spacer or linker domain between the binding domain and the transmembrane domain. Where included, such a spacer or linker domain may position the binding domain away from the host cell surface to further enable proper cell-to-cell contact, binding, and activation. An extracellular spacer domain is generally located between the extracellular binding domain and the transmembrane domain of the CAR. The length of the extracellular spacer may be varied to optimize target molecule binding based on the selected target molecule, selected binding epitope, binding domain size and affinity (see, e.g., Guest et al., *J. Immunother.* 28:203-11, 2005; PCT Publication No. WO 2014/031687). In certain embodiments, an extracellular spacer domain is an immunoglobulin hinge region (e.g., IgG1, IgG2, IgG3, IgG4, IgA, IgD). An immunoglobulin hinge region may be a wild type immunoglobulin hinge region or an altered wild type immunoglobulin hinge region. An altered IgG₄ hinge region is described in PCT Publication No. WO 2014/031687, which hinge region is incorporated herein by reference in its entirety. In some embodiments, an extracellular spacer domain comprises a modified IgG₄ hinge region having an amino acid sequence of ESKY-GPPCPPCP (SEQ ID NO: 118).

[0118] Other examples of hinge regions that may be used in the CARs described herein include the hinge region from the extracellular regions of type 1 membrane proteins, such as CD8a, CD4, CD28 and CD7, which may be wild-type or variants thereof. In some embodiments, an extracellular spacer domain comprises all or a portion of an immunoglobulin Fc domain selected from: a CH1 domain, a CH2 domain, a CH3 domain, or combinations thereof (see, e.g., PCT Publication WO2014/031687, which spacers are incorporated herein by reference in their entirety). In yet further embodiments, an extracellular spacer domain may comprise a stalk region of a type II C-lectin (the extracellular domain located between the C-type lectin domain and the transmembrane domain). Type II C-lectins include CD23, CD69, CD72, CD94, NKG2A, and NKG2D.

[0119] CARs of the present disclosure comprise a transmembrane domain that connects and is positioned between the extracellular domain and the intracellular signaling domain. The transmembrane domain typically ranges in length from about 15 amino acids to about 30 amino acids. The transmembrane domain is a hydrophobic alpha helix that transverses the host cell membrane and anchors the CAR in the host cell membrane. The transmembrane domain may be directly fused to the binding domain or to the extracellular spacer domain if present. In certain embodiments, the transmembrane domain is derived from an inte-

gral membrane protein (e.g., receptor, cluster of differentiation (CD) molecule, enzyme, transporter, cell adhesion molecule, or the like). The transmembrane domain can be selected from the same molecule as the extracellular domain or the intracellular signaling domain. In some embodiments, the transmembrane domain and the extracellular domain are each selected from different molecules. In some embodiments, the transmembrane domain and the intracellular signaling domain are each selected from different molecules. In yet other embodiments, the transmembrane domain, the extracellular domain, and the intracellular signaling domain are each selected from different molecules.

[0120] Exemplary transmembrane domains for use in CARs of the present disclosure include a CD28, CD2, CD4, CD8a, CD5, CD3e, CD38, CD3ζ, CD9, CD16, CD22, CD25, CD27, CD33, CD37, CD40, CD45, CD64, CD79a, CD79B, CD80, CD86, CD95 (Fas), CD134 (OX40), CD137 (4-1BB), CD150 (SLAMF1), CD152 (CTLA4), CD154 (CD40L), CD200R, CD223 (LAG3), CD270 (HVEM), CD272 (BTLA), CD273 (PD-L2), CD274 (PD-L1), CD278 (ICOS), CD279 (PD-1), CD300, CD357 (GITR), A2aR, DAP10, FcRα, FcRβ, FcRγ, Fyn, GAL9, KIR, Lck, LAT, LRP, NKG2D, NOTCH1, NOTCH2, NOTCH3, NOTCH4, PTCH2, ROR2, Ryk, Slp76, SIRPα, pTα, TCRα, TCRβ, TIM3, TRIM, LPA5, and Zap70 transmembrane domain. An exemplary CD8a transmembrane domain with CD8a hinge region comprises an amino acid sequence of SEQ ID NO: 52.

[0121] The intracellular signaling domain of a CAR is an intracellular effector domain and is capable of transmitting functional signals to a cell in response to binding of the extracellular domain of the CAR to a target molecule and activates at least one of the normal effector functions or responses of the immune cell, e.g., a T cell engineered to express the CAR. In some embodiments, the CAR induces a function of a T cell such as cytolytic activity or T helper activity, such as secretion of cytokines or other factors. The intracellular signaling domain may be any portion of an intracellular signaling molecule that retains sufficient signaling activity. In some embodiments, the intracellular signaling domain is obtained from an antigen receptor component (e.g., TCR) or costimulatory molecule. In some embodiments, a full-length intracellular signaling domain of an antigen receptor or costimulatory molecule is used. In some embodiments, a truncated portion of an intracellular signaling domain of an antigen receptor or costimulatory molecule is used, provided that the truncated portion retains sufficient signal transduction activity. In further embodiments, an intracellular signaling domain is a variant of a full length or truncated portion of an intracellular signaling domain of an antigen receptor co stimulatory molecule, provided that the variant retains sufficient signal transduction activity (i.e., is a functional variant).

[0122] In certain embodiments, the intracellular signaling domain of a CAR comprises an immunoreceptor tyrosine-based activation motif (ITAM) containing signaling domain. An ITAM containing signaling domain generally contains at least one (one, two, three, four, or more) ITAMs, which refer to a conserved motif of YXXL/I-X₆₋₈-YXXL/I. An ITAM containing signaling domain may initiate T cell activation signaling following antigen binding or ligand engagement. ITAM-signaling domains include, for example, intracellular signaling domains of CD3γ, CD3δ, CD3ε, CD3ζ, CD5, CD22, CD79a, CD278 (ICOS), DAP10, DAP12, FcRγ, and

CD66d. Exemplary CD3ζ signaling domains that may be used in CARs of the present disclosure comprise an amino acid sequence of SEQ ID NO: 56 or SEQ ID NO: 122.

[0123] CAR intracellular signaling domains optionally comprise a costimulatory signaling domain, which, when activated in conjunction with a primary or classic (e.g., ITAM-driven) activation signal, promotes or enhances T cell response, such as T cell activation, cytokine production, proliferation, differentiation, survival, effector function, or combinations thereof. Costimulatory signaling domains for use in CARs include, for example, CD27, CD28, CD40L, GITR, NKG2C, CARD1, CD2, CD7, CD27, CD30, CD40, CD54 (ICAM), CD83, CD134 (OX-40), CD137 (4-1BB), CD150 (SLAMF1), CD152 (CTLA4), CD223 (LAG3), CD226, CD270 (HVEM), CD273 (PD-L2), CD274 (PD-L1), CD278 (ICOS), DAP10, LAT, LFA-1, LIGHT, NKG2C, NKD2C, SLP76, TRIM, ZAP70, or any combination thereof. In a particular embodiment, the costimulatory signaling domain comprises a OX40, CD2, CD27, CD28, ICAM-1, LFA-1 (CD11a/CD18), ICOS (CD278), or 4-1BB (CD137) signaling domain. An exemplary 4-1BB costimulatory signaling domain comprises an amino acid sequence of SEQ ID NO: 54. In certain embodiments, a CAR comprises one, two, or more costimulatory signaling domains.

[0124] In some embodiments, a CAR of the present disclosure is a first generation CAR, a second generation CAR, or a third generation CAR. A first generation CAR generally has an intracellular signaling domain comprising an intracellular signaling domain of CD3ζ, FcγRI, or other ITAM-containing activating domain to provide a T cell activation signal. Second generation CARs further comprise a costimulatory signaling domain (e.g., a costimulatory signaling domain from an endogenous T cell costimulatory receptor, such as CD28, 4-1BB, or ICOS). Third generation CARs comprise an ITAM-containing activating domain, a first costimulatory signaling domain and a second costimulatory signaling domain.

[0125] In some embodiments, one or more of the extracellular domain, the binding domain, the linker, the transmembrane domain, the intracellular signaling domain, or the costimulatory domain comprises junction amino acids. “Junction amino acids” or “junction amino acid residues” refer to one or more (e.g., about 2-20) amino acid residues between two adjacent domains, motifs, regions, modules, or fragments of a protein, such as between a binding domain and an adjacent linker, between a transmembrane domain and an adjacent extracellular or intracellular domain, or on one or both ends of a linker that links two domains, motifs, regions, modules, or fragments (e.g., between a linker and an adjacent binding domain or between a linker and an adjacent hinge). Junction amino acids may result from the construct design of a fusion protein (e.g., amino acid residues resulting from the use of a restriction enzyme site or self-cleaving peptide sequences during the construction of a polynucleotide encoding a fusion protein). For example, a transmembrane domain of a fusion protein may have one or more junction amino acids at the amino-terminal end, carboxy-terminal end, or both.

[0126] In specific embodiments, the transgene encodes an anti-CD19 CAR molecule. In such embodiments, the extracellular binding domain may include an scFv as detailed in Table 4, having an anti-CD19 light chain variable region of SEQ ID NO: 46, a G4S linker of SEQ ID NO: 48, and an anti-CD19 heavy chain variable region of SEQ ID NO: 50.

An anti-CD19 CAR useful in the context of the present description may include a CD8a hinge and transmembrane domain of SEQ ID NO: 52. In certain embodiments, the anti-CD19 CAR includes a 4-1BB costimulatory domain according to SEQ ID NO: 54 and a CD3 ζ effector domain according to SEQ ID NO: 56. In a specific embodiment, an anti-CD19 CAR encoded by the transgene has an amino acid sequence according to SEQ ID NO: 42 or SEQ ID NO: 42 absent the signal peptide of SEQ ID NO: 44.

[0127] In other embodiments, the transgene encodes an anti-CD19 CAR molecule and extracellular binding domain may include an scFv having an anti-CD19 light chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 46, and an anti-CD19 heavy chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%,

98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 50. An anti-CD19 CAR useful in the context of the present description may include a CD8a hinge and transmembrane domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 52. In certain embodiments, the anti-CD19 CAR includes a 4-1BB costimulatory domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 54 and a CD3 ζ effector domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 56. In a specific embodiment, an anti-CD19 CAR encoded by the transgene has an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 42 or SEQ ID NO: 42 absent the signal peptide of SEQ ID NO: 44.

TABLE 4

Anti-CD19 CAR molecule	
Full molecule	
Nucleotide	Amino acid (signal peptide underlined)
ATGGCCTTACCAGTGACCGCCTTG	MALPVTALLLPLALLHAARPDIQM
CTCCTGCCGCTGGCCTTGCTGCTCC	TQTSSLSASLGDRVTISCRASQDISK
ACGCCGCCAGGCCGGACATCCAGA	YLNWYQQKPDGTVKLLIYHTSRLHS
TGACACAGACTACATCCTCCTGT	GVPSRFSGSGGTDYSLTISNLEQEDI
CTGCCTCTCTGGGAGACAGAGTCA	ATYFCQQGNTLPYTFGGGKLEITG
CCATCAGTTGCAGGGCAAAGTCAGG	GGGSGGGSGGGSEVKLQESGPG
ACATTAGTAAATATTTAAATTGGT	LVAPSQSLSVTCTVSGVSLPDYGVS
ATCAGCAGAAACCAGATGGAACATG	WIRQPPRKGLEWLGVIWGSETTYYN
TTAAACTCCTGATCTACCATACATC	SALKSRLTI IKDNSKSQVFLKMNSLQ
AAGATTACACTCAGGAGTCCCATC	TDDTAIYYCAKHYYGGSYAMDY
AAGGTTCACTGTCAGTGGGTCTGG	WGQTSVTVSSTTTPAPRPPTPAPTI
AACAGATTATTCTCTACCATTAG	ASQPLSLRPEACRPAAGAVHTRGL
CACCTGGAGCAAGAAGATATTGC	DFACDIYIWAPLAGTCGVLLLSLVIT
CACCTACTTTTGCCAACAGGGTAA	LYCKRGRKLLYIFKQPFMRPVQTT
TACGCTTCGTACACGTTTCGGAGG	QEEDGCS CRFP EEEEGGCELVRKFSR
GGGACCAAGCTGGAGATCACAG	SADAPAYKQGQNQLYNELNLGRRE
GTGGAAGTGGATCGGCGGTGGTG	EYDVLDRRGRDP EMMGGKPRKNP
GGTCGGGTGGCGCGGATCTGAGG	QEGLYNELQKDKMAEAYSEI GMKG
TGAACTGCAAGAGTCAGGACCTG	ERRRGKGDGLYQGLSTATKDTYD
GCCTGGTGGCGCCCTCACAGAGCC	ALHMQALPPR (SEQ ID NO: 42)
TGTCGCTCACATGCACTGTCTCAG	
GGGTCTCATTACCCGACTATGGTG	
TAAGCTGGATTGCGCCAGCCTCCAC	
GAAAGGGTCTGGAGTGGCTGGGA	
GTAATATGGGTTAGTGAACCCACA	
TACTATAATTGAGCTCTCAAATCC	
AGACTGACCATCATCAAGGACAAAC	
TCCAAGAGCCAAGTTTTCTTAAAA	
ATGAACAGCTGCAAACTGATGAC	
ACAGCCATTACTACTGTGCCAAA	
CATTATTACTACGGTGGTAGCTAT	
GCTATGGACTACTGGGCCAAGGA	
ACCTCAGTCACCGTCTCCTCAACC	
ActACaCCAGCaCct aGACCACCAAC	
ACctGCGCCaACCATCGCaTCGCAG	
CCaCTGTCTCTCGCCCGAGGCaT	
GCCGCGCAGCaGctGGGGGCGCAG	
TGCACACaAGGGGGCTGGACTTCG	
CaTGATATCTACATCTGGGCaCC	
aTTGGCaGGGACTTGTGGGTCTCT	
CTCCTGTCACTGGTTATCACCCCTTT	
ACTGCAAAAGGGGCGAAAAGAAA	
CTCCTGTATATATTCAAACAACCA	
TTTATGAGACCACTACAACTACT	
CAAGAGGAAGATGGCTGTAGCTGC	
CGATTTCCAGAAGAAGAAGAGG	
AGGATGTGAACAGAGTGAAGTT	
CAGCAGGAGCGCAGACCCCCCGC	

TABLE 4-continued

Anti-CD19 CAR molecule	
GTACAAGCAGGGCCAGAACCAGCT CTATAACGAGCTCAATCTAGGACG AAGAGAGGAGTACGATGTTTTGGA CAAGAGACGTGGCCGGGACCCCTGA GATGGGGGGAAGCCGAGAAGGA AGAACCCCTCAGGAAGGCCTGTACA ATGAACTGCAGAAAGATAAGATG GCGGAGGCCTACAGTGAGATTGGG ATGAAAGGCGAGCGCCGAGGGG CAAGGGGCACGATGGCCTTTACCA GGGTCTCAGTACAGCCACCAAGGA CACCTACGACGCCCTTCACATGCA GGCCCTGCCCCCTCGC (SEQ ID NO: 41)	
CD8a signal peptide	
Nucleotide	Amino acid
ATGGCCTTACCAGTGACCGCCTTG CTCCTGCCGCTGGCCTGCTGCTCC ACGCCGCCAGGCCG (SEQ ID NO: 43)	MALPVTALLLPLALLLHAARP (SEQ ID NO: 44)
Anti-CD19 light chain variable region	
Nucleotide	Amino acid
GACATCCAGATGACACAGACTACA TCCTCCCTGTCTGCCTCTCTGGGAG ACAGAGTCACCATCAGTTGCAGGG CAAGTCAGACATTAGTAAATATT TAAATTGGTATCAGCAGAAACCAG ATGGAAGTGTAAACTCCTGATCT ACCATACATCAAGATTACACTCAG GAGTCCCATCAAGTTTCAGTGGCA GTGGGTCTGGAACAGATTATTCTC TCACCATTAGCAACCTGGAGCAAG AAGATATTGCCACTTACTTTTGCCA ACAGGGTAATACGCTTCCGTACAC GTTCGGAGGGGGACCAAGCTGG AGATCACA (SEQ ID NO: 45)	DIQMTQTSSLSASLGDRVTISCRAS QDISKYLNWYQQKPDGTVKLLIYHT SRLHSGVPSRFGSGSGTDYSLTISN LEQEDIATYFCQQGNTLPYTFGGGT KLEIT (SEQ ID NO: 46)
G ₄ S Linker	
Nucleotide	Amino acid
GGTGgaGGTGGaTCGGGCGGTGGT GGGTCGGGTGGCGGCGGATCT (SEQ ID NO: 47)	GGGGSGGGSGGGGS (SEQ ID NO: 48)
Anti-CD19 heavy chain variable region	
Nucleotide	Amino acid
GAGGTGAAACTGCAGGAGTCAGG ACCTGGCCTGGTGGCGCCCTCACA GAGCCTGTCCGTCACATGCACTGT CTCAGGGGTCTCATTACCCGACTA TGGTGTAAAGCTGGATTGCCAGCC TCCACGAAAGGGTCTGGAGTGGCT GGGAGTAATATGGGGTAGTGAAAC CACATACTATAATTAGCTCTCAA ATCCAGACTGACCATCATCAAGGA CAACTCCAAGAGCCAAGTTTCTT AAAAATGAACAGTCTGCAAACTGA TGACACAGCCATTACTACTGTGC CAAACATTATTACTACGGTGGTAG CTATGCTATGGACTACTGGGGCCA AGGAACCTCAGTCACCGTCTCCTC A (SEQ ID NO: 49)	EVKLQESGPGLVAPSQSLSVTCTVS GVS LPDYGVSWIRQPPRKLEWLGV IWGSETTYNSALKSRLTIKDNSKS QVFLKMNSLQTD DTAIYYCAKHYY YGGSYAMDYWGQGSVT VSS (SEQ ID NO: 50)

TABLE 4-continued

Anti-CD19 CAR molecule	
CD8a Hinge and Transmembrane	
Nucleotide	Amino acid
ACCACTACACCAGCACCTAGACCA	TTTPAPRPPTPAPTIASQPLSLRPEAC
CCAACACCTGCGCCACCATCGCA	RPAAGGAVHTRGLDFACDIYIWAPL
TCGCAGCCACTGTCTCTGCGCCCA	AGTCGVLILLSLVITLYC (SEQ ID NO:
GAGGCATGCCGCCAGCAGCTGGG	52)
GGCGCAGTGCACACAGGGGGCT	
GGACTTCGCATGTGATATCTACAT	
CTGGGCACCAATTGGCAGGGACTTG	
TGGGGTCCTTCTCCTGTCACTGGTT	
ATCACCCCTTTACTGC (SEQ ID NO:	
51)	
4-1BB costimulatory domain	
Nucleotide	Amino acid
AAACGGGGCAGAAAGAACTCCT	KRGRKKLLYIFKQPFMRPVQTQEE
GTATATATTCAAACAACATTTAT	DGCSCRFPEEEEGGCEL (SEQ ID NO:
GAGACCAGTACAACTACTCAAGA	54)
GGAAGATGGCTGTAGCTGCCGATT	
TCCAGAAGAAGAAGAGGAGGAT	
GTGAAGT (SEQ ID NO: 53)	
CD3 zeta effector domain #1	
Nucleotide	Amino acid
AGAGTGAAATTTCAGCAGGAGCGC	RVKFSRSADAPAYKQGQNQLYNEL
AGACGCCCCCGCGGTACAAGCAGGG	NLGRREEYDVLDRRRGRDPEMGGK
CCAGAACCACTCTATAACGAGCT	PRRKNPQEGLYNELQDKMAEAYS
CAATCTAGACGAAGAGAGGAGT	EIGMKGERRRRGKHDGLYQGLSTA
ACGATGTTTTGGACAAGAGACGTG	TKDTYDALHMQALPPR (SEQ ID NO:
GCCGGGACCCTGAGATGGGGGGA	56)
AAGCCGAGAAGGAAGAACCCCTCA	
GGAAGGCCCTGTACAATGAAGTGA	
GAAAGATAAGATGGCGGAGGCCT	
ACAGTGAGATTGGGATGAAAGGC	
GAGCGCCGGAGGGGCAAGGGGCA	
CGATGGCCTTTACCAGGGTCTCAG	
TACAGCCACCAAGGACCTTACGA	
CGCCCTTACATGCAGGCCCTGCC	
CCCTCGC (SEQ ID NO: 55)	

[0128] In other embodiments, the transgene encodes an anti-BCMA CAR molecule. In such embodiments, the extracellular binding domain may include an scFv as detailed in Table 5, having an anti-BCMA light chain variable region of SEQ ID NO: 62, a linker of SEQ ID NO: 64, and an anti-BCMA heavy chain variable region of SEQ ID NO: 66. An anti-BCMA CAR useful in the context of the present description may include a CD8a hinge and transmembrane domain of SEQ ID NO: 52. In certain embodiments, the anti-BCMA CAR includes a 4-1BB costimulatory domain according to SEQ ID NO: 54 and a CD3 ζ effector domain according to SEQ ID NO: 122. In a specific embodiment, an anti-BCMA CAR encoded by the transgene has an amino acid sequence according to SEQ ID NO: 58 or SEQ ID NO: 58 absent the signal peptide of SEQ ID NO: 60.

[0129] In some embodiments, the transgene encodes an anti-BCMA CAR molecule. In such embodiments, the extracellular binding domain may include an scFv having an anti-BCMA light chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%,

97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 62, and an anti-BCMA heavy chain variable region that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 66. An anti-BCMA CAR useful in the context of the present description may include a CD8a hinge and transmembrane domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 52. In certain embodiments, the anti-BCMA CAR includes a 4-1BB costimulatory domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 54 and a CD3 ζ effector domain that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 122. In a specific embodiment, an anti-BCMA CAR encoded by the transgene has an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 58 or SEQ ID NO: 58 absent the signal peptide of SEQ ID NO: 60.

TABLE 5

Anti-BCMA CAR molecule	
Full molecule	
Nucleotide	Amino acid (signal peptide underlined)
ATGGCACTCCCGTCACCGCCCTT	MALPVTALLPLALLLHAARPDIVL
CTCTTGCCCCCTCGCCCTGCTGCTGC	TQSPPSLAMSLGKRATISCRASESVTI
ATGCTGCCAGGCCCCGACATTGTGC	LGSHLIHWYQQKPGQPPTLLIQLASN
TCACTCAGTCACCTCCCAGCCTGG	VQTGVPARFSGSGSRDFTLTIDPVE
CCATGAGCCTGGGAAAAGGGCC	EDDVAVYYCLQSRITPRTFGGKTL
ACCATCTCCTGTAGAGCCAGTGAG	EIKGSTSGSGKPGSGEGSTKGQIQLV
TCCGTCACAATCTTGGGGAGCCAT	QSGPELKKPGETVKISCKASGYFTD
CTTATTCACTGGTATCAGCAGAAG	YSINWVKRAPGKGLKWMGWINTET
CCCGGCGAGCCTCCAACCTTCTT	REPAYAYDFRGRFAFSLETSASTAY
ATTCACTCGCGTCAAACGTCCAG	LQINNLKYEDTATYFCALDYSYAM
ACGGGTGTACCTGCCAGATTTTCT	DYWGQGSTSVTVSSAAATTPAPRPP
GGTAGCGGGTCCCGCACTGATTTT	TPAPTIASQPLSLRPEACRPAAGGAV
ACACTGACCATAGATCCAGTGGAA	HTRGLDFACDIYIWAPLAGTCGVLL
GAAGACGATGTGGCGGTGATTAT	LSLVITLYCKRGRKLLYIFKQPFMR
TGCTTGACAGACAGAACGATTCTT	PVQTTQEEEDGSCSRFPFEEEGGCELR
CGCACATTTGGTGGGGGTACTAAG	VKFSRSADAPAYQQGQNQLYNELN
CTGGAGATTAAAGGAAGCACGTCC	LGRRREYDVLDRRGRDPEMGGKP
GGCTCAGGGAAGCCGGGCTCCGGC	RRKNPQEGLYNELQDKMAEAYSEI
GAGGGAAGCACGAAGGGGCAAAT	GMKGERRRGKGGHDGLYQGLSTATK
TCAGCTGGTCCAGAGCGGACCTGA	DTYDALHMQALPPR (SEQ ID NO:
GCTGAAAAAACC CGCGAGACTGT	58)
TAAGATCAGTTGTAAGCATCTGG	
CTATACCTTCACCGACTACAGCAT	
AAATTGGGTGAAACGAGCCCTGG	
AAAGGGCTCAAATGGATGGGTG	
GATCAATACCGAACTAGGAGCC	
TGCTTATGCATATGACTTCCGCGG	
GAGATTGCCTTTTCACTCGAGAC	
ATCTGCCTCTACTGTACTCTCAA	
ATAAACAACTCAAGTATGAAGAT	
ACAGCCACTTACTTTTGCGCCCTCG	
ACTATAGTTACGCCATGGACTACT	
GGGGACAGGGAACCTCCGTTACCG	
TCAGTTCCGCGGCCAACCACAA	
CACCTGCTCCAAGGCCCCCACAC	
CCGCTCCAATATAGCCAGCCAAC	
CATTGAGCCTCAGACCTGAAGCTT	
GCAGGCCCCGAGCAGGAGGCGCC	
GTCCATACGCGAGGCCTGGACTTC	
GCGTGTGATATTTATATTTGGGCA	
CCTTTGGCCGGAACATGTGGGGTG	
TTGCTTCTCTCCCTTGTGATCACTC	
TGTATTGTAAGCGCGGAGAAAGA	
AGCTCCTGTACATCTTCAAGCAGC	
CTTTTATGCGACCTGTGCAACCA	
CTCAGGAAGAAGATGGGTGTTTAT	
GCCGCTTCCCCGAGGAGGAAGAAG	
GAGGGTGTGAAGTGAAGGTGAAT	
TTTCTAGAAGCGCCGATGCTCCCG	
CATATCAGCAGGTCAGAATCAGC	
TCTACAATGAATGAATCTCGGCA	
GGCGAGAAGAGTACGATGTTCTGG	
ACAAGAGACGGGGCAGGGATCCC	
GAGATGGGGGGAAGCCCCGGAG	
AAAAAATCCTCAGGAGGGGTGTA	
CAATGAGCTGCAGAAGGACAAGA	
TGGCTGAAGCCTATAGCGAGATCG	
GAATGAAAGGCGAAGACGCAGA	
GGCAAGGGCATGACGGTCTGTAC	
CAGGGTCTCTCTACAGCCACCAAG	
GACACTTATGATGCGTTGCATATG	
CAAGCCTTGCCACCCCGC (SEQ ID	
NO: 57)	

TABLE 5-continued

Anti-BCMA CAR molecule	
CD8a signal peptide	
Nucleotide	Amino acid
ATGGCACTCCCGTCACCGCCCTT CTCTTGCCCTCGCCTGCTGCTGC ATGCTGCCAGGCC (SEQ ID NO: 59)	MALPVTALLLPALLLHAARP (SEQ ID NO: 60)
Anti-BCMA light chain variable region	
Nucleotide	Amino acid
GACATTGTGCTCACTCAGTCACCT CCCAGCCTGGCCATGAGCCTGGGA AAAAGGGCCACCATCTCCTGTAGA GCCAGTGAGTCCGTCACAATCTTG GGGAGCCATCTTATTCAGTGGTAT CAGCAGAAGCCCGGCAGCCTCCA ACCTTCTTATTACAGTCGCGTCAA ACGTCCAGACGGGTACCTGCCA GATTTTCTGGTAGCGGTCCTCCGA CTGATTTTACACTGACCATAGATC CAGTGAAGAAGACGATGTGGCC GTGTATTATTGTCTGAGAGCAGA ACGATTCCTCGCACATTTGGTGGG GGTACTAAGCTGGAGATTAAG (SEQ ID NO: 61)	DIVLTQSPPSLMSLGKRATISCRAS ESVTILGSHLIHWYQQKQPPTLLI QLASNVQTGVPARFSGSGSRDFTL TIDPVEEDDVAVYCLQSRITPRTFG GGTKLEIK (SEQ ID NO: 62)
218 Linker	
Nucleotide	Amino acid
GGAAGCACGTCCGGCTCAGGGAA GCCGGGCTCCGGCGAGGGAAGCA CGAAGGGG (SEQ ID NO: 63)	GSTSGSGKPGSGEGSTKG (SEQ ID NO: 64)
Anti-BCMA heavy chain variable region	
Nucleotide	Amino acid
CAAATTCAGCTGGTCCAGAGCGGA CCTGAGCTGAAAAACCCGGCGAG ACTGTTAAGATCAGTTGTAAAGCA TCTGGCTATACCTTCACCGACTAC AGCATAAATTGGGTGAAACGAGCC CCTGGAAGGGCCTCAATGGATG GGTTGGATCAATACCGAACTAGG GAGCCTGCTTATGCATATGACTTC CGCGGGAGATTGCGCTTTTCACTC GAGACATCTGCCTCTACTGCTTAC CTCCAAATAACAACCTCAAGTAT GAAGATACAGCCACTTACTTTTGC GCCCTCGACTATAGTTACGCCATG GACTACTGGGACAGGGAACCTCC GTTACCGTCAGTTCC (SEQ ID NO: 65)	QIQLVQSGPELKKPGETVKISCKASG YFTDYSINWVKRAPGKGLKWMG WINTETREPAYAYDFRGRFAFSLETS ASTAYLQINNLYEDTATYFCALDY SYAMDYWGQGSTSVTVSS (SEQ ID NO: 66)
CD8a Hinge and Transmembrane	
Nucleotide	Amino acid
ACCACAACCTGCTCCAAGGCC CCCACACCGCTCCAACCTATAGCC AGCCAACCATGAGCCTCAGACCT GAAGCTTGAGCCCGCAGCAGGA GGCGCGTCCATACGCGAGGCGTG GACTTCGCGTGTGATATTTATATTT GGGCACCTTTGGCCGAACATGTG GGGTGTGCTTCTCTCCCTGTGAT CACTCTGTATTGT (SEQ ID NO: 119)	TTTPAPRPPTPAPTIASQPLSLRPEAC RPAAGGAVHTRGLDFACDIYIWAPL AGTCGVLLLSLVITLYC (SEQ ID NO: 52)

TABLE 5-continued

Anti-BCMA CAR molecule	
4-1BB costimulatory domain	
Nucleotide	Amino acid
AAGCGCGGAGAAAGAAGCTCCT	KRGRKLLYIFKQPFMRPVQTQEE
GTACATCTTCAAGCAGCCTTTTATG	DGCCRFPPEEEGGCEL (SEQ ID NO: 54)
CGACCTGTGCAAAACCACTCAGGAA	
GAAGATGGGTGTTTCATGCCGCTTC	
CCCGAGGAGGAAGAAGGAGGGTG	
TGAACTG (SEQ ID NO: 120)	
CD3 zeta effector domain #2	
Nucleotide	Amino acid
AGGGTGAAATTTTCTAGAAGCGCC	RVKFSRSADAPAYQQGQNQLYNEL
GATGCTCCCGCATATCAGCAGGGT	NLGRREEYDVLDKRRGRDPEMGGK
CAGAATCAGCTCTACAATGAATTG	PRRKNPQEGLYNELQDKMAEAYS
AATCTCGGCAGCGAGAAGAGTAC	EIGMKGERRRKGHDGLYQGLSTA
GATGTTCTGGACAAGAGACGGGGC	TKDITYDALHMQALPPR (SEQ ID NO: 122)
AGGGATCCCGAGATGGGGGAAA	
GCCCCGGAGAAAAATCCTCAGGA	
GGGGTTGTACAATGAGCTGCAGAA	
GGACAAGATGGCTGAAGCCTATAG	
CGAGATCGGAATGAAAGGCGAAA	
GACGCAGAGGCAAGGGCATGAC	
GGTCTGTACCAGGGTCTCTCTACA	
GCCACCAAGGACACTTATGATCGC	
TTGCATATGCAAGCCTTGCCACCC	
CGC (SEQ ID NO: 121)	

[0130] CARs of the present disclosure may comprise polynucleotide sequences derived from any mammalian species, including humans, primates, cows, horses, goats, sheep, dogs, cats, mice, rats, rabbits, guinea pigs, pigs, transgenic species thereof, or any combination thereof. In some embodiments, the chimeric antigen receptor is murine, chimeric, human, or humanized.

Methods for Producing Lentiviral Vectors

[0131] Methods for producing the LVV disclosed are provided herein. LVV have been developed and based on the retroviral genome by combining its components into recombinant plasmid DNA vectors. The plasmid DNA vectors can then be transfected into producer cell lines to transfer genes required for lentiviral particle production. LVV packaging systems generally comprise a transfer plasmid encoding the transgene of interest, an envelope plasmid (e.g., VSV-G), and packaging plasmid(s). Second generation LVV packaging systems contain a single packaging plasmid encoding Gag, Pol, Rev, and Tat genes and a separate Env plasmid. In some embodiments, the methods described herein utilize a third generation vector system for producing LVV. Third generation vector production systems improve upon the safety of 2nd generation LVV packaging systems and are typically four plasmid systems that include a transgene plasmid combined with three packaging plasmids, VSV-G, GagPol, and Rev. Thus, Rev and GagPol are separated into two plasmids. Tat is also eliminated from 3rd generation LVV packaging systems by the addition of a chimeric 5' LTR fused to a heterologous promoter (e.g., CMV or RSV promoter) on the transfer plasmid. The gag gene encodes a Gag polyprotein precursor comprising structural proteins of the lentivirus, including matrix, capsid, and nucleocapsid. The pol gene encodes a Pol polyprotein precursor supplying

enzymatic functions of the lentivirus essential for replication, including protease, reverse transcriptase, and integrase. The rev gene encodes the Rev protein which binds to the Rev Response Element (RRE) to allow nuclear export of unspliced and singly spliced HIV RNA during viral replication. The Gag and Pol polyprotein precursors are cleaved during production of viral particles.

[0132] Producer cells that may be used for making LVV of the present disclosure include human embryonic kidney (HEK) 293 cells and derivatives thereof. Producer cells may be an adherent cell line, such as HEK293T producer cells, or a suspension cell line, such as HEK293T/17 SF producer cells.

[0133] The transgene plasmid is combined with the three packaging plasmids, GagPol plasmid, VSV-G env plasmid, and Rev plasmid, and used to transfect producer cells, such as HEK293 producer cells. In some embodiments, the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid. In some embodiments, the ratios of the packaging plasmids are determined by mass. In some embodiments, the mass of each of the transgene plasmid and GagPol plasmid are higher than the mass of each of the VSV-G env plasmid and Rev plasmid. In some embodiments, the defined ratio of transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid is at about 1:1:1:1 to about 5:4:1:1. In some embodiments, the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 2:1:1:1 to about 5:4:1:1. In some embodiments, the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 2:1:1:1. In some embodiments, the producer cells are transfected

with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 3:1:1:1. In some embodiments, the the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 3.125:3.125:2.5:1.25. In some embodiments, the the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 4:2:1:1. In some embodiments, the the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 5:4:1:1. Productive lentiviral particles are harvested from the producer cell culture media. Exemplary materials and methods for producing LVV particles are described in *Production of Lentiviral Vectors*, Merten et al., Molecular Therapy—Methods & Clinical Development (2016), 3, 16017, the contents of which are incorporated herein by reference. Third generation production systems are used for research and development and clinical purposes. Schematic representations of helper plasmids suitable for use in a third generation LVV production system are provided in FIG. 1.

[0134] The transgene plasmid encodes the only genetic materials that is transferred to the lymphocytes (e.g., T cells

encodes a first signal peptide, a polynucleotide that encodes a targeting protein, a polynucleotide that encodes one of an internal ribosomal entry site (IRES), a furin cleavage site, or a viral 2A peptide, a polynucleotide that encodes a second signal peptide, and a polynucleotide that encodes a mutated VSV-G envelope protein. In certain embodiments, the polynucleotide encoding the mutant VSV-G envelope protein is positioned 5' to the polynucleotide encoding the targeting protein. In other embodiments, the polynucleotide encoding the mutant VSV-G envelope protein is positioned 3' to the polynucleotide encoding the targeting protein. The polynucleotide that encodes the targeting protein and the polynucleotide that encodes the mutated VSV-G are separated in the tandem cassette by the polynucleotide that encodes the IRES, furin cleavage site, or viral 2A peptide, which allows for co-expression of the two proteins from a single mRNA. In certain embodiments, a viral 2A peptide is a porcine teschovirus-1 (P2A), *Thosea asigna* virus (T2A), equine rhinitis A virus (E2A), foot-and-mouth disease virus (F2A), or a variant thereof. Specific examples of amino acid and nucleotide sequences for the viral 2A peptides are provided in Table 7.

TABLE 7

2A sequences		
T2A peptide from <i>Thosea asigna</i> virus capsid protein	EGRGSLLTCGD VEENPGP (SEQ ID NO: 79)	GAGGGCAGGGGAAGTCTTC TAACATGCGGGGACGTGGA GGAAATCCCGGCCCC (SEQ ID NO: 80)
P2A peptide from porcine teschovirus-1 polyprotein	ATNFSLLKQAG DVEENPGP (SEQ ID NO: 81)	GCCACGAACCTCTCTGT AAAGCAAGCAGGAGACGT GGAAGAAACCCCGTCTT (SEQ ID NO: 82)
F2A peptide from foot-and-mouth disease virus polyprotein	VKQTLNFDLLK LAGDVESNPGP (SEQ ID NO: 83)	GTGAAACAGACTTTGAATT TTGACCTTCTCAAGTTGGC GGGAGACGTCGAGTCCAAC CCTGGGCC (SEQ ID NO: 84)
E2A peptide from equine rhinitis A virus polyprotein	QCTNYALLKLA GDVESNPGP (SEQ ID NO: 85)	CAGTGTAATAATTATGCTC TCTTGAAATTGGCTGGAGA TGTTGAGAGCAACCCAGGT CCC (SEQ ID NO: 86)

or NK cells) targeted by the resulting LVV and comprises a lentiviral backbone containing the transgene expression cassette flanked by cis-acting elements for encapsidation, reverse transcription and integration. In specific embodiments, the transgene plasmid includes a deletion in the U3 element of the 3' LTR that causes loss of transcriptional capacity of the viral LTR once transferred into the target cells. The expression cassette included in the transgene plasmid may encode a single heterologous protein (e.g., a single CAR as described herein) or multiple heterologous proteins (e.g., multiple CARs as described herein) for introduction into and expression by the target cell.

[0135] In some embodiments of a four plasmid LVV system, the VSV-G env plasmid includes a tandem expression cassette that encodes a mutated VSV-G envelope protein and a targeting protein as disclosed herein. In specific embodiments, the tandem expression cassette included in the VSV-G env plasmid includes a polynucleotide that

[0136] In certain embodiments, the present disclosure provides a method for producing a lymphocyte targeted lentiviral vector according to any one of the preceding claims, the method comprising: transfecting a producer cell with a GagPol plasmid, a Rev plasmid, a transgene plasmid, and a VSV-G env plasmid, wherein; the GagPol plasmid comprises a one or more polynucleotides encoding a lentiviral gag gene and a lentiviral pol gene and is capable of expressing a lentiviral gag polyprotein and a lentiviral pol polyprotein within the producer cell;

[0137] the Rev plasmid comprises a polynucleotide encoding a lentiviral rev gene and is capable expressing a lentiviral rev protein within the producer cell; the transgene plasmid comprises an expression cassette comprising a polynucleotide encoding a CAR; and the VSV-G env plasmid comprises a tandem expression cassette, wherein the tandem expression cassette comprises a polynucleotide encoding a mutated VSV-G

envelope protein as disclosed herein and a polynucleotide encoding a lymphocyte targeting protein as disclosed herein and the VSV-G env plasmid is capable of expressing the mutated VSV-G envelope protein and the lymphocyte targeting protein within the producer cell; culturing the producer cell in a culture medium; and harvesting lentiviral vector from the culture medium.

[0138] In some embodiments, the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid. In some embodiments, the defined ratio of transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 1:1:1:1 to about 5:4:1:1. In some embodiments, the the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 2:1:1:1 to about 5:4:1:1. In some embodiments, the the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 3:1:1:1. In some embodiments, the the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 3.125:3.125:2.5:1.25. In some embodiments, the the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 4:2:1:1. In some embodiments, the the producer cells are transfected with a defined ratio of the

transgene plasmid, GagPol plasmid, VSV-G env plasmid, and Rev plasmid at about 5:4:1:1.

[0139] In one embodiment, the tandem expression cassette of the VSV-G env plasmid encodes an anti-CD3 targeting protein and a mutated VSV-G envelope protein. For example, in a specific embodiment, the tandem expression cassette may comprise a polynucleotide that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 67, which encodes an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 68. Such an embodiment is described in more detail in Table 8. In another embodiment, the tandem expression cassette may comprise a polynucleotide that has about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 67 absent the signal peptide of SEQ ID NO: 31, which encodes an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 68 absent the signal peptide of SEQ ID NO: 32. As illustrated in Table 8, the anti-CD3 targeting protein is separated from the mutated VSV-G envelope protein via a P2A self-cleaving peptide. In certain embodiments, that P2A peptide may be encoded by a nucleic acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 69, and the P2A peptide may have an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 70.

TABLE 8

Anti-CD3 targeting molecule co-expressed with mutant VSV-G	
Full molecule	
Nucleotide	Amino acid (signal peptide underlined)
ATGGCCTTACCACTGACCGCCTTG	MALPVTALLLPLALLLHAARPDIQM
CTCCTGCCGCTGGCCTTGCTGCTCC	TQTSSLSASLGDRVTISCRASQDIR
ACGCCGCCAGGCGGACATCCAGA	NYLNWYQKPKDGTVKLLIYYTSRL
TGACCCAGACCACTCCTCCTGT	HSGVPSKFSGSGSGTDYSLTISNLEQ
CTGCCTCTCTGGGAGACAGAGTCA	EDIATYFCQGGNTLPWTFAGGTLKE
CATCAGTTGCAGGGCAAGTCAGG	IKRAGGSGGGSGGGSGGGSEVQLQ
ACATTAGAAATTATTAACTGGT	QSGPELVKPGASKMISCKASGYSFT
ATCAACAGAAACAGATGGAACGTG	GYTMNWVKQSHGKNLEWMGLINP
TTAAACTCCTGATCTACTACACATC	YKGVSTYNQKFKDKATLTVDKSSST
AAGATTACACTCAGGAGTCCCATC	AYMELLSLTSEDSAVYYCARSGYY
AAAGTTCACTGGCAGTGGGTCTGG	GDSDWYFDVWGAGTTVTVSSTTP
AACAGATTATTCTCTACCATTAG	APRPPTPAPTIASQPLSLRPEACRPAA
CAACCTGGAGCAAGAGGATATTGC	GGAVHTRGLDFACDIYIWAPLAGTC
CACCTACTTTTGCCAAACAGGGTAA	GVLLLSLVI TLYCGSGATNFSLLKQA
TACGCTTCCGTGGACGTTTCGCTGG	GDVEENPGPMKCLLYLAFPLFIGNVC
AGGCACCAAGCTGGAATCAAAC	KFTIVFPHNQKGNWKNVPSNYHYCP
GGGCTGGAGGCGGTAGTGGCGGTG	SSSDLNWNHNDLIGTALQVKMPQSH
GATCAGGTGGAGGCGCGGTGGC	KAIQADGWMCHASKWNTTCDFRW
GGATCTGAGGTGCAGCTCCAGCAG	YGPKYITHSIRSFTPSVEQCKESIEQT
TCTGGACCTGAGCTGGTGAAGCCT	KQGTWLNPGFPPQSCGYATVTDAAE
GGAGCTTCAATGAAGATATCCTGC	AVIVQVTPHHVLVDEYTGWVDSQ
AAGGCTTCTGGTTACTCATCTACTG	FINGKCSNYICPTVHNSSTWHSYDK
GCTACACCATGAAGTGGTGAAGC	VKGLCDSNLISMDITFFSEDELSSSL
AGAGTCATGGAAGAACCTTGAGT	GKEGTGFRSNFYAYETGGKACKMQ
YGATGGGACTTATTAATCCTTACA	YCKHWGVRLPSGVWFEMADKDLF
AAGGTGTTAGTACCTACAACAGA	AAARFPECPEGSSISAPQSSTVDVSLI
AGTTCAAGGACAAGGCCACATTAA	QDVERILDYSLCQETWSKIRAGLPIS
CTGTAGACAAGTCATCCAGCACAG	PVDLSYLAPKNPGTGPAFTIINGTLK
CCTCATGGAACCTCCTCAGTCTGA	YFETRYIRVDIAAPILSRMVGMSGT
CATCTGAGGACTCTGCAGTCTATT	TTEAELWDDWAPYEDVEIGPNQVL

TABLE 8-continued

Anti-CD3 targeting molecule co-expressed with mutant VSV-G	
ACTGTGCAAGATCGGGGTACTACG	RTSSGYKFPLYMIGHGMLDSLHLS
GTGATAGTGACTGGTACTTCGATG	SKAQVFEHPHIQDAASQLPDDSLFF
TCTGGGGCCGACGGGACCACGGTCA	GDTGLSKNPIELVEGWFSWKSSTIAS
CCGTCTCCTCAACCACTACACCAG	FFFIIGLIIGLFLVLRVGIHLCKIKKHT
CACCTAGACCACCAACACCTGCGC	KKRQIYTDIEMNRLGK (SEQ ID NO:
CAACCATCGCATCGCAGCCACTGT	68)
CTCTGCGCCAGAGGCATGCCGGC	
CAGCAGCTGGGGGCGCAGTGACACA	
CAAGGGGGCTGGACTTCGCATGTG	
ATATCTACATCTGGGCACCATTTGG	
CAGGGACTTGTGGGGTCTTCTCC	
TGTCACTGGTTATCACCCCTTTACTG	
CGGCAGCGGCGCCCAACTTCTC	
CCTGCTGAAGCAGGCCGGCGACGT	
GGAGAAAACCTGGCCCCATGAA	
GTGTCTGCTGTACCTGGCGTTCTTG	
TTTATCGGGTGAACTGCAAGTTC	
ACTATCGTGTTCGCCACAACCAA	
AAGGGCAACTGGAAAAACGTGCCT	
TCAAATTACCATTATTGCCCCAGC	
AGCTCGGACCTGAACTGGCACAA	
GACCTCATTTGGAACCGCGCTGCAG	
GTGAAGATGCCACAGAGCCACAA	
GGCTATCCAGGCTGACGGATGGAT	
GTGCCACGCGTCAAAATGGGTGAC	
TACCTGCGATTTCCGCTGGTACGG	
ACCAAATACATCACGCACAGCAT	
CAGATCATTCACCCCGTCAGTGA	
ACAATGCAAAGAATCCATCGAACA	
GACTAAGCAGGGAACCTGGCTGAA	
CCCTGGATTTCCGCCGCGTCTG	
TGGGTACGCAACCGTGACCGATGC	
AGAGGCCGTGATCGTGCAAGTCA	
GCCGCATCACGTGCTTGTGGACGA	
GTACACCGGAGAATGGGTGATTTC	
CCAGTTCATCAACGCCAAGTGCTC	
CAACTACATTTGCCCAACCGTGCA	
CAACAGCACTACTTGGCATAGCGA	
CTACAAAGTGAAGGGTCTGTGTGA	
TTCCAACTGATCTCCATGGATATC	
ACTTTCTTCTCGGAAGACGGCGAA	
CTGTCTCTACTGGGCAAGAAGGA	
ACTGGGTTTCGCTCAAATTACTTCG	
CCTACGAACTGGAGGAAAAGCCT	
GCAAGATGCAGTACTGCAAGCACT	
GGGGCGTGAGACTACCCAGCGGTG	
TCTGGTTTCGAGATGGCCGATAAGG	
ACCTGTTTGCAGCAGCGAGATTCC	
CGGAATGCCCTGAGGGATCGAGCA	
TCTCCGCTCCAAGCCAACTTCAG	
TGGACGTGAGCCTGATCCAGGACG	
TGGAACGGATTCTCGACTACTCGC	
TGTGCCAGGAGACCTGGTCGAAGA	
TCAGAGCGGGACTGCCCATCTCAC	
CGGTGGACCTGTCTTACCTGGCGC	
CAAAGAATCCGGGCACTGGACCGG	
CGTTCACCATCATCAACGGCACCC	
TCAAATACTTCGAGACGCGGTACA	
TCCGGGTGGACATCGCAGCTCCGA	
TCCTCTCCCGATGGTGGGAATGA	
TCTCGGGACTACTACCGAAGCCG	
AGCTCTGGGACGACTGGGCACCTT	
ACGAGGATGTGAGATCGGACCTA	
ACGGAGTGCTCCGGACCTCCTCCG	
GGTACAAGTTCCTCTGTACATGA	
TCGGCCATGGCATGCTGGACTCGG	
ATCTGCATCTGTCTCCAAAGCAC	
AGGTGTTTGAACACCCACACATTC	
AAGACGCCGCCAGCCAGCTGCCGG	
ACGATGAGTCGCTGTTCTTCGGAG	
ACACGGGCTTGTCAAAGAATCCA	
TCGAGCTGGTGAAGGATGGTTTT	
CATCTGGAAAAGCAGCATCGCTT	
CATTCTTCTCATCATTGGCCTGAT	
CATCGGCCTATTTCTAGTCTGCCG	

TABLE 8-continued

Anti-CD3 targeting molecule co-expressed with mutant VSV-G	
GTGGGAATTCATCTGTGCATCAAG CTCAAGCACACTAAGAAGCGGCAA ATCTACACTGATATCGAGATGAAT CGCCTGGGCAAG (SEQ ID NO: 67)	
CD8a signal peptide	
Nucleotide	Amino acid
ATGGCCTTACCAGTGACCGCCTTG CTCCTGCCGCTGGCCTTGCTGCTCC ACGCCGCCAGGCCG (SEQ ID NO: 31)	MALPVTALLPLALLLHAARP (SEQ ID NO: 32)
Anti-CD3 Light chain variable region	
Nucleotide	Amino acid
GACATCCAGATGACCCAGACCACC TCCTCCCTGTCTGCCTCTCTGGGAG ACAGAGTCACCATCAGTTGAGGG CAAGTCAGGACATTAGAAATTATT TAAACTGGTATCAACAGAAACCAG ATGGAACTGTTAACTCCTGATCT ACTACACATCAAGATTACACTCAG GAGTCCCATCAAAGTTCAGTGGCA GTGGGTCTGGAACAGATTATTCTC TCACCATTAGCAACCTGGAGCAAG AGGATATTGCCACTTACTTTTGCCA ACAGGGTAATACGCTTCCGTGGAC GTTGCTTGAGGACCAAGCTGGA AATCAAACGGGCT (SEQ ID NO: 33)	DIQMTQTSSLSASLGDRVTISCRAS QDIRNYLNWYQQKPDGTVKLLIYYT SRLHSGVPSKFSGSGSGTDYSLTISN LEQEDIATYFCQQGNTLPWTFAGGT KLEIKRA (SEQ ID NO: 34)
G ₃ S linker	
Nucleotide	Amino acid
GGAGGCGGTAGTGGCGGTGGATCA GGTGGAGGCAGCGGTGGCGGATCT (SEQ ID NO: 35)	GGGSGGSGGGSGGGS (SEQ ID NO: 36)
Anti-CD3 heavy chain variable region	
Nucleotide	Amino acid
GAGGTGCAGCTCCAGCAGTCTGGA CCTGAGCTGGTGAAGCCTGGAGCT TCAATGAAGATATCCTGCAAGGCT TCTGTTACTCATTCACTGGCTACA CCATGAAGTGGGTGAAGCAGAGTC ATGGAAAGAACCTTGAGTGGATGG GACTTATTAATCCTTACAAAGGTG TTAGTACCTACAACAGAAAGTTCA AGGACAAGGCCACATTAAGTGTAG ACAAGTCATCCAGCACAGCCTACA TGGAATCCTCAGTCTGACATCTG AGGACTCTGCAGTCTATTACTGTG CAAGATCGGGTACTACGGTGATA GTGACTGGTACTTCGATGTCTGGG GCGCAGGGACCACGGTCACCGTCT CCTCA (SEQ ID NO: 37)	EVQLQQSGPELVKPGASMKISKAS GYSFTGYTMNWVKQSHGKNLEWM GLINPYKGVSTYNQKFKDKATLTVD KSSSTAYMELLSLTSEDSAVYYCAR SGYYGDSWYFDVWGAGTTVTVSS (SEQ ID NO: 38)
CD8a hinge and transmembrane	
Nucleotide	Amino acid
ACCACTACACCAGCACCTAGACCA CCAACACCTGCGCCAACCATCGCA TCGCAGCCACTGTCTCTGCGCCA GAGGCATGCCGCCAGCAGCTGGG GGCGCAGTGCACACAAGGGGCT GGACTTCGATGTGATATCTACAT	TTTPAPRPPTPAPTIASQPLSLRPEAC RPAAGGAVHTRGLDFACDIYWAPL AGTCGVLLLSLVITLYC (SEQ ID NO: 40)

TABLE 8-continued

Anti-CD3 targeting molecule co-expressed with mutant VSV-G	
CTGGGCACCATTTGGCAGGGACTTG TGGGGTCTTCTCTCTGTCACTGGTT ATCACCCCTTACTGC (SEQ ID NO: 39)	
P2A self-cleaving peptide	
Nucleotide	Amino acid
GGCAGCGGCGCCACCAACTTCTCC CTGCTGAAGCAGGCCGGCGACGTG GAAGAAAACCTGGCCCC (SEQ ID NO: 69)	GS GATNFSLLKQAGDVEENPGP (SEQ ID NO: 70)
VSV-G signal sequence	
Nucleotide	Amino acid
ATGAAGTGTCTGCTGTACCTGGCG TTCCTGTTTATCGGGGTGAACTGC (SEQ ID NO: 71)	MKCLLYLAFLEFIGVNC (SEQ ID NO: 72)
Mutant VSV-G (Trop-002)	
Nucleotide	Amino acid
AAAGTTACATATCGTGTTCGCGAC AACCAAAAGGGCAACTGGAAAAA CGTGCCCTTCAAATTACCATTTATGC CCCAGCAGCTCGGACCTGAACCTGG CACAATGACCTCATTTGAACCGCG CTGCAGGTGAAGATGCCACAGAGC CACAAGGCTATCCAGGCTGACGGA TGGATGTGCCACGCGTCAAAATGG GTGATACCTGCGGATTTCCGCTGG TACGGAACCAATACATCACGAC AGCATCAGATCATTACCCCGTCA GTGGAACAATGCAAGAATCCATC GAACGAGCTAAGCAGGGAACCTG GCTGAACCTGGATTTCCGCGCGCA GTCGCTGGGTACGCAACCGTGAC CGATCGAGAGGCGGTGATCGTGCA AGTCACGCGCATCACGTGCTTGT GGACGAGTACACCGGAGAATGGG TCGATTCACGATTCATCAACGCA AGTGCTCCAATACATTTGCCCAA CCGTGCACAACGCACTACTTGGC ACAGCGACTACAAAGTGAAGGGTC TGTGTGATTCCAACTGATCTCCAT GGATATCACTTTCTTCTCGGAAGA CGGCGAAGTGTCTCACTGGGCAA AGAAGGAAGTGGGTTTCGCTCAA TTACTTCGCTACGAACTGGAGG AAAAGCCTGCAAGATGCAGTACTG CAAGCACTGGGCGTGAGACTACC CAGCGGTGCTGTTTCGAGATGGC CGATAAGGACCTGTTTGCAGCAGC GAGATTCGCGAATGCCCTGAGGG ATCGAGCATCTCCGCTCAAGCCA AACTTCAGTGGACGTGAGCCTGAT CCAGGACGTGGAACGGATTCTCGA CTACTCGCTGTGCCAGGAGACCTG GTCGAAGATCAGAGCGGACTGCC CATCTACCGGTGGACCTGTCTTA CCTGGCGCAAAGAATCCGGGCAC TGGACCGGCGTTCAACATCATCAA CGGCACCTCAAATACTTCGAGAC GCGGTACATCCGGGTGGACATCGC AGCTCCGATCTCTCCCGATGGT GGGAATGATCTCGGGACTACTAC CGAAGCCGAGCTCTGGGACGACTG GGACCTTACGAGGATGTCCGAGAT CGGACCTAACCGAGTGCTCCGAC CTCTCCGGGTACAAGTTCCCTCTG TACATGATCGGCATGGCATGCTG	KFTIVFPHNQKGNWKNVPSNYHYCP SSSDLNWHNDLIGTALQVKMPQSH KAIQADGWMCHASKWVTTCDFRW YGPKYITHSIRSFTPSVEQCKESIEQT KQGTWLNPGFPPQSCGYATVTDAAE AVIVQVTPHHVLVDEYTG EWVDSQ PINGKCSNYICPTVHNSTTWHSDYK VKGLCDSNLISMDITFFSEDGELSSL GKEGTGFRSNYFAYETGGKACKMQ YCKHWGVRLPSGVWFEMADKDLF AAARFPECPEGSSISAPSQTSVDVSLI QDVERILDYSLCQETWSKIRAGLPIS PVDLSYLAPKNPGTGPAFTIINGTLK YFETRYIRVDIAAPILSRMVMISGT TTEAELWDDWAPYEDVEIGPNGVL RTSSGYKFPLYMIGHGMLDSDLHLS SKAQVFEHPHIQDAASQLPDDESLEFF GDTGLSKNPIELVEGWFSWKSSIAS FFFIIGLIIGLFLVLRVGIHLCKLKH KKRQIYTDIEMNRLGK (SEQ ID NO: 74)

TABLE 8-continued

Anti-CD3 targeting molecule co-expressed with mutant VSV-G
GACTCGGATCTGCATCTGTCGTCC AAAGCACAGGTGTTTGAACACCCA CACATTCAGACGCCGCCAGCCAG CTGCCGGACGATGAGTCGCTGTTC TTCGGAGACACGGGCTTGTCAAAG AATCCCATCGAGCTGGTGAAGGA TGGTTTTTCATCCTGGAAAAGCAGC ATCGCTTCATTCTTCTTCATCATTG GCCTGATCATCGGCCTATTCTAGT CCTGCGGGTGGGAATTCATCTGTG CATCAAGCTAAGCACACTAAGAA GCGGCAAACTACACTGATATCGA GATGAATCGCCTGGGCAAG (SEQ ID NO: 73)

[0140] In another embodiment, the tandem expression cassette of the VSV-G env plasmid encodes a CD80 targeting protein and a mutated VSV-G envelope protein. For example, in a specific embodiment, the tandem expression cassette may comprise a polynucleotide according to SEQ ID NO: 75, which encodes an amino acid sequence of SEQ ID NO: 76. Such an embodiment is described in more detail in Table 9. As illustrated in Table 9, the CD80 targeting protein is separated from the mutated VSV-G envelope protein via a P2A self-cleaving peptide encoded by SEQ ID NO: 69 and having an amino acid sequence according to SEQ ID NO: 70.

[0141] In some embodiments, where the tandem expression cassette of the VSV-G env plasmid encodes a CD80 targeting protein and a mutated VSV-G envelope protein, the

tandem expression cassette may include a polynucleotide that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 75, which encodes an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 76. In certain embodiments, the CD80 targeting protein is separated from the mutated VSV-G envelope protein via a P2A self-cleaving peptide encoded a nucleic acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 69, and the P2A self-cleaving peptide includes an amino acid that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 70.

TABLE 9

CD80 T cell targeting molecule co-expressed with mutant VSV-G	
Full molecule	
Nucleotide	Amino acid (signal sequence underlined)
ATGGGTCATACACGCCGCCAAGGA	MGHTRRQGTSPSKCPYLNFQQLLVL
ACCTCACCATCTAAGTGCCCATAT	AGLSHFCSGVIHVTKEVKEVATLSC
CTGAATTTCTTTCAACTTCTCGTGC	GHNVSVEELAQTIRIYWQKEKKMVL
TGGCGGGGCTCAGTCATTCTGCA	TMMSGDMNIWPEYKNRTIFDITNNL
GTGGGGTCATTACGTTACTAAAG	SIVILALRPSDEGTYECVVLKYEKDA
AGGTCAAGGAGGTCGCAACATTGA	FKREHLAEVTLSTKADFPPTPSISDFEI
GTTGTGGCCATAACGTATCAGTTG	PTSNIRRIICSTSGGFPEPHLSWLENG
AAGAACTCGCGCAGACACGGATT	EELNAINTTVSQDPETELYAVSSKLD
ACTGGCAAAGGAAAAGAAAGATG	FNMTTNHSMCLIKYGHRLVNQTFN
GTGTTGACAATGATGACGGGTGAC	WNNTKQEHFPDNLPSWAILTISVN
ATGAACATTTGGCCAGAGTACAAA	GIFVICCLTYCFAPRCREGSGATNFS
AATCGAACGATATTCGATATAACC	LLKQAGDVEENPGPMKCLLYLALFLF
AATAACTTGTCCATAGTAATACTT	IGVNCKFTIVFPHNQKGNWKNVPSN
GCCTTGCGACCTTCTGACGAGGGA	YHYCPSSDLNWNHNDLIGTALQVK
ACGTATGAATGTGTAGTGCTTAAG	MPQSHKAIQADGWMCHASKVVT
TATGAAAAGATGCCCTTAAGCGG	CDFRWYGPKYIHTSIRSFPTPSVEQCK
GACACTTGGCTGAGGTTACACTC	ESIEQTKQGTWLNPGFPFPQSCGYAT
TCCGTTAAGGCGGACTTTCCTACG	VTDAAEAVIVQVTPHHVLVDEYTG
CCGCTATATCCGACTTCGAGATA	WVDSQFINGKCSNYICPTVHNSTTW
CCCACTTCTAACATTGACGCATC	HSYKVKGLCDNLISMDITFFSEDEG
ATTTGCTCAACCTCAGGTGGTTTCC	ELSSLGKEGTGFRSNYPAYETGGKA
CAGAGCCTCACTTGAGCTGGCTGG	CKMQYCKHWGVRLPSPGVWFEMAD
AGAATGGCGAAGAACTTAACGCA	KDLFAARFPCEPESGISAPSTSV
ATCAATACCACGGTGTCCCAAGAC	DVSLIQDVERILDYSLCQETWSKIRA
CCGAGACAGAGCTGTACGCCGTG	GLPISVDLSYLAPKNPGTGPAPTIIIN
TCATCCAACTGGATTTTAACATG	GTLKYFETRYIRVDIAAPILSRMVMG
ACGACAAATCATAGTTTCATGTGT	ISGTTTEAELWDDWAPYEDVEIGPN
CTGATCAAAATATGGGCATCTCAGG	GVLRTSSGKYKFLPYMIGHMLDSL

TABLE 9-continued

CD80 T cell targeting molecule co-expressed with mutant VSV-G	
GTGAATCAGACTTTTAATTGGAAC	HLSSKAQVFEHPHIQDAASQLPDDE
ACTACCAACAAGAGCACTTCCCA	SLFFGDTGLSKNPIELVEGWFSWKS
GATAATCTGTTGCCAGCTGGGCG	SIASFFFIIGLIIGLFLVLRVGIHLCIKL
ATAACTCTTATCTCCGTCACCGGT	KHTKKRQIYTDIEMNRLGK
ATCTTCGTAATTGCTGCCTCACCT	(SEQ ID NO: 76)
ATTGTTTCGCGCTCGATGCCGAG	
AAGGCAGCGGCCCACTTCT	
CCCTGCTGAAGCAGGCCGCGACG	
TGGAAGAAAACCTGGCCCCATGA	
AGTGCTGCTGTACCTGGCGTTCT	
GTTTATCGGGTGAACTGCAAGTT	
CACTATCGTGTTCGCCACAACCA	
AAAGGGCAACTGGAACAACTGTC	
CTTCAAATTACCATTATTGCCCCAG	
CAGCTCGGACCTGAACTGGCACA	
TGACCTCATTTGGAACCGCGCTGCA	
GGTGAAGATGCCACAGAGCCACA	
AGGCTATCCAGGCTGACGGATGGA	
TGTGCCACGCGTCAAAATGGGTGA	
CTACCTGCGATTTCGCTGGTACG	
GACCAAAATACATCAGCACAGCA	
TCAGATCATTCACCCCGTCAGTGG	
AACAATGCAAGAATCCATCGAAC	
AGACTAAGCAGGGAACCTGGCTGA	
ACCCCTGGATTTCCGCCGCGAGTCGT	
GTGGGTACGCAACCGTGACCGATG	
CAGAGGCCGTGATCGTGCAAGTCA	
CGCCGCATCACGTGCTTGTGGACG	
AGTACACCGGAGAATGGGTGAT	
CCCAGTTCATCAACGGCAAGTGT	
CCAACTACATTTGCCCAACCGTGC	
ACAACAGCACTACTTGGCACAGCG	
ACTACAAAGTGAAGGTCTGTGTG	
ATTCCAACCTGATCTCCATGGATA	
TCACTTTCTTCTCGGAAGACGGCG	
AACTGTCTCACTGGGCAAGAAG	
GAACCTGGGTTTCGCTCAAATTACT	
TCGCCTACGAAACTGGAGGAAAAG	
CCTGCAAGATGCAGTACTGCAAGC	
ACTGGGGCGTGAGACTACCCAGCG	
GTGTCTGGTTCGAGATGGCCGATA	
AGGACCTGTTTGACGACGAGAT	
TCCCGGAATGCCCTGAGGGATCGA	
GCATCTCCGCTCCAAGCAAACCT	
CAGTGGACGTGAGCCTGATCCAGG	
ACGTGGAACGGATTCTGACTACT	
CGCTGTGCCAGGAGACCTGGTGA	
AGATCAGAGCGGGAATGCCATCT	
CACCGGTGGACCTGTCTACCTGG	
CGCCAAAGAATCCGGGCACTGGAC	
CGGCGTTCACCATCATCAACGGCA	
CCCTCAAATACTTCGAGACGCGGT	
ACATCCGGGTGGACATCGCAGCTC	
CGATCCTCTCCCGGATGGTGGGAA	
TGATCTCGGGGACTACTACCGAAG	
CCGAGCTCTGGGACGACTGGGCAC	
CTTACGAGGATGTCGAGATCGGAC	
CTAACGGAGTGCTCCGGACCTCCT	
CCGGGTACAAGTTCCTCTGTACA	
TGATCGGCCATGGCATGCTGGACT	
CGGATCTGCATCTGTCGTCCAAAG	
CACAGGTGTTTGAACACCCACACA	
TTCAAGACGCCGCCAGCTGTC	
CGGACGATGAGTCGCTGTTCTTCG	
GAGACACGGGCTTGTCAAAGAATC	
CCATCGAGCTGGTGGAAGGATGGT	
TTTCATCCTGGAAGACGAGATCG	
CTTCATTCTTCTTCATCATTGGCCT	
GATCATCGGCCTATTCTAGTCCTG	
CGGGTGGGAATTATCTGTGCATC	
AAGCTCAAGCACACTAAGAAGCG	
GCAAATCTACACTGATATCGAGAT	
GAATCGCCTGGGCAAG	
(SEQ ID NO: 75)	

TABLE 9-continued

CD80 T cell targeting molecule co-expressed with mutant VSV-G	
CD80 signal peptide	
Nucleotide	Amino acid
ATGGGTCATACACGCCCAAGGA	MGHTRRQGTSPSKCPYLNFFQLLV
ACCTCACCATCTAAGTGCCCATAT	AGLSHFCSGV (SEQ ID NO: 4)
CTGAATTTCTTTCAACTTCTCGTGC	
TGGCGGGGCTCAGTCATTCTGCA	
GTGGGGTC (SEQ ID NO: 3)	
CD80 extracellular domain (ECD)	
Nucleotide	Amino acid
ATTCACGTTACTAAAGAGGTCAAG	IHVTKEVKVATLSCGHNVSVLELA
GAGGTCGCAACATTGAGTTGTGGC	QTRIVWQKEKKMVLTMMSGDMNI
CATAACGTATCAGTTGAAGAACTC	WPEYKNRTIFDITNNLSIVILALRPSD
GCGCAGACACGGATTACTGGCAA	EGTYECVVLKYEKDAFKREHLAEV
AAGGAAAAAGAGATGGTGTGAC	TLSVKADFPPTPSISDFEIPTSNIRRIICS
AATGATGAGCGGTGACATGAACAT	TSGGFPEPHLSWLENGEELNAINTTV
TTGGCCAGAGTACAAAATCGAAC	SQDPETELYAVSSKLDFNMTTNHSF
GATATTCGATATAACCAATAACTT	MCLIKYGHRLRVNQTFNWNNTTKQEH
GTCCATAGTAATACTTGCTTGCG	FPDNLLPS (SEQ ID NO: 6)
ACCTTCTGACGAGGGAACGTATGA	
ATGTGTAGTGCTTAAGTATGAAAA	
AGATGCCCTTAAAGCGGAACACTT	
GGCTGAGGTACACTCTCCGTAA	
GGCGGACTTTCCTACGCCGTCTAT	
ATCCGACTTCGAGATACCACTTC	
TAACATTCGACGCATCATTTGCTC	
AACCTCAGGTGGTTTCCAGAGCC	
TCACCTGAGCTGGCTGGAGAATGG	
CGAAGAACTTAACGCAATCAATAC	
CACGGTGTCCCAAGACCCGAGAC	
AGAGCTGTACGCCGTGTCATCAA	
ACTGGATTTTAAACATGACGACAAA	
TCATAGTTTCATGTGTCTGATCAAA	
TATGGGCATCTCAGGTGAATCAG	
ACTTTTAATTGGAACACTACCAAA	
CAAGAGCACTTCCAGATAATCTG	
TTGCCAAGC (SEQ ID NO: 5)	
CD80 transmembrane and intracellular domain	
Nucleotide	Amino acid
TGGGCGATAACTTTATCTCCGTC	WAITLISVNGIFVICCLTYCFAPRCRE
AACGGTATCTTCGTAATTGTGCGC	(SEQ ID NO: 8)
TCACCTATGTTTCGCGCTCGATG	
CCGAGAA (SEQ ID NO: 7)	
P2A self-cleaving	
Nucleotide	Amino acid
GGCAGCGCGCCACCAACTTCTCC	GSBATNFSLLKQAGDVEENPGP
CTGCTGAAGCAGGCCGCGACGTG	(SEQ ID NO: 70)
GAAGAAAACCTGGCCCC (SEQ ID NO: 69)	
VSV-G signal sequence	
Nucleotide	Amino acid
ATGAAGTGCTGTGTACCTGGCG	MKCLLYLAFLEFIGVNC (SEQ ID NO: 72)
TTCTGTGTTATCGGGTGAATGC	
(SEQ ID NO: 71)	
Mutant VSV-G	
Nucleotide	Amino acid
AAGTTCACTATCGTGTTCGCGAC	KFTIVFPHNQKGNWKNVPSNYHYCP
AACCAAAAGGGCAACTGGAAAAA	SSSDLNWHNDLIGTALQVKMPQSH
CGTGCCCTCAAATTACCATTATTGC	KAIQADGWMCHASKWVTTCDFRW

TABLE 9-continued

CD80 T cell targeting molecule co-expressed with mutant VSV-G	
CCCAGCAGCTCGGACCTGAACTGG	YGPKYITHSIRSFTPSVEQCKESIEQT
CACAATGACCTCATTGGAACCGCG	KQGTWLNPGFPPQSCGYATVTDAAE
CTGCAGGTGAAGATGCCACAGAGC	AVIVQVTPHHVLVDEYTGWVDSQ
CACAAGGCTATCCAGGCTGACGGA	FINGKCSNYICPTVHNSSTTWHSDYK
TGGATGTGCCACGCGTCAAAATGG	VKGLCDSNLISMDITFFSEDGELSSL
GTGACTACCTGCGATTTCGCTGG	GKEGTGFRSNYFAYETGGKACKMQ
TACGGACCAAAATACATCACGCAC	YCKHWGVRLP SGVWFEMADKDLF
AGCATCAGATCATTCAACCCGTCA	AAARFPECPEGSSISAPSQTSVDVSLI
GTGGAACAATGCAAGAATCCATC	QDVERILDYSLCQETWSKIRAGLPIS
GAACAGACTAAGCAGGGAACCTG	PVDLSYLAPKNPGTGPAFTIINGTLK
GCTGAACCCCTGGATTTCGCGCGCA	YFETRYIRVDIAAPILSRMVGMI SGT
GTCTGTGTGGGTACGCAACCGTGAC	TTEAELWDDWAPYEDVEIGPNGVL
CGATCGACAGGCGCGTGATCGTGCA	RTSSGYKFPLYMIGHGMLDSLHL
AGTCACGCCGCGATCACGTGCTTGT	SKAQVFEHPHIQDAASQLPDDSLFF
GGACGAGTACACCGGAGATGGG	GDTGLSKNPIELVEGWFSWSSIAS
TGATTCCAGTTCATCAACGGCA	FFFIIGLIIGLFLVLRVGIHLCKIKHT
AGTGCTCCAATACTACATTTGCCCAA	KKRQIYTDIEMNRLGK (SEQ ID NO:
CCGTGCACAAACGCACTACTTGGC	74)
ACAGCGACTACAAAGTGAAGGGTC	
TGTGTGATTCCAACCTGATCTCCAT	
GGATATCACTTTCTTCTCGGAAGA	
CGGCGAAGTGTCTCTCACTGGGCAA	
AGAAGGAAGTGGGTTTCGCTCAA	
TTACTTCGCCTACGAACTGGAGG	
AAAAGCCTGCAAGATGCAGTACTG	
CAAGCACTGGGCGTGAGACTACC	
CAGCGGTGTCTGGTTCGAGATGGC	
CGATAAGGACCTGTTTGCAGCAGC	
GAGATTCCCGGAATGCCCTGAGGG	
ATCGAGCATCTCCGCTCCAAGCCA	
AACCTTCAGTGGACGTGAGCCTGAT	
CCAGGACGTGGAACGGATTCTCGA	
CTACTCGCTGTGCCAGGAGACCTG	
GTGGAAGATCAGAGCGGACTGCC	
CATCTCACCGGTGGACCTGTCTTA	
CCTGGCGCCAAAGAATCCGGGCAC	
TGGACCGCGCTTCAACATCATCAA	
CGGCACCTCAAATACTTCGAGAC	
GCGGTACATCCGGGTGGACATCGC	
AGCTCCGATCTCTCCCGGATGGT	
GGGAATGATCTCGGGGACTACTAC	
CGAAGCCGAGCTCTGGGACGACTG	
GGCACCTTACGAGGATGTCGAGAT	
CGGACCTAACGGAGTGCTCCGGAC	
CTCCTCCGGGTACAAGTTCCTCTG	
TACATGATCGGCCATGGCATGCTG	
GACTCGGATCTGCATCTGTCTGTC	
AAAGCACAGGTGTTTGAACACCCA	
CACATTCAAGACGCCGCCAGCCAG	
CTGCCGACGATGAGTCGCTGTTC	
TTCCGAGACACGGGCTTGTCAAAG	
AATCCCATCGAGCTGGTGAAGGA	
TGGTTTTCATCTTGAAAAGCAGC	
ATCGCTTCATTCTTCTCATCATTG	
GCCTGATCATCGGCCTATTTCTAGT	
CCTGCGGGTGGGAATTCATCTGTG	
CATCAAGCTCAAGCACACTAAGAA	
GCGGCAATCTACACTGATATCGA	
GATGAATCGCCTGGGCAAG (SEQ	
ID NO: 73)	

[0142] In another embodiment, the tandem expression cassette of the VSV-G env plasmid encodes a CD80 targeting protein, an anti-CD3 targeting protein, and a mutated VSV-G envelope protein. For example, in a specific embodiment, the tandem expression cassette may comprise a CD80 targeting polynucleotide according to SEQ ID NO: 1, which encodes an amino acid sequence of SEQ ID NO: 2. In a specific embodiment, the tandem expression cassette may comprise an anti-CD3 targeting polynucleotide according to SEQ ID NO: 9, which encodes an amino acid sequence of SEQ ID NO: 10. Such an embodiment is described in more

detail in Table 3. As illustrated in Table 3, the CD80 targeting protein is separated from the anti-CD3 targeting protein via a P2A self-cleaving peptide encoded by SEQ ID NO: 29 and having an amino acid sequence according to SEQ ID NO: 30.

[0143] In some embodiments, where the tandem expression cassette of the VSV-G env plasmid encodes a CD80 targeting protein, anti-CD3 targeting protein, and a mutated VSV-G envelope protein, the tandem expression cassette may include a polynucleotide encoding the CD80 targeting

protein and anti-CD3 targeting protein that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 21, which encodes an amino acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 22. In certain embodiments, the CD80 targeting protein is separated from the anti-CD3 targeting protein via a P2A self-cleaving peptide encoded a nucleic acid sequence that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 29, and the P2A self-cleaving peptide includes an amino acid that has at least about 75%, 80%, 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, 99.5%, or 100% identity to SEQ ID NO: 30.

[0144] In some embodiments, wherein the tandem expression cassette of the VSV-G env plasmid encodes a CD80 targeting protein, anti-CD3 targeting protein, and a mutated VSV-G envelope protein, the CD80 targeting protein and anti-CD3 targeting protein are separated from the mutated VSV-G envelope protein via a 2A self-cleaving peptide.

[0145] In some embodiments of a four plasmid LVV system, the GagPol plasmid includes a tandem expression cassette that encodes the Gag polyprotein precursor and Pol polyprotein precursor and a targeting protein as disclosed herein. In specific embodiments, the tandem expression cassette included in the GagPol plasmid includes a polynucleotide that encodes a first signal peptide, a polynucleotide that encodes a targeting protein, a polynucleotide that encodes one of an internal ribosomal entry site (IRES), a furin cleavage site, or a viral 2A peptide, a polynucleotide that encodes a second signal peptide, and a polynucleotide that encodes Gag polyprotein precursor and Pol polyprotein precursor. In certain embodiments, the polynucleotide encoding the Gag polyprotein precursor and Pol polyprotein precursor is positioned 5' to the polynucleotide encoding the targeting protein. In other embodiments, the polynucleotide encoding the Gag polyprotein precursor and Pol polyprotein precursor is positioned 3' to the polynucleotide encoding the targeting protein. The polynucleotide that encodes the targeting protein and the polynucleotide that encodes the Gag polyprotein precursor and Pol polyprotein precursor are separated in the tandem cassette by the polynucleotide that encodes the IRES, furin cleavage site, or viral 2A peptide, which allows for co-expression of Gag polyprotein precursor and Pol polyprotein precursor and the targeting protein from a single mRNA. In certain embodiments, a viral 2A peptide is a porcine teschovirus-1 (P2A), *Thosea asigna* virus (T2A), equine rhinitis A virus (E2A), foot-and-mouth disease virus (F2A), or a variant thereof. Specific examples of amino acid and nucleotide sequences for the viral 2A peptides are provided in Table 7.

[0146] In some embodiments, the GagPol plasmid containing the tandem expression cassette encoding Gag polyprotein precursor and Pol polyprotein precursor and the targeting protein is transfected into producer cells at a higher molar concentration than the VSV-G env plasmid. In some embodiments, the GagPol plasmid containing the tandem expression cassette encoding Gag polyprotein precursor and Pol polyprotein precursor and the targeting protein is transfected into producer cells at a lower molar concentration than the VSV-G env plasmid.

[0147] In one embodiment, the tandem expression cassette of the GagPol plasmid encodes an anti-CD3 targeting protein and Gag polyprotein precursor and Pol polyprotein precursor. In some embodiments, the anti-CD3 targeting protein is separated from the Gag polyprotein precursor and Pol polyprotein precursor via a P2A self-cleaving peptide.

[0148] In another embodiment, the tandem expression cassette of the GagPol plasmid encodes a CD80 targeting protein and Gag polyprotein precursor and Pol polyprotein precursor. In some embodiments, the CD80 targeting protein is separated from Gag polyprotein precursor and Pol polyprotein precursor via a P2A self-cleaving peptide.

[0149] In some embodiments of a four plasmid LVV system, the Rev plasmid includes a tandem expression cassette that encodes the Rev protein and a targeting protein as disclosed herein. In specific embodiments, the tandem expression cassette included in the Rev plasmid includes a polynucleotide that encodes a first signal peptide, a polynucleotide that encodes a targeting protein, a polynucleotide that encodes one of an internal ribosomal entry site (IRES), a furin cleavage site, or a viral 2A peptide, a polynucleotide that encodes a second signal peptide, and a polynucleotide that encodes a Rev protein. In certain embodiments, the polynucleotide encoding the Rev protein is positioned 5' to the polynucleotide encoding the targeting protein. In other embodiments, the polynucleotide encoding the Rev protein is positioned 3' to the polynucleotide encoding the targeting protein. The polynucleotide that encodes the targeting protein and the polynucleotide that encodes the Rev protein are separated in the tandem cassette by the polynucleotide that encodes the IRES, furin cleavage site, or viral 2A peptide, which allows for co-expression of the Rev protein and the targeting protein from a single mRNA. In certain embodiments, a viral 2A peptide is a porcine teschovirus-1 (P2A), *Thosea asigna* virus (T2A), equine rhinitis A virus (E2A), foot-and-mouth disease virus (F2A), or a variant thereof. Specific examples of amino acid and nucleotide sequences for the viral 2A peptides are provided in Table 7.

[0150] In some embodiments, the Rev plasmid containing the tandem expression cassette encoding the Rev protein and the targeting protein is transfected into producer cells at a higher molar concentration than the VSV-G env plasmid. In some embodiments, the Rev plasmid containing the tandem expression cassette encoding the Rev protein and the targeting protein is transfected into producer cells at a lower molar concentration than the VSV-G env plasmid.

[0151] In one embodiment, the tandem expression cassette of the Rev plasmid encodes an anti-CD3 targeting protein and a Rev protein. In some embodiments, the anti-CD3 targeting protein is separated from the Rev protein via a P2A self-cleaving peptide.

[0152] In another embodiment, the tandem expression cassette of the GagPol plasmid encodes a CD80 targeting protein and a Rev protein. In some embodiments, the CD80 targeting protein is separated from the Rev protein via a P2A self-cleaving peptide.

[0153] Examples of mutated VSV-G sequences that may be used in the four plasmid, 3rd generation lentiviral vector system of the present disclosure are provided in Table 10.

TABLE 10

WT VSV-G reference and mutant sequences	
WT VSV-G	
Nucleotide	Amino acid (signal sequence underlined)
ATGAAGTGTCTGCTGTACCTGGCGT	MKCLLYLAFLFIGVNCFTIVFPHN
TCCTGTTTATCGGGGTGAAC TGCAA	QKGNWKNVPSNYHYCPSSSDLNW
GTTCATATCGTGTTCGCGACAAC	HNDLIGTALQVKMPKSHKAIQADG
CAAAAGGGCAACTGGAAAACGTG	WMCHASKWVTTCDFRWYGPKYIT
CCTTCAAATTACCATTTATGCCCCA	HSIRSFTPSVEQCKESI BQTKQGTWL
GCAGCTCGGACCTGAAC TGGCACA	NPGFPPQSCGYATVTD AEAVIVQVT
ATGACCTCATTTGGAACCGCGCTGCA	PHHVLVDEYTG EWVDSQFINGKCS
GGTGAAGATGCCAAAGAGCCACAA	NYICPTVHNS TTWHS DYKVKGLCD
GGCTATCCAGGCTGACGGATGGAT	SNLISMDITFFSEDELSSLGKEGTG
GTGCCACGCGTCAAAATGGGTGAC	FRSNYFAYETGGKACKMQYCKHW
TACCTGCGATTTCCGCTGGTACGGA	GVRLPSGVWFEMADKDLFAAARFP
CCAAAATACATCACGCACAGCATC	ECPEGSSISAPSQTSVDVSLIQDVERI
AGATCATTCACCCGCTCAGTGGAAC	LDYSLCQETWSKIRAGLPISPDLS
AATGCAAGAATCCATCGAACAGA	YLAPKNPGTGPAFTI INGTLKYFETR
CTAAGCAGGGAACCTGGCTGAACC	YIRVDIAAPILSRMVGMISGTTTERE
CTGGATTTCCGCCGAGTCTGTGTG	LWDDWAPYEDVEIGPNGLVRTSSG
GTACGCAACCGTGACCGATGCAGA	YKFPLYMIGHGMLSDHLHSSKAQ
GGCCGTGATCGTGCAAGTCACGCC	VFEHPHIQDAASQLPDDESLFFGDT
GCATCACGTGCTTGTGGACGAGTAC	GLSKNPIELVEGWFSWKSIAFFFF
ACCGGAGAAATGGGTGATTCACAG	IIGLIIIGLFLVLRVGIHLCKLKHTKK
TTTCATCAACGGCAAGTGCTCCAAC	RQIYTDIEMNRLGK (SEQ ID NO: 78)
ACATTTGCCCAACCGTGCAACAG	
CAC TACTTGGCACAGCGACTACAA	
AGTGAAGGTCGTGTGATTCACAC	
CTGATCTCCATGGATATCACTTTCT	
TCTCGGAAGACGGCGAAGTGTCTC	
ACTGGGCAAGAAGGAATCGGGTT	
TCGCTCAAATTACTTCGCTACGAA	
ACTGGAGGAAAAGCCTGCAAGATG	
CAGTACTGCAAGCACTGGGGCGTG	
AGACTACCCAGCGGTGTCTGGTTCTG	
AGATGGCCGATAAGGACCTGTTTGC	
AGCAGCGAGATTCCTCGAATGCC	
TGAGGGATCGAGCATCTCCGCTCCA	
AGCCAAACTTCAGTGGACGTGAGC	
CTGATCCAGGACGTGGAACGGATT	
CTCGACTACTCGCTGTGCCAGGAGA	
CCTGGTCGAAGATCAGAGCGGGAC	
TGCCCATCTCACCGGTGGACCTGTC	
CTACCTGGCGCCAAAGAATCCGGG	
CAC TGGACCGCGTTTACCATCATC	
AACGGCACCTCAAATACTTCGAG	
ACGCGGTACATCCGGGTGGACATC	
GCAGCTCCGATCCTCTCCCGGATGG	
TGGGAATGATCTCGGGGACTACTAC	
CGAACGCGAGCTCTGGGACGACTG	
GGCACCTTACGAGGATGTCGAGAT	
CGGACCTAACGGAGTGCTCCGGAC	
CTCCTCCGGGTACAAGTTCCCTCTG	
TACATGATCGGCCATGGCATGCTGG	
ACTCGGATCTGCATCTGCTGCTCAA	
AGCACAGGTGTTTGAACACCCACA	
CATTCAAGACGCCCGCAGCCAGCT	
GCCGGACGATGAGTCGCTGTTCTTC	
GGAGACACGGGCTTGTCAAAGAAT	
CCCATCGAGCTGGTGAAGGATGG	
TTTTCATCTTGAAAAGCAGCATCG	
CTTCATTCTTCTCATCATGGCCTG	
ATCATCGGCCTATTTCTAGTCTGTC	
GGGTGGGAATTCATCTGTGCATCAA	
GCTCAAGCACACTAAGAAGCGGCA	
AATCTA CACTGATATCGAGATGAAT	
CGCCTGGGCAAG (SEQ ID NO: 77)	

TABLE 10-continued

WT VSV-G reference and mutant sequences	
WT VSV-G Signal Sequence	
Nucleotide	Amino acid
ATGAAGTGTCTGCTGTACCTGGCGT	MKCLLYLAFLFIGVNC
TCCTGTTTATCGGGTGAAGTGC	(SEQ ID NO: 88)
(SEQ ID NO: 87)	
WT VSV-G without signal sequence	
Nucleotide	Amino acid
AAGTTCACATATCGTGTTCGCGACA	KFTIVFPHNQKGNWKNVPSNYHYC
ACCAAAAGGGCAACTGGAAAAACG	PSSDLNWHNDLIGTALQVKMPKS
TGCTTCAAATTACCATTTATGCCC	HKAIQADGWMCHASKWVTTCDFR
CAGCAGCTCGGACCTGAACTGGCA	WYGPKYITHSIRSFTPSVEQCKESIE
CAATGACCTCATTTGGAACCGCGCTG	QTKQGTWLNPGFPPQSCGYATVTD
CAGGTGAAGATGCCAAGAGCCAC	AEAVIVQVTPHHVLVDEYTGWVD
AAGGCTATCCAGGCTGACGGATGG	SQFINGKCSNYICPTVHNSTTWHS
ATGTGCCACGCGTCAAAATGGGTG	YKVKGLCDNLISMDITFFSEDGEL
ACTACCTGCGATTTCCGCTGGTACG	SSLGKEGTGFRSNYPAYETGGKAC
GACCAAAATACATCACGCACAGCA	KMQYCKHWGVRLPSPGVWFEMAD
TCAGATCATTCACCCCGTCAGTGGA	KDLFAAARFPECPEGSSISAPSQTSV
ACAATGCAAGAATCCATCGAACA	DVSLIQDVERILDYSLCQETWSKIR
GACTAAGCAGGGAACCTGGCTGAA	AGLPISPDLSYLAPKNPOTGPAFTI
CCCTGGATTTCGCGCGCAGTCGTGT	INGTLKYFETRYIRVDIAAPILSRMV
GGGTACGCAACCGTGACCGATGCA	GMISGTTTERELWDDWAPYEDVEI
GAGGCGGTGATCGTGCAAGTCACG	GPNGVLRSSGYKFPLYMIGHGML
CCGCATCACGTGCTTGTGGACGAGT	DSDLHLSSKAQVFEHPHIQDAASQL
ACACCGGAGAATGGGTGCGATTCCC	PDDESLFFGDTGLSKNPIELVEGWF
AGTTCATCAACGGCAAGTGCTCCAA	SSWKSSIASFFFIIGLIIGLFLVLRVGI
CTACATTTGCCCAACCGTGCAACAAC	HLCKLKHKKRQIYTDIEMNRLGK
AGCACTACTTGGCACAGCGACTAC	(SEQ ID NO: 90)
AAAGTGAAGGGTCTGTGTGATTCCA	
ACCTGATCTCCATGGATATCACTTT	
CTTCTCGGAAGACGCGCAACTGTCC	
TCACTGGGCAAGAAGAACTGGG	
TTTCGCTCAAATTACTTCGCCTACG	
AAACTGGAGGAAAAGCCTGCAAGA	
TGCAGTACTGCAAGCACTGGGGCG	
TGAGACTACCCAGCGGTGTCTGGTT	
CGAGATGGCCGATAAGGACCTGTTT	
GCAGCAGCGAGATTCGCGAATGC	
CCTGAGGGATCGAGCATCTCCGCTC	
CAAGCCAACTTCAGTGGACGTGA	
GCCTGATCCAGGACGTGGAACGGA	
TTCTCGACTACTCGCTGTGCCAGGA	
GACCTGGTCGAAGATCAGAGCGGG	
ACTGCCCCTCTCACCGGTGGACCTG	
TCCTACCTGGCGCCAAAGAATCCGG	
GCACTGGACCGCGTTCACCATCAT	
CAACGGCACCCCTCAAATACTTCGAG	
ACGCGGTACATCCGGGTGGACATC	
GCAGCTCCGATCCTCTCCCGGATGG	
TGGGAATGATCTCGGGGACTACTAC	
CGAACGCGAGCTCTGGGACGACTG	
GGCACCTTACGAGGATGTCGAGAT	
CGGACCTAACGGAGTGCTCCGGAC	
CTCCTCCGGGTACAAGTTCCCTCTG	
TACATGATCGGCCATGGCATGTCTG	
ACTCGGATCTGCATCTGTCGTCCAA	
AGCACAGGTGTTTGAACACCCACA	
CATTCAAGACGCGCCAGCCAGCT	
GCCGGACGATGAGTCGCTGTTCTTC	
GGAGACACGGGCTTGTCAAAGAAT	
CCCATCGAGCTGGTGAAGGATGG	
TTTTTCATCTGGAAAAGCAGCATCG	
CTTCATTCTTCTTCATCATTGGCCTG	
ATCATCGGCCTATTCTAGTCTCTGC	
GGGTGGGAATTCATCTGTGCATCAA	
GCTCAAGCACACTAAGAAGCGGCA	
AATCTACACTGATATCGAGATGAAT	
CGCCTGGGCAAG	
(SEQ ID NO: 89)	

TABLE 10-continued

WT VSV-G reference and mutant sequences	
Trop-002 Mutated VSV-G	
Nucleotide	Amino Acid
AAGTTCACATATCGTGTTCCTCGCAC	KFTIVFPHNQKGNWKNVPSNYHYC
ACCAAAGGGCACTGGAAAAACG	PSSSDLNWHNDLIGTALQVKMPQS
TGCCCTCAAATTACCATATTGCCC	HKAIQADGWMCHASKWVTTCDPR
CAGCAGCTCGGACCTGAACTGGCA	WYGPXYITHSIRSFTPSVEQCKESIE
CAATGACCTCATTGGAACCGCGCTG	QTKQGTWLNPGFPPQSCGYATVTD
CAGGTGAAGATGCCACAGAGCCAC	AEAVIVQVTPHHVLVDEYTGWVD
AAGGCTATCCAGGCTGACGGATGG	SQFINGKCSNYICPTVHNSTTWHSD
ATGTGCCACGCGTCAAATGGGTG	YKVKGLCDNLSMDITFFSEDGEL
ACTACCTGCGATTTCCGCTGGTACG	SSLGKEGTGFRSNFYAYETGGKAC
GACCAAAATACATCACGCACAGCA	KMQYCKHWGVRLP SGVWFEMAD
TCAGATCATTCAACCGCTCAGTGGA	KDLFAARFPECPEGSSISAPQTSV
ACAATGCAAGAATCCATCGAACA	DVSLIQDVERILDYSLCQETWSKIR
GACTAAGCAGGGAACCTGGCTGAA	AGLPISPVDLSYLAPKNPGTGPAFTI
CCCTGGATTTCCGCGCAGTCGTGT	INGTLKYFETRYIRVDIAAPILSRMV
GGGTACGCAACCGTGACCGATGCA	GMISGTTTEAELWDDWAPYEDVEI
GAGGCCGTGATCGTGCAAGTCACG	GPNGVLR TSSGYKPLYMIGHGML
CCGCATCAGTGCTTGTGGACGAGT	DSDLHLSSKAQVFEHPIQDAASQL
ACACCGGAGAATGGGTGCGATTCCC	PDDESLFFGDTGLSKNPIELVEGWF
AGTTCATCAACGGCAAGTGCTCCAA	SSWKSSIASFFFIIGLII GLFLVLRVGI
CTACATTTGCCCAACCGTGCACAAC	HLCKLKHKKRQIYTDIEMNRLGK
AGCACTACTTGGCAGCGACTAC	(SEQ ID NO: 74)
AAAGTGAAGGGTCTGTGATTCCA	
ACCTGATCTCCATGGATATCACTTT	
CTTCTCGGAAGACGGCGAACTGTCC	
TCACTGGGCAAAGAAGGAAC TGGG	
TTTCGCTCAAATTACTTCGCTACG	
AAACTGGAGGAAAAGCCTGCAAGA	
TGCAGTACTGCAAGCACTGGGGCG	
TGAGACTACCCAGCGGTGTCTGGTT	
CGAGATGGCCGATAAGGACCTGTTT	
GCAGCAGCGAGATTCCCGGAATGC	
CCTGAGGGATCGAGCATCTCCGCTC	
CAAGCCAACTTCAGTGACGTGA	
GCCTGATCCAGGACGTGGAACGGA	
TTCTCGACTACTCGCTGTGCCAGGA	
GACCTGGTCGAAGATCAGAGCGGG	
ACTGCCCATCTCACCGGTGGACCTG	
TCCTACTGGCGCCAAAGAA TCCGG	
GCACTGGACCGCGCTTACCATCAT	
CAACGGCACCTCAAATACTTCGAG	
ACGCGGTACATCCGGGTGGACATC	
GCAGCTCCGATCCTCTCCGGATGG	
TGGGAATGATCTCGGGGACTACTAC	
CGAAGCCGAGCTCTGGGACGACTG	
GGCACCTTACGAGGATGTCGAGAT	
CGGACCTAACGGAGTGCTCCGGAC	
CTCTCCGGGTACAAGTTCCCTCTG	
TACATGATCGGCCATGGCATGCTGG	
ACTCGGATCTGCATCTGTCGTCAA	
AGCACAGGTGTTTGAACACCCACA	
CATTCAAGACGCCGCCAGCCAGCT	
GCCGGACGATGAGTCGCTGTTCTTC	
GGAGACACGGGCTTGTCAAAGAAT	
CCCATCGAGCTGGTGGAAGGATGG	
TTTTCATCTCGAAAAGCAGCATCG	
CTTCATTCTTCTTCATATTGGCCTG	
ATCATCGGCCTATTCTAGTCCTGC	
GGGTGGGAATTATCTGTCATCAA	
GCTCAAGCACACTAAGAAGCGGCA	
AATCTACACTGATATCGAGATGAAT	
CGCCTGGGCAAG (SEQ ID NO: 73)	
Trop-002 Mutated VSV-G	
Nucleotide	Amino acid
AAGTTCACATATCGTGTTCCTCGCAC	KFTIVFPHNQKGNWKNVPSNYHYC
ACCAAAGGGCACTGGAAAAACG	PSSSDLNWHNDLIGTALQVKMPQS
TGCCCTCAAATTACCATATTGCCC	HKAIQADGWMCHASKWVTTCDPR
CAGCAGCTCGGACCTGAACTGGCA	WYGPXYITHSIRSFTPSVEQCKESIE
CAATGACCTCATTGGAACCGCGCTG	QTKQGTWLNPGFPPQSCGYATVTD

TABLE 10-continued

WT VSV-G reference and mutant sequences	
CAGGTGAAGATGCCACAGAGCCAC	AEAVIVQVTPHHVLVDEYTGWVD
AAGGCTATCCAGGCTGACGGATGG	SQFINGKCSNYICPTVHNSTTWHSD
ATGTGCCACGCGTCAAAATGGGTG	YKVKGLCDNLISMDITFFSEDGEL
ACTACCTGCGATTTCCGCTGGTACG	SSLGKEGTGFRSNYFAYETGGKAC
GACCAAAATACATCACGCACAGCA	KMQYCKHWGVRLP SGVWFEMAD
TCAGATCATTCACCCCGTCAGTGGA	KDLFAAARFPECPEGSSISAPSQTSV
ACAATGCAAGAATCCATCGAACA	DVSLIQDVERILDYSLCQETWSKIR
GACTAAGCAGGGAACCTGGCTGAA	AGLPISPVDLSYLAPKNPGTGPAFTI
CCCTGGATTTCCGCGCAGTCGTGT	INGTLKYFETRYIRVDIAAPILSRMV
GGGTACGCAACCGTGACCGATGCA	GMISSGTTTEAELWDDWAPYEDVEI
GAGGCCGTGATCGTGCAAGTCACG	GPNGVLR TSSGYKPLYMIGHGML
CCGCATCATCGTGCTGTGGACGAGT	DSDLHLSSKAQVFEHPHIQDAASQL
ACACCGGAGAATGGGTGCGATTTCC	PDDESLFFGDTGLSKNPIELVEGWF
AGTTTCATCAACGGCAAGTGCTCAA	SSWKSSIASFFFIIGLIIGLFLVLRVGI
CTACATTTGCCCAACCGTGCACAAC	HLCKLKHKKRQIYTDIEMNRLGK
AGCACTACTTGGCATAGCGACTACA	(SEQ ID NO: 74)
AAGTGAAGGCTCTGTGTGATTCCAA	
CCTGATCTCCATGGATATCACTTTC	
TTCTCGGAAGACGGCGAACTGTCCT	
CAC TGGGCAAAGAAGGAAC TGGGT	
TTCTGCTCAAATTACTTCGCCTACGA	
AACTGGAGGAAAAGCCTGCAAGAT	
GCAGTACTGCAAGCACTGGGGCGT	
GAGACTACCCAGCGGTGTCTGGTTC	
GAGATGGCCGATAAGGACCTGTTT	
GCAGCAGCGAGATTCCCGGAATGC	
CCTGAGGGATCGAGCATCTCCGCTC	
CAAGCCAACTTCAGTGGACGTGA	
GCCTGATCCAGGACGTGGAACGGA	
TTCTCGACTACTCGTGTGCCAGGA	
GACCTGGTCGAAGATCAGAGCGGG	
ACTGCCCATCTCACCGGTGGACCTG	
TCCTACTTGGCGCCAAAGATCCGG	
GCACTGGACCGCGCTTACCATCAT	
CAACGGCACCCCTCAAATACTTCGAG	
ACGCGGTACATCCGGGTGGACATC	
GCAGTCCGATCCTCTCCGGATGG	
TGGGAATGATCTCGGGGACTACTAC	
CGAAGCCGAGCTCTGGGACGACTG	
GGCACCTTACGAGGATGTCGAGAT	
CGGACCTAACGGAGTGCTCCGGAC	
CTCCTCCGGGTACAAGTTCCTCTG	
TACATGATCGGCCATGGCATGCTGG	
ACTCGGATCTGCATCTGTCGTCAA	
AGCACAGGTGTTTGAACACCCACA	
CATTCAAGACGCCGCCAGCCAGCT	
GCCGGACGATGAGTCGCTGTTCTTC	
GGAGACACGGGCTTGTCAAAGAA	
CCCATCGAGCTGGTGAAGGATGG	
TTTTTCATCTGAAAAGCAGCATCG	
CTTCATTCTTCTCATCATTTGGCCTG	
ATCATCGGCCTATTTCTAGTCCTGC	
GGGTGGGAATTTCATCTGTCATCAA	
GCTCAAGCACACTAAGAAGCGGCA	
AATCTACACTGATATCGAGATGAAT	
CGCCTGGGCAAGTAG (SEQ ID NO:	
91)	
Trop-051 Mutated VSV-G	
Nucleotide	Amino acid
AAGTTCATATCGTGTTCGACACA	KFTIVFPHNQKGNWKNVPSNYHYC
ACCAAAAGGGCAACTGGAACACG	PSSSDLNWNHNDLIGTALQVKMPKS
TGCCCTTCAAATTACCATTATTGCCC	HKAIQADGWMCHASKWVTTCDFR
CAGCAGCTCGGACCTGAACTGGCA	WYGPKYITHSIRSFTPSVEQCKESIE
CAATGACCTCATTGGAACCGCGCTG	QTKQGTWLNPGFPFPQSCGYATVTD
CAGGTGAAGATGCCAAAGAGCCAC	AEAVIVQVTPHHVLVDEYTGWVD
AAGGCTATCCAGGCTGACGGATGG	SQFINGKCSNYICPTVHNSTTWHSD
ATGTGCCACGCGTCAAAATGGGTG	YKVKGLCDLSALASMDITFFSEDGEL
ACTACCTGCGATTTCCGCTGGTACG	SSLGKEGTGFRSNYFAYETGGKAC
GACCAAAATACATCACGCACAGCA	KMQYCKHWGVRLP SGVWFEMAD
TCAGATCATTCACCCCGTCAGTGGA	KDLFAAARFPECPEGSSISAPSQTSV
ACAATGCAAGAATCCATCGAACA	DVSLIQDVERILDYSLCQETWSKIR
GACTAAGCAGGGAACCTGGCTGAA	AGLPISPVDLSYLAPKNPGTGPAFTI

TABLE 10-continued

WT VSV-G reference and mutant sequences	
CCCTGGATTTCGCCGCGAGTCGTGT	INGTLKYPETRYIRVDIAAPILSRMV
GGGTACGCAACCGTGACCGATGCA	GMISGTTAARELWDDWAPYEDVEI
GAGGCCGTGATCGTGCAAGTCACG	GPNGVLRVTSSGYKFPLYMIGHGML
CCGCATCAGTGCTTGTGGACGAGT	DSDLHLSSKAQVFEHPHIQDAASQL
ACACCGGAGAAATGGGTCGATTCCC	PDDESLFFGDTGLSKNPIELVEGWF
AGTTTCATCAACGGCAAGTGCTCCAA	SSWKSSIASFFFIIGLIIGLFLVLRVGI
CTACATTTGCCCAACCGTGCAACAAC	HLCIKLKHKKRQIYTDIEMNRLGK
AGCACTACTTGGCACAGCGACTAC	(SEQ ID NO: 93)
AAAGTGAAGGGTCTGTGTGATTCCG	
CCCTGGCCTCCATGGATATCACTTT	
CTTCTCGGAAGACGGCGAACTGTCC	
TCACTGGGCAAAGAAGGAACTGGG	
TTTCGCTCAAATTACTTCGCCTACG	
AAACTGGAGGAAAAGCCTGCAAGA	
TGCAGTACTGCAAGCACTGGGGCG	
TGAGACTACCCAGCGGTGTCTGGTT	
CGAGATGGCCGATAAGGACCTGTTT	
GCAGCAGCGAGATTCGGGAATGC	
CCTGAGGGATCGAGCATCTCCGCTC	
CAAGCCAACTTCAGTGACGTGA	
GCCTGATCCAGGACGTGGAACGGA	
TTCTCGACTACTCGCTGTGCCAGGA	
GACCTGGTCAAGATCAGAGCGGG	
ACTGCCCATCTCACCGGTGGACCTG	
TCCTACCTGGCGCCAAAGAATCCGG	
GCACTGGACCGCGCTTACCATCAT	
CAACGGCACCCCTCAAATACCTCGAG	
ACGCGGTACATCCGGGTGGACATC	
GCAGCTCCGATCCTCTCCGGATGG	
TGGGAATGATCTCGGGACTACTGC	
CGCCCGCGAGCTCTGGGACGACTG	
GGCACCTTACGAGGATGTCGAGAT	
CGGACCTAACGGAGTGCTCCGGAC	
CTCCTCCGGGTACAAGTTCCCTCTG	
TACATGATCGGCCATGGCATGCTGG	
ACTCGGATCTGCATCTGTCTGCTCAA	
AGCACAGGTGTTTGAACACCCACA	
CATTCAAGACGCGCCAGCCAGCT	
GCCGGACGATGAGTCGTGTTCTTC	
GGAGACACGGGCTTGTCAAAGAAT	
CCCATCGAGCTGGTGGAGGATGG	
TTTTTCATCTGAAAGCAGCATCG	
CTTCATTCTTCTTCATCATTTGGCCTG	
ATCATCGGCCTATTTCTAGTCTCTGC	
GGGTGGGAATTCATCTGTGCATCAA	
GCTCAAGCACACTAAGAAGCGGCA	
AATCTACACTGATATCGAGATGAAT	
CGCCTGGGCAAGTAG (SEQ ID NO: 92)	
Trop-052 Mutated VSV-G	
Nucleotide	Amino acid
AAGTTCACTATCGTGTTCGCGACA	KFTIVFPHNQKGNWKNVPSNYHYC
ACCAAAAGGGCAACTGGAAAAACG	PSSDLNWHNDLIGTALQVKMPKS
TGCCTTCAAATTACCATTTATGCCC	HKAIQADGWMCHASKWVTTCDPR
CAGCAGCTCGGACCTGAACTGGCA	WYGPKYITHSIRSFTPSVEQCKESIE
CAATGACCTCATTTGGAACCGCGCTG	QTKQGTWLNPGFPPQSCGYATVTD
CAGGTGAAGATGCCAAAGAGCCAC	AEAVIVQVTPHHVLVDEYTGWVD
AAGGCTATCCAGGCTGACGGATGG	SQFINGKCSNYICPTVHNSTTWHSD
ATGTGCCACGCGTCAAAATGGGTG	YKVKGLCDNLISMDITPFSDEGEL
ACTACCTCGGATTTCCGCTGGTACG	SSLGKEGTGFRSNFYAYETGGKAC
GACCAAAATACATCACGCACAGCA	KMQYCKHWGVRLPSPGVWFEMAD
TCAGATCATTCACCCCGTCAGTGGA	KDLFAARFPECEPGSSISAPSQTSV
ACAATGCAAGAATCCATCGAACA	DVSLIQDVERILDYSLCQETWSKIR
GACTAAGCAGGGAACCTGGCTGAA	AGLPISPVDLSYLAPKNPGTGPAFTI
CCCTGGATTTCGCCGCGAGTCGTGT	INGTLKYPETRYIRVDIAAPILSRMV
GGGTACGCAACCGTGACCGATGCA	GMISGTTWARELWDDWAPYEDVEI
GAGGCCGTGATCGTGCAAGTCACG	GPNGVLRVTSSGYKFPLYMIGHGML
CCGCATCAGTGCTTGTGGACGAGT	DSDLHLSSKAQVFEHPHIQDAASQL
ACACCGGAGAAATGGGTCGATTCCC	PDDESLFFGDTGLSKNPIELVEGWF
AGTTTCATCAACGGCAAGTGCTCCAA	SSWKSSIASFFFIIGLIIGLFLVLRVGI
CTACATTTGCCCAACCGTGCAACAAC	HLCIKLKHKKRQIYTDIEMNRLGK
AGCACTACTTGGCACAGCGACTAC	(SEQ ID NO: 95)

TABLE 10-continued

WT VSV-G reference and mutant sequences	
AAAGTGAAGGGTCTGTGTGATTCCA ACCTGATCTCCATGGATATCACTTT CTTCTCGGAAGACGGCGAACTGTCC TCACTGGGCAAAGAAGGAACCTGGG TTTCGCTCAAATTACTTCGCCTACG AAACTGGAGGAAAAGCCTGCAAGA TGCAGTACTGCAAGCACTGGGGCG TGAGACTACCCAGCGGTGTCTGGTT CGAGATGGCCGATAAGGACCTGTTT GCAGCAGCGAGATTCCTGGGAATGC CCTGAGGGATCGAGCATCTCCGCTC CAAGCCAACTTCAGTGGACGTGA GCCTGATCCAGGACGTGGAACGGA TTTCGACTACTCGCTGTGCCAGGA GACCTGGTCGAAGATCAGAGCGGG ACTGCCCCATCTCACCGGTGGACCTG TCCTACCTGGCGCCAAAGAATCCGG GCACTGGACCGCGCTTACCATCAT CAACGGCACCCCTCAAATACTTCGAG ACGCGGTACATCCGGGTGGACATC GCAGCTCCGATCTCTCCCGATGG TGGGAATGATCTCGGGGACTACTTG GGCCCGCGAGCTCTGGGACGACTG GGCACCTTACGAGGATGTCGAGAT CGGACCTAACGGAGTGCTCCGGAC CTCCTCCGGGTACAAGTTCCTCTG TACATGATCGGCCATGGCATGTCTG ACTCGGATCTGCATCTGTCGTCCAA AGCACAGGTGTTTGAACACCCACA CATTCAAGACGCCCGCAGCCAGCT GCCGGACGATGAGTCGTGTTCTTC GGAGACACGGGCTTGTCAAAGAAT CCCATCGAGCTGGTGAAGGATGG TTTTCATCTGGAAAAGCAGCATCG CTTCATTCTTCTTCATCATTTGGCCTG ATCATCGGCCTATTCTAGTCTCTGC GGGTGGGAATTCATCTGTGCATCAA GCTCAAGCACACTAAGAAGCGGCA AATCTACACTGATATCGAGATGAAT CGCCTGGGCAAGTAG (SEQ ID NO: 94)	
Trop-055 Mutated VSV-G	
Nucleotide	Amino acid
AAGTTCATATCGTGTTCCTGCACG CCGCCAAGGGCAACTGGAAAAACG TGCCTTCAAATTACCATATTGCCC CAGCAGCTCGGACCTGAACCTGGCA CAATGACCTCATTTGAACCGCGCTG CAGGTGAAGATGCCAAAGAGCCAC AAGGCTATCCAGGCTGACGGATGG ATGTGCCACGCGTCAAATGGGTG ACTACCTGCGATTTCCGCTGGTACG GACCAAAATACATCACGCACAGCA TCAGATCATTCACCCCGTCAGTGGA ACAATGCAAGAATCCATCGAACA GACTAAGCAGGGAACCTGGCTGAA CCCTGGATTTCCGCGCAGTCTGT GGGTACGCAACCGTGACCGATGCA GAGGCCGTGATCGTGCAAGTCACG CCGCATCAGTGCTTGTGGACGAGT ACACCGGAGAATGGGTGCGATTCCC AGTTCATCACCGGCAAGTGCTCAA CTACATTTGCCCAACCGTGCACAAC AGCACTACTTGGCACAGCGACTAC AAAGTGAAGGGTCTGTGTGATTCCG CCCTGATCTCCATGGATATCACTTT CTTCTCGGAAGACGGCGAACTGTCC TCACTGGGCAAAGAAGGAACCTGGG TTTCGCTCAAATTACTTCGCCTACG AAACTGGAGGAAAAGCCTGCAAGA TGCAGTACTGCAAGCACTGGGGCG TGAGACTACCCAGCGGTGTCTGGTT	KFTIVFPHAAKGNWKNVPSNYHYC PSSSDLNWHNDLIGTALQVKMPKS HKAIQADGWMCHASKWVTTCDFR WYGPYIITHSIRSFTPSVEQCKESIE QTKQGTWLNPGFPPQSCGYATVTD AEAVIVQVTPHHVLVDEYTGWEVD SQFINGKCSNYICPTVHNSTTWHSD YKVKGLCDSALISMDITFFSEDGEL SSLGKEGTGFRSNYFAYETGGKAC KMQYCKHWGVRLP SGVWFEMAD KDLFAAARFPECPEGSSISAPSQTSV DVSLIQDVERILDYSLCQETWSKIR AGLPISPVDL SYLAPKNPGTGPAFTI INGTLKYFETRYIRVDIAAPILSRMV GMISGTTTERELWDDWAPYEDVEI GPNGLVRTSSGYKFPLYMIGHGML DSDLHLSSKAQVFEHPHIQDAASQL PDDESLEFFGDTGLSKNPIELVEGWF SSWKSSIASFFFIIGLIIGLFLVLRVGI HLCIKLKHKKRQIYTDIEMNRLGK (SEQ ID NO: 97)

TABLE 10-continued

WT VSV-G reference and mutant sequences	
CGAGATGGCCGATAAGGACCTGTTT GCAGCAGCGAGATTCCCGGAATGC CCTGAGGGATCGAGCATCTCCGCTC CAAGCCAACTTCAGTGGACGTGA GCCTGATCCAGGACGTGGAACGGA TTCTCGACTACTCGCTGTGCCAGGA GACCTGGTCGAAGATCAGAGCGGG ACTGCCCATCTCACCGGTGGACCTG TCCTACCTGGCGCCAAAGAATCCGG GCACTGGACCGCGCTTACCATCAT CAACGGCACCTCAAATACTTCGAG ACGCGGTACATCCGGGTGGACATC GCAGTCCGATCCTCTCCGGATGG TGGGAATGATCTCGGGACTACTAC CGAACGCGAGCTCTGGGACGACTG GGCACCTTACGAGGATGTCGAGAT CGGACCTAACGGAGTGCTCCGGAC CTCTCCGGGTACAAGTTCCTCTG TACATGATCGGCCATGGCATGCTGG ACTCGGATCTGCATCTGTCGTCAA AGCACAGGTGTTTGAACACCCACA CATTCAAGACGCCGCCAGCCAGCT GCCGGACGATGAGTCGCTGTTCTTC GGAGACACGGGCTTGTCGAAGAAT CCCATCGAGCTGGTGAAGGATGG TTTTCATCCTGGAAGAAGCAGCATCG CTTCATTCTTCTCATATTGGCCTG ATCATCGGCCTATTCTAGTCTCTGC GGGTGGGAATTATCTGTGCATCAA GCTCAAGCACACTAAGAAGCGGCA AATCTACACTGATATCGAGATGAAT CGCCTGGGCAAGTAG (SEQ ID NO: 96)	
Trop-056 Mutated VSV-G	
Nucleotide	Amino acid
AAGTTCACTATCGTGTTCGACACA ACACCACACAAAGGGCAACTGGA AAAACGTGCCTTCAAATTACCATTA TTGCCCCAGCAGCTCGGACCTGAAC TGGCACAATGACCTCATTGGAACCG CGCTGCAGGTGAAGATGCCAAAGA GCCACAAGGCTATCCAGGCTGACG GATGGATGTGCCACGCGTCAAAAT GGGTGACTACCTGCGGATTTCGCTG GTACGGACCAAATACATCACGCA CAGCATCAGATCATTCACCCCGTCA GTGGAACAATGCAAAGAATCCATC GAACAGACTAAGCAGGGAACCTGG CTGAACCCCTGGATTTCGCCCGCAGT CGTGTGGGTACGCAACCGTGACCG ATGCAGAGGCCGTGATCGTGCAAG TCACGCCGCATCACGTGCTTGGA CGAGTACACCGGAGAATGGGTGGA TTCCAGTTTCATCAACGGCAAGTGC TCCAACATACATTTGCCAACCGTGC ACAACAGCACTACTTGGCACAGCG ACTACAAAGTGAAGGCTGTGTGTG ATTCCAACCTGATCTCCATGGATAT CACTTCTTCTCGGAAGACGGCGAA CTGTCTCACTGGGCAAAGAAGGA ACTGGGTTTCGCTCAAATTACTTCG CCTACGAACTGGAGGAAAGCCT GCAAGATGCAGTACTGCAAGCACT GGGGCGTGAGACTACCCAGCGGTG TCTGGTTCGAGATGGCCGATAAGG ACCTGTTTGCAGCAGCGAGATTCCC GGAATGCCCTGAGGGATCGAGCAT CTCCGCTCCAAGCCAACTTCAGTG GACGTGAGCCTGATCCAGGACGTG GAACGGATTCTCGACTACTCGCTGT GCCAGGAGACCTGGTCAAGATCA GAGCGGGACTGCCCATCTCACCGGT	KPTIVFPHNTTQKNWKNVPSNYH YCPSSSDLNWHNDLIGTALQVKMP KSHKAIQADGWMCHASKWVTTCD FRWYGPKYITHSIRSFTPSVECKES IEQTKQGTWLNPGFPPQSCGYATVT DAEAVIVQVTPHHVLVDEYTG EWV DSQFINGKCSNYICPTVHNSTTWH DYKVKGLCDSNLISMDITFFSEGE LSSLGKEGTGFRSNYFAYETGGKA CKMQYCKHWGVRLLPSGVWFEMA DKDLFAAARFPECPEGSSISAPSQTS VDVS LIQDVERILDYSLCQETWSKI RAGLPI SPVDLSYLAPKNPGTGPAF TIINGTLKYFETRYIRVDIAAPILSR MVGMISGTTTERELWDDWAPYED VEIGPNGLVLRSSGYKFPLYMIGHG MLDSDLHLSSKAQVFEHPHIQDAA SQLPDDES LFFGDTGLSKNPIELVEG WFSWKSSIASFFFIIGLIIGLFLVLR VGIHLCKIKHTKKRQIYTDIEMNR LGK (SEQ ID NO: 99)

TABLE 10-continued

WT VSV-G reference and mutant sequences	
GGACCTGTCTACCTGGCGCCAAAG AATCCGGGCACTGGACCGCGTTC ACCATCATCAACGGCACCCCTCAAAT ACTTCGAGACGCGGTACATCCGGGT GGACATCGCAGCTCCGATCTCTCTCC CGGATGGTGGGAATGATCTCGGGG ACTACTACCGAACGCGAGCTCTGG GACGACTGGGCACCTTACGAGGAT GTCGAGATCGGACCTAACGGAGTG CTCCGGACCTCCTCGGGTACAAGT TCCCTCTGTACATGATCGGCCATGG CATGCTGGACTCGGATCTGCATCTG TCGTCCAAGCACAGGTGTTTGAAC ACCCACACATTCAAGACGCCGCCA GCCAGCTGCCGGACGATGAGTCGC TGTTCTTCGGAGACACGGGCTTGTC AAAGAATCCCATCGAGCTGGTGGA AGGATGGTTTTTCATCCTGGAAAAGC AGCATCGCTTCATTCTTCTCATCAT TGGCCTGATCATCGGCCTATTTCTA GTCTCGCGGTGGGAATTCATCTGT GCATCAAGCTCAAGCACACTAAGA AGCGGCAAAATCTACACTGATATCG AGATGAATCGCCTGGGCAAGTAG (SEQ ID NO: 98)	
Trop-058 Mutated VSV-G	
Nucleotide	Amino acid
AAGTTCACATATCGTGTTCGCGACA ACCAAAAGGGCAACTGGAAAAACG TGCCTTCAAATTACCATTATTGCCC CAGCAGCTCGGACCTGAAGTGGCA CAATGACCTCATTTGGAACCGCGCTG CAGGTGAAGATGCCAGGCGGCAGC AAGAGCCACAAGGCTATCCAGGCT GACGGATGGATGTGCCACGCGTCA AAATGGGTGACTACCTGCGATTTC GCTGTGACGGACCAAAATACATCA CGCACAGCATCAGATCATTACCCC GTCAGTGAACAAATGCAAGAAATC CATCGAACAGACTAAGCAGGGAAC CTGGCTGAACCTGGATTTCGCGCG CAGTCGTGTGGGTACGCAACCGTG ACCGATGCAGAGGCGGTGATCGTG CAAGTCACGCCGCATCAGTGCTTG TGGACGAGTACACCGAGAATGGG TCGATTCCAGTTCATCAACGGCAA GTGCTCCAACACTAATTGGCCAAAC GTGCACAACAGCACTACTTGGCAC AGCGACTACAAGTGAAGGGTCTG TGTGATTCCAACCTGATCTCCATGG ATATCACTTTCTTCTCGGAAGACGG CGAAGTGTCTCACTGGGCAAGA AGGAAGTGGGTTTCGCTCAAATTAC TTCGCCTACGAACTGGAGGAAAA GCCTGCAAGATGCAGTACTGCAAG CACTGGGCGGTGAGACTACCCAGC GGTGCTGTGGTTCGAGATGGCCGATA AGGACCTGTTTGACAGCAGCAGATT CCCAGGATGCCCTGAGGGATCGAG CATCTCCGCTCCAAGCCAACTTCA GTGGACGTGAGCCTGATCCAGGAC GTGGAACGGATTCTCGACTACTCGC TGTGCCAGGAGACCTGGTCAAGA TCAGAGCGGGACTGCCCATCTCACC GGTGGACCTGTCTACCTGGCGCCA AAGAATCCGGGCACTGGACCGCG TTCACCATCATCAACGGCACCCCTCA AATACTTCGAGACGCGGTACATCCG GGTGGACATCGCAGCTCCGATCCTC TCCCGGATGGTGGGAATGATCTCGG GGACTACTACCGAACGCGAGCTCT GGGACGACTGGGCACCTTACGAGG	KFTIVFPHNQKGNWKNVPSNYHYC PSSSDLNWHNDLIGTALQVKMPGG SKSHKAIQADGWMCHASKWVTTT DFRWYGPKYITHSIRSFTPSVEQCK ESIEQTKQGTWLNPGFPQSCGYAT VTDAEAVIVQVTPHHVLVDEYTGE WVDSQFINGKCSNYICPTVHNSTT WHSYKVKGLCDNLISMDITFFSE DGELSSLGKEGTGFRSNYPAYETG GKACKMQYCKHWGVRLP SGVWFE MADKDLFAAARFPECPEGSSISAPS QTSVDVSLIQDVERILDYSLCQETW SKIRAGLPISPDLSYLAPKNPGTGP AFTIINGTLKYFETRYIRVDIAAPILS RMVGMISGTTTERELWDDWAPYE DVEIGPNGLVLTSSGYKFPLYMIGH GMLDSDLHLSSKAQVFEHPHIQDA ASQLPDDESLLFFGDTGLSKNP IELVE GWFFSWKSSIASFFFIIGLIIGLFLVL RVGIHLCKIKLHKKRQIYTDIEMN RLGK (SEQ ID NO: 101)

TABLE 10-continued

WT VSV-G reference and mutant sequences	
ATGTCGAGATCGGACCTAACGGAG TGCTCCGGACCTCCTCCGGGTACAA GTTCCCTCTGTACATGATCGGCCAT GGCATGCTGGACTCGGATCTGCATC TGTCGTCAAAGCACAGGTGTTTGA ACACCCACACATTCAGACGCCGC CAGCCAGCTGCCGGACGATGAGTC GCTGTTCTTCGGAGACACGGGCTTG TCAAAGAATCCCATCGAGCTGGTG GAAGGATGGTTTTTCATCTGGAAAA GCAGCATCGCTTCATTCTTCTCAT CATTGGCCTGATCATCGGCCTATTT CTAGTCCTGCGGGTGGGAATTCATC TGTGCATCAAGCTCAAGCACACTAA GAAGCGGCAAATCTACACTGATAT CGAGATGAATCGCTGGGCAAGTA G (SEQ ID NO: 100)	
Trop-061 Mutated VSV-G	
Nucleotide	Amino acid
AAGTTCATATCGTGTTCGCGAC GGCGGAAGCAACGGCGGGAGCCA AAAGGGCAACTGGAAAAACGTGC CTTCAAATTACCATTATTGCCCCA GCAGCTCGGACCTGAAGTGGCAC AATGACCTCATTGGAACCGCGCT GCAGGTGAAGATGGCAAAGAGCC ACAAGGCTATCCAGCTGACGGA TGGATGTGCCACGCGTCAAAATG GGTGACTACCTGCGATTTCCGCTG GTACGGACCAAATACATCACGC ACAGCATCAGATCATTACCCCGT CAGTGGAAACAATGCAAAGAATCC ATCGAACAGACTAAGCAGGGAAC CTGGCTGAACCTGGATTTCGGCC GCAGTCGTGTGGGTACGCAACCG TGACCGATGCAGAGCCGTGATC GTGCAAGTCACGCCGATCACGT GCTTGTGGACGAGTACACCGGAG AATGGGTCGATTCCAGTTCATCA ACGGCAAGTGCTCCAACTACATTT GCCCAACCGTGACACACAGCACT ACTTGGCACAGCGACTACAAAGT GAAGGGTCTGTGTGATTCCAACCT GATCTCCATGGATATCACTTTCTT CTCGGAAGACGGCGAACTGTCTC CACTGGGCAAAGAAGGAATGGG TTTCGCTCAAATTACTTCGCCTAC GAAACTGGAGGAAAAGCCTGCAA GATGCAGTACTGCAAGCACTGGG GCGTGAGACTACCCAGCGGTGTC TGGTTCGAGATGGCCGATAAGGA CCTGTTTGCAGCAGCGAGATTTCC GGAATGCCCTGAGGGATCGAGCA TCTCCGCTCCAAGCCAACTTCAG TGGACGTGAGCCTGATCCAGGAC GTGGAACGGATTCTCGACTACTCG CTGTGCGAGGAGACCTGGTCGAA GATCAGAGCGGACTGCCCATCT CACCGTGGACCTGTCTTACCTGG CGCCAAAGAATCCGGGCACTGGA CCGCGGTTACCATCATCAACGGC ACCCTCAAATACTTCGAGACGCG GTACATCCGGGTGGACATCGCAG CTCCGATCCTCTCCCGGATGGTGG GAATGATCTCGGGGACTACTACC GAACGCGAGCTCTGGGACGACTG GGCACCTTACGAGGATGTCGAGA TCGGACCTAACGGAGTGCTCCGG ACCTCCTCCGGGTACAAGTTCCTT CTGTACATGATCGGCATGGCATG CTGGACTCGGATCTGCATCTGTCG TCCAAAGCACAGGTGTTTGAAACA	
KFTIVFPHGSSNGGSQKGNWKNVPS NYHYCPSSDLNWHNDLIGTALQVK MPKSHKAIQADGWMCHASKWVTTT DFRWYGPKYI THSIRSFTPSVEQCKES IEQTKQGTWLNPGFPPQSCGYATVTD AEAVIVQVTPHHVLVDEYTGWEVDS QFINGKCSNYICPTVHNSTTWHSYK VKGLCDSNLSMDITFFSEDGELSSLG KEGTGFRSNFYFAYETGGKACKMQYC KHWGVRLPSGVWFEMADKDLFAAA RFPECPEGSSISAPQTSVDVSLIQDVE RILDYSLCQETWSKIRAGLPISPDLS YLAPKNPGTGPAFTIINGTLKYFETRY IRVDAAPILSRMVGMI SGTTTTERELW DDWAPYEDVEIGPNGVLRITSSGYKFP LYMI GHGMLSDLHLSSKAQVFEHP HIQDAASQLPDDESLEFFGDTGLSKNPI ELVEGWPFSSWKSSIASFFFIIGLIIGLFL VLRVGIHLCLKHKTKKQIYTDIEM NRLGK (SEQ ID NO: 103)	

TABLE 10—continued

WT VSV-G reference and mutant sequences
CCCACACATTCAAGACGCCGCCA GCCAGCTGCCGACGATGAGTCG CTGTTCTTCGGAGACACGGGCTTG TCAAAGAATCCCATCGAGCTGGT GGAAGGATGGTTTTCATCCTGGA AAAGCAGCATCGCTTCATTCTTCT TCATCATTGGCCTGATCATCGGCC TATTTCTAGTCCTGCGGGTGGGAA TTCATCTGTGCATCAAGCTCAAGC AACTAAGAAGCGGCAAACTCTAC ACTGATATCGAGATGAATCGCCT GGGCAAGTAG (SEQ ID NO: 102)

[0154] In other embodiments, the methods for producing LVV according to the present disclosure may utilize a five plasmid system. Where a five plasmid system is used, the LVV production system utilizes four plasmids from a third generation vector system (i.e., a transgene plasmid, a GagPol plasmid, an Env plasmid, and a VSV-G env plasmid). In addition, a five plasmid system as described herein includes a fifth plasmid that encodes a targeting protein (the “targeting protein plasmid”). The GagPol and Env packaging plasmids included in a five plasmid system may be standard packaging plasmids as described in *Production of Lentiviral Vectors*, Merten et al., *Molecular Therapy—Methods & Clinical Development* (2016), 3, 16017. The transgene plasmid includes an expression cassette that encodes one or more CAR transgenes according to the present disclosure, and the VSV-G env plasmid includes an expression cassette that encodes a mutant VSV-G as described herein.

[0155] In some embodiment, the fifth plasmid comprises an expression cassette encoding a CD80 targeting protein. In some embodiments, the fifth plasmid comprises an expression cassette encoding an anti-CD3 targeting protein. In some embodiments, the fifth plasmid comprises an expression cassette encoding a CD80 targeting protein and an anti-CD3 targeting protein. Examples of CD80 targeting protein and anti-CD3 targeting proteins that may be used in the fifth plasmid are provided in Tables 1-3.

[0156] In certain embodiments, the present disclosure provides a method for producing a lymphocyte targeted lentiviral vector according to any one of the preceding claims, the method comprising: transfecting a producer cell with a GagPol plasmid, a Rev plasmid, a transgene plasmid, a VSV-G env plasmid, and a lymphocyte targeting protein plasmid; the GagPol plasmid comprises a one or more polynucleotides encoding a lentiviral gag gene and a lentiviral pol gene and is capable of expressing a lentiviral gag protein and a lentiviral pol protein within the producer cell; the Rev plasmid comprises a polynucleotide encoding a lentiviral rev gene and is capable of expressing a lentiviral rev protein within the producer cell; the transgene plasmid comprises an expression cassette comprising a polynucleotide encoding a CAR; and the VSV-G env plasmid comprises a polynucleotide encoding a mutated VSV-G envelope protein as described herein and is capable of expressing the mutated VSV-G envelope protein within the producer cell; and the lymphocyte targeting protein plasmid comprises a polynucleotide encoding a lymphocyte targeting protein as described herein and is capable of expressing the lymphocyte targeting protein within the producer cell; culturing the

producer cell in a culture medium; and harvesting lentiviral vector from the culture medium.

[0157] In some embodiments, the producer cells are transfected with a defined ratio of the transgene plasmid, GagPol plasmid, VSV-G env plasmid, Rev plasmid, and lymphocyte targeting plasmid. The lymphocyte targeting plasmid may be at a ratio of about 0.25 to about 5 relative to the VSV-G env plasmid (by mass). For example, using the defined ratios for the four packaging plasmid system (transgene plasmid, GagPol plasmid, VSV-G env plasmid, Rev plasmid) described herein, the ratio of the lymphocyte targeting protein plasmid can be adjusted relative to the VSV-G env plasmid. For example, the lymphocyte targeting plasmid may be at a ratio of about 0.25:1, 0.5:1, 1:1, 1.25:1, 1.5:1, 1.75:1, 2:1, 2.25:1, 2.5:1, 2.75:1, 3:1, 3.25:1, 3.5:1, 3.75:1, 4:1, 4.25:1, 4.5:1, 4.75:1 or 5:1 relative to the VSV-G env plasmid.

[0158] In embodiments of LVV production methods that utilize a combination of five plasmids, the fifth plasmid is a targeting protein plasmid and includes an expression cassette that encodes one or more targeting proteins. For example the targeting protein plasmid may include an expression cassette that includes a polynucleotide that encodes a CD80 targeting protein, an anti-CD3 targeting protein, or both a CD80 targeting protein and an anti-CD3 targeting protein. In certain embodiments, a targeting protein vector plasmid an expression cassette comprising a polynucleotide according to SEQ ID NO: 1, which encodes a CD80 targeting protein according to SEQ ID NO: 2. In other embodiments, a targeting protein plasmid includes an expression cassette comprising a polynucleotide according to SEQ ID NO: 9, which encodes an anti-CD3 targeting protein according to SEQ ID NO: 10. In other embodiments, a targeting protein plasmid includes an expression cassette comprising a polynucleotide according to SEQ ID NO: 116, which encodes an anti-CD3 targeting protein according to SEQ ID NO: 117. In other embodiments, a targeting protein plasmid includes a tandem expression cassette that comprises a polynucleotide according to SEQ ID NO: 21, which encodes a polypeptide according to SEQ ID NO: 22. Examples of mutated VSV-G sequences that may be used in the five plasmid, 3rd generation lentiviral vector system of the present disclosure are provided in Table 10.

[0159] In the four plasmid packaging system, the concentration of the lymphocyte targeting protein on the surface of the transduced cells may be associated with the concentration of the plasmid containing the targeting protein. For example if the lymphocyte targeting protein plasmid plasmid is contained in the VSV-G env plasmid, the concentra-

tion of the lymphocyte targeting protein on the surface of the transduced cells may be associated with the concentration of the VSV-G env plasmid. In another example, if the lymphocyte targeting protein plasmid is contained in the GagPol plasmid, the concentration of the lymphocyte targeting protein on the surface of the transduced cells may be associated with the concentration of the GagPol plasmid. In another example, if the lymphocyte targeting protein plasmid is contained in the Rev plasmid, the concentration of the lymphocyte targeting protein on the surface of the transduced cells may be associated with the concentration of the Rev plasmid. In the five plasmid packaging system, the concentration of the lymphocyte targeting protein on the surface of the transduced cells may be associated with the concentration of the plasmid containing the targeting protein.

[0160] In the four plasmid packaging system, the transduction efficiency may be associated with the concentration of the plasmid containing the targeting protein. For example if the lymphocyte targeting protein plasmid is contained in the VSV-G env plasmid, the transduction efficiency may be associated with the concentration of the VSV-G env plasmid. In another example, if the lymphocyte targeting protein plasmid is contained in the GagPol plasmid, the transduction efficiency may be associated with the concentration of the GagPol plasmid. In another example, if the lymphocyte targeting protein plasmid is contained in the Rev plasmid, the transduction efficiency may be associated with the concentration of the Rev plasmid. In the five plasmid packaging system, the transduction efficiency may be associated with the concentration of the lymphocyte targeting protein plasmid.

[0161] Following transfection of producer cells, the lentiviral particles product may be harvested from the cell supernatant or culture media. Downstream processes of LVV to maximize LVV recovery while minimizing components which may negatively impact efficacy or safety are known in the art. A typical process, involves sequential purification steps comprising removing cells and their debris followed by enrichment of LVV and the removal of host cell or serum proteins, nucleic acids and lipids. The LVV product may be further concentrated prior to exchanging into a suitable formulation buffer for stability and then finally undergoing sterile filtration prior to storage or application. A clarification step may be initially performed on the harvested LVV supernatants to remove large impurities such as aggregates and cell debris. In some embodiments, the clarification step comprises centrifugation and/or conventional flow filtration. In some embodiments, the clarification step includes a nuclease digestion step. In some embodiments, the LVV product undergoes further purification steps, including for example, ion exchange chromatography (e.g., anion exchange chromatography). The LVV may also be concentrated, e.g., by tangential flow filtration or ultrafiltration/diafiltration. In some embodiments, harvesting LVV from culture medium comprises centrifugation. In some embodiments, harvesting LVV from culture medium comprises anion exchange chromatography. In some embodiments, harvesting LVV from culture medium comprises anion exchange chromatography and tangential flow filtration.

[0162] In some embodiments, the lentiviral vectors described herein, which incorporate a mutated VSV-G env and one or more lymphocyte targeting proteins, are capable of producing a high titer LVV product, as compared to

standard LVV incorporating another fusogenic env protein (e.g., cocl env, paramyxovirus env, truncated VSV-G env). As used herein viral titer refers to infectious virus particle titer as measured by transducing units (TU) per mL. In some embodiments, the titer of LVV described herein is at least $1e7$ TU/mL, at least $1e8$ TU/mL, at least $1e9$ TU/mL, at least $1e10$ TU/mL, at least $1e11$ TU/mL, or at least $1e12$ TU/mL as measured in concentrated LVV product. In some embodiments, the titer of LVV described herein is about $1e7$ TU/mL to about $1e12$ TU/mL in concentrated LVV product.

Engineered Lymphocytes

[0163] The LVV described herein can be used to modify targeted lymphocytes (e.g., T cells, B cells, or NK cells) to express a CAR encoded by a transgene carried by the LVV. In certain embodiments, the engineered lymphocytes have been transduced by an LVV according to the present disclosure by contacting the lymphocytes with the LVV of the present disclosure. In such embodiments, the engineered lymphocytes (e.g., T cells, B cells, or NK cells) express a CAR encoded by the transgene carried by the LVV. These lymphocytes are also referred to herein as “CAR modified lymphocytes.” In specific embodiments, the engineered T cells are referred to herein as “CAR modified T cells.” In other specific embodiments, the engineered NK cells are referred to herein as “CAR modified NK cells.” In specific embodiments, the engineered T cells are referred to herein as “CAR modified B cells.” In specific embodiments the CAR modified T cell expresses a CAR encoded by a transgene carried by an LVV described herein and is selected from naïve T cells (CD45RA+, CCR7+, CD62L+, CD27+, CD45RO-), central memory T cells (CD45RA-, CD45RO+, CD62L+, CCR7+, CD27+), effector memory T cells (CD45RA-, CD45RO+, CCR7-, CD62L-, CD27-), $\gamma\delta$ T cells, mucosal-associated invariant T (MAIT) cells, Tregs, natural killer T cells, and tissue resident T cells.

[0164] Transduction of targeted lymphocytes (e.g., T cells, B cells, or NK cells) by LVV described herein may be performed ex vivo or in vivo.

[0165] In certain embodiments, the lymphocytes (e.g., T cells, B cells, or NK cells) can be primary cells or cell lines derived from human, mouse, rat, or other mammals. If obtained from a mammal, a lymphocyte can be obtained from numerous sources, including blood, bone marrow, lymph node, thymus, or other tissues or fluids. A lymphocyte composition (e.g., T cell composition, B cell composition, or NK cell composition) may be enriched or purified. T cell lines are well known in the art, some of which are described in Sandberg et al., *Leukemia* 21:230, 2000. In certain embodiments, the T cells lack endogenous expression of a TCR α gene, TCR β gene, or both. Such T cells may naturally lack endogenous expression of TCR α and β chains, or may have been modified to block expression (e.g., T cells from a transgenic mouse that does not express TCR α and β chains or cells that have been manipulated to inhibit expression of TCR α and β chains) or to knockout a TCR α chain, a TCR β chain, or both genes.

[0166] In some embodiments, prior to genetic modification of the lymphocytes (e.g., T cells, B cells, or NK cells) with a polynucleotide encoding a CAR molecule, a source of lymphocytes (e.g., T cells or NK cells) may be obtained from a subject (e.g., whole blood, peripheral blood mononuclear cells (PBMCs), bone marrow, lymph node tissue, cord blood, thymus tissue, tissue from a site of infection,

ascites, pleural effusion, spleen tissue). The lymphocytes, T cells, B cells, or NK cells can be enriched or isolated from a sample taken from the subject using methods known in the art. For example, T cells can be enriched in a whole blood sample taken from a subject using known methods for hypotonic lysis of red blood cells or T cells can be isolated from a sample taken from a subject using known gradient sedimentation, e.g., Ficoll® reduction) techniques. Alternatively, T cells can be isolated from whole blood using magnetic antibody-labeled beads followed by column separation. Specific host cell subsets can be collected in accordance with known techniques and enriched or depleted by known techniques, such as affinity binding to antibodies, flow cytometry and/or immunomagnetic selection. In some embodiments, after enrichment and/or depletion steps, the T cells are placed in contact with LVV as disclosed such that T cells obtained from the subject are transduced by the LVV and express the CAR encoded by the transgene carried by the LVV. In some embodiments, after enrichment and/or depletion steps, the NK cells are placed in contact with LVV as disclosed such that NK cells obtained from the subject are transduced by the LVV and express the CAR encoded by the transgene carried by the LVV. In some embodiments, after enrichment and/or depletion steps, the B cells are placed in contact with LVV as disclosed such that B cells obtained from the subject are transduced by the LVV and express the CAR encoded by the transgene carried by the LVV.

[0167] Generally, resting T cells, such as resting CD4 and CD8 lymphocytes, resting B cells, and resting NK cells are refractory to genetic transduction by lentiviral vectors. Resting T cells, also known as quiescent T cells or naïve T cells, refer to T cells that are not mitotically active or have not been exposed to a cognate antigen presented on an antigen presenting cell, such as a macrophage or dendritic cell. An example of a marker for resting T cells is CD28. Alternatively, markers that are expressed on activated T cells but not resting T cells include, for example, 4-1BB, PD-1, and HLA-DR. Similarly, resting B cells and resting NK cells, refer to B cells and NK cells that are not mitotically active or have not been exposed to a cognate antigen, respectively. Examples of markers for resting B cells include CD21 and CD23, and the absence of CD80, CD86, CD95, or CD25. CD137 and GITR, which are expressed on activated NK cells, are absent on resting NK cells. As a result, in order to facilitate transduction of T cells, B cells, or NK cells using a lentiviral vector, the T cells, B cells, or NK cells are typically activated in vitro using stimulation reagents before genetic modification via a lentiviral vector can occur. Following stimulation and transduction, the genetically modified cells are then typically expanded in vitro and subsequently reintroduced into a patient. However, LVV described herein are capable of transducing resting T cells, B cells, and/or resting NK cells. In certain embodiments, methods of transducing T cells with an LVV according to the present description include, placing the LVV in contact with a population of T cells, wherein in the T cells are not activated and the T cells are not exposed to an exogenous stimulating agent during the transduction process. In some such embodiments, the methods of transducing T cells wherein the T cells are not activated and the T cells are not exposed to an exogenous stimulating agent during the transduction process comprises transducing T cells with a LVV comprising a CD80 targeting protein, an anti-CD3 targeting protein, or both a CD80 targeting protein and an anti-CD3

targeting protein. Examples of exogenous T cell stimulating agents include an anti-CD3 antibody or antigen binding fragment thereof (i.e., anti-CD3 antibody or antigen binding fragment thereof that is not part of the LVV), an anti-CD28 antibody or antigen binding fragment thereof (i.e., anti-CD28 antibody or antigen binding fragment thereof that is not part of the LVV), anti-CD2 antibody or antigen binding fragment thereof, IL-2, IL-7, IL-15, PHA, or any combination thereof. Exogenous stimulating agents may be contacted to the T cell in soluble form or immobilized on a solid substrate, such as a bead or cell culture plate. In some embodiments, an anti-CD3 antibody or antigen binding fragment thereof, an anti-CD28 antibody or antigen binding fragment thereof, or both the anti-CD3 antibody and anti-CD28 antibody or antigen binding fragments thereof are contacted to the T cells in soluble form or immobilized on a solid substrate. Examples of anti-CD3 antibodies include OKT3, UCTH1, and BW264/56. An example of an anti-CD2 antibody is LT2. An example of an anti-CD28 antibody is 15E8. In certain embodiments, methods of transducing NK cells with an LVV according to the present description include, placing the LVV in contact with a population of NK cells, wherein in the NK cells are not activated and the NK cells are not exposed to an exogenous stimulating agent during the transduction process. Examples of exogenous NK cell stimulating agents include anti-CD2 antibody or antigen binding fragment thereof, anti-CD335 antibody or antigen binding fragment thereof, IL-2, IL-15, IL-12, IL-18, IL-21, or any combination thereof. In certain embodiments, methods of transducing B cells with an LVV according to the present description include, placing the LVV in contact with a population of B cells, wherein in the B cells are not activated and the B cells are not exposed to an exogenous stimulating agent during the transduction process. Examples of exogenous B cell stimulating agents include CD154 and mixed Ig F(ab)₂.

[0168] In some embodiments, a lentiviral vector having a CD80 targeting protein and an anti-CD3 targeting protein is capable of efficiently transducing both CD4 and CD8 T cells as compared to standard LVV.

[0169] In some embodiments, a lentiviral vector having a CD80 targeting protein and an anti-CD3 targeting protein is capable of activating T cells without exogenous stimulation (e.g., T cell stimulating agent) to a comparable level as T cells transduced with standard LVV that are treated with exogenous stimulation. Examples of exogenous T cell stimulating agents include an anti-CD3 antibody or antigen binding fragment thereof (i.e., anti-CD3 antibody or antigen binding fragment thereof that is not part of the LVV), an anti-CD28 antibody or antigen binding fragment thereof (i.e., anti-CD28 antibody or antigen binding fragment thereof that is not part of the LVV), anti-CD2 antibody or antigen binding fragment thereof, IL-2, IL-7, IL-15, PHA, or any combination thereof.

[0170] In certain embodiments, methods of transducing T cells with an LVV according to the present description exhibit enhanced transduction of CD4 T cells compared to standard LVV. In some such embodiments, the methods of enhancing transduction of CD4 T cells in a mixed population of T cells (e.g., composed of CD4 and CD8 T cells) comprises transducing the mixed population of T cells with a LVV according to the present description comprising a CD80 targeting protein.

[0171] In certain embodiments, methods of transducing T cells with an LVV according to the present description efficiently transduce both CD4 and CD8 T cells. In some such embodiments, the methods of transducing T cells wherein both CD4 and CD8 T cells are efficiently transduced comprises transducing T cells with a LVV comprising a CD80 targeting protein and an anti-CD3 targeting protein.

[0172] In certain embodiments, an LVV according to the present description is capable of transducing T cells under “stressed” conditions, e.g., at low multiplicity of infection (MOI) or without treatment with exogenous IL-2 during transduction. In some embodiments, the LVV comprises a CD80 targeting protein and an anti-CD3 targeting protein. In some embodiments, the LVV according to the present description is capable of transducing T cells at a LVV concentration about 5 to about 25x lower than a non-lymphocyte targeting LVV.

[0173] The expression of a CAR molecule on host cells may be assessed by methods known in the art, such as quantitative PCR or flow cytometry following staining with a fluorescently labeled antigen for the CAR binding domain.

[0174] The expression of a CAR molecule on host cells may be functionally characterized according to any of a large number of art-accepted methodologies for assaying host T cell activity, including determination of T cell binding, activation or induction and also including determination of T cell responses that are antigen-specific. Examples include determination of T cell proliferation, T cell cytokine release, antigen-specific T cell stimulation, CTL activity (e.g., by detecting ^{51}Cr or Europium release from pre-loaded target cells, induction of caspase activity in target cells, extracellular release of lactate dehydrogenase by target cells), changes in T cell phenotypic marker expression, and other measures of T cell functions. Assaying host NK cell activity may also be assayed using similar methodologies. Procedures for performing these and similar assays are may be found, for example, in Lefkovits (*Immunology Methods Manual: The Comprehensive Sourcebook of Techniques*, 1998). See, also, *Current Protocols in Immunology*; Weir, *Handbook of Experimental Immunology*, Blackwell Scientific, Boston, MA (1986); Mishell and Shigii (eds.) *Selected Methods in Cellular Immunology*, Freeman Publishing, San Francisco, CA (1979); Green and Reed, *Science* 281:1309 (1998) and references cited therein. Cytokine levels may be determined according to methods known in the art, including for example, ELISA, ELISPOT, intracellular cytokine staining, flow cytometry, and any combination thereof (e.g., intracellular cytokine staining and flow cytometry). Immune cell proliferation and clonal expansion resulting from an antigen-specific elicitation or stimulation of an immune response may be determined by isolating lymphocytes, such as circulating lymphocytes in samples of peripheral blood cells or cells from lymph nodes, stimulating the cells with antigen, and measuring cytokine production, cell proliferation and/or cell viability, such as by incorporation of tritiated thymidine or non-radioactive assays, such as MTT assays and the like.

Methods of & Compositions for Treatment

[0175] In one aspect, the present disclosure provides methods of treating a disease in a subject comprising administering to the subject an effective amount of an LVV as described herein, a CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) as described herein, or pharmaceutical

compositions thereof. In another aspect, the methods of treating a disease in a subject according to the present description include administering to the subject an effective amount of an LVV as described herein, a CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) as described herein, or pharmaceutical compositions thereof, in combination with one or more additional therapeutic agents.

[0176] Diseases that may be treated with the LVV or CAR modified lymphocytes (e.g., T cells, B cells, or NK cells) according to the present description include cancers. Adoptive immune and gene therapies are promising treatments for various types of cancer (Morgan et al., *Science* 314:126, 2006; Schmitt et al., *Hum. Gene Ther.* 20:1240, 2009; June, J. Clin. Invest. 117:1466, 2007) and infectious disease (Kitchen et al., *PLoS One* 4:38208, 2009; Rossi et al., *Nat. Biotechnol.* 25:1444, 2007; Zhang et al., *PLoS Pathog.* 6:e1001018, 2010; Luo et al., *J. Mol. Med.* 89:903, 2011).

[0177] A wide variety of cancers, including solid tumors and leukemias are amenable to the compositions and methods disclosed herein. Exemplary cancers that may be treated using the receptors, modified host cells, and composition described herein include adenocarcinoma of the breast, prostate, and colon; all forms of bronchogenic carcinoma of the lung; myeloid leukemia; melanoma; hepatoma; neuroblastoma; papilloma; apudoma; choristoma; branchioma; malignant carcinoid syndrome; carcinoid heart disease; and carcinoma (e.g., Walker, basal cell, basosquamous, Brown-Pearce, ductal, Ehrlich tumor, Krebs 2, Merkel cell, mucinous, non-small cell lung, oat cell, papillary, scirrhous, bronchiolar, bronchogenic, squamous cell, and transitional cell). Additional types of cancers that may be treated using the receptors, modified host cells, and composition described herein include histiocytic disorders; malignant histiocytosis; leukemia; Hodgkin's disease; immunoproliferative small; non-Hodgkin's lymphoma; plasmacytoma; multiple myeloma; chronic myeloid leukemia (CML); acute myeloid leukemia (AML); plasmacytoma; reticuloendotheliosis; melanoma; chondroblastoma; chondroma; chondrosarcoma; fibroma; fibrosarcoma; giant cell tumors; histiocytoma; lipoma; liposarcoma; mesothelioma; myxoma; myxosarcoma; osteoma; osteosarcoma; chordoma; cranio-pharyngioma; dysgerminoma; hamartoma; mesenchymoma; mesonephroma; myosarcoma; ameloblastoma; cementoma; odontoma; teratoma; thymoma; trophoblastic tumor. Further, the following types of cancers are also contemplated as amenable to treatment using the receptors, modified host cells, and composition described herein: adenoma; cholangioma; cholesteatoma; cyclindroma; cystadenocarcinoma; cystadenoma; granulosa cell tumor; gynandroblastoma; hepatoma; hidradenoma; islet cell tumor; Leydig cell tumor; papilloma; sertoli cell tumor; theca cell tumor; leiomyoma; leiomyosarcoma; myoblastoma; myoma; myosarcoma; rhabdomyoma; rhabdomyosarcoma; ependymoma; gangli-neuroma; glioma; medulloblastoma; meningioma; neurilemmoma; neuroblastoma; neuroepithelioma; neurofibroma; neuroma; paraganglioma; paraganglioma nonchromaffin. The types of cancers that may be treated also include angiokeratoma; angiolymphoid hyperplasia with eosinophilia; angioma sclerosing; angiomatosis; glomangioma; hemangioendothelioma; hemangioma; hemangiopericytoma; hemangiosarcoma; lymphangioma; lymphangiomyoma; lymphangiosarcoma; pinealoma; carcinosarcoma; chondrosarcoma; cystosarcoma phylloides; fibrosarcoma; hemangiosarcoma; leiomyosarcoma; leukosarcoma; liposar-

coma; lymphangiosarcoma; myosarcoma; myxosarcoma; ovarian carcinoma; rhabdomyosarcoma; sarcoma; neoplasms; neurofibromatosis; cervical dysplasia, and peritoneal cancer.

[0178] Examples of hyperproliferative disorders amenable to therapy using the receptors, modified host cells, and composition described herein include B-cell cancers (B-cell malignancies), including B-cell lymphomas (such as various forms of Hodgkin's disease, non-Hodgkin's lymphoma (NHL) or central nervous system lymphomas), leukemias (such as acute lymphoblastic leukemia (ALL), chronic lymphocytic leukemia (CLL), Hairy cell leukemia, B cell blast transformation of chronic myeloid leukemia, acute myeloid leukemia (AML), chronic myeloid leukemia, and myelomas (such as multiple myeloma). Additional B cell cancers that may be treated using the receptors, modified host cells, and composition described herein include small lymphocytic lymphoma, B-cell prolymphocytic leukemia, lymphoplasmacytic lymphoma, splenic marginal zone lymphoma, plasma cell myeloma, solitary plasmacytoma of bone, extraosseous plasmacytoma, extra-nodal marginal zone B-cell lymphoma of mucosa-associated (MALT) lymphoid tissue, nodal marginal zone B-cell lymphoma, follicular lymphoma, mantle cell lymphoma, diffuse large B-cell lymphoma, mediastinal (thymic) large B-cell lymphoma, intravascular large B-cell lymphoma, primary effusion lymphoma, Burkitt's lymphoma/leukemia, B-cell proliferations of uncertain malignant potential, lymphomatoid granulomatosis, and post-transplant lymphoproliferative disorder.

[0179] The CAR modified lymphocytes included in the compositions disclosed herein and administered to a subject may include, CAR modified T cells, e.g., CD4+ T cells, CD8+ T cells, Natural Killer T cells, gamma delta T cells, or MAIT cells; CAR modified B cells, or CAR modified NK cells. In certain embodiments, methods of treating a subject comprise administering an effective amount of an LVV as described herein or CAR modified lymphocytes (i.e., recombinant cells that express one or more CARs) according to the present disclosure. In some embodiments, CAR modified T cells are administered to a subject. In some embodiments, CAR modified NK cells are administered to a subject. In some embodiments, CAR modified B cells are administered to a subject. The CAR modified lymphocytes (e.g., T cells, B cells, or NK cells) may be xenogeneic, syngeneic, allogeneic, or autologous to the subject.

[0180] Pharmaceutical compositions including the LVV or CAR engineered lymphocytes may be administered in a manner appropriate to the disease or condition to be treated (or prevented) as determined by persons skilled in the medical art. An appropriate dose, suitable duration, and frequency of administration of the compositions will be determined by such factors as the condition of the patient, size, weight, body surface area, age, sex, type and severity of the disease, particular therapy to be administered, particular form of the active ingredient, time and the method of administration, and other drugs being administered concurrently. The present disclosure provides pharmaceutical compositions comprising LVV or CAR modified lymphocytes and a pharmaceutically acceptable carrier, diluent, or excipient. Suitable excipients include water, saline, dextrose, glycerol, or the like and combinations thereof. Other suitable infusion medium can be any isotonic medium formulation, including saline, Normosol R (Abbott), Plasma-Lyte A (Baxter), 5% dextrose in water, or Ringer's lactate.

[0181] A treatment effective amount of engineered lymphocytes (e.g., T cells, B cells, or NK cells) in a pharmaceutical composition is at least one cell (for example, one CAR modified T cell) and is more typically greater than 10^2 cells, for example, up to 10^6 , up to 10^7 , up to 10^8 cells, up to 10^9 cells, up to 10^{10} cells, or up to 10^{11} cells or more. In certain embodiments, the cells are administered in a range from about 10^6 to about 10^{10} cells/m², preferably in a range of about 10^7 to about 10^9 cells/m². The number of cells will depend upon the ultimate use for which the composition is intended as well the type of cells included therein. For example, a composition comprising T cells modified to contain a CAR will comprise a T cell population containing from about 5% to about 95% or more of such cells. In certain embodiments, a composition comprising CAR modified T cells comprises a cell population comprising at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more of such cells. In certain embodiments, a composition comprising CAR modified NK cells comprises a cell population comprising at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more of such cells. In certain embodiments, a composition comprising CAR modified B cells comprises a cell population comprising at least 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95% or more of such cells. For uses provided herein, the lymphocytes (e.g., T cells, B cells, or NK cells) are generally in a volume of a liter or less, 500 mls or less, 250 mls or less, or 100 mls or less. Hence the density of the desired cells is typically greater than 104 cells/ml and generally is greater than 10^7 cells/ml, generally 10^8 cells/ml or greater. The lymphocytes (e.g., T cells, B cells, or NK cells) may be administered as a single infusion or in multiple infusions over a range of time. Repeated infusions of CAR modified lymphocytes (e.g., T cells, B cells, or NK cells) may be separated by days, weeks, months, or even years if relapses of disease or disease activity are present. A clinically relevant number of immune cells can be apportioned into multiple infusions that cumulatively equal or exceed 10^6 , 10^7 , 10^8 , 10^9 , 10^{10} , or 10^{11} cells. A preferred dose for administration of a host cell comprising a recombinant expression vector as described herein is about 10^7 cells/m², about 5×10^7 cells/m², about 10^8 cells/m², about 5×10^8 cells/m², about 10^9 cells/m², about 5×10^9 cells/m², about 10^{10} cells/m², about 5×10^{10} cells/m², or about 10^{11} cells/m².

[0182] The LVV and/or CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) compositions as described herein may be administered to a subject intravenously, intraperitoneally, intratumorally, into the bone marrow (e.g., intraosseous administration), into the lymph node (intranodally), and/or into cerebrospinal fluid.

[0183] The LVV vectors and/or CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) compositions may be administered to a subject in combination with one or more additional therapeutic agents. Examples of therapeutic agents that may be administered in combination with the LVV vectors or CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) according to the present description include radiation therapy, adoptive cellular immunotherapy agent (e.g., recombinant TCR, enhanced affinity TCR, CAR, TCR-CAR, scTCR fusion protein, dendritic cell vaccine), antibody therapy, immune checkpoint molecule inhibitor therapy, UV light therapy, electric pulse therapy, high inten-

sity focused ultrasound therapy, oncolytic virus therapy, or a pharmaceutical therapy, such as a chemotherapeutic agent, a therapeutic peptide, a hormone, an aptamer, antibiotic, anti-viral agent, anti-fungal agent, anti-inflammatory agent, a small molecule therapy, or any combination thereof.

[0184] Radiation therapy includes external beam radiation therapy (e.g., conventional external beam radiation therapy, stereotactic radiation, 3-dimensional conformal radiation therapy, intensity-modulated radiation therapy, volumetric modulated arc therapy, particle therapy, proton therapy, and auger therapy), brachytherapy, systemic radioisotope therapy, intraoperative radiotherapy, or any combination thereof.

[0185] Exemplary antibodies that may be used in conjunction with the LVV or CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) compositions described herein include rituximab, pertuzumab, trastuzumab, alemtuzumab, Ibritumomab tiuxetan, Brentuximab vedotin, cetuximab, bevacizumab, abciximab, adalimumab, alefacept, basilizimab, belimumab, bezlotoxumab, canakinumab, certolizumab pegol, daclizumab, denosumab, efalizumab, golimumab, olaratumab, palivizumab, panitumumab, and tocilizumab.

[0186] Exemplary inhibitors of immune checkpoint molecules that may be used in conjunction with the LVV or CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) compositions described herein include checkpoint inhibitors targeting PD-L1, PD-L2, CD80, CD86, B7-H3, B7-H4, HVEM, adenosine, GAL9, VISTA, CEACAM-1, CEACAM-3, CEACAM-5, PVRL2, PD-1, CTLA-4, BTLA, KIR, LAG3, TIM3, A2aR, CD244/2B4, CD160, TIGIT, LAIR-1, PVRIG/CD112R, or any combination thereof. In certain embodiments, an immune checkpoint inhibitor may be an antibody, a peptide, an RNAi agent, or a small molecule. An antibody specific for CTLA-4 may be ipilimumab or tremelimumab. An antibody specific for PD-1 may be pidlizumab, nivolumab, or pembrolizumab. An antibody specific for PD-L1 may be durvalumab, atezolizumab, or avelumab.

[0187] Exemplary chemotherapeutics that may be used in conjunction with the LVV or CAR modified lymphocyte (e.g., T cell, B cell, or NK cell) compositions described herein may include an alkylating agent, a platinum based agent, a cytotoxic agent, an inhibitor of chromatin function, a topoisomerase inhibitor, a microtubule inhibiting drug, a DNA damaging agent, an antimetabolite (such as folate antagonists, pyrimidine analogs, purine analogs, and sugar-modified analogs), a DNA synthesis inhibitor, a DNA interactive agent (such as an intercalating agent), and a DNA repair inhibitor.

[0188] As referred to herein, a chemotherapeutic includes non-specific cytotoxic agents that inhibit mitosis or cell division, as well as molecularly targeted therapy that blocks the growth and spread of cancer cells by targeting specific molecules that are involved in tumor growth, progression, and metastasis (e.g., oncogenes). Exemplary non-specific chemotherapeutics for use in conjunction with the expression cassette compositions described herein may include an alkylating agent, a platinum based agent, a cytotoxic agent, an inhibitor of chromatin function, a topoisomerase inhibitor, a microtubule inhibiting drug, a DNA damaging agent, an antimetabolite (such as folate antagonists, pyrimidine analogs, purine analogs, and sugar-modified analogs), a

DNA synthesis inhibitor, a DNA interactive agent (such as an intercalating agent), hypomethylating agent, and a DNA repair inhibitor.

[0189] Examples of chemotherapeutic agents considered for use in combination therapies contemplated herein include vemurafenib, dabrafenib, trametinib, cobimetinib, anastrozole (Arimidex®), bicalutamide (Casodex®), bleomycin sulfate (Blenoxane®), busulfan (Myleran®), busulfan injection (Busulfex®), capecitabine (Xeloda®), N4-pentoxycarbonyl-5-deoxy-5-fluorocytidine, carboplatin (Paraplatin®), carmustine (BiCNU®), chlorambucil (Leukeran®), cisplatin (Platinol®), cladribine (Leustatin®), cyclophosphamide (Cytoxan® or Neosar®), cytarabine, cytosine arabinoside (Cytosar-U®), cytarabine liposome injection (DepoCyt®), dacarbazine (DTIC-Dome®), dactinomycin (Actinomycin D, Cosmegen), daunorubicin hydrochloride (Cerubidine®), daunorubicin citrate liposome injection (DaunoXome®), dexamethasone, docetaxel (Taxotere®), doxorubicin hydrochloride (Adriamycin®, Rubex®), etoposide (Vepesid®), fludarabine phosphate (Fludara®), 5-fluorouracil (Adrucil®, Efudex®), flutamide (Eulexin®), tezacetibine, Gemcitabine (difluorodeoxycytidine), hydroxyurea (Hydrea®), Idarubicin (Idamycin®), ifosfamide (IFEX®), irinotecan (Camptosar®), L-asparaginase (ELSPAR®), leucovorin calcium, melphalan (Alkeran®), 6-mercaptopurine (Purinethol®), methotrexate (Folex®), mitoxantrone (Novantrone®), mylotarg, paclitaxel (Taxol®), phoenix (Yttrium90/MX-DTPA), pentostatin, polifeprosan 20 with carmustine implant (Gliadel®), fadbra tamoxifen citrate (Nolvadex®), teniposide (Vumon®), 6-thioguanine, thiotepa, tirapazamine (Tirazone®), topotecan hydrochloride for injection (Hycamptin®), vinblastine (Velban®), vincristine (Oncovin®), ibrutinib, venetoclax, crizotinib, alectinib, brigatinib, ceritinib, and vinorelbine (Navelbine®).

[0190] Exemplary alkylating agents for use in combination therapies contemplated herein include nitrogen mustards, ethylenimine derivatives, alkyl sulfonates, nitrosoureas and triazenes; uracil mustard (Aminouracil Mustard®, Chlorethaminacil®, Demethylodopan®, Desmethyldopan®, Haemanthamine®, Nordopan®, Uracil nitrogen Mustard®, Uracillost®, Uracilmustaza®, Uramustine®, Uramustine®), chlormethine (Mustargen®), cyclophosphamide (Cytoxan®, Neosar®, Clafen®, Endoxan®, Procytox®, Revimmune™), ifosfamide (Mitoxana®), melphalan (Alkeran®), Chlorambucil (Leukeran®), pipobroman (Amedel®, Vercyte®), triethylenemelamine (Hemel®, Hexalen®, Hexastat®), triethylenethiophosphoramine, Temozolomide (Temodar®), thiotepa (Thiopex®), busulfan (Busilvex®, Myleran®), carmustine (BiCNU®), lomustine (CeeNU®), streptozocin (Zanosar®), and Dacarbazine (DTIC-Dome®). Additional exemplary alkylating agents for use in combination therapies contemplated herein include, without limitation, Oxaliplatin (Eloxatin®); Temozolomide (Temodar® and Temodal®); Dactinomycin (also known as actinomycin-D, Cosmegen®); Melphalan (also known as L-PAM, L-sarcosylsin, and phenylalanine mustard, Alkeran®); Altretamine (also known as hexamethylmelamine (HMM), Hexalen®); Carmustine (BiCNU®); Bendamustine (Treanda®); Busulfan (Busulfex® and Myleran®); Carboplatin (Paraplatin®); Lomustine (also known as CCNU, CeeNU®); Cisplatin (also known as CDDP, Platinol® and Platinol®-AQ); Chlorambucil (Leukeran®); Cyclophosphamide (Cytoxan® and Neosar®); Dacarbazine (also known

as DTIC, DIC and imidazole carboxamide, DTIC-Dome®; Altretamine (also known as hexamethylmelamine (HMM), Hexalen®); Ifosfamide (Ifex®); Prednumustine; Procarbazine (Matulane®); Mechlorethamine (also known as nitrogen mustard, mustine and mechlorethamine hydrochloride, Mustargen®); Streptozocin (Zanosar®); Thiotepe (also known as thiophosphamide, TESP and TSPA, Thioplex®); Cyclophosphamide (Endoxan®, Cytosan®, Neosar®, Procytox®, Revimmune®); and Bendamustine HCl (Treanda®).

[0191] Exemplary platinum based agents for use in combination therapies contemplated herein include carboplatin, cisplatin, oxaliplatin, nedaplatin, picoplatin, satraplatin, phenanthriplatin, and triplatin tetranitrate.

[0192] Exemplary hypomethylating agents for use in combination therapies include azacitidine and decitabine.

[0193] Exemplary molecularly targeted inhibitors for use in combination therapies contemplated herein include small molecules that target molecules involved in cancer cell growth and survival, including for example, receptor tyrosine kinase inhibitors, RAF inhibitors, BCL-2 inhibitors, ABL inhibitors, TRK inhibitors, c-KIT inhibitors, c-MET inhibitors, CDK4/6 inhibitors, FAK inhibitors, FGFR inhibitors, FLT3 inhibitors, IDH1 inhibitors, IDH2 inhibitors, PDGFRA inhibitors, and RET inhibitors.

[0194] Exemplary molecularly targeted therapy includes hormone antagonists, signal transduction inhibitors, gene expression inhibitors (e.g., translation inhibitors), apoptosis inducers, angiogenesis inhibitors (e.g., a VEGF pathway inhibitor), tyrosine kinase inhibitors (e.g., an EGF/EGFR pathway inhibitor), growth factor inhibitors, GTPase inhibitors, serine/threonine kinase inhibitors, transcription factor inhibitors, inhibitors of driver mutations associated with cancer, B-Raf inhibitors, RAF inhibitors, a MEK inhibitors, mTOR inhibitors, adenosine pathway inhibitors, EGFR inhibitors, PI3K inhibitors, BCL2 inhibitors, VEGFR inhibitors, MET inhibitors, MYC inhibitors, BCR-ABL inhibitors, ABL inhibitors, HER2 inhibitors, H-RAS inhibitors, K-RAS inhibitors, PDGFR inhibitors, ALK inhibitors, ROS1 inhibitors, BTK inhibitors, TRK inhibitors, c-KIT inhibitors, c-MET inhibitors, CDK4/6 inhibitors, FAK inhibitors, FGFR inhibitors, FLT3 inhibitors, IDH1 inhibitors, IDH2 inhibitors, PARP inhibitors, PARP inhibitors, PDGFRA inhibitors, and RET inhibitors.

[0195] Exemplary angiogenesis inhibitors include, without limitation A6 (Angstrom Pharmaceuticals), ABT-510 (Abbott Laboratories), ABT-627 (Atrasentan) (Abbott Laboratories/Xinlay), ABT-869 (Abbott Laboratories), Actimid (CC4047, Pomalidomide) (Celgene Corporation), AdGVPEDE.11D (GenVec), ADH-1 (Exherin) (Adherex Technologies), AEE788 (Novartis), AG-013736 (Axitinib) (Pfizer), AG3340 (Prinomastat) (Agouron Pharmaceuticals), AGX1053 (AngioGenex), AGX51 (AngioGenex), ALN-VSP (ALN-VSP 02) (Alnylam Pharmaceuticals), AMG 386 (Amgen), AMG706 (Amgen), Apatinib (YN968D1) (Jiangsu Hengrui Medicine), AP23573 (Ridaforolimus/MK8669) (Ariad Pharmaceuticals), AQ4N (Novavea), ARQ 197 (ArQule), ASA404 (Novartis/Antisoma), Atiprimod (Callisto Pharmaceuticals), ATN-161 (Attenuon), AV-412 (Aveo Pharmaceuticals), AV-951 (Aveo Pharmaceuticals), Avastin (Bevacizumab) (Genentech), AZD2171 (Cediranib/Recentin) (AstraZeneca), BAY 57-9352 (Telaatinib) (Bayer), BEZ235 (Novartis), BIBF1120 (Boehringer Ingelheim Pharmaceuticals), BIBW 2992 (Boehringer Ingelheim Pharma-

ceuticals), BMS-275291 (Bristol-Myers Squibb), BMS-582664 (Brivanib) (Bristol-Myers Squibb), BMS-690514 (Bristol-Myers Squibb), Calcitriol, CCI-779 (Torisel) (Wyeth), CDP-791 (ImClone Systems), Ceflatonin (Homoharringtonine/HHT) (ChemGenex Therapeutics), Celebrex (Celecoxib) (Pfizer), CEP-7055 (Cephalon/Sanofi), CHIR-265 (Chiron Corporation), NGR-TNF, COL-3 (Metastat) (Collagenex Pharmaceuticals), Combretastatin (Oxigene), CP-751,871 (Figitumumab) (Pfizer), CP-547,632 (Pfizer), CS-7017 (Daiichi Sankyo Pharma), CT-322 (Angiocept) (Adnexus), Curcumin, Dalteparin (Fragmin) (Pfizer), Disulfiram (Antabuse), E7820 (Eisai Limited), E7080 (Eisai Limited), EMD 121974 (Cilengitide) (EMD Pharmaceuticals), ENMD-1198 (EntreMed), ENMD-2076 (EntreMed), Endostar (Simcere), Erbitux (ImClone/Bristol-Myers Squibb), EZN-2208 (Enzon Pharmaceuticals), EZN-2968 (Enzon Pharmaceuticals), GC1008 (Genzyme), Genistein, GSK1363089 (Foretinib) (GlaxoSmithKline), GW786034 (Pazopanib) (GlaxoSmithKline), GT-111 (Vascular Biogenics Ltd.), IMC-1121B (Ramucirumab) (ImClone Systems), IMC-18F1 (ImClone Systems), IMC-3G3 (ImClone LLC), INCB007839 (Incyte Corporation), INGN 241 (Introgen Therapeutics), Iressa (ZD1839/Gefitinib), LBH589 (Faridak/Panobinostat) (Novartis), Lucentis (Ranibizumab) (Genentech/Novartis), LY317615 (Ezrastaurin) (Eli Lilly and Company), Macugen (Pegaptanib) (Pfizer), MEDI522 (Abrigin) (MedImmune), MLN518 (Tandutinib) (Millennium), Neovastat (AE941/Benefin) (Aeterna Zentaris), Nexavar (Bayer/Onyx), NM-3 (Genzyme Corporation), Noscipine (Cougar Biotechnology), NPI-2358 (Nereus Pharmaceuticals), OSI-930 (OSI), Palomid 529 (Paloma Pharmaceuticals, Inc.), Panzem Capsules (2ME2) (EntreMed), Panzem NCD (2ME2) (EntreMed), PF-02341066 (Pfizer), PF-04554878 (Pfizer), PI-88 (Progen Industries/Medigen Biotechnology), PKC412 (Novartis), Polyphenon E (Green Tea Extract) (Polypheno E International, Inc.), PPI-2458 (Praecis Pharmaceuticals), PTC299 (PTC Therapeutics), PTK787 (Vatalanib) (Novartis), PXD101 (Belinostat) (CuraGen Corporation), RAD001 (Everolimus) (Novartis), RAF265 (Novartis), Regorafenib (BAY73-4506) (Bayer), Revlimid (Celgene), Retaane (Alcon Research), SN38 (Liposomal) (Neopharm), SNS-032 (BMS-387032) (Sunesis), SOM230 (Pasireotide) (Novartis), Squalamine (Genaera), Suramin, Sutent (Pfizer), Tarceva (Genentech), TB-403 (Thrombogenics), Tempostatin (Collard Biopharmaceuticals), Tetrathiomolybdate (Sigma-Aldrich), TG100801 (TargeGen), Thalidomide (Celgene Corporation), Tinzaparin Sodium, TKI258 (Novartis), TRC093 (Traccon Pharmaceuticals Inc.), VEGF Trap (Aflibercept) (Regeneron Pharmaceuticals), VEGF Trap-Eye (Regeneron Pharmaceuticals), Veglin (VasGene Therapeutics), Bortezomib (Millennium), XL184 (Exelixis), XL647 (Exelixis), XL784 (Exelixis), XL820 (Exelixis), XL999 (Exelixis), ZD6474 (AstraZeneca), Vorinostat (Merck), and ZSTK474.

[0196] Exemplary B-Raf inhibitors include vemurafenib, dabrafenib, and encorafenib.

[0197] Exemplary MEK inhibitors include binimetinib, cobimetinib, refametinib, selumetinib, and trametinib.

[0198] Exemplary BTK inhibitors include ibrutinib, Loxo-305, tirabrutinib, GDC-0853, acalabrutinib, ONO-4059, spebrutinib, BGB-3111, HM71224, and M7583.

[0199] Exemplary TRK inhibitors include entrectinib, larotrectinib, CH7057288, ONO-7579, LOXO-101, lestaurtinib, and LOXO-195.

[0200] Exemplary c-KIT inhibitors include imatinib, sunitinib, and ponatinib.

[0201] Exemplary c-MET inhibitors include capmatinib, crizotinib, tivantinib, onartuzumab, INCB28060, AMG-458, savolitinib, and tepotinib.

[0202] Exemplary CDK4/6 inhibitors include palbociclib, ribociclib, abemaciclib, and trilaciclib.

[0203] Exemplary FAK inhibitors include defactinib, GSK2256098, BI853520, and PF-00562271.

[0204] Exemplary FGFR inhibitors include erdafitinib, pemigatinib, infigratinib, rogaratinib, AZD4547, BGJ398, FP-1039, and ARQ 087.

[0205] Exemplary FLT-3 inhibitors include quizartinib, crenolanib, gilteritinib, midostaurin, and lestaurtinib.

[0206] Exemplary IDH1 inhibitors include ivosidenib, BAY-1436032, and AGI-5198.

[0207] An exemplary IDH2 inhibitor includes enasidenib.

[0208] Exemplary PARP inhibitors include talazoparib, niraparib, rucaparib, olaparib, veliparib, CEP 9722, E7016, AG014699, MK4827, BMN-673, and Pamiparib (BGB-290).

[0209] Exemplary PDGFRA inhibitors include imatinib, regorafenib, crenolanib, and olaratumab.

[0210] Exemplary pan-RAF inhibitors include belvarafenib, LXH254, LY3009120, INU-152, and HM95573.

[0211] Exemplary RET inhibitors include lenvatinib, Alecitinib, vandetanib, cabozantinib, BLU-667, and LOXO-292.

[0212] Exemplary ROS1 inhibitors include ceritinib, lorlatinib, entrectinib, crizotinib, TPX-0005, and DS-6051b.

[0213] Exemplary Vascular Endothelial Growth Factor (VEGF) receptor inhibitors include, but are not limited to, Bevacizumab (Avastin®), axitinib (Inlyta®); Brivanib alaninate (BMS-582664, (S)—((R)-1-(4-(4-Fluoro-2-methyl-1H-indol-5-yloxy)-5-methylpyrrolo[2,1-f][1,2,4]triazin-6-yloxy)propan-2-yl)2-aminopropanoate); Sorafenib (Nexavar®); Pazopanib (Votrient®); Sunitinib malate (Sutent®); Cediranib (AZD2171, CAS 288383-20-1); Vargatef (BIBF1120, CAS 928326-83-4); Foretinib (GSK1363089); Telatinib (BAY57-9352, CAS 332012-40-5); Apatinib (YN968D1, CAS 811803-05-1); Imatinib (Gleevec®); Ponatinib (AP24534, CAS 943319-70-8); Tivozanib (AV951, CAS 475108-18-0); Regorafenib (BAY73-4506, CAS 755037-03-7); Vatalanib dihydrochloride (PTK787, CAS 212141-51-0); Brivanib (BMS-540215, CAS 649735-46-6); Vandetanib (Caprelsa® or AZD6474); Motesanib diphosphate (AMG706, CAS 857876-30-3, N-(2,3-dihydro-3,3-dimethyl-1H-indol-6-yl)-2-[(4-pyridinylmethyl)amino]-3-pyridinecarboxamide, described in PCT Publication No. WO 02/066470); Dovitinib dilactate (TK1258, CAS 852433-84-2); Linfanib (ABT869, CAS 796967-16-3); Cabozantinib (XL184, CAS 849217-68-1); Lestaurtinib (CAS 111358-88-4); N-[5-[[5-(1,1-Dimethylethyl)-2-oxazolyl]methyl]thio]-2-thiazolyl]-4-piperidinecarboxamide (BMS38703, CAS 345627-80-7); (3R,4R)-4-Amino-1-((4-((3-methoxyphenyl)amino)pyrrolo[2,1-f][1,2,4]triazin-5-yl)methyl)piperidin-3-ol (BMS690514); N-(3,4-Dichloro-2-fluorophenyl)-6-methoxy-7-[(3 α ,5 β ,6 α)-octahydro-2-methylcyclopenta[c]pyrrol-5-yl]methoxy]-4-quinazolinamine (XL647, CAS 781613-23-8); 4-Methyl-3-[[1-methyl-6-(3-pyridinyl)-1H-pyrazolo[3,4-d]pyrimidin-4-yl]amino]-N-[3-(trifluoromethyl)phenyl]-benzamide (BH712, CAS 940310-85-0); and Aflibercept (Eylea®).

[0214] Exemplary EGF pathway inhibitors include, without limitation tyrphostin 46, EKB-569, erlotinib (Tarceva®),

gefitinib (Iressa®), erbitux, nimotuzumab, lapatinib (Tykerb®), cetuximab (anti-EGFR mAb), ¹⁸⁸Re-labeled nimotuzumab (anti-EGFR mAb), and those compounds that are generically and specifically disclosed in WO 97/02266, EP 0 564 409, WO 99/03854, EP 0 520 722, EP 0 566 226, EP 0 787 722, EP 0 837 063, U.S. Pat. No. 5,747,498, WO 98/10767, WO 97/30034, WO 97/49688, WO 97/38983 and WO 96/33980. Exemplary EGFR antibodies include, but are not limited to, Cetuximab (Erbitux®); Panitumumab (Vectibix®); Matuzumab (EMD-72000); Trastuzumab (Herceptin®); Nimotuzumab (hR3); Zalutumumab; TheraCIM h-R3; MDX0447 (CAS 339151-96-1); and ch806 (mAb-806, CAS 946414-09-1). Exemplary Epidermal growth factor receptor (EGFR) inhibitors include, but not limited to, Erlotinib hydrochloride (Tarceva®); ceritinib; brigatinib; osimertinib; icotinib; Gefitinib (Iressa®); N-[4-[(3-Chloro-4-fluorophenyl)amino]-7-[(3"S)-tetrahydro-3-furanyl]oxy]-6-quinazolinyl]-4(dimethylamino)-2-butenamide, Tovok®; Vandetanib (Caprelsa®); Lapatinib (Tykerb®); (3R,4R)-4-Amino-1-((4-((3-methoxyphenyl)amino)pyrrolo[2,1-f][1,2,4]triazin-5-yl)methyl)piperidin-3-ol (BMS690514); Canertinib dihydrochloride (CI-1033); 6-[4-[(4-Ethyl-1-piperazinyl)methyl]phenyl]-N-[(1R)-1-phenylethyl]-7H-Pyrrolo[2,3-d]pyrimidin-4-amine (AEE788, CAS 497839-62-0); Mubritinib (TAK165); Pelitinib (EKB569); Afatinib (BIBW2992); Neratinib (HKI-272); N-[4-[[1-[(3-Fluorophenyl)methyl]-1H-indazol-5-yl]amino]-5-methylpyrrolo[2,1-f][1,2,4]triazin-6-yl]-carbamate acid, (3S)-3-morpholinylmethyl ester (BMS599626); N-(3,4-Dichloro-2-fluorophenyl)-6-methoxy-7-[(3 α ,5 β ,6 α)-octahydro-2-methylcyclopenta[c]pyrrol-5-yl]methoxy]-4-quinazolinamine (XL647, CAS 781613-23-8); 4-[4-[(1R)-1-Phenylethyl]amino]-7H-pyrrolo[2,3-d]pyrimidin-6-yl]-phenol (PKI166, CAS 187724-61-4); rocletitinib.

[0215] Exemplary mTOR inhibitors include, without limitation, rapamycin (Rapamune®), and analogs and derivatives thereof; SDZ-RAD; Temsirolimus (Torisel®); also known as CCI-779; Ridaforolimus (formally known as deferolimus, (1R,2R,4S)-4-[(2R)-2-[(1R,9S,12S,15R,16E,18R,19R,21R,23S,24E,26E,28Z,30S,32S,35R)-1,18-dihydroxy-19,30-dimethoxy-15,17,21,23,29,35-hexamethyl-2,3,10,14,20-pentaoxo-11,36-dioxo-4-azatricyclo[30.3.1.0^{4,9}]hexatriaconta-16,24,26,28-tetraen-12-yl]propyl]-2-methoxycyclohexyl dimethylphosphinate, also known as AP23573 and MK8669, and described in PCT Publication No. WO 03/064383); Everolimus (Afinitor® or RAD001); Rapamycin (AY22989, Sirolimus®); Simapimod (CAS 164301-51-3); (5-{2,4-Bis[(3S)-3-methylmorpholin-4-yl]pyrido[2,3-d]pyrimidin-7-yl}-2-methoxyphenyl)methanol (AZD8055); 2-Amino-8-[trans-4-(2-hydroxyethoxy)cyclohexyl]-6-(6-methoxy-3-pyridinyl)-4-methyl-pyrido[2,3-d]pyrimidin-7(8H)-one (PF04691502, CAS 1013101-36-4); and N²-[1,4-dioxo-[[4-(4-oxo-8-phenyl-4H-1-benzopyran-2-yl)morpholinium-4-yl]methoxy]butyl]-L-arginylglycyl-L- α -aspartyl-L-serine-, inner salt (SF1126, CAS 936487-67-1).

[0216] Exemplary Phosphoinositide 3-kinase (PI3K) inhibitors include, but are not limited to, duvelisib, idelalisib, 4-[2-(1H-Indazol-4-yl)-6-[[4-(methylsulfonyl)piperazin-1-yl]methyl]thieno[3,2-d]pyrimidin-4-yl]morpholine (also known as GDC 0941 and described in PCT Publication Nos. WO 09/036082 and WO 09/055730); 2-Methyl-2-[4-[3-methyl-2-oxo-8-(quinolin-3-yl)-2,3-dihydroimidazo[4,5-c]quinolin-1-yl]phenyl]propionitrile (also known as BEZ

235 or NVP-BEZ 235, and described in PCT Publication No. WO 06/122806); 4-(trifluoromethyl)-5-(2,6-dimorpholino-pyrimidin-4-yl)pyridin-2-amine (also known as BKM120 or NVP-BKM120, and described in PCT Publication No. WO2007/084786); Tozasertib (VX680 or MK-0457, CAS 639089-54-6); (5Z)-5-[[4-(4-Pyridinyl)-6-quinolinyl]methylene]-2,4-thiazolidinedione (GSK1059615, CAS 958852-01-2); (1E,4S,4aR,5R,6aS,9aR)-5-(Acetyloxy)-1-[(di-2-propenylamino)methylene]-4,4a,5,6,6a,8,9,9a-octahydro-11-hydroxy-4-(methoxymethyl)-4a,6a-dimethyl-cyclopenta [5,6]naphtho[1,2-c]pyran-2,7,10(1H)-trione (PX866, CAS 502632-66-8); and 8-Phenyl-2-(morpholin-4-yl)-chromen-4-one (LY294002, CAS 154447-36-6).

[0217] Exemplary Protein Kinase B (PKB) or AKT inhibitors include, but are not limited to, 8-[4-(1-Aminocyclobutyl)phenyl]-9-phenyl-1,2,4-triazolo[3,4-f][1,6]naphthyrin-3(2H)-one (MK-2206, CAS 1032349-93-1); Perifosine (KRX0401); 4-Dodecyl-N-1,3,4-thiadiazol-2-yl-benzenesulfonamide (PHT-427, CAS 1191951-57-1); 4-[2-(4-Amino-1,2,5-oxadiazol-3-yl)-1-ethyl-7-[(3S)-3-piperidinylmethoxy]-1H-imidazo[4,5-c]pyridin-4-yl]-2-methyl-3-butyn-2-ol (GSK690693, CAS 937174-76-0); 8-(1-Hydroxyethyl)-2-methoxy-3-[(4-methoxyphenyl)methoxy]-6H-dibenzo[b,d]pyran-6-one (palomid 529, P529, or SG-00529); Tricirbine (6-Amino-4-methyl-8-(β-D-ribofuranosyl)-4H,8H-pyrrolo[4,3,2-de]pyrimido[4,5-c]pyridazine); (αS)-α-[[[5-(3-Methyl-1H-indazol-5-yl)-3-pyridinyl]oxy]methyl]-benzeneethanamine (A674563, CAS 552325-73-2); 4-[(4-Chlorophenyl)methyl]-1-(7H-pyrrolo[2,3-d]pyrimidin-4-yl)-4-piperidinamine (CCT128930, CAS 885499-61-6); 4-(4-Chlorophenyl)-4-[4-(1H-pyrazol-4-yl)phenyl]-piperidine (AT7867, CAS 857531-00-1); and Archexin (RX-0201, CAS 663232-27-7).

[0218] In certain embodiments, a tyrosine kinase inhibitor used in combination with the LVV or CAR modified T cells is an anaplastic lymphoma kinase (ALK) inhibitor. Exemplary ALK inhibitors include crizotinib, ceritinib, alectinib, brigatinib, dalantercept, entrectinib, and lorlatinib.

[0219] In certain embodiments where the LVV or CAR modified lymphocytes (e.g., T cells or NK cells) are administered in combination with one or more additional therapies, the LVV, CAR modified lymphocytes (e.g., T cells or NK cells), or one or more additional therapies may be administered at a dose that might otherwise be considered sub-therapeutic if administered as a monotherapy. Combination therapy includes administration of an LVV or CAR modified lymphocyte (e.g., T cell or NK cell) composition as described herein before an additional therapy (e.g., 1 day to 30 days or more before the additional therapy), concurrently with an additional therapy (on the same day), or after an additional therapy (e.g., 1 day-30 days or more after the additional therapy). In certain embodiments, the LVV or CAR modified lymphocytes (e.g., T cells or NK cells) are administered concurrently with the one or more additional therapies. In further embodiments, the LVV or CAR modified lymphocytes (e.g., T cells or NK cells) are administered 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, or 30 days before or after administration of the one or more additional therapies. In still further embodiments, the LVV or CAR modified lymphocytes (e.g., T cells or NK cells) are administered within 4 weeks, within 3 weeks, within 2 weeks, or within 1 week before or after administration of the one or more additional therapies. Where the one or more additional

therapies involves multiple doses, the LVV or CAR modified lymphocytes (e.g., T cells or NK cells) may be administered before or after the initial dose of the one or more additional therapies, after the final dose of the one or more additional therapies, or in between multiple doses of the one or more additional therapies.

[0220] Subjects that can be treated by the compositions and methods of the present disclosure include animals, such as humans, primates, cows, horses, sheep, dogs, cats, mice, rats, rabbits, guinea pigs, or pigs. The subject may be male or female, and can be any suitable age, including infant, juvenile, adolescent, adult, and geriatric subjects.

EXAMPLES

Example 1: Methods for Generating Lentiviral Vectors with Defined Tropism

[0221] Self-inactivating lentiviral vectors (LVV) as described herein were produced using a third generation production system. An expression plasmid that contains the gene sequences desired for delivery by the LVV was combined at defined ratios with three packaging plasmids, VSV-G, GagPol, and Rev, and used to transfect HEK293 producer cells. Productive viral particles were harvested from the HEK293 culture media 2-3 days later.

[0222] The tropism of the resulting LVV was tested to determine whether it can be redirected using a non-viral targeting protein as described herein expressed from one of the three packaging plasmids or by addition of a fifth plasmid to the transfection mix. The LVV vector includes a mutated VSV-G envelope protein, where the mutated VSV-G includes mutations that abolish binding of the mutated VSV-G to its native receptor such that viral entry occurs via the non-viral, tropism defining targeting protein. Mutations in the LDL-receptor binding site of VSV-G were found to abolish binding to its natural receptor. VSV-G bearing a variety of mutations (Trop-002 (SEQ ID NO: 74), Trop-051 (SEQ ID NO: 93), Trop-052 (SEQ ID NO: 95), Trop-055 (SEQ ID NO: 97), or Trop-061 (SEQ ID NO: 103)) was expressed together with CD80 (SEQ ID NO: 2) on the LVV. CD80 is the natural ligand to T cell surface receptors, CD28 and CTLA4, and provides tropism to T cells and not B cells. Binding to the natural receptor for VSV-G is assessed by transduction of Raji B cells. Several mutations in VSV-G prohibit transduction of Raji B cells yet transduction of Jurkat T cells is maintained (FIGS. 2A-2B).

[0223] In a first method, a tropism-defining targeting protein derived from CD80 (SEQ ID NO: 76) was expressed from the VSV-G packaging plasmid. CD80 is the natural ligand to T cell surface receptors, CD28 and CTLA4, and provides tropism to T cells. A codon optimized CD80 molecule (encoding SEQ ID NO: 2) was cloned downstream of the mutated VSV-G (Trop-002, encoding SEQ ID NO: 74), which contains mutations to abolish LDL receptor binding and was used to package a green fluorescent protein (GFP)-expressing LVV. CD80 and VSV-G were expressed at relatively equivalent levels on the surface of HEK293 producer cells, suggesting comparable decoration of the LVV particles (FIG. 3A). LVV generated with this approach were capable of transducing T cells as assessed by GFP expression (FIG. 3B).

[0224] In a second method, the tropism-defining CD80 targeting protein (encoding SEQ ID NO: 2) was cloned into the Rev packaging plasmid and transfected at a lower molar

concentration to the VSV-G env (encoding Trop-002 (SEQ ID NO: 74)) plasmid. The modified plasmids was used to package a GFP-expressing LVV. Assessment of VSV-G and CD80 expression on the HEK-293 producer cells showed relative expression levels trended with the molar concentrations of the packaging plasmids. LVV generated with these packaging approaches were capable of transducing T cells as assessed by GFP expression (FIG. 4). Reduced expression of CD80 was observed in LVVs packaged with CD80 targeting protein contained in the Rev packaging plasmid due to the lower ratio of the Rev plasmid.

[0225] In a third method, the tropism defining CD80 targeting protein (encoding SEQ ID NO: 2) was added as an additional plasmid during transfection of HEK293 cells intended to produce LVV (Trop-002 VSV-G env). The amount of the CD80 plasmid used during LVV production was associated with the expression on HEK293 producer cells and is likely representative of decoration of the LVV particles. LVV generated with this packaging approach were capable of transducing T cells as assessed by GFP expression and transduction efficiency was associated with CD80 levels on the LVV particles (FIG. 5).

[0226] Regardless of the method selected, high titer, tropism-redirection LVV can be generated with adherent or suspension HEK293 producer cells and concentrated by centrifugation or anion exchange chromatography (AEX) followed by tangential flow filtration (TFF) (FIGS. 6A-6C). Concentration by AEX/TFF resulted in LVV preparations with a high level of purity and recovery (FIG. 6D).

Example 2: CD4 T Cells Enriched by CD80 LVV

[0227] The tropism-defining targeting molecule precisely defines the specific cell type transduced with this LVV. The ligand for CD80, CD28, is naturally expressed on T cells. However, a higher frequency of CD4 T cells express CD28 in human peripheral blood compared to CD8 T cells (80% vs 50%). When a LVV was packaged to decorate the Trop-002 VSV-G envelope with a CD80 targeting protein (SEQ ID NO: 76) was used to transduce human PBMC, a higher composition of transduced CD4 T cells was observed compared to transduced CD8 T cells and was maintained for up to a week in culture (FIG. 13).

Example 3: Methods for Ex Vitro Generation of Tumor-Reactive T Cells Engineered to Express a Chimeric Antigen Receptor (CAR) Using a Lentiviral Vector Containing T Cell-Specific Tropism

[0228] Compared to standard lentiviral vectors (LVV) that use common third generation packaging systems, LVV packaged to redirect tropism as provided herein may have improved transduction efficiencies and specificity. The levels of available viral particles are likely increased when transducing a specific cell type as part of a heterogeneous mixture, as the LVV particles are not taken up by other, non-desired cell types. Whether improved transduction of T cells with a CAR molecule is possible in a heterogeneous peripheral blood mononuclear cell (PBMC) mixture was evaluated. A representative anti-BCMA CAR (SEQ ID NO: 58) used to treat Multiple Myeloma or a representative anti-CD19 CAR (SEQ ID NO: 42) used to treat non-Hodgkin Lymphoma was used. GFP reporter was used as control. The LVV was packaged to decorate the envelope

with anti-CD3 targeting protein and CD80 targeting protein and redirect tropism toward the T cell surface molecules, CD3, CD28 and CTLA4. The CD80 targeting protein was cloned into the Trop-002 VSV-G env packaging plasmid (SEQ ID NO: 76) and the anti-CD3 targeting protein (encoding SEQ ID NO: 10) was expressed from a fifth packaging plasmid.

[0229] The efficiency of T cell transduction within PBMC obtained from healthy donors leukapheresis after informed consent was tested. The PBMC were enriched using red cell lysis and matched donors were transduced with standard LVV or LVV with T cell redirected tropism (anti-CD3 and CD80) at LVV concentrations 5 to 25 times lower than typically used for CAR T cell transduction. Expression of an anti-BCMA CAR was assessed by flow cytometry after staining with a fluorescently labeled BCMA molecule to specifically detect anti-BCMA CAR T cells. Even with extremely low viral levels, T cells transduced using the T cell tropic LVV showed higher levels of transduction compared to standard LVV (FIG. 7).

[0230] In the same study whether anti-BCMA CAR T cells can be generated without antibodies to activate T cells was evaluated. T cell activating antibodies (e.g., anti-CD28 and anti-CD3) are typically required for transduction using standard LVV during CAR T cell generation. It was confirmed that standard LVV indeed requires activation using antibodies to transduce normal human primary T cells (FIG. 7). By contrast, a T cell redirected LVV (anti-CD3 and CD80) can cause transduction even in the absence of prior T cell activation (without the presence of IL-2 and exogenous activating antibodies anti-CD3 and anti-CD28) (FIG. 7). A consequence of T cell activation is the subsequent cellular expansion when cultured in media containing IL-2. PBMCs transduced in the presence or absence of exogenous activating anti-CD3 and anti-CD28 antibodies after standard LVV or T cell redirected LVV (anti-CD3 and CD80) showed that T cell redirected LVV was uniquely able to expand T cells even in the absence of antibody activation (FIG. 8). In another example, transduction of an anti-CD19 CAR also was achieved without prior T cell activation when using the T cell redirected LVV (anti-CD3 and CD80) (FIG. 9).

[0231] T cell activation and transduction efficiency can be dictated by the selection of anti-CD3 antibodies used to generate the T cell redirected LVV. Examples of envelope-bound anti-CD3 binding proteins having 12F6 binding domains in VH (SEQ ID NO:113)-VL (SEQ ID NO:115) orientation and VL (SEQ ID NO:115)-VH (SEQ ID NO:113) orientation are shown to cause a variable amount of T cell transduction as well as T cell activation (FIG. 10).

Example 4: T Cell Specificity of T Cell Targeted Lentiviral Vector

[0232] The T cell specificity of an LVV packaged to decorate the envelope (Trop-002 VSV-G env) with an anti-CD3 targeting protein (SEQ ID NO: 10) and CD80 targeting protein (SEQ ID NO: 2) was confirmed by transduction of a T cell line and a panel of B cell lines (Raji, Ramos, Jeko-1, NALM-6). Transduction of these cell lines assessed by flow cytometry for a GFP transgene was compared to standard LVV. Both standard LVV and a T cell-targeted LVV effectively transduced the T cell line (Jurkat). The standard LVV effectively transduced all the B cell lines, confirming these

lines are sensitive to LVV transduction, but no substantial transduction by the T cell-targeting LVV was detected in any of the B cell lines (FIG. 14).

Example 5: Tumor-Specific T Cells Generated Using a T Cell Targeted Lentiviral Vector

[0233] Using methods described herein, T cells expressing an anti-BCMA CAR (SEQ ID NO: 58) were generated by transducing PBMC from healthy donors with LVVs having Trop-002 VSV-G env and CD80 targeting protein and anti-CD3 targeting protein. The CD80 targeting protein was cloned into the Trop-002 VSV-G env packaging plasmid (SEQ ID NO: 76) and the anti-CD3 targeting protein (encoding SEQ ID NO: 10) was expressed from a fifth packaging plasmid to produce the T cell targeting LVVs. The T cell transductions were performed in the absence of any T cell stimulation typically required for efficient LVV-mediated gene transfer (exogenous anti-CD3 and anti-CD28).

[0234] First, the activity of anti-BCMA CAR T cells generated by T cell-redirection LVV was examined compared to standard LVV. Anti-tumor activity was assessed by intracellular cytokine staining for interferon-gamma and tumor necrosis factor-alpha after co-culture with a BCMA-negative cell line (Nalm-6) or BCMA-positive cell line (RPMI-8826). Anti-BCMA CAR T cells exhibited increased expression of T cell effector cytokines after culture with BCMA-positive cell lines that is not observed with BCMA-negative cell lines whether generated by standard LVV or T cell-redirection LVV (FIG. 11).

Example 6: Methods of T Cell-Specific In Vivo Delivery of Genetic Material Using a T Cell Tropic Lentiviral Vector

[0235] The efficient and specific transduction of T cells with a T cell targeted LVV could permit safe and effective delivery of genetic material to T cells in vivo. A humanized mouse model infused intravenously with a GFP-expressing, T cell targeted LVV was used to test this hypothesis. LVVs having Trop-002 mutated VSV-G and coated with anti-CD3 targeting protein and CD80 targeting protein were produced. The CD80 targeting protein was cloned into the Trop-002 VSV-G env packaging plasmid (SEQ ID NO: 76) and the anti-CD3 targeting protein (encoding SEQ ID NO: 10) was

expressed from a fifth packaging plasmid to produce the T cell targeting LVVs. Immunocompromised NCG mice were humanized via intravenous injection of human PBMC and supported by daily intraperitoneal administration of recombinant IL-2 for 4 days. One day after PBMC administration, the T cell targeted lentiviral vector is administered intravenously to the mice. Peripheral blood and splenic cells were harvested on day 7, and T cell specificity was assessed by flow cytometry. GFP positive cells were only detected in human T cells (CD3+) and not human B cells (CD20+), confirming T cell specificity of LVV transduction (FIG. 12A). GFP was detected in both CD8+ and CD8- (CD4) T cells, indicating that the T cell specific LVV transduces both CD4 and CD8 T cells in vivo (FIG. 12B).

[0236] Specific T cell transduction has a key safety advantage to other in vivo CAR delivery approaches by minimizing risk of transducing tumor cells. Transduction of T cells without transduction of tumor cells is evaluated using PBMCs from chronic lymphocytic leukemia patients with detectable CD19 tumor cells. The PBMC are transduced with T cell targeted LVV and transduced cells are assessed using flow cytometry.

[0237] The various embodiments described above can be combined to provide further embodiments. All of the U.S. patents, U.S. patent application publications, U.S. patent applications, foreign patents, foreign patent applications and non-patent publications referred to in this specification and/or listed in the Application Data Sheet, including but not limited to U.S. Provisional Patent Application No. 63/154,639, filed on Feb. 26, 2021, are incorporated herein by reference, in their entirety. Aspects of the embodiments can be modified, if necessary to employ concepts of the various patents, applications and publications to provide yet further embodiments.

[0238] These and other changes can be made to the embodiments in light of the above-detailed description. In general, in the following claims, the terms used should not be construed to limit the claims to the specific embodiments disclosed in the specification and the claims, but should be construed to include all possible embodiments along with the full scope of equivalents to which such claims are entitled. Accordingly, the claims are not limited by the disclosure.

SEQUENCE LISTING

```
Sequence total quantity: 123
SEQ ID NO: 1          moltype = DNA  length = 819
FEATURE              Location/Qualifiers
misc_feature          1..819
                        note = CD80 Targeting Full molecule
source                1..819
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 1
atgggtcata cagcgccga aggaacctca ccatctaagt gccatatct gaatttcttt 60
caacttctcg tctgtgcggg gctcagtcac ttctgcagtg ggttcattca cgttactaaa 120
gaggtcaagg aggtcgcaac attgagttgt ggccataacg tatcagttga agaactcgcg 180
cagacacgga ttacttgcca aaaggaaaag aagatggtgt tgacaatgat gagcggtgac 240
atgaacattt ggccagagta caaaaatcga acgatattcg atataacca taacttgttc 300
atagtaatac ttgccttgcg accttctgac gagggaaact atgaatgtgt agtgccttaag 360
tatgaaaaag atgcctttta gcgggaacac ttggctgagg ttacactctc cgtaaggcg 420
gactttccta cgccgtctat atccgaattc gagataccca cttctaacat tcgacgcac 480
atttgtcaa cctcaggtgg ttcccagag cctcacttga gctggctgga gaatggcgaa 540
gaacttaacg caatcaatac caggtgttc caagaccgg agacagagct gtacgccgtg 600
```

-continued

```

tcacccaac tggattttta catgacgaca aatcatagtt tcatgtgtct gatcaaatat 660
gggcatctca gggatgaatca gacttttaat tggaaacacta ccaacaaga gcacttocca 720
gataatctgt tgccaagctg ggcgataact cttatctccg tcaacgggtat cttcgtaatt 780
tgctgctca cctattgttt cgcgcctcga tgccgagaa 819

```

```

SEQ ID NO: 2      moltype = AA length = 273
FEATURE          Location/Qualifiers
REGION           1..273
                  note = CD80 Targeting Full molecule
REGION           1..35
                  note = MISC_FEATURE - signal peptide
source           1..273
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 2
MGHTRRQGTG PSKCPYLNFF QLLVLGLSLH FCSGVIHVTK EVKEVATLSC GHNVSVLELA 60
QTRIYWQKEK KMLVTMMSGD MNIWPEYKNR TIFDITNNLS IVILALRPSD EGTYESVVLK 120
YEKDAFKREH LAEVLTSVKA DFPTPSISDF EIPTSNIIRI ICSTSGGFPE PHLWSLENGE 180
ELNAINTTYS QDPETELYAV SSKLDFNMTT NHSPMCLIKY GHLRVNQTFN WNTTKQEHFP 240
DNLPSWAIT LISVNGIFVI CCLTYCFAPR CRE 273

```

```

SEQ ID NO: 3      moltype = DNA length = 105
FEATURE          Location/Qualifiers
misc_feature      1..105
                  note = CD80 signal peptide
source           1..105
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 3
atgggtcata cagcccgcca aggaacctca ccatctaagt gcccatatct gaatttcctt 60
caacttctcg tgctggcggg gctcagtcac ttctgcagtg gggtc 105

```

```

SEQ ID NO: 4      moltype = AA length = 35
FEATURE          Location/Qualifiers
REGION           1..35
                  note = CD80 signal peptide
source           1..35
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 4
MGHTRRQGTG PSKCPYLNFF QLLVLGLSLH FCSGV 35

```

```

SEQ ID NO: 5      moltype = DNA length = 633
FEATURE          Location/Qualifiers
misc_feature      1..633
                  note = CD80 extracellular domain (ECD)
source           1..633
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 5
attcacgtta ctaagagagt caaggagggt gcaacattga gttgtggcca taacgtatca 60
gttgaagaac tcgcgacgac acggatttac tggcaaaagg aaaagaagat ggtgttgaca 120
atgatgagcg gtgacatgaa catttggcca gagtacaaaa atcgaacgat attcgatata 180
accaataact tgtccatagt aatacttgcc ttgcgacctt ctgacgaggg aacgtatgaa 240
tgtgtagtcg ttaagtatga aaaagatgcc tttaagcggg aacacttgcc tgagggttaca 300
ctctccgtta aggcggactt tcctacgcgg tctatatccg acttcgagat acccacttct 360
aacattcgac gcacatcttg ctcaacctca ggtggtttcc cagagcctca cttgagctgg 420
ctggagaatg gcgaagaact taacgcaatc aataccacgg tgtcccaaga cccggagaca 480
gagctgtacg ccgtgtcatc caaactggat tttaacatga cgacaaatca tagtttcatg 540
tgtctgatca aatatgggca tctcagggtg aatcagactt ttaattggaa cactacaaaa 600
caagagcact tcccagataa tctgttgcca agc 633

```

```

SEQ ID NO: 6      moltype = AA length = 211
FEATURE          Location/Qualifiers
REGION           1..211
                  note = CD80 extracellular domain (ECD)
source           1..211
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 6
IHVTKVKEV ATLSCGHNV VEEAQTRIY WQKEKMMVLT MMSGDMNIWP EYKNRTIFDI 60
TNNLSIVILA LRPSDEGTYE CVVLKYEKDA FKREHLAEVT LSVKADFPPT SISDFEIPTS 120
NIIRIICSTS GGFPEPHLSW LENGEELNAI NTTVSQDPET ELYAVSSKLD FNMNTNHSFM 180
CLIKYGLHRV NQTFNWNNTK QEHFPDNLPS S 211

```

```

SEQ ID NO: 7      moltype = DNA length = 81

```

-continued

FEATURE
misc_feature
Location/Qualifiers
1..81
note = CD80 transmembrane and intracellular domain
source
1..81
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 7
 tgggcgataa ctcttatctc cgtcaacggt atcttcgtaa tttgctgect cacctattgt 60
 ttccgcgcctc gatgccgaga a 81

SEQ ID NO: 8
 FEATURE
 REGION
 Location/Qualifiers
 1..27
 note = CD80 transmembrane and intracellular domain
 source
 1..27
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 8
 WAITLISVNG IFVICCLTYC FAPRCRE 27

SEQ ID NO: 9
 FEATURE
 misc_feature
 Location/Qualifiers
 1..1011
 note = Anti-CD3 (UCTH1) Full molecule
 source
 1..1011
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 9
 atggccttac cagtgaccgc cttgtcctcg ccgctggcct tgetgtctca cgccgccagg 60
 ccggacatcc agatgaccga gaccacctcc tccctgtctg cctctctggg agacagagtc 120
 accatcagtt gcagggcaag tcaggacatt agaaattatt taaactggta tcaacagaaa 180
 ccagatggaa ctgttaaact cctgatctac tacacatcaa gattacactc aggagtccca 240
 tcaaagtcca gtggcagtggt gtctggaaca gattattctc tcaccattag caacctggag 300
 caagaggata ttgccactta cttttgccaa cagggttaata cgcttcctg gacgttcgct 360
 ggaggcacca agctggaaat caaacgggct ggaggcggta gtggcggtag atcaggtgga 420
 ggcagcggtg gcggatctga ggtgcagctc cagcagctcg gacctgagct ggtgaagcct 480
 ggagcttcaa tgaagatata ctgcaaggct tctggttact cattcaactgg ctacaccatg 540
 aactgggtga agcagagtc tggaaagaac cttgagtggg tgggacttat taatccttac 600
 aaaggtgtta gtacctacaa ccagaagtcc aaggacaagg ccacattaac ttagtagacaag 660
 tcatccagca cagcctacat ggaactcctc agtctgacat ctgaggactc tgcagtcctat 720
 tactgtgcaa gatcggggta ctacgggtgat agtgactggg acttcgatgt ctggggcgca 780
 gggaccacgg tcaccgtctc ctcaaccact acaccagcac ctagaccacc aacacctgcg 840
 ccaaccatcg catcgcagcc actgtctctg cgcccagagg catgccggcc agcagctggg 900
 ggcgccagtc acacaagggg gctggacttc gcatgtgata tctacatctg ggcaccattg 960
 gcagggaact gtggggctct tctcctgtca ctgggtatca ccctttactg c 1011

SEQ ID NO: 10
 FEATURE
 REGION
 Location/Qualifiers
 1..337
 note = Anti-CD3 (UCTH1) Full molecule
 REGION
 1..21
 note = MISC_FEATURE - signal peptide
 source
 1..337
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 10
 MALPVTALLL PLALLLHAAR PDIQMTQTTS SLSASLGDRV TISCRASQDI RNYLNWYQQK 60
 PDGTVKLLIY YTSRLHSGVP SKFSGSGSGT DYSLTISNLE QEDIATYFCQ QGNTLPWTFA 120
 GGTKLEIKRA GGGSGGSGG GSGGGSEVQL QQSGPELVKP GASKKISCKA SGYSFTGYTM 180
 NWVKQSHGKN LEWMGLINPY KGVSTYNQKF KDKATLTVDK SSSTAYMELL SLTSEDSAVY 240
 YCARSGYYGD SDWYFDVWGA GTTVTVSSST TPAPRPPTFA PTIASQPLSL RPEACRPAAG 300
 GAVHTRGLDF ACDIYIWAPL AGTCGVLLLS LVITLYC 337

SEQ ID NO: 11
 FEATURE
 misc_feature
 Location/Qualifiers
 1..63
 note = CD8a signal peptide
 source
 1..63
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 11
 atggccttac cagtgaccgc cttgtcctcg ccgctggcct tgetgtctca cgccgccagg 60
 ccg 63

SEQ ID NO: 12
 FEATURE
 Location/Qualifiers
 moltype = AA length = 21

-continued

REGION 1..21
 note = CD8a signal peptide
 source 1..21
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 12
 MALPVTALLL PLALLLHAAR P 21

SEQ ID NO: 13 moltype = DNA length = 327
 FEATURE Location/Qualifiers
 misc_feature 1..327
 note = Anti-CD3 (UCTH1) Light chain variable region
 source 1..327
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 13
 gacatccaga tgacccagac cacctcctcc ctgtctgcct ctctgggaga cagagtcacc 60
 atcagttgca gggcaagtca ggacattaga aattatttaa actgggtatca acagaaacca 120
 gatggaactg ttaaaactcct gatctactac acatcaagat tacactcagg agtcccatca 180
 aagttcagtg gcagtggggc tggaaacagat tattctctca ccattagcaa cctggagcaa 240
 gaggatattg ccacttactt ttgccaacag ggtaatacgc ttccgtggac gttcgtctga 300
 ggcaccaagc tggaaatcaa acgggct 327

SEQ ID NO: 14 moltype = AA length = 109
 FEATURE Location/Qualifiers
 REGION 1..109
 note = Anti-CD3 (UCTH1) Light chain variable region
 source 1..109
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 14
 DIQMTQTSS LSASLGDRVT ISCRASQDIR NYLNWYQQKP DGTVKLLIYY TSRLHSGVPS 60
 KFSGSGSGTD YSLTISNLEQ EDIATYFCQQ GNTLPWTFAG GTKLEIKRA 109

SEQ ID NO: 15 moltype = DNA length = 48
 FEATURE Location/Qualifiers
 misc_feature 1..48
 note = G3S linker
 source 1..48
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 15
 ggaggcggta gtggcgggtg atcaggtgga ggcagcgggtg gcggatct 48

SEQ ID NO: 16 moltype = AA length = 16
 FEATURE Location/Qualifiers
 REGION 1..16
 note = G3S linker
 source 1..16
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 16
 GGGSGGSGG GSGGGS 16

SEQ ID NO: 17 moltype = DNA length = 366
 FEATURE Location/Qualifiers
 misc_feature 1..366
 note = Anti-CD3 (UCTH1) heavy chain variable region
 source 1..366
 mol_type = other DNA
 organism = synthetic construct
 SEQUENCE: 17
 gaggtgcagc tccagcagtc tggacctgag ctggtgaagc ctggagcttc aatgaagata 60
 tcctgcaagg cttctgggta ctcattcact ggctacacca tgaactgggt gaagcagagt 120
 catggaaaaga accttgagtg gatgggactt attaatcctt acaaggtgt tagtacctac 180
 aaccagaagt tcaaggacaa ggccacatta actgtagaca agtcatccag cacagcctac 240
 atggaactcc tcagtctgac atctgaggac tctgcagtct attactgtgc aagatcgggg 300
 tactacgggtg atagtgactg gtacttcgat gtctggggcg cagggaaccac ggtaaccgtc 360
 tcctca 366

SEQ ID NO: 18 moltype = AA length = 122
 FEATURE Location/Qualifiers
 REGION 1..122
 note = Anti-CD3 (UCTH1) heavy chain variable region
 source 1..122
 mol_type = protein

-continued

organism = synthetic construct

SEQUENCE: 18
EVQLQQSGPE LVKPGASKMI SCKASGYSFT GYTMNWVKQS HGKNLEWMGL INPYKGVSTY 60
NQKFKDKATL TVDKSSSTAY MELLSLTSED SAVVYCARGS YYGSDSDWYFD VWGAGTTVTV 120
SS 122

SEQ ID NO: 19 moltype = DNA length = 207
FEATURE Location/Qualifiers
misc_feature 1..207
 note = CD8a hinge and transmembrane
source 1..207
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 19
accactacac cagcacctag accaccaaca cctgcgcca ccatcgcatc gcagccactg 60
tctctgcgcc cagaggcatg ccggccagca gctgggggag cagtgcacac aagggggctg 120
gacttcgcat gtgatatcta catctgggca ccattggcag ggacttgttg ggtccttctc 180
ctgtcactgg ttatcacctt ttactgc 207

SEQ ID NO: 20 moltype = AA length = 69
FEATURE Location/Qualifiers
REGION 1..69
 note = CD8a hinge and transmembrane
source 1..69
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 20
TTTPAPRPPT PAPTASQPL SLRPEACRPA AGGAVHTRGL DFACDIYIWA PLAGTCGVLL 60
LSLVITLYC 69

SEQ ID NO: 21 moltype = DNA length = 1896
FEATURE Location/Qualifiers
misc_feature 1..1896
 note = Co-expressed CD80 and anti-CD3 T cell targeting
 molecules
source 1..1896
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 21
atgggtcata cagcgcgcca aggaacctca ccatctaagt gcccatatct gaatttcttt 60
caacttctcg tgctggcggg gctcagtcct ttctgcagtg gggtcattca cgttactaaa 120
gagggtcaagg aggtcgcaac attgagttgt ggccataacg tatcagttga agaactcgcg 180
cagacacgga tttactggca aaaggaaaag aagatggtgt tgacaatgat gagcggtagc 240
atgaacattt ggccagagta caaaaatcga acgatattcg atataaccaa taacttgttc 300
atagtaatac ttgccttgcg accttctgac gagggaaacgt atgaatgtgt agtgcttaag 360
tatgaaaaag atgcctttaa gcgggaacac ttggctgagg ttacactctc cgttaaggcg 420
gactttccta cgccgtctat atccgacttc gagataccca cttctaactc tcgacgcac 480
atgtgtcaa cctcaggtgg ttcccagag cctcacttga gctggctgga gaatggcgaa 540
gaacttaacg caatcaatac caccgtgttc caagaccgag agacagagct gtacgcctg 600
tcacccaac tggattttaa catgacgaca aatcatagtt tcatgtgtct gatcaaatat 660
gggcatctca gggtagaatc gacttttaac tggaaacta ccaaacaga gcacttccca 720
gataatctgt tgccaagctg ggcgataact cttatctccg tcaacggtat cttcgtaatt 780
tgctgcctca cctattgttt cgccgctcga tgccgagaag gcagcgcgcg caccaacttc 840
tccctgtcga agcaggcgcg cgacgtggaa gaaaaccctg gcccacatggc cttaccagtg 900
accgccttgc tccgctcgct ggccttgcgt ctccacgccc ccaggccgga catccagatg 960
accagacca cctctcctct gtctgcctct ctgggagaca gagtcaccat cagttgcagg 1020
gcaagtccag acattagaaa ttatttaaac tggatcaac agaaaccaga tggaaactgt 1080
aaactcctga tctactacac atcaagatta cactcaggag tcccatcaaa gttcagtgcc 1140
agtgggtctg gaacagatta ttctctcacc attagcaacc tggagcaaga ggatattgcc 1200
acttactttt gccaacaggg taatacgtct ccgtggagct tcgctggagg caccagctg 1260
gaaatcaaac gggctggagg cggtagtggc ggtgagtcag gtggaggcag cggtggcgga 1320
tctgaggtgc agctccagca gtctggacct gagctgggtg agcctggagc ttcaatgaag 1380
atatcctgca aggtctctgg ttactcattc actggctaca ccatgaactg ggtgaagcag 1440
agtcatggaa agaaccttga gtggatggga cttattaatc cttacaaagg tgtagtagc 1500
tacaaccaga agttcaagga caaggccaca ttaactgtag acaagtcac cagcacagcc 1560
tacatggaac tctcagctct gacatctgag gactctgcag tctattactg tgcaagatcg 1620
gggtactacg gtgatatgta ctgggtacttc gatgtctggg gcgcagggac caggtgcacc 1680
gtctcctcaa ccaactacac agcacctaga ccaccaacac ctgcgccaac catcgcatcg 1740
cagccactgt ctctgcgccc agaggcatgc cggccagcag ctggggggcg agtgcacaca 1800
agggggctgg acttcgcatg tgatatctac atctgggcac cattggcagg gacttgtggg 1860
gtccttctcc tgtcactggg taccacctt tactgc 1896

SEQ ID NO: 22 moltype = AA length = 632
FEATURE Location/Qualifiers
REGION 1..632
 note = Co-expressed CD80 and anti-CD3 T cell targeting

-continued

```

REGION          molecules
                1..35
                note = MISC_FEATURE - signal peptide
source          1..632
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 22
MGHTRRQGTG PSKCPYLNFF QLLVLVLAGLSH FCSGVIHVTK EVKEVATLSC GHNVSVVEELA 60
QTRIWQKEK KMLVTMMSGD MNIWPEYKNR TIFDITNNLS IVILALRPSD EGTYESVVLK 120
YEKDAFKREH LAEVTLSVKA DFPTPSISDF EIPTSNIIRI ICSTSGGFPE PHLSWLENGE 180
ELNAINTTVS QDPETELYAV SSKLDFNMTT NHSMCLIKY GHLRVNQTFN WNTTKQEHFP 240
DNLLPSWAIT LISVNGIFVI CCLTYCFAPR CREGSGATNF SLLKQAGDVE ENPGPMALPV 300
TALLPLALL LHAARPDQM TQTSSLSAS LGDRVTISCR ASQDIRNYLN WYQKPDGTV 360
KLLIYTSRL HSGVPSKFSG SSGGTDYSLT ISNLEQEDIA TYFCQQGNTL PWTGAGGKTL 420
ETKRAGGSG GSGGGGSGG SEVQLQQSGP ELVKPGASMK ISCKASGYSF TGYTMNWVKQ 480
SHGNLEWNG LINPYKGVST YNQKFKDKAT LTVDKSSSTA YMELLSLTSE DSAVYYCARS 540
GYYGSDWYF DVWGAGTTVT VSSTTTPAPR PPTPAPTAS QPLSLRPEAC RPAAGGAVHT 600
RGLDFACDIY IWAPLAGTCG VLLLSLVITL YC 632

SEQ ID NO: 23      moltype = DNA length = 105
FEATURE           Location/Qualifiers
misc_feature      1..105
                  note = CD80 signal peptide
source           1..105
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 23
atgggtcata cagcgcccca aggaacctca ccatctaagt gcccatatct gaatttcttt 60
caacttctcg tgcggcggg gctcagtcac ttctgcagtg gggtc 105

SEQ ID NO: 24      moltype = AA length = 35
FEATURE           Location/Qualifiers
REGION           1..35
                  note = CD80 signal peptide
source           1..35
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 24
MGHTRRQGTG PSKCPYLNFF QLLVLVLAGLSH FCSGV 35

SEQ ID NO: 25      moltype = DNA length = 633
FEATURE           Location/Qualifiers
misc_feature      1..633
                  note = CD80 extracellular domain (ECD)
source           1..633
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 25
attcacgtta ctaaaagaggt caaggaggtc gcaacattga gttgtggcca taacgtatca 60
gttgaagaac tcgcgcagac acggatttac tggcaaaagg aaaagaagat ggtgttgaca 120
atgatgagcg gtgacatgaa catttggcca gagtacaaaa atcgaacgat attcgatata 180
accaataact tgtccatagt aatacttgcc ttgcgacctt ctgacgaggg aacgtatgaa 240
tgtgtagtgc ttaagtatga aaaagatgcc tttaagcggg aacacttggc tgaggttaca 300
ctctccgtta aggcggactt tcctacgcgc tctatatccg acttcgagat acccacttct 360
aacattcgac gcatcatttg ctcaacctca ggtggtttcc cagagcctca cttgagctgg 420
ctggagaatg gcgaagaact taacgcaatc aataaccacg tgtcccaaga cccggagaca 480
gagctgtacg ccgtgtcatc caaactggat ttaacatga cgacaaatca tagtttcatg 540
tgtctgatca aatatgggca tctcagggtg aatcagactt ttaattggaa cactacaaa 600
caagagcact tcccagataa tctgttgcca agc 633

SEQ ID NO: 26      moltype = AA length = 211
FEATURE           Location/Qualifiers
REGION           1..211
                  note = CD80 extracellular domain (ECD)
source           1..211
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 26
IHVTKEVKEV ATLSGHNVS VEELAQTRIY WQKEKKMVL TMMSGDMNIWP EYKNRTIFDI 60
TNNLSIVILA LRPSDEGTYE CVVLKYEKDA FKREHLAEV LSVKADFPPT SISDFEIPTS 120
NIRRIICSTS GGFPEPHLSW LENGEELNAI NTVSQDPET ELYAVSSKLD FNMTTNHSFM 180
CLIKYGHRLV NQTFNWNNTK QEHPDNLLP S 211

SEQ ID NO: 27      moltype = DNA length = 81
FEATURE           Location/Qualifiers
misc_feature      1..81

```

-continued

source note = CD80 transmembrane and intracellular domain
 1..81
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 27
 tgggcgataa ctcttatctc cgctcaacgg atcttcgtaa tttgctgctt cacctattgt 60
 ttcgcgctc gatgccgaga a 81

SEQ ID NO: 28 moltype = AA length = 27
 FEATURE Location/Qualifiers
 REGION 1..27
 note = CD80 transmembrane and intracellular domain
 source 1..27
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 28
 WAITLISVNG IFVICCLTYC FAPRCRE 27

SEQ ID NO: 29 moltype = DNA length = 66
 FEATURE Location/Qualifiers
 misc_feature 1..66
 note = P2A self-cleaving peptide
 source 1..66
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 29
 ggcagcggcg ccaccaactt ctcctgctg aagcaggccg gcgacgtgga agaaaaccct 60
 ggcccc 66

SEQ ID NO: 30 moltype = AA length = 22
 FEATURE Location/Qualifiers
 REGION 1..22
 note = P2A self-cleaving peptide
 source 1..22
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 30
 GSGATNFSLL KQAGDVEENP GP 22

SEQ ID NO: 31 moltype = DNA length = 63
 FEATURE Location/Qualifiers
 misc_feature 1..63
 note = CD8a signal peptide
 source 1..63
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 31
 atggccttac cagtgaaccg cttgctctg ccgctggcct tgctgtctca cgccgccagg 60
 ccg 63

SEQ ID NO: 32 moltype = AA length = 21
 FEATURE Location/Qualifiers
 REGION 1..21
 note = CD8a signal peptide
 source 1..21
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 32
 MALPVTALLL PLALLHAAR P 21

SEQ ID NO: 33 moltype = DNA length = 327
 FEATURE Location/Qualifiers
 misc_feature 1..327
 note = Anti-CD3 Light chain variable region
 source 1..327
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 33
 gacatccaga tgaccagac cacctcctcc ctgtctgctt ctctgggaga cagagtcacc 60
 atcagttgca gggcaagtca ggacattaga aattatttaa actgggtatca acagaaacca 120
 gatggaaactg ttaaaactcct gatctactac acatcaagat tacactcagg agtcccatca 180
 aagttcagtg gcagtggttc tggaacagat tattctctca ccattagcaa cctggagcaa 240
 gaggatattg ccacttaact ttgccaacag ggtaatacgc ttccgtggac gttcgctgga 300
 ggcaccaagc tggaatcaa acgggct 327

SEQ ID NO: 34 moltype = AA length = 109

-continued

FEATURE	Location/Qualifiers	
REGION	1..109	
	note = Anti-CD3 Light chain variable region	
source	1..109	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 34		
DIQMTQTSS LSASLGDRVT	ISCRASQDIR NYLNWYQQKP DGTVKLLIYY TSRLHSGVPS	60
KFSGSGSGTD YSLTISNLEQ	EDIATYFCQQ GNTLPWTFAG GTKLEIKRA	109
SEQ ID NO: 35	moltype = DNA length = 48	
FEATURE	Location/Qualifiers	
misc_feature	1..48	
	note = G3S linker	
source	1..48	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 35		
ggagggcggtg	gtggcggtgg atcaggtgga ggcagcggtg gcggatct	48
SEQ ID NO: 36	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = G3S linker	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 36		
GGGSGGSGG	GS	16
SEQ ID NO: 37	moltype = DNA length = 366	
FEATURE	Location/Qualifiers	
misc_feature	1..366	
	note = Anti-CD3 heavy chain variable region	
source	1..366	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 37		
gaggtgcagc tccagcagtc	tggacctgag ctggtgaagc ctggagcttc aatgaagata	60
tcttgcaagg cttctgggta	ctcattcact ggctacacca tgaactgggt gaagcagagt	120
catggaaaga accttgagtg	gatgggactt attaatcctt acaaaggtgt tagtacctac	180
aaccagaagt tcaaggacaa	ggccacatta actgtagaca agtcatccag cacagcctac	240
atggaaactcc tcagtctgac	atctgaggac tctgcagtct attactgtgc aagatcgggg	300
tactacggtg atagtgcactg	gtacttcgat gtctggggcg caggggaccac ggtaaccgtc	360
tcctca		366
SEQ ID NO: 38	moltype = AA length = 122	
FEATURE	Location/Qualifiers	
REGION	1..122	
	note = Anti-CD3 heavy chain variable region	
source	1..122	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 38		
EVQLQQSGPE LVKPGASMKI	SCKASGYSTF GYTMNWVKQS HGKNLEWMGL INPYKGVSTY	60
NQKFKDKATL TVDKSSSTAY	MELLSLTSED SAVYYCARSG YYGDSWDWYFD VWGAGTTVTV	120
SS		122
SEQ ID NO: 39	moltype = DNA length = 207	
FEATURE	Location/Qualifiers	
misc_feature	1..207	
	note = CD8a hinge and transmembrane	
source	1..207	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 39		
accactacac cagcacctag	accaccaaca cctgcgcgcaa ccatcgcctc gcagccactg	60
tctctgcgcc cagagggcatg	ccggccagca gctgggggag cagtgcacac aagggggctg	120
gacttcgcat gtgatattcta	catctgggca ccattggcag ggacttgttg ggtccttctc	180
ctgtcactgg ttatcacccct	ttactgc	207
SEQ ID NO: 40	moltype = AA length = 69	
FEATURE	Location/Qualifiers	
REGION	1..69	
	note = CD8a hinge and transmembrane	
source	1..69	

-continued

```

mol_type = protein
organism = synthetic construct

SEQUENCE: 40
TTTPAPRPPT PAPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDIYIWA PLAGTCGVLL 60
LSLVITLYC 69

SEQ ID NO: 41      moltype = DNA length = 1458
FEATURE           Location/Qualifiers
misc_feature       1..1458
                   note = Anti-CD19 CAR molecule
source            1..1458
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 41
atggccttac cagtgaccgc cttgctcctg ccgctggcct tgetgctcca cgccgccagg 60
ccggacatcc agatgacaca gactacatcc tccctgtctg cctctctggg agacagagtc 120
accatcagtt gcaggggcaag tcaggacatt agtaaatatt taaattggta tcagcagaaa 180
ccagatggaa ctgttaaaact cctgatctac catacatcaa gattacactc aggagtccca 240
tcaaggttca gtggcagtggt gctggaaca gattattctc tcaccattag caacctggag 300
caagaagata ttgccactta cttttgccaa cagggttaata cgcttccgta caggttcgga 360
ggggggacca agctggagat cacagggtga ggtggatcgg gcggtgggtg gtcgggtggc 420
ggcgggatct aggtgaaact gcaggagtca ggactggcc tgggtggcgc ctcacagagc 480
ctgtccgta catgcactgt ctcagggttc tcattaccgc actatgggtg aagctggatt 540
cgccagcctc cagcaaaagg tctggagtgg ctgggagtaa tatgggtag tgaaccaca 600
tactataatt cagctctcaa atccagactg accatcatca aggacaactc caagagccaa 660
gttttcttaa aaatgaacag tctgcaaaact gatgacacag ccatttacta ctgtgccaaa 720
cattattact acggtggtag ctatgctatg gactactggg gccaaagAAC ctcagtcacc 780
gtctcctcaa ccactacacc agcacctaga ccaccaacac ctgcgccaac catcgcatcg 840
cagccactgt ctctgcgccc agaggcatgc cggccagcag ctgggggcgc agtgccacaca 900
agggggctgg acttcgatg tgatatctac atctgggcac cattggcagg gacttgtggg 960
gtccttctcc tgtcactggt taccacctt tactgcaaac ggggcagaaa gaaactctg 1020
tatatatcca aacaaccatt tatgagacca gtacaaaacta ctcaagagga agatggctgt 1080
agctgccgat ttccagaaga agaagaagga ggatgtgaac tgagagtga gttcagcagg 1140
agcgacagcg ccccgcgctg caagcagggc cagaaccagc tctataacga gctcaatcta 1200
ggacgaagag aggagtacga tgttttggac aagagacgtg gccgggaccc tgagatgggg 1260
ggaaagccga gaaggaagaa cctcaggaag ggcctgtaca atgaactgca gaaagataag 1320
atggcggagg cctacagtga gattgggatg aaagcgagc gccggagggg caagggggc 1380
gatggccttt accagggtct cagtacagcc accaaggaca cctacgagc ccttcacatg 1440
caggccctgc cccctcgc 1458

SEQ ID NO: 42      moltype = AA length = 486
FEATURE           Location/Qualifiers
REGION           1..486
                   note = Anti-CD19 CAR molecule
REGION           1..21
                   note = MISC_FEATURE - signal peptide
source            1..486
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 42
MALPVTALLL PLALLLHAAR PDIQMTQTTS SLSASLGDRV TISCRASQDI SKYLNWYQQK 60
PDGTVKLLIY HTSRLHSGVP SRFSGSGSGT DYSLTISNLE QEDIATYFCQ QGNTLPYTFG 120
GGTKLEITGG GSGGGGSGG GGSEVKLQES GPGLVAPSQS LSVTCTVSGV SLPDYGVSWI 180
RQPPRKGLEW LGVIWGSETT YYNSALKSRL TIIKDNSKSQ VFLKMNSLQT DDTAIYYCAK 240
HYYYGGSYAM DYWGQTSVT VSSTTTAPR PPTPAPTAS QPLSLRPEAC RPAAGGAVHT 300
RGLDFACDIY IWAPLAGTCG VLLLSLVITL YCKRGRKKLL YIFKQPFMRP VQTQEDGDC 360
SCRFPEEEEG GCELRVKFSR SADAPAYKQG QNQLYNELNL GRREEYDVL D KRRGRDPEMG 420
GKPRRKNPQE GLYNELQKDK MAEAYSEIGM KGERRRGKGH DGLYQGLSTA TKDTYDALHM 480
QALPPR 486

SEQ ID NO: 43      moltype = DNA length = 63
FEATURE           Location/Qualifiers
misc_feature       1..63
                   note = CD8a signal peptide
source            1..63
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 43
atggccttac cagtgaccgc cttgctcctg ccgctggcct tgetgctcca cgccgccagg 60
ccg 63

SEQ ID NO: 44      moltype = AA length = 21
FEATURE           Location/Qualifiers
REGION           1..21
                   note = CD8a signal peptide
source            1..21

```

-continued

```

mol_type = protein
organism = synthetic construct

SEQUENCE: 44
MALPVTALLL PLALLLHAAR P 21

SEQ ID NO: 45      moltype = DNA length = 321
FEATURE           Location/Qualifiers
misc_feature       1..321
                   note = Anti-CD19 light chain variable region
source            1..321
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 45
gacatccaga tgacacagac tacatcctcc ctgtctgcct ctctgggaga cagagtcacc 60
atcagttgca gggcaagtca ggacattagt aaatatTTaa attggatca gcagaaacca 120
gatggaactg ttaaaactcct gatctaccat acatcaagat tacactcagg agtcccatca 180
aggttcagtg gcagtgggtc tggaaacagat tattctctca ccattagcaa cctggagcaa 240
gaagatatgt ccacttactt ttgccaacag ggtaatacgc ttccgtacac gttcggaggg 300
gggaccaagc tggagatcac a 321

SEQ ID NO: 46      moltype = AA length = 107
FEATURE           Location/Qualifiers
REGION            1..107
                   note = Anti-CD19 light chain variable region
source            1..107
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 46
DIQMTQTSS LSASLGDRVT ISCRASQDIS KYLNWYQQKP DGTVKLLIYH TSRLHSGVPS 60
RFSGSGSGTD YSLTISNLEQ EDIATYFCQQ GNTLPYTFGG GTKLEIT 107

SEQ ID NO: 47      moltype = DNA length = 45
FEATURE           Location/Qualifiers
misc_feature       1..45
                   note = G4S Linker
source            1..45
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 47
ggtggagggtg gatcggggcgg tgggtgggtcg ggtggcggcg gatct 45

SEQ ID NO: 48      moltype = AA length = 15
FEATURE           Location/Qualifiers
REGION            1..15
                   note = G4S Linker
source            1..15
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 48
GGGSGGGGS GGGGS 15

SEQ ID NO: 49      moltype = DNA length = 360
FEATURE           Location/Qualifiers
misc_feature       1..360
                   note = Anti-CD19 heavy chain variable region
source            1..360
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 49
gaggtgaaac tgcaggagtc aggacctggc ctggtggcgc cctcacagag cctgtccgtc 60
acatgcactg tctcaggggt ctcatcacc gactatgggt taagctggat tcgccagcct 120
ccacgaaagg gtctggagtg gctgggagta atatggggta gtgaaaccac atactataat 180
tcagctctca aatccagact gaccatcatc aaggacaact ccaagagcca agttttctta 240
aaaatgaaca gtctgcaaac tgatgacaca gccatttact actgtgcca acattattac 300
tacggtggtg gctatgctat ggcaactcgg ggccaaggaa cctcagtcac cgtctcctca 360

SEQ ID NO: 50      moltype = AA length = 120
FEATURE           Location/Qualifiers
REGION            1..120
                   note = Anti-CD19 heavy chain variable region
source            1..120
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 50
EVKLQESGPG LVAPSQSLSV TCTVSGVSLP DYGVSWIRQP PRKGLEWLGW IWGSETTYYN 60
SALKSRLTII KDNKSQVFL KMNSLQTDIT AIYYCAKHYY YGGSYAMDYW GQGTSVTVSS 120

```

-continued

SEQ ID NO: 51 moltype = DNA length = 207
FEATURE Location/Qualifiers
misc_feature 1..207
 note = CD8a Hinge and Transmembrane
source 1..207
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 51
accactacac cagcacctag accaccaaca cctgcgccaa ccctcgcacg gcagccactg 60
tctctgcgcc cagaggcatg ccggccagca gctgggggcg cagtgcacac aagggggctg 120
gacttcgcat gtgatatcta catctgggca ccattggcag ggacttgtgg ggtccttctc 180
ctgtcactgg ttatcacctt ttactgc 207

SEQ ID NO: 52 moltype = AA length = 69
FEATURE Location/Qualifiers
REGION 1..69
 note = CD8a Hinge and Transmembrane
source 1..69
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 52
TTTPAPRPPT PPTIASQPL SLRPEACRPA AGGAVHTRGL DFACDIYIWA PLAGTCGVLL 60
LSLVITLYC 69

SEQ ID NO: 53 moltype = DNA length = 126
FEATURE Location/Qualifiers
misc_feature 1..126
 note = 4-1BB costimulatory domain
source 1..126
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 53
aaacggggca gaaagaaact cctgtatata ttcaacaac catttatgag accagtacaa 60
actactcaag aggaagatgg ctgtagctgc cgatttcag aagaagaaga aggaggatgt 120
gaactg 126

SEQ ID NO: 54 moltype = AA length = 42
FEATURE Location/Qualifiers
REGION 1..42
 note = 4-1BB costimulatory domain
source 1..42
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 54
KRGRKKLLYI FKQPFMRPVQ TTQEEDGCSC RFPEEEEGGC EL 42

SEQ ID NO: 55 moltype = DNA length = 336
FEATURE Location/Qualifiers
misc_feature 1..336
 note = CD3 zeta effector domain no. 1
source 1..336
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 55
agagtgaagt tcagcaggag cgcagacgcc cccgcgtaca agcagggcca gaaccagctc 60
tataacgagc tcaatctagg acgaagagag gactacgatg ttttgacaa gagacgtggc 120
cgggaccctg agatgggggg aaagccgaga aggaagaacc ctcaggaagg cctgtacaat 180
gaactgcaga aagataagat ggcggaggcc tacagtgaga ttgggatgaa aggcgagcgc 240
cggaggggca aggggcacga tggcctttac cagggtctca gtacagccac caaggacacc 300
tacgaagccc ttcacatgca ggcctgccc cctcgc 336

SEQ ID NO: 56 moltype = AA length = 112
FEATURE Location/Qualifiers
REGION 1..112
 note = CD3 zeta effector domain no. 1
source 1..112
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 56
RVKFSRSADA PAYKQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN 60
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR 112

SEQ ID NO: 57 moltype = DNA length = 1479
FEATURE Location/Qualifiers
misc_feature 1..1479

-continued

```

source          note = Anti-BCMA CAR full molecule
                1..1479
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 57
atggcactcc ccgtcaccgc ccttctcttg cccctcgccc tgctgctgca tgctgccagg 60
cccgacattg tgctcactca gtcacctccc agcctggcca tgagcctggg aaaaagggcc 120
accatctcct gtagagccag tgagtcgctc acaatcttgg ggagccatct tattcaactgg 180
tatcagcaga agcccgggca gccccaacc cttcttatto agctcgctgc aaacgtccag 240
acgggtgtac ctgccagatt ttctggtagc gggccccgca ctgattttac actgaccata 300
gatccagtg aagaagacga tgtggccgtg tattattgtc tgcagagcag aacgattcct 360
cgcacatttg gtgggggtac taagctggag attaagggaa gcacgtccgg ctgagggaa 420
ccgggctccg gcgaggggaag cacgaagggg caaattcagc tgggtccagag cggacctgag 480
ctgaaaaaac ccgpcgagac tgttaagatc agttgttaag catctggcta taccttcacc 540
gactacagca taaattgggt gaaacgagcc cctggaaaagg gcctcaaatg gatgggttgg 600
atcaataaccg aaactaggga gccctgcttat gcatactgact tccgcgggag attcgccctt 660
tcaactcaga catctgcctc tactgcttac ctccaaataa acaacctcaa gtatgaagat 720
acagccactt acttttgcgc cctcgactat agttacgcca tggactactg gggacaggga 780
acctccgcta ccgtcagttc cgcggccgca accacaacac ctgctccaag gcccccaca 840
cccgctccaa ctatagccag ccaaccattg agcctcagac ctgaagcttg caggcccgcga 900
gcaggaggcg ccgtccatcc gcgagggcctg gacttcgctg gtgatattta tatttgggca 960
cctttggccg gaacatgtgg ggtgttgctt ctctcccttg tgatcactct gtattgtaag 1020
cgcggggaaa agaagctcct gtacatcttc aagcagcctt ttatgcgacc tgtgcaaacc 1080
actcaggaag aagatgggtg ttcatgcgcg ttccccgagg aggaagaagg aggggtgtgaa 1140
ctgaggggtg aattttctag aagcgcgcat gctcccgcat atcagcaggg tcagaatcag 1200
ctctacaatg aattgaatct cggcaggcga gaagagtagc atgttctgga caagagacgg 1260
ggcagggatc ccgagatggg gggaaaagccc cggagaaaaa atcctcagga ggggttgtac 1320
aatgagctgc agaaggacaa gatggctgaa gcctatagcg agatcggaat gaaagcgcaa 1380
agacgcagag gcaaggggca tgacggtctg taccagggtc tctctacage caccaaggac 1440
acttatgatg cgttgcatat gcaagccttg ccaccccgc 1479

SEQ ID NO: 58      moltype = AA length = 493
FEATURE           Location/Qualifiers
REGION            1..493
                  note = Anti-BCMA CAR full molecule
REGION            1..21
                  note = MISC_FEATURE - signal peptide
source            1..493
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 58
MALPVTALLL PLALLLHAAR PDIVLTQSPP SLAMSLGKRA TISCRASESV TILGSHLIHW 60
YQOKPGQPPT LLIQLASNVQ TGVPAREFSGS GSRTDFTLTI DPVEEDDVAV YYCLQSR TIP 120
RTFGGGTKLE IKGSTSGSGK PGSGEGSTKG QIQLVQSGPE LKKPGETVKI SCKASGYTFT 180
DYSINWVKRA PGKGLKMWGW INTETREPAY AYDFRGRFAF SLETSASTAY LQINNLYKED 240
TATYFCALDY SYAMDYWGQG TSVTVSSAAA TTTPAPRPPT PAPTIASQPL SLRPEACRPA 300
AGGAVHTRGL DFACDIYIWA PLAGTCGVLL LSLVITLYCK RGRKKLLYIF KQPFMRPVQT 360
TQEDGDCSCR FPEEEEGGCE LRVKFSRSAD APAYQQGQNG LYNELNLGRR EEYDVLDKRR 420
GRDPEMGGKP RRKNPQEGLY NELQKDKMAE AYSEIGMKGE RRRGKGHDGL YQGLSTATKD 480
TYDALHMQAL PPR 493

SEQ ID NO: 59      moltype = DNA length = 63
FEATURE           Location/Qualifiers
misc_feature      1..63
                  note = CD8a signal peptide
source            1..63
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 59
atggcactcc ccgtcaccgc ccttctcttg cccctcgccc tgctgctgca tgctgccagg 60
ccc 63

SEQ ID NO: 60      moltype = AA length = 21
FEATURE           Location/Qualifiers
REGION            1..21
                  note = CD8a signal peptide
source            1..21
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 60
MALPVTALLL PLALLLHAAR P 21

SEQ ID NO: 61      moltype = DNA length = 333
FEATURE           Location/Qualifiers
misc_feature      1..333
                  note = Anti-BCMA light chain variable region

```

-continued

```

source                1..333
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 61
gacattgtgc tcaactcagtc acctcccagc ctggccatga gcctgggaaa aagggccacc 60
atctcctgta gagccagtgta gtccgctcaca atcttgggga gccatcttat tcaactggat 120
cagcagaagc ccgggcagccc tccaaccctt cttattcagc tcgcgtcaaa cgtccagacg 180
gggtgtacctg ccagattttc tggtagcggg tcccgcactg attttacact gaccatagat 240
ccagtgggaag aagacgatgt ggccgtgtat tattgtctgc agagcagaac gattcctcgc 300
acatttggtg ggggtactaa gctggagatt aag                                     333

SEQ ID NO: 62          moltype = AA length = 111
FEATURE               Location/Qualifiers
REGION                1..111
                      note = Anti-BCMA light chain variable region
source                1..111
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 62
DIVLTQSPSPS LAMSLGKRAT ISCRASESVT ILGSHLIHWY QQKPGQPPTL LIQLASNVQT 60
GVPARFSGSG SRTDFTLTID PVEEDDAVAV YCLQSRITPR TFGGGTKLEI K          111

SEQ ID NO: 63          moltype = DNA length = 54
FEATURE               Location/Qualifiers
misc_feature          1..54
                      note = 218 Linker
source                1..54
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 63
ggaagcacgt ccggtccagg gaagccgggc tccggcgagg gaagcacgaa gggg          54

SEQ ID NO: 64          moltype = AA length = 18
FEATURE               Location/Qualifiers
REGION                1..18
                      note = 218 Linker
source                1..18
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 64
GSTSGSGKPG SGEGSTKG                                     18

SEQ ID NO: 65          moltype = DNA length = 351
FEATURE               Location/Qualifiers
misc_feature          1..351
                      note = Anti-BCMA heavy chain variable region
source                1..351
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 65
caaattcagc tggccagag cggacctgag ctgaaaaaac ccggcgagac tgttaagatc 60
agttgtaaag catctggcta taccttcacc gactacagca taaattgggt gaaacgagcc 120
cctggaaagg gcctcaaatg gatgggttgg atcaataaccg aaactaggga gcctgcttat 180
gcatatgact tccgcgggag attcgccttt tcaactcgaga catctgcctc tactgcttac 240
ctccaaataa acaacctcaa gtatgaagat acagccactt acttttgcgc cctcgactat 300
agttacgcca tggactactg gggacaggga acctccgtta ccgtcagttc c          351

SEQ ID NO: 66          moltype = AA length = 117
FEATURE               Location/Qualifiers
REGION                1..117
                      note = Anti-BCMA heavy chain variable region
source                1..117
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 66
QIQLVQSGPE LKKPGETVKI SCKASGYTFT DYSINWVKRA PGKGLKWMGW INTETREPAY 60
AYDFRGRFAP SLETSASTAY LQINNPKYED TATYFCALDY SYAMDYWGQG TSVTVSS    117

SEQ ID NO: 67          moltype = DNA length = 2610
FEATURE               Location/Qualifiers
misc_feature          1..2610
                      note = Anti-CD3 targeting molecule co-expressed with mutant
                      VSV-G Full molecule
source                1..2610
                      mol_type = other DNA
                      organism = synthetic construct

```

-continued

SEQUENCE: 67

```

atggccttac cagtgaccgc cttgtcctcg ccgctggcct tgetgctcca cgcgcacagg 60
cgggacatcc agatgaccca gaccacctcc tccctgtctg cctctctggg agacagagtc 120
accatcagtt gcagggcaag tcaggacatt agaaattatt taaactggta tcaacagaaa 180
ccagatggaa ctgttaaaact cctgatctac tacacatcaa gattacactc aggagtccca 240
tcaaagttca gtggcagtggt gtctggaaca gattattctc tcaccattag caacctggag 300
caagaggata ttgccactta cttttgcca cagggttaata cgcttccgtg gacgttcgct 360
ggaggcacca agctgggaaat caaacgggct ggaggcggtg gtggcggtgg atcaggtgga 420
ggcagcggtg gcggatctga ggtgcagctc cagcagctcg gacctgagct ggtgaagcct 480
ggagcttcaa tgaagatata ttgcaaggct tctggttact cattcactgg ctacaccatg 540
aactgggtga agcagagtcg tggaaagaac cttgagtgga tgggacttat taatccttac 600
aaaggtgtta gtacctacaa ccagaagtcc aaggacaagg ccacattaac tgtagacaag 660
tcatccagca cagcctacat ggaactctc agtctgacat ctgaggactc tgcagcttat 720
tactgtgcaa gatcggggta ctacgggtgat agtgactggt acttcgatgt ctggggcgca 780
gggaccacgg tcaccgtctc ctcaaccact acaccagcac ctagaccacc aacacctgcg 840
ccaaccatcg catcgacgc actgtctctg cgcccagagg catgcccggc agcagctggg 900
ggcgacgtgc acacaagggg gctggacttc gcatgtgata tctacatctg ggcaccattg 960
gcagggaact gtggggctct tctcctgtca ctggttatca ccctttactg cggcagcgcg 1020
gccaccaact tctcctgtct gaagcaggcc ggcgacgtgg aagaaaaacc tggcccatg 1080
aagtgtctgc tgtacctggc gttcctggtt atcgggggtga actgcaagt cactatcgtg 1140
tttcgcgaca accaaaaggg caactggaaa aactgcgctt caaattacca ttattgcccc 1200
agcagctcgg acctgaactg gcacaatgac ctcatggaa ccgcgctgca ggtgaagatg 1260
ccacagagcc acaaggctat ccaggctgac ggtatggatg gccacgcgtc aaaatgggtg 1320
actacctcgc atttcctgtg gtacggacca aaatacatca cgcacagcat cagatcattc 1380
accccgctag tggaaacaat caaagaatcc atcgaaacga ctaagcaggg aacctggctg 1440
aacctgtgat ttccgcgcga gtctgtgtgg tacgcaaccg tgaccgatgc agaggccgtg 1500
atcgtgcaag tcacgcccga tcacgtgctt gtggacgagt acaccggaga atgggtcgat 1560
tccaggttca tcaacggcaa gtgtcccaac tacatttgcc caaccgtgca caacagcact 1620
acttggcata gcgactacaa agtgaagggt ctgtgtgatt ccaacctgat ctccatggat 1680
atcactttct tctcggaaga cggcgaactg tctcactcgg gcaaaagaag aactgggttt 1740
cgctcaaat accctgccta cgaactggga ggaagcct gcaagatgca gtactgcaag 1800
cactggggcg tgagactacc cagcgtgtgc tgggtcgaga tggccgataa ggacctgtt 1860
gcagcagcga gattcccga atgccctgag ggtatcgaga tctccgctcc aagccaaact 1920
tcagtggacg tgagcctgat ccaggacgtg gaacggattc tcgactactc gctgtgccag 1980
gagacctggt cgaagatcag agcgggactg cccatctcac cgggtggacct gtcctacctg 2040
gcgcaaaaga atccgggcac tggaccggcg ttcaccatca tcaacggcac cctcaaatat 2100
ttcgagacgc ggtacatccg ggtggacatc gcagctccga tctctccc gattggtgga 2160
atgatctcgg ggactactac cgaagccgag ctctgggacg actgggcacc ttacagggat 2220
gtcgagatcg gacctaacgg agtgcctcgg acctcctccg ggtacaagt cctctgtgac 2280
atgatcgcc atggctgctg ggaactcgat ctgcatctgt cgtccaaagc acaggtgttt 2340
gaacaccac acattcaaga cgcgcacagc cagctgcggg acgatgagtc gctgtcttc 2400
ggagacacgg gctgtgcaaa gaatccatc gagctgggtg aaggatggtt ttcactctg 2460
aaaagcgaga tgcgttcatt cttcttcac attggcctga tcatcgccct atttctagt 2520
ctgcgggtgg gaattcatct gtgcacaaag ctcaagcaca ctaagaagcg gcaaatctac 2580
actgatatcg agatgaatcg cctgggcaag 2610

```

SEQ ID NO: 68

moltype = AA length = 870

FEATURE

Location/Qualifiers

REGION

1..870

note = Anti-CD3 targeting molecule co-expressed with mutant
VSV-G Full molecule

REGION

1..21

note = MISC_FEATURE - signal peptide

source

1..870

mol_type = protein

organism = synthetic construct

SEQUENCE: 68

```

MALPVTALLL PLALLLHAAR PDIQMTQTTS SLSASLGDRV TISCRASQDI RNYLNWYQQK 60
PDGTVKLLIY YTSRLHSGVP SKFSGSGSGT DYSLTISNLE QEDIATYFCQ QGNTLPWTF 120
GGTKLEIKRA GGGSGGGSGG GSGGGSEVQL QSGPELVKP GASKMISCKA SGYSFTGYTM 180
NWWKQSHGKN LEWMGLINPY KGVSTYNQKF KDKATLTVDK SSSTAYMELL SLTSEDSAVY 240
YCARSYYGD SDWYFDVWGA GTTVTVSST TPAPRPPTPA PTIASQPLSL RPEACRPAG 300
GAVHTRGLDF ACDIYIWAFL AGTCGVLLLS LVITLYCGSG ATNFSLLKQA GDVEENPGPM 360
KCLLYLAFIF IGVNCKFTIV PPHNQKGNWK NVPSNYHYCP SSDLNWNHD LIGTALQVKM 420
PQSHKAIQAD GWMCHASKWV TTCDFRWYGP KYIHSIRSF TPSVEQKES IEQTKQGTWL 480
NPGFPPQSCG YATVTDAAV IVQVTPHHVL VDEYTGWVD SQFINGKCSN YICPTVHNST 540
TWHSDYKVKG LCDNSLISMD ITFFSEDGEL SSLKGEGTGF RSNYFAYETG GKACKMQYCK 600
HWGVRLPSCV WPEMADKDLF AARFPECPE GSSISAPSQT SVDVSLIQDV ERILDYSLCQ 660
ETWSKIRAGL PISPVDSLVL APKNPGTGPA FTIINGTLKY PETRYIRVDI AAPILSRMVG 720
MISGTTTEAE LWDDWAPYED VEIGPNGVLR TSSGYKFLY MIGHGMLDSD LHLSSKAQVF 780
EHPHIQDAAS QLPDDESLEFF GDTGLSKNPI ELVEGWFWSSW KSSIASFFFI IGLIIGLFLV 840
LRVGIHLCKI LKHTKKRQIY TDIEMNRLGK 870

```

SEQ ID NO: 69

moltype = DNA length = 66

FEATURE

Location/Qualifiers

misc_feature

1..66

-continued

```

source                note = P2A self-cleaving peptide
                      1..66
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 69
ggcagcggcg ccaccaactt ctccctgctg aagcaggccg gcgacgtgga agaaaaccct 60
ggcccc                                                    66

SEQ ID NO: 70          moltype = AA length = 22
FEATURE               Location/Qualifiers
REGION                1..22
                      note = P2A self-cleaving peptide
source                1..22
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 70
GSGATNFSLL KQAGDVEENP GP                                22

SEQ ID NO: 71          moltype = DNA length = 48
FEATURE               Location/Qualifiers
misc_feature          1..48
                      note = VSV-G signal sequence
source                1..48
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 71
atgaagtgtc tgctgtacct ggcgttctctg tttatcgggg tgaactgc 48

SEQ ID NO: 72          moltype = AA length = 16
FEATURE               Location/Qualifiers
REGION                1..16
                      note = VSV-G signal sequence
source                1..16
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 72
MKCLLYLAFI FIGVNC                                16

SEQ ID NO: 73          moltype = DNA length = 1485
FEATURE               Location/Qualifiers
misc_feature          1..1485
                      note = Mutant VSV-G (Trop-002)
source                1..1485
                      mol_type = other DNA
                      organism = synthetic construct

SEQUENCE: 73
aagttcacta tcgtgtttcc gcacaaccaa aagggaactt ggaaaaacgt gccttcaaat 60
taccattatt gccccagcag ctcggaacctg aactggcaca atgacctcat tggaaaccgcg 120
ctgcaggtga agatgccaca gagccacaag gctatccagg ctgacggatg gatgtgccac 180
gcgtcaaaaat gggtgactac ctgcgatttc cgtctgtacg gacccaaaata catcacgcac 240
agcatcagat cattcacccc gtcagtggaa caatgcaaa aatccatcga acagactaag 300
cagggaacct ggctgaaccc ttgatttccg ccgcagtcgt gtgggtacgc aaccgtgacc 360
gatgcagagg ccgtgatcgt gcaagtcacg ccgcacacag tgcttgtgga cgagtacacc 420
ggagaatggg tcgattccca gttcatcaac ggcaagtgc ccaactacat ttgcccaccc 480
gtgcacaaca gcaactactt gcacagcgac tacaaagtga agggctctgt tgattccaac 540
ctgatctcca tggatatacc tttcttctcg gaagacggcg aactgtcttc actgggcaaa 600
gaaggaactg ggtttcgttc aaattacttc gcctacgaaa ctggaggaaa agcctgcaag 660
atgcagtagt gcaagcactg gggcgtgaga ctaccacagc gtgtctgggt cgagatggcc 720
gataaggacc tgtttgacgc agcgagattc ccggaatgcc ctgagggatc gagcatctcc 780
gctccaagcc aaacttcagt ggacgtgagc ctgatccagg acgtggaacg gattctcgac 840
tactcgctgt gccaggagac ctggctcgaag atcagagcgg gactgcccat ctcccggtg 900
gacctgtcct acctggcgcc aaagaatccg ggcactggac cggcggttcac catcatcaac 960
ggcaccctca aatacttcga gacgcggtac atccgggttg acatcgagc tccgactctc 1020
tcccggtatg tgggaatgat ctcggggact actaccgaa cagagctctg ggacgactgg 1080
gcaccttacg aggatgtcga gatcggaact aacggagtgc tccggacctc ctccgggtac 1140
aagttccctc tgtacatgat cggccatggc atgctggact cggatctgca tctgtcgtcc 1200
aaagcacagg tgtttgaaac cccacacatt caagacgccg ccagccagct gccggacgat 1260
gagtcgctgt tcttcggaga cacgggcttg tcaagaatc ccatcgagct ggtggaagga 1320
tggttttcat cctggaaaag cagcatcgct tcattcttct tcatcattgg cctgatcatc 1380
ggcctatttc tagtcctgcg ggtgggaatt catctgtgca tcaagctcaa gcacactaag 1440
aagcggcaaa tctacactga tatcgagatg aatcgctcgg gcaag 1485

SEQ ID NO: 74          moltype = AA length = 495
FEATURE               Location/Qualifiers
REGION                1..495
                      note = Mutant VSV-G (Trop-002)

```

-continued

source 1..495
mol_type = protein
organism = synthetic construct

SEQUENCE: 74
KFTIVFPHNQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA LQVKMPQSHK AIQADGWMCH 60
ASKWVTTDCF RWYGPXYITH SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQYT PHHVLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDNS 180
LISMDITFFS EDGELSSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRVDIAAPIL SRMVGMISSG TTEAELWDDW 360
APYEDVEIGP NGVLRSTSGY KFLPYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

SEQ ID NO: 75 moltype = DNA length = 2418
FEATURE Location/Qualifiers
misc_feature 1..2418
note = CD80 T cell targeting molecule co-expressed with
mutant VSV-G Full molecule

source 1..2418
mol_type = other DNA
organism = synthetic construct

SEQUENCE: 75
atgggtcata caccgcgcga aggaacctca ccatctaagt gcccatatct gaatttcttt 60
caacttctcg tgcctggcggg gctcagtcac ttctgcagtg gggcatttca cgtaactaaa 120
gaggtcaagg aggtcgcaac attgagttgt ggccataacg tatcagttga agaactcgcg 180
cagacacgga ttacttgcca aaagggaaaag aagatgggtg tgacaatgat gagcgggtgac 240
atgaacattt ggcagagta caaaaatcga acgatattcg atataaccaa taacttgctc 300
atagtaatac ttgccttgcg acctctgac gagggaaacg atgaatgtgt agtgcttaag 360
tatgaaaaag atgcctttaa gcgggaacac ttggctgagg ttacactctc cgtaaggcgc 420
gactttccta cgccgtctat atccgacttc gagataccca cttctaaccat tcgacgcac 480
atgtgctcaa cctcaggtgg ttcccagag cctcacttga gctggctgga gaattggcga 540
gaacttaacg caatcaatac caccgtgttc caagaccggg agacagagct gtacgcgcgt 600
tcataccaac tggattttaa atgcagcaca aatcatagtt tcatgtgtct gatcaaatat 660
gggcatctca gggatgaatc gacttttaac tggaaacact ccaaaacaaga gcacttccca 720
gataatctgt tgccaagctg ggcgataact cttatctccg tcaacgggat cttcgtaatt 780
tgctgctcga cctattgttt cgccgctcga tgccgagaag gcagcggcgc caccacattc 840
tccctgctga agcaggccgg cgacgtggaa gaaaaccctg gcccacatga gtgtctgctg 900
tacctggcgt tcctgtttat cggggtgaaac tgcaagtcca ctatcgtgtt tccgcacaac 960
caaaagggca actggaaaaa cttgccttca aattaccatt attgcccag cagctcggac 1020
ctgaactggc acaatgacct cattggaacc gcgctgcagg tgaagatgcc acagagccac 1080
aaggctatcc aggtgcagcg atggatgtgc caccgctcaa aatgggtgac tacctgcgat 1140
ttccgctggt accgaccaaa atacatcacg cacagcatca gatcattcac cccgtcagtg 1200
gaacaatgca aagaatccat cgaacagact aagcagggaa cctggctgaa cctgggattt 1260
ccgcgcgagt cgtgtgggta cgcaaccgtg accgatgcag aggcctgat cgtgcaagtc 1320
acgcgcgcat acgtgcttgt ggcagagtag accggagaat gggctgattc ccagttcatc 1380
aacggcaagt gctccaacta catttgccca accgtgcaca acagcactac ttggcacagc 1440
gactacaaag tgaagggtct gtgtgattcc aacctgatct ccatggatat cactttcttc 1500
tcggaagacg gcgaactgtc ctcaactggc aaagaaggaa ctgggttttc ctcaaattac 1560
ttcgctacg aaactggagg aaaagcctgc aagatgcagt actgcaagca ctggggcgtg 1620
agactaccca gcggtgtctg gtcgagatg gccgataagg acctgtttgc agcagcgaga 1680
ttcccggaat gccctgaggg atcgagcatc tccgctccaa gccaaacttc agtgagcgtg 1740
agcctgaccc aggcagtgga acggattctc gactactcgc tgtgccagga gacctggctg 1800
aagatcagag cgggactgcc catctcaccg gtggacctgt cctacctggc gccaaagaat 1860
ccgggcactg gaccggcgtt caccatcatc aacggcaccg tcaataactt cgagacgcgg 1920
tacatccggg tggacatcgc agctccgac ctctcccgga ttgtgggaat gatctcgggg 1980
actactaccg aagccgagct ctgggacgac tgggcacatt acgaggatgt cgagatcgga 2040
cctaaccggg tgcctccggg ctctccggg tacaagtctc ctctgtacat gatcgccat 2100
ggcatgctgg actcggatct gcatctgtcg tccaaagcac aggtgtttga acaccacac 2160
attcaagacg ccgccagcca gctgccggac gatgagtcgc tgttcttcgg agacacgggc 2220
ttgtcaaaag atcccatcga gctgggtgaa ggatgggttt catcctggaa aagcagcatc 2280
gcttcattct tcttcatcat tggcctgac atcgccctat ttctagtcct gcgggtggga 2340
attcatctgt gcatcaagct caagcacact aagaagcggc aaatctacac tgatcgcag 2400
atgaatcgcc tgggcaag 2418

SEQ ID NO: 76 moltype = AA length = 806
FEATURE Location/Qualifiers
REGION 1..806
note = CD80 T cell targeting molecule co-expressed with
mutant VSV-G Full molecule

REGION 1..35
note = MISC_FEATURE - signal sequence

source 1..806
mol_type = protein
organism = synthetic construct

SEQUENCE: 76

-continued

MGHTRRQGT	PSKCPYLNF	QLLVLAGLSH	FCSGVIHVTK	EVKEVATLSC	GHNVSVEELA	60
QTRIYWQKEK	KMVLTMMSGD	MNIWPEYKNR	TIFDITNNLS	IVILALRPSD	EGTYECVVLK	120
YEKDAFKREH	LAEVTLVSKA	DFPTPSISDF	EIPTSNIIRI	ICSTSGGFPE	PHLSWLENGE	180
ELNAINTTVS	QDPETELYAV	SSKLDPMNTT	NHSFMCLIKY	GHLRVNQTFN	WNTTKQEHFP	240
DNLLPSWAIT	LISVNGIFVI	CCLTYCFAPR	CREGSGATNF	SLLKQAGDVE	ENPGPMKCLL	300
YLAFLEFIGVN	CKFTIVFPFN	QKGNWKNVPS	NYHYCPSSSD	LNWHNDLIGT	ALQVKMPQSH	360
KAIQADGWMC	HASKWVTTC	FRWYGPYIT	HSIRSFTPSV	EQCKESIEQT	KQGTWLNPGF	420
PPQSCGYATV	TDAAEAVIVQV	TPHHVLVDEY	TGEWVDSQFI	NGKCSNYICP	TVHNSTTWS	480
DYKVKGLCDS	NLISMDITFF	SEDGELSSSLG	KEGTGFRSNY	FAYETGGKAC	KMQYCKHWGV	540
RLPSGVWFEM	ADKDLFAAR	FPECPEGSSI	SAPSQTSVDV	SLIQDVERIL	DYSLCQETWS	600
KIRAGLPISP	VDLSYLAPKN	PGTGPAFTII	NGTLKYFETR	YIRVDIAAPI	LSRMVGMISG	660
TTTEAELWDD	WAPYEDVEIG	PNGVLRITSSG	YKFPLYMIGH	GMLDSDLHLS	SKAQVFEHPH	720
IQDAASQLPD	DESLFFGDTG	LSKNPIELVE	GWFSWKKSSI	ASFFFIIGLI	IGLFLVLRVG	780
IHLCKIKLHT	KKRQIYTDIE	MNRLGK				806

SEQ ID NO: 77 moltype = DNA length = 1533
 FEATURE Location/Qualifiers
 misc_feature 1..1533
 note = WT VSV-G
 source 1..1533
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 77

atgaagtgtc	tgtgtgacct	ggcggttctcg	tttatcgggg	tgaactgcaa	gttccactatc	60
gtgtttccgc	acaacccaaa	gggcaactgg	aaaaacgtgc	cttcaaatta	ccattattgc	120
cccagcagct	cgacacctgaa	ctggcaccaat	gacctcattg	gaaccgcgct	gcagggtgaag	180
atgccaaaaga	gccacaaggc	tatccaggct	gacggatgga	tgtgccacgc	gtcaaaatgg	240
gtgactacct	gcgatttccg	ctggtacgga	ccaaaataca	tcacgcacag	catcagatca	300
ttcaccccg	cagtgggaaca	atgcaaaaga	tccatcgaa	agactaagca	gggaacctgg	360
ctgaacctcg	gatttccgcc	gcagtcgtgt	gggtacgcaa	ccgtgaccga	tcagaggcc	420
gtgactgtgc	aagtcacgcc	gcacacgtg	ctgtgggacg	agtacaccgg	agaatgggtc	480
gattcccgat	tcatacaacg	caagtgtctc	aactacattt	gccaacccgt	gcacaacagc	540
actacttggc	acagcgacta	caaagtgaag	ggtctgtgtg	attccaacct	gatctccatg	600
gatatactt	tcttctcgga	agacggcgaa	ctgtcctcac	tgggcaaaaga	aggaaactggg	660
tttcgctcaa	attacttcgc	ctacgaaact	ggaggaaaag	cctgcaagat	gcagactatgc	720
aagcactggg	gcgtgagact	accacgcggt	gtctggttcg	agatggccga	taaggacctg	780
tttgacgag	cgagattccc	ggaatgccct	gagggatcga	gcatactccg	tccaagccaa	840
acttcagtg	acgtgagcct	gatccaggac	gtggaaacga	ttctcgacta	ctcgtgtgtc	900
caggagacct	ggtcgaagat	cagagcggga	ctgcccatct	caccggtgga	cctgtcctac	960
ctggcgccaa	agaatccggg	actgggaccg	gcgttcacca	tcatacaacg	caccctcaaa	1020
tacttcgaga	cgcggtacat	ccgggtggac	atcgacagtc	cgatcctctc	ccggatgggtg	1080
ggaatgatct	cggggactac	taccgaacgc	gagctctggg	acgactgggc	accttacgag	1140
gatgtcgaga	tcggactcaa	cggagtgtct	cggacctcct	ccgggtacaa	gttcctctctg	1200
tacatgatcg	gccatggcat	gctggactcg	gatctgcac	tgtcgtccaa	agcacagggtg	1260
tttgaaaccc	cacacattca	agacgcccgc	agccagctgc	cggacgatga	gtcgtgtgtc	1320
ttcgagagca	cgggcttctg	aaagaatccc	atcgagctgg	tggaaaggatg	gttttcatcc	1380
tggaaaagca	gcacgcgttc	attcttcttc	atcattggcc	tgatcatcgg	cctatttcta	1440
gtcctgcggg	tgggaattca	tctgtgcac	aagctcaagc	acactaagaa	gcgggcaaatc	1500
tacactgata	tcgagatgaa	tcgcctgggc	aag			1533

SEQ ID NO: 78 moltype = AA length = 511
 FEATURE Location/Qualifiers
 REGION 1..511
 note = WT VSV-G
 REGION 1..16
 note = MISC_FEATURE - signal sequence
 source 1..511
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 78

MKCLLYLAF	FIGVNCFTI	VFPNQHKNW	KNVPSNYHYC	PSSSDLNWHN	DLIGTALQVK	60
MPKSHKAIQA	DGWMCHASKW	VTTCDFRWYG	PKYITHSIRS	FTPSVEQCKE	SIEQTKQGTW	120
LNPGFPQQSC	GYATVTDAAE	VIVQVTPHHV	LVDEYTGWEV	DSQFINGKCS	NYICPTVHNS	180
TTWHSYDKVK	GLCDNLISM	DITFFSEDEG	LSSLGKEGTG	FRSNYFAYET	GGKACKMQYC	240
KHWGVRLP	SGVLFEMADKDL	FAAARFPECP	EGSSISAPSQ	TSVDVSLIQD	VERILDYSLC	300
QETWSKIRAG	LPISPVDLSY	LAPKNPGTGP	AFTIINGTLK	YFETRYIRVD	IAAPILSRMV	360
GMISGTTTER	ELWDDWAPYE	DVEIGPNGLV	RTSSGYKPLP	YMIGHGMLDS	DLHLSSKAQV	420
FEHPHQDAA	SQLPDDESLF	FQDGLSKNPN	IELVEGWFS	WKSSIASFFF	IIGLIIGLFL	480
VLRVGIHLCT	KLKHTKKRQI	YTDIEMNRLG	K			511

SEQ ID NO: 79 moltype = AA length = 18
 FEATURE Location/Qualifiers
 REGION 1..18
 note = T2A peptide from Thosea asigna virus capsid protein
 source 1..18
 mol_type = protein

-continued

```

organism = synthetic construct

SEQUENCE: 79
EGRGSLTTCG DVEENPGP                                     18

SEQ ID NO: 80      moltype = DNA  length = 54
FEATURE           Location/Qualifiers
misc_feature       1..54
                   note = T2A peptide from Thosea asigna virus capsid protein
source            1..54
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 80
gagggcaggg gaagtcttct aacatgcggg gacgtggagg aaaatcccg cccc       54

SEQ ID NO: 81      moltype = AA  length = 19
FEATURE           Location/Qualifiers
REGION            1..19
                   note = P2A peptide from porcine teschovirus-1 polyprotein
source            1..19
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 81
ATNFSLLKQA GDVEENPGP                                     19

SEQ ID NO: 82      moltype = DNA  length = 57
FEATURE           Location/Qualifiers
misc_feature       1..57
                   note = P2A peptide from porcine teschovirus-1 polyprotein
source            1..57
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 82
gccacgaact tctctctgtt aaagcaagca ggagacgtgg aagaaaacc cggtcct       57

SEQ ID NO: 83      moltype = AA  length = 22
FEATURE           Location/Qualifiers
REGION            1..22
                   note = F2A peptide from foot-and-mouth disease virus
                   polyprotein
source            1..22
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 83
VKQTLNFDLL KLAGDVESNP GP                                  22

SEQ ID NO: 84      moltype = DNA  length = 66
FEATURE           Location/Qualifiers
misc_feature       1..66
                   note = F2A peptide from foot-and-mouth disease virus
                   polyprotein
source            1..66
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 84
gtgaaacaga ctttgaattt tgacctcttc aagttggcgg gagacgtcga gtccaaccct   60
gggccc                                           66

SEQ ID NO: 85      moltype = AA  length = 20
FEATURE           Location/Qualifiers
REGION            1..20
                   note = E2A peptide from equine rhinitis A virus polyprotein
source            1..20
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 85
QCTNYALLKL AGDVESNPGP                                     20

SEQ ID NO: 86      moltype = DNA  length = 60
FEATURE           Location/Qualifiers
misc_feature       1..60
                   note = E2A peptide from equine rhinitis A virus polyprotein
source            1..60
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 86
cagtgacta attatgctct cttgaaattg gctggagatg ttgagagcaa cccaggtccc   60

```

-continued

SEQ ID NO: 87 moltype = DNA length = 48
 FEATURE Location/Qualifiers
 misc_feature 1..48
 note = WT VSV-G Signal Sequence
 source 1..48
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 87
 atgaagtgtc tgctgtacct ggcgttctcg tttatcgggg tgaactgc 48

SEQ ID NO: 88 moltype = AA length = 16
 FEATURE Location/Qualifiers
 REGION 1..16
 note = WT VSV-G Signal Sequence
 source 1..16
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 88
 MKCLLYLAPL FIGVNC 16

SEQ ID NO: 89 moltype = DNA length = 1485
 FEATURE Location/Qualifiers
 misc_feature 1..1485
 note = WT VSV-G without signal sequence
 source 1..1485
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 89
 aagttcacta tcgtgtttcc gcacaaccaa aaggccaact ggaaaaacgt gccttcaaat 60
 taccattatt gccccagcag ctccggacctg aactggcaca atgacctcat tggaaacgcg 120
 ctgcagggtga agatgccaaa gagccacaag gctatccagg ctgacggatg gatgtgccac 180
 gcgtcaaaat gggtgactac ctgcgatttc cgttggtacg gacccaaaata catcacgcac 240
 agcatcagat cattcaccoc gtcagtggaa caatgcaaa aatccatcga acagactaag 300
 caggggaacct ggctgaaccc tggatttccg ccgcagtcgt gtgggtacgc aaccgtgacc 360
 gatgcagagg ccgtgatcgt gcaagtcacg ccgcacacac tgcttctgga cgagtacacc 420
 ggagaatggg tcgattccca gtccatcaac ggcaagtgc ccaactacat ttgcccaacc 480
 gtgcacaaca gcactacttg gcacagcgac tacaaagtga agggctctgtg tgattccaac 540
 ctgatctcca tggatatcac ttctctctcg gaagacggcg aactgtcctc actgggcaaa 600
 gaaggaaactg ggtttcgctc aaattacttc gcctacgaaa ctggaggaaa agcctgcaag 660
 atgcagtagt gcaagcactg gggcgtaga ctaccacagc gtgtctgggt cgagatggcc 720
 gataaggacc tgtttgagc agcgagattc ccggaatgcc ctgagggatc gagcatctcc 780
 gctccaagcc aaacttcagt ggcagtgagc ctgatccagg acgtggaacg gattctcgac 840
 tactcgctgt gccaggagac ctggctgaag atcagagcgg gactgcccac ctcaccgggtg 900
 gacctgtcct acctggcgcc aaagaatccg ggcactggac cggcggtcac catcatcaac 960
 ggcaccctca aatacttcga gacgcggtag atccgggttg acatcgagc tccgactctc 1020
 tcccggatgg tgggaatgat ctccgggact actaccgaac gcgagctctg ggacgactgg 1080
 gcaccttacg aggatgtcga gatcggacct aacggagtgc tccggacctc ctccgggtac 1140
 aagttccctc tgtacatgat gggccatggc atgctggact cggatctgca tctgtcgtcc 1200
 aaagcacagg tgtttgaaca ccacacatt caagacgccc ccagccagct gccggacgat 1260
 gactcgctgt tcttcggaga caccgggcttg tcaaagaatc ccacgagct ggtggaagga 1320
 tgggtttcat cctggaaaag cagcatcgct tcattcttct tcatcattgg cctgacatc 1380
 ggcctatttc tagtctcgcg ggtgggaatt catctgtgca tcaagctcaa gcacactaag 1440
 aagcggcaaa tctacactga tatcgagatg aatcgctcgg gcaag 1485

SEQ ID NO: 90 moltype = AA length = 495
 FEATURE Location/Qualifiers
 REGION 1..495
 note = WT VSV-G without signal sequence
 source 1..495
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 90
 KPTIVFPHNQ KGNWKNVPSN YHYCPSSSDL NWNHDLIGTA LQVKMPKSHK AIQADGWMCH 60
 ASKWVTTCDP RWYGPKYITH SIRSFPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
 DAEAVIVQVT PHHLVDEYT GEWVDSQFIN GKCSNYICPT VHNSTTWHS D YKVKGLCDSN 180
 LISMDITFFS EDGELSSLGK EGTGFRSNFY AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
 DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
 DLSYLAPKNP GTGPAFTIIN GTLKYFETRY IRVDIAAPIL SRMVGMSGT TTERELWDDW 360
 APYEDVEIGP NGVLTSSGY KFPLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
 ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
 KRQIYTDIEM NRLGK 495

SEQ ID NO: 91 moltype = DNA length = 1488
 FEATURE Location/Qualifiers
 misc_feature 1..1488

-continued

```

note = Trop-002 Mutated VSV-G
source      1..1488
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 91
aagttcacta tcgtgtttcc gcacaaccaa aagggaact ggaaaaacgt gccttcaaat 60
taccattatt gccccagcag ctccggacctg aactggcaca atgacctcat tggaaaccgcg 120
ctgcaggtga agatgccaca gagccacaag gctatccagg ctgacggatg gatgtgccac 180
gcgtcaaaat gggtgactac ctgcgatttc cgctggtagc gaccaaata catcacgcac 240
agcatcagat cattcacccc gtcagtggaa caatgcaaag aatccatcga acagactaag 300
cagggaaacct ggtgaaaccc tggatttccg ccgcagtcgt gtgggtacgc aaccgtgacc 360
gatgcagagg ccgtgatcgt gcaagtcacg ccgcacacg tgcttgtaga cgagtacacc 420
ggagaatggg tcgattccca gtccatcaac ggcaagtgtc ccaactacat ttgcccaacc 480
gtgcacaaca gcactacttg gcatagcgac tacaaagtga agggctctgt tgattccaac 540
ctgatctcca tggatatcac ttcttctcgc gaagacggcg aactgtcctc actgggcaaa 600
gaaggaaactg ggtttcgctc aaattacttc gcctacgaaa ctggaggaaa agcctgcaag 660
atgcagtagt gcaagcactg gggcgtgaga ctaccacagc gtgtctggtt cgagatggcc 720
gataaggacc tgtttgcagc agcgagattc ccggaatgcc ctgagggatc gagcatctcc 780
gctccaagcc aaacttcagt ggcagtggag ctgatccagg acgtggaacg gattctcgac 840
tactcgctgt gccaggagag ctggctgaaag atcagagcgg gactgcccat ctcccggtg 900
gacctgtcct acctggcgcc aaagaatccg ggcactggac cggcggtcac catcatcaac 960
ggcaccctca aatacttcga gacgcggtac atccgggtgg acatcgagc tccgatctc 1020
tcccggatgg tgggaatgat ctccgggact actaccgaag ccgagctctg ggacgactgg 1080
gcaccttacg aggatgtcga gatccgacct aacggagtgc tccggacctc ctccgggtac 1140
aagttccctc tgtacatgat cggccatggc atgctggact cggatctgca tctgtcgtcc 1200
aaagcacagg tgtttgaaca ccacacatt caagacgccc ccagccagct gccggacgat 1260
gagtcgctgt tcttcggaga caccggcttg tcaagaatc ccatcgagct ggtggaagga 1320
tggttttcat cctggaaaag cagcatcgct tcattcttct tcatcattgg cctgatcatc 1380
ggcctatttc tagtctcgcg ggtgggaatt catctgtgca tcaagctcaa gcacactaag 1440
aagcggcaaa tctacactga tatcgagatg aatcgctcgg gcaagtag 1488

```

```

SEQ ID NO: 92      moltype = DNA length = 1488
FEATURE            Location/Qualifiers
misc_feature        1..1488
                    note = Trop-051 Mutated VSV-G
source              1..1488
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 92
aagttcacta tcgtgtttcc gcacaaccaa aagggaact ggaaaaacgt gccttcaaat 60
taccattatt gccccagcag ctccggacctg aactggcaca atgacctcat tggaaaccgcg 120
ctgcaggtga agatgccaaa gagccacaag gctatccagg ctgacggatg gatgtgccac 180
gcgtcaaaat gggtgactac ctgcgatttc cgctggtagc gaccaaata catcacgcac 240
agcatcagat cattcacccc gtcagtggaa caatgcaaag aatccatcga acagactaag 300
cagggaaacct ggtgaaaccc tggatttccg ccgcagtcgt gtgggtacgc aaccgtgacc 360
gatgcagagg ccgtgatcgt gcaagtcacg ccgcacacg tgcttgtaga cgagtacacc 420
ggagaatggg tcgattccca gtccatcaac ggcaagtgtc ccaactacat ttgcccaacc 480
gtgcacaaca gcactacttg gcacagcgac tacaaagtga agggctctgt tgattccgcc 540
ctggcctcca tggatatcac ttcttctcgc gaagacggcg aactgtcctc actgggcaaa 600
gaaggaaactg ggtttcgctc aaattacttc gcctacgaaa ctggaggaaa agcctgcaag 660
atgcagtagt gcaagcactg gggcgtgaga ctaccacagc gtgtctggtt cgagatggcc 720
gataaggacc tgtttgcagc agcgagattc ccggaatgcc ctgagggatc gagcatctcc 780
gctccaagcc aaacttcagt ggcagtggag ctgatccagg acgtggaacg gattctcgac 840
tactcgctgt gccaggagac ctggctgaaag atcagagcgg gactgcccat ctcccggtg 900
gacctgtcct acctggcgcc aaagaatccg ggcactggac cggcggtcac catcatcaac 960
ggcaccctca aatacttcga gacgcggtac atccgggtgg acatcgagc tccgatctc 1020
tcccggatgg tgggaatgat ctccgggact actgccgccc gcgagctctg ggacgactgg 1080
gcaccttacg aggatgtcga gatccgacct aacggagtgc tccggacctc ctccgggtac 1140
aagttccctc tgtacatgat cggccatggc atgctggact cggatctgca tctgtcgtcc 1200
aaagcacagg tgtttgaaca ccacacatt caagacgccc ccagccagct gccggacgat 1260
gagtcgctgt tcttcggaga caccggcttg tcaagaatc ccatcgagct ggtggaagga 1320
tggttttcat cctggaaaag cagcatcgct tcattcttct tcatcattgg cctgatcatc 1380
ggcctatttc tagtctcgcg ggtgggaatt catctgtgca tcaagctcaa gcacactaag 1440
aagcggcaaa tctacactga tatcgagatg aatcgctcgg gcaagtag 1488

```

```

SEQ ID NO: 93      moltype = AA length = 495
FEATURE            Location/Qualifiers
REGION              1..495
                    note = Trop-051 Mutated VSV-G
source              1..495
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 93
KFTIVFPNHQ KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA LQVKMPKSHK AIQADGWMCH 60
ASKWVTTCDF RWYGPKYITH SIRSFPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DABEIVQVT PHHLVD EYT GEWVDSQFIN GKCSNYICT VHNSTTWHSD YKVKGLCDSA 180

```

-continued

LASMDITFFS	EDGELSSLGK	EGTGFRSNYP	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TAARELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 94 moltype = DNA length = 1488
 FEATURE Location/Qualifiers
 misc_feature 1..1488
 note = Trop-052 Mutated VSV-G
 source 1..1488
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 94

aagttcacta	tcggtgtttcc	gcacaaccaa	aagggcaact	ggaaaaacgt	gccttcaaat	60
taccattatt	gccccagcag	ctcggacctg	aactggcaca	atgacctcat	tggaaaccgcg	120
ctgcagggtga	agatgccaaa	gagccacaag	gctatccagg	ctgacggatg	gatgtgccac	180
gcgtcaaaat	gggtgactac	ctgcgatttc	cgtcggtacg	gacaaaata	catcacgcac	240
agcatcagat	cattcacccc	gtcagtgga	caatgcaaa	aatccatcga	acagactaag	300
cagggaaacct	ggctgaaccc	tggatttccg	cgcagtcgt	gtgggtacgc	aaccgtgacc	360
gatgcagagg	cgtgatcgt	gcaagtcacg	cgcacacg	tgcttgtgga	cgagtacacc	420
ggagaatggg	tcgattccca	gttcatcaac	ggcaagtgc	ccaactacat	ttgcccacc	480
gtgcacaaca	gcactacttg	gcacagcgac	tacaaagtga	agggctctgtg	tgattccaac	540
ctgatctcca	tggatatcac	tttcttctcg	gaagacggcg	aactgtcctc	actgggcaaa	600
gaaggaaactg	ggtttcgctc	aaattacttc	gcctacgaaa	ctggaggaaa	agcctgcaag	660
atgcagtact	gcaagcactg	gggcgtgaga	ctaccacagc	gtgtctggtt	cgagatggcc	720
gataaggacc	tggttgcagc	agcgagattc	ccggaatgcc	ctgagggatc	gagcatctcc	780
gctccaagcc	aaacttcagt	ggacgtgagc	ctgatccagg	acgtggaacg	gattctcgac	840
tactcgctgt	gccaggagac	ctggtcgaag	atcagagcgg	gactgcccat	ctcacccgtg	900
gacctgtcgt	acctggcgcc	aaagaatccg	ggcactggac	cggcgttcac	catcatcaac	960
ggcacccctca	aataacttga	gacgcgggtac	atccgggtgg	acatcgcagc	tccgatcctc	1020
tcccggtatg	tgggaatgat	ctcggggact	acttgggccc	gcgagctctg	ggacgactgg	1080
gcacttaacg	aggatgtcga	gatcggacct	aacggagtcg	tccggacctc	ctccgggtac	1140
aaagttccctc	tgtacatgat	cgcccatggc	atgctggact	cggatctgca	tctgtcgtcc	1200
aaagcacagcg	tggttgaaca	cccacacatt	caagacgccg	ccagccagct	gccggacgat	1260
gagtcgctgt	tcttcggaga	cacgggcttg	tcaaagaatc	ccatcgagct	ggtggaagga	1320
tgggtttcat	cctggaaaag	cagcatcgct	tcattctctt	tcattcttgg	cctgatcatc	1380
ggcctatttc	tagtcctgcg	ggtgggaatt	catctgtgca	tcaagctcaa	gcacactaag	1440
aagcggcaaa	tctacactga	tatcgagatg	aatcgctcgg	gcaagtag		1488

SEQ ID NO: 95 moltype = AA length = 495
 FEATURE Location/Qualifiers
 REGION 1..495
 note = Trop-052 Mutated VSV-G
 source 1..495
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 95

KFTIVFPHNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	LQVKMPKSHK	AIQADGWMCH	60
ASKWVTTDCF	RWYGPKYITH	SIRSFTPSVE	QCKESIEQTK	QGTWLNPGFP	PQSCGYATVT	120
DAEAVIVQVT	PHHVLVDEYT	GEWVDSQFIN	GKCSNYICPT	VHNSTTWHS	YKVKGLCDN	180
LISMDITFFS	EDGELSSLGK	EGTGFRSNYP	AYETGGKACK	MQYCKHWGVR	LPSGVWFEMA	240
DKDLFAAARF	PECPEGSSIS	APSQTSVDVS	LIQDVERILD	YSLCQETWSK	IRAGLPISPV	300
DLSYLAPKNP	GTGPAFTIIN	GTLKYFETRY	IRVDIAAPIL	SRMVGMISGT	TWARELWDDW	360
APYEDVEIGP	NGVLTSSGY	KFPLYMIGHG	MLDSLHLSS	KAQVFEHPHI	QDAASQLPDD	420
ESLFFGDTGL	SKNPIELVEG	WFSSWKSSIA	SFFFIIGLII	GLFLVLRVGI	HLCIKLKHTK	480
KRQIYTDIEM	NRLGK					495

SEQ ID NO: 96 moltype = DNA length = 1488
 FEATURE Location/Qualifiers
 misc_feature 1..1488
 note = Trop-055 Mutated VSV-G
 source 1..1488
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 96

aagttcacta	tcggtgtttcc	gcacgcccgc	aagggcaact	ggaaaaacgt	gccttcaaat	60
taccattatt	gccccagcag	ctcggacctg	aactggcaca	atgacctcat	tggaaaccgcg	120
ctgcagggtga	agatgccaaa	gagccacaag	gctatccagg	ctgacggatg	gatgtgccac	180
gcgtcaaaat	gggtgactac	ctgcgatttc	cgtcggtacg	gacaaaata	catcacgcac	240
agcatcagat	cattcacccc	gtcagtgga	caatgcaaa	aatccatcga	acagactaag	300
cagggaaacct	ggctgaaccc	tggatttccg	cgcagtcgt	gtgggtacgc	aaccgtgacc	360
gatgcagagg	cgtgatcgt	gcaagtcacg	cgcacacg	tgcttgtgga	cgagtacacc	420
ggagaatggg	tcgattccca	gttcatcaac	ggcaagtgc	ccaactacat	ttgcccacc	480
gtgcacaaca	gcactacttg	gcacagcgac	tacaaagtga	agggctctgtg	tgattccgcc	540

-continued

```

ctgatctcca tggatatcac tttcttctcg gaagacggcg aactgtctct actgggcaaa 600
gaaggaactg gggttcgctc aaattacttc gcctacgaaa ctggaggaaa agcctgcaag 660
atgcagtact gcaagcactg gggcgtgaga ctaccacgag ctgtctgtgt cgagatggcc 720
gataaggacc tgtttgcagc agcgagattc ccggaatgcc ctgagggatc gagcatctcc 780
gctccaagcc aaacttcagt ggacgtgagc ctgatccagg acgtggaacg gattctcgac 840
tactcgctgt gccaggagac ctggtcgaag atcagagcgg gactgcccat ctcccggtg 900
gacctgtctc acctggcgcc aaagaatccg ggcactggac cggcgttcac catcatcaac 960
ggcacccctc aataactcga gacgcggtag atccgggtgg acatcgagc tccgatctc 1020
tcccggatgg tgggaatgat ctccgggact actaccgaac gcgagctctg ggacgactgg 1080
gcacettacg aggatgtcga gatcggacct aacggagtgc tccggacctc ctccgggtac 1140
aagttccctc tgtacatgat cggccatggc atgctggact cggatctgca tctgtcgtcc 1200
aaagcacagc tgtttgaaca cccacacatt caagacgccg ccagccagct gccggacgat 1260
gagtcgctgt tcttcggaga cacgggcttg tcaaagaatc ccatcgagct ggtggaagga 1320
tggttttcat cctggaaaag cagcatcgct tcatctctc tcatcattgg cctgatcatc 1380
ggcctatttc tagtcctcgc ggtgggaatt catctgtgca tcaagctcaa gcacactaag 1440
aagcggcaaa tctacactga tatcgagatg aatcgctcgg gcaagtag 1488

```

```

SEQ ID NO: 97      moltype = AA length = 495
FEATURE
REGION            1..495
                  note = Trop-055 Mutated VSV-G
source            1..495
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 97
KFTIVFPHAA KGNWKNVPSN YHYCPSSSDL NWHNDLIGTA LQVKMPKSHK AIQADGWMCH 60
ASKWVTTDCF RWYGPKYITH SIRSFTPSVE QCKESIEQTK QGTWLNPGFP PQSCGYATVT 120
DAEAVIVQVT PHHVLVDYET GERVDSQFIN GKCSNYICPT VHNSTTWHSY YKVKGLCDSA 180
LISMDITFFS EDGELSSLGK EGTGFRSNYF AYETGGKACK MQYCKHWGVR LPSGVWFEMA 240
DKDLFAAARF PECPEGSSIS APSQTSVDVS LIQDVERILD YSLCQETWSK IRAGLPISPV 300
DLSYLAPKNP GTGPAFTIIN GTLKYPETRY IRVDIAAPIL SRMVGMSIGT TTERELWDDW 360
APYEDVEIGP NGVLRITSSGY KPFLYMIGHG MLDSDLHLSS KAQVFEHPHI QDAASQLPDD 420
ESLFFGDTGL SKNPIELVEG WFSWKSSIA SFFFIIGLII GLFLVLRVGI HLCIKLKHTK 480
KRQIYTDIEM NRLGK 495

```

```

SEQ ID NO: 98      moltype = DNA length = 1494
FEATURE
misc_feature       1..1494
                  note = Trop-056 Mutated VSV-G
source            1..1494
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 98
aagttcacta tcgtgtttcc gcacaaacac acacaaaagg gcaactggaa aaacgtgcct 60
tcaaattacc attattgccc cagcagctcg gacctgaact ggcacaaatga cctcatttga 120
accgcgctgc aggtgaagat gccaaagagc cacaaggcta tccaggctga cggatggatg 180
tgccacgcgt caaaatgggt gactacctgc gatttccgct ggtagcgacc aaaatacatc 240
acgcacagca tcagatcatt caccctcgta gtggaacaat gcaaagaatc catcgaaacg 300
actaagcagg gaacctggct gaaccttgga tttccgccc agtcgtgtgg gtacgcaacc 360
gtgaccgatg cagaggccgt gatcgtgcaa gtcacgcgc atcacgtgct tgtggacgag 420
tacaccggag aatgggtcga ttcccagttc atcaacggca agtgctccaa ctacatttgc 480
ccaaccgtgc acaacagcac tacttggcac agcgactaca aagtgaaggg tctgtgtgat 540
tccaacctga tctccatgga tatcacttcc tttctggaag acggcgaaact gtctcactg 600
ggcaaaagaag gaactgggtt tcgctcaaat tacttcgctc acgaaactgg aggaaaagcc 660
tgcaagatgc agtactgcaa gcaactgggc gtgagactac ccagcgggtg ctggttcgag 720
atggccgata aggacctgtt tgcagcagcg agattcccgg aatgccctga gggatcgagc 780
atctccgctc caagccaaac ttcagtgagc gtgagcctga tccaggacgt ggaacggatt 840
ctcgactact cgtgtgcga ggagacctgg tcgaagatca gagcgggact gccatctca 900
ccggtggacc tgtctacct ggcgcacaa aatccgggca ctggaccggc gttcaccatc 960
atcaacggca cctcaataa cttcgagacg cggtagatcc ggggtggacat cgcagctccg 1020
atcctctccc ggtatgggtg aatgatctcg gggactacta ccgaacgcga gctctgggac 1080
gactgggacg cttacgagga tgtcgagatc ggacctaacy gagtgtctcg gacctctcc 1140
gggtacaagt tccctctgta catgatcgcc catggcatgc tggactcgga tctgcatctg 1200
tcgtccaaag cacaggtgtt tgaacacca cacattcaag acgcccgcag ccagctgccg 1260
gacgatgagt cgtgttctt cggagacacg ggctgtgcaa agaattccat cgagctgggtg 1320
gaaggatggt tttcatctg gaaaagcagc atcgcttcat tcttcttcat cattggcctg 1380
atcatcgccc tatttctagt cctgcgggtg ggaattcatc tgtgcatcaa gctcaagcac 1440
actaagaagc ggcaaatcta cactgatata gagatgaatc gcctgggcaa gtag 1494

```

```

SEQ ID NO: 99      moltype = AA length = 497
FEATURE
REGION            1..497
                  note = Trop-056 Mutated VSV-G
source            1..497
                  mol_type = protein
                  organism = synthetic construct

```

-continued

SEQUENCE: 99

KFTIVFPHNT	TQKGNWKNVP	SNHYHCPSSS	DLNWHNDLIG	TALQVKMPKS	HKAIQADGWM	60
CHASKWVTTT	DFRWYGPYKI	THSIRSFTPS	VEQCKESIEQ	TKQGTWLNPG	FPPQSCGYAT	120
VTDAEAVIVQ	VTPHHVLVDE	YTGWEVDSQF	INGKCSNYIC	PTVHNSTTWH	SDYKVKGLCD	180
SNLISMDITF	FSEDELSSSL	GKEGTGFRSN	YFAYETGGKA	CKMQYCKHWG	VRLPSGVWFE	240
MADKDLFAAA	RFPECPEGSS	ISAPSQTSVD	VSLIQDVERI	LDYSLCQETW	SKIRAGLPIS	300
PVDLSYLAPK	NPGTGPAFTI	INGTLKYFET	RYIRVDIAAP	ILSRMVGMI	GTTERELWD	360
DWAPYEDVEI	GPNGVLRTSS	GYKFPLYMIG	HGMLDSDLHL	SSKAQVFEHP	HIQDAASQLP	420
DDESLFFGDT	GLSKNPIELV	EGWFSSWKSS	IASFFFIIGL	IIGLFLVLRV	GIHLCKIKLH	480
TKKRQIYTDI	EMNRLGK					497

SEQ ID NO: 100 moltype = DNA length = 1497
 FEATURE Location/Qualifiers
 misc_feature 1..1497
 note = Trop-058 Mutated VSV-G
 source 1..1497
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 100

aagttcacta	tcgtgtttcc	gcacaaccaa	aagggaact	ggaaaaacgt	gccttcaaat	60
taccattatt	gccccagcag	ctcggacctg	aactggcaca	atgacctcat	tggaaaccgcg	120
ctgcagggtg	agatgccagg	cgccagcaag	agccacaagg	ctatccaggc	tgacggatgg	180
atgtgccacg	cgtcaaaatg	ggtgactacc	tcgcatttcc	gctggtagcg	acccaaaatac	240
atcacgcaca	gcatacagatc	attcaccccc	tcagtggaac	aatgcaaaag	atccatcgaa	300
cagactaagc	agggaaacctg	gctgaaccct	ggatttccgc	cgcagtcgtg	tgggtacgca	360
accgtgacgc	atgcagaggg	cgtgatcggt	caagtcacgc	cgcatacagc	gcttgtggac	420
gagtacaccg	gagaatgggt	cgattccccg	ttcatcaacg	gcaagtgctc	caactacatt	480
tgcccaaccg	tgcaacaacg	cactacttgg	cacagcgact	acaaagtga	gggtctgtgt	540
gattccaacc	tgcattccat	ggatactact	ttcttctcgg	aagacggcga	actgtcctca	600
ctgggcaaaag	aaggaaactgg	gtttcgctca	aattacttcg	cctacgaaac	tggaggaaaa	660
gacctgcaaga	tgcatgactg	caagcactgg	ggcgtgagac	taccagcgcg	tgtctggttc	720
gagatggcgc	ataaggacct	gtttgcagca	gcgagattcc	cggaaatgcc	tgagggatcg	780
agcatctccg	ctccaagcca	aacttcagtg	gacgtgagcc	tgatccagga	cgtggaaacgg	840
attctcgact	actcgctgtg	ccaggagacc	tggtcgaaga	tcagagcggg	actgcccatc	900
tcaccgggtg	acctgtccta	cctggcgcca	aagaatccgg	gcactggacc	ggcggtccac	960
atcatcaacg	gcacccctcaa	atacttcgag	acgcgggtaca	tcgggggtgga	catcgagcgt	1020
ccgatcctct	cccggatggt	gggaatgatc	tcgggggacta	ctaccgaacg	cgagctctgg	1080
gacgactggg	cacettacga	ggatgtcgag	atcggacctc	acggagtgct	ccggacctcc	1140
tcgggggtaca	agttccctct	gtacatgatc	ggccatggca	tgctggactc	ggatctgcat	1200
ctgtcgtcca	aagcacagggt	gtttgaacac	ccacacattc	aagacgcgcg	cagccagctg	1260
cgggacgatg	agtcgctggt	cttcggagac	acgggcttgt	caaagaatcc	catcgagctg	1320
gtggaaggat	gggttttcac	ctggaaaagc	agcatcgctt	cattctctct	catcattggc	1380
ctgatcatcg	gcctatttct	agtcctgcgg	gtgggaattc	atctgtgcat	caagctcaag	1440
cacactaaga	agcgggcaat	ctacactgat	atcgagatga	atcgccctggg	caagtag	1497

SEQ ID NO: 101 moltype = AA length = 498
 FEATURE Location/Qualifiers
 REGION 1..498
 note = Trop-058 Mutated VSV-G
 source 1..498
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 101

KFTIVFPHNQ	KGNWKNVPSN	YHYCPSSSDL	NWHNDLIGTA	LQVKMPGGSK	SHKAIQADGW	60
MCHASKWVTT	CDFRWYGPYK	ITHSIRSFTP	SVEQCKESIE	QTKQGTWLN	GFPPQSCGYA	120
TVTDAEAVIV	QVTPHHVLVD	EYTGWEVDSQ	FINGKCSNYI	CPTVHNSTTW	HSDYKVKGLC	180
DSNLISMDIT	FFSEDELSSS	LGKEGTGFRS	NYFAYETGGK	ACKMQYCKHW	GVRLPSGVWF	240
EMADKDLFAA	ARFPECPEGS	SISAPSQTSV	DVSLIQDVER	ILDYSLCQET	WSKIRAGLPI	300
SPVDLSYLAP	KNPGTGPAFT	IINGTLKYFE	TRYIRVDIAA	PILSRMVGMI	SGTTERELW	360
DDWAPYEDVE	IGPNGVLRTS	SGYKFPLYMI	HGMLDSDLH	LSSKAQVFEH	PHIQDAASQL	420
PDDESLFFGD	TGLSKNPIEL	VEGWSSWKS	SIASFFFIIG	LIIGLFLVLR	VGIHLCKIKL	480
HTKKRQIYTD	IEMNRLGK					498

SEQ ID NO: 102 moltype = DNA length = 1506
 FEATURE Location/Qualifiers
 misc_feature 1..1506
 note = Trop-061 Mutated VSV-G
 source 1..1506
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 102

aagttcacta	tcgtgtttcc	gcacggcgga	agcaacggcg	ggagccaaaa	gggcaactgg	60
aaaaaacgtg	cttcaaatga	ccattattgc	cccagcagct	cggacctgaa	ctggcacaat	120
gacctcattg	gaaccgcgct	gcaggatga	atgccaaaag	gccacaaggc	tatccaggct	180
gacggatgga	tgtgccacgc	gtcaaaatgg	gtgactacct	gcgatttccg	ctggtagcga	240
ccaaaataca	tcacgcacag	catcagatca	ttcacccccg	cagtggaaac	atgcaaaaga	300

-continued

```

tccatcgaaac agactaagca gggaaacctgg ctgaaccctg gatttccgcc gcagtcgtgt 360
gggtacgcaa ccgtagacga tgcagaggcc gtgatcgtgc aagtcacgcc gcatacgtg 420
cttgtggacg agtacaccgg agaattgggtc gattcccagt tcatcaacgg caagtgtctc 480
aactacattt gcccaacctg gcacaacagc actacttggc acagcgacta caaagtgaag 540
ggtctgtgtg attccaacct gatctccatg gatatacatt tcttctcgga agacggcgaa 600
ctgtcctcac tgggcaaaga aggaactggg ttctgctcaa attacttcgc ctacgaaact 660
ggaggaaaag cctgcaagat gcagtactgc aagcactggg gcgtgagact acccagcggg 720
gtctgggttc agatggcgga taaggacctg ttgcagcag cgagattccc ggaatgccct 780
gagggatcga gcatactccg tccaagccaa acttcagtgg acgtgagcct gatccaggac 840
gtggaacgga ttctcgacta ctctgtgtgc caggagacct ggtcgaagat cagagcggga 900
ctgcccctct caccggtgga cctgtcctac ctggcgccaa agaattccgg cactggaccg 960
gcgttcacca tcatcaacgg caccctcaaa tacttcgaga cgcggtacat ccgggtggac 1020
atcgacagtc cgactctctc ccggtggtg ggaatgatct cggggactac tacccaacgc 1080
gagctctggg agcactgggc accttacgag gatgtcgaga tcggacctaa cggagtgtct 1140
cggacctctc ccgggtacaa gttccctctg tacatgatcg gccatggcat gctggactcg 1200
gatctgcata tgtcgtccaa agcacagggtg ttgaaacacc cacacattca agacgccgcc 1260
agccagctgc cggacgatga gtcgtgttc ttccgagaca cgggctgtgc aaagaatccc 1320
atcgagctgg tggaaaggatg gttttcatcc tggaaaagca gcatacgttc attcttcttc 1380
atcattggcc tgatcatcgg cctatttcta gtctgcggg tgggaattca tctgtgcata 1440
aagctcaagc acactaagaa gcggcaaatc tacactgata tcgagatgaa tcgcctgggc 1500
aagtag

```

```

SEQ ID NO: 103      moltype = AA length = 501
FEATURE            Location/Qualifiers
REGION            1..501
                  note = Trop-061 Mutated VSV-G
source            1..501
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 103
KFTIVFPHGG SNGGSQKGNW KNPVSNYHYC PSSSDLNWHN DLIGTALQVK MPKSHKAIQA 60
DGWMCHASKW VTTCDFRWYG PKYITHSIRS FTPSVEQCKE SIEQTKQGTW LNPGFPPQSC 120
GYATVTDAAE VIVQVTPHHV LVDEYTGGEV DSQFINGKCS NYICPTVHNS TTWHSYKVK 180
GLCDNLISM DITFSEDEGE LSSLGKEGTG FRSNYFAYET GKGACKMQYC KHGVRPLPSG 240
VWFEMADKDL FAAARFPECP EGSSISAPSQ TSVDVSLIQD VERILDYSLC QETWSKIRAG 300
LPISPVLDLSY LAPKNPGTGP AFTIINGTLK YFETRYIRVD IAAPILSRMV GMISGTTTER 360
ELWDDWAPYE DVEIGPNGLV RTSSGYKFPL YMIGHGMLDS DLHLSSKAQV FEHPHIQDAA 420
SQLPDDESLF FGDGTLSKNP IELVEGWFSF WKSSIASFFF IIGLIIGLFL VLRVGIHLCI 480
KLKHTKKRQI YTDIEMNRLG K 501

```

```

SEQ ID NO: 104      moltype = AA length = 19
FEATURE            Location/Qualifiers
REGION            1..19
                  note = PDGFR stalk
source            1..19
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 104
AVGQDTQEVV VVPHSLPFK 19

```

```

SEQ ID NO: 105      moltype = AA length = 49
FEATURE            Location/Qualifiers
REGION            1..49
                  note = PDGFR stalk
source            1..49
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 105
ASAKPTTTPA PRPPTPAPTI ASQPLSLRPE AARPAAGGAV HTRGLDFAK 49

```

```

SEQ ID NO: 106      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION            1..17
                  note = flexible linker
source            1..17
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 106
GAPGSGGGGS GGGGSAS 17

```

```

SEQ ID NO: 107      moltype = AA length = 31
FEATURE            Location/Qualifiers
REGION            1..31
                  note = IgG4 hinge domain
source            1..31
                  mol_type = protein

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 107
ESKYGPPCPP CPAVGQDTQE VIVVPHSLPF K                               31

SEQ ID NO: 108                moltype = AA  length = 39
FEATURE                        Location/Qualifiers
REGION                        1..39
                                note = tetrameric coiled coil linker
source                        1..39
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 108
ASGGGGSGEL AAIKQELAAI KKELAAIKWE LAAIKQGAG                       39

SEQ ID NO: 109                moltype = AA  length = 14
FEATURE                        Location/Qualifiers
REGION                        1..14
                                note = dimeric coiled coil linker
source                        1..14
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 109
ASESKYGPPC PPCP                                                    14

SEQ ID NO: 110                moltype = AA  length = 20
FEATURE                        Location/Qualifiers
REGION                        1..20
                                note = Ig kappa leader sequence
source                        1..20
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 110
METDTLLLWV LLLWVPGSTG                                             20

SEQ ID NO: 111                moltype =   length =
SEQUENCE: 111
000

SEQ ID NO: 112                moltype = DNA  length = 357
FEATURE                        Location/Qualifiers
misc_feature                  1..357
                                note = Anti-CD3 (12F6) heavy chain variable region
source                        1..357
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 112
caagtgcagc tccagcagag cggcgctgag ctggcccgcc cggcgccag cgtgaagatg  60
agctgtaaag ccagcgggcta tacatttacc agctacacca tgcactgggt caagcagcgg 120
cctggccagg gcctggaatg gattggatat atcaacccca gcagcgggcta caccaagtac 180
aaccagaaat tcaaggacaa ggccaccctg accgccgaca agagctcctc aacagcctac 240
atgcaactga gcagcctgac cagcgaggat agcgccgtgt actactgcgc cagatggcag 300
gactacgacg tgtacttcga ctactggggc caaggcaca cactgaccgt gtccagc    357

SEQ ID NO: 113                moltype = AA  length = 119
FEATURE                        Location/Qualifiers
REGION                        1..119
                                note = Anti-CD3 (12F6) heavy chain variable region
source                        1..119
                                mol_type = protein
                                organism = synthetic construct

SEQUENCE: 113
QVQLQQSGAE LARPGASVKM SCKASGYTFT SYTMHWVKQR PGQGLEWIGY INPSSGYTKY  60
NQKFKDKATL TADKSSSTAY MQLSSLTSED SAVYYCARWQ DYDVYFDYWG QGTTLTVSS 119

SEQ ID NO: 114                moltype = DNA  length = 321
FEATURE                        Location/Qualifiers
misc_feature                  1..321
                                note = Anti-CD3 (12F6) light chain variable region
source                        1..321
                                mol_type = other DNA
                                organism = synthetic construct

SEQUENCE: 114
cagatcgtgc tgagccagtc cccagccatc ctgtctgccg gccctggcga gaaggtgacc  60
atgacctgca gagcctcttc ttctgtttcc tacatgcact ggtatcagca aaagcccgcc 120
agctctccta agccttggat ctacgccaca agcaacctgg ctacggcgct gcctgctcgc 180
ttcagcggca gcggcagcgg caccagctac agcctgacca tcagcagagt ggaagccgag 240

```

-continued

gagcgccgcca cctactactg ccagcagtggt tcctctaata ctccaacatt cgcgcgcgccg 300
 accaagctgg aaacccaaaag a 321

SEQ ID NO: 115 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Anti-CD3 (12F6) light chain variable region
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 115
 QIVLSQSPA I LSASPGKEVT MTCRASSSVS YMHVYQQKPG SSPKPWIYAT SNLASGVPAR 60
 FSGSGSGTSY SLTISRVEAE DAATYYCQW SSNPPTFGGG TKLETKR 107

SEQ ID NO: 116 moltype = DNA length = 996
 FEATURE Location/Qualifiers
 misc_feature 1..996
 note = Anti-CD3 (12F6) Full molecule
 source 1..996
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 116
 atggccttac cagtgacgcg cttgtctctg ccgctggcct tgtgtctcca cgccgcccagg 60
 ccgcaagtgc agctccagca gagcgcgctg gagctggccc ggcccggcgc cagcgtgaag 120
 atgagctgta aagccagcgg ctatacattt accagctaca ccatgacctg ggtcaagcag 180
 cggcctggcc agggcctgga atggattgga tatatcaacc ccagcagcgg ctacaccaag 240
 tacaaccaga aattcaagga caaggccacc ctgaccgccc acaagagctc ctcaacagcc 300
 tacatgcaac tgagcagcct gaccagcgag gatagcgccg tgtactactg cgccagatgg 360
 caggactacg acgtgtactt cgactactgg ggccaaggca caaactgac cgtgtccagc 420
 ggaggcggtg gtggcggtgg atcaggtgga ggcagcggtg gcgcatctca gatcgtgctg 480
 agccagtcgc cagccatctt gtctgccagc cctggcgaga aggtgacctt gacctgcaga 540
 gctctttctt ctgtttccta catgcactgg tatcagcaaa agcccggcag ctctcctaag 600
 ccttggtact acgcccacaag caacctggct agcggcgctg ctgctcgctt cagcggcagc 660
 ggcagcgcca ccagctacag cctgaccatc agcagagtgg aagccgagga cgccgccacc 720
 tactactgcc agcagtggtc ctctaatact ccaacattcg gcggcgccac caagctggaa 780
 accaaaagaa ccaactacac agcacctaga ccaccaacac ctgcccgaac catcgcatcg 840
 cagccactgt ctctgcgcc agaggcatgc cgccagcag ctgggggcgc agtgacaca 900
 agggggcctg acttcgcatg tgatatctac atctgggcac cattggcagg gaettgtggg 960
 gtctctctcc tgtcactggt taccaccctt tactgc 996

SEQ ID NO: 117 moltype = AA length = 332
 FEATURE Location/Qualifiers
 REGION 1..332
 note = Anti-CD3 (12F6) Full molecule
 REGION 1..21
 note = MISC_FEATURE - signal peptide
 source 1..332
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 117
 MALPVTALLL PLALLLHAAR PQVQLQQSGA ELARPGASVK MSCKASGYTF TSYTMHWVKQ 60
 RPSQGLEWIG YINPSSGYTK YNQKFKDKAT LTADKSSSTA YMLSSLTSE DSAVYYCARW 120
 QDYDVFYDYW QGQTTLTVSS GGGSGGGSGG GSGGGSQIVL SQSPAILSAS PGEKVTMTCR 180
 ASSSVSYMHW YQKPGSSPK PWIYATSNLA SGVPARFSGS GSGTSYSLTI SRVEAEDAAT 240
 YYCQWSSNP PTFGGGKLE TKRTTTPAPR PPTPAPTIAS QPLSLRPEAC RPAAGGAVHT 300
 RGLDFACDIY IWAPLAGTCG VLLLSLVITL YC 332

SEQ ID NO: 118 moltype = AA length = 12
 FEATURE Location/Qualifiers
 REGION 1..12
 note = IgG4 hinge region
 source 1..12
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 118
 ESKYGPPCPP CP 12

SEQ ID NO: 119 moltype = DNA length = 207
 FEATURE Location/Qualifiers
 misc_feature 1..207
 note = CD8a Hinge and Transmembrane
 source 1..207
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 119
 accacaacac ctgctccaag gccccccaca cccgctccaa ctatagccag ccaaccattg 60

-continued

```

agcctcagac ctgaagcttg cagggccgca gcaggaggcg cgtccatcac gcgaggcctg 120
gacttcgcgt gtgatattta tatttgggca cctttggccg gaacatgtgg ggtgttgctt 180
ctctcccttg tgatcactct gtattgt 207

```

```

SEQ ID NO: 120      moltype = DNA  length = 126
FEATURE            Location/Qualifiers
misc_feature        1..126
                    note = 4-1BB costimulatory domain
source              1..126
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 120
aagcgcgga gaaagaagct cctgtacatc ttcaagcagc cttttatgcg acctgtgcaa 60
accactcagg aagaagatgg gtgttcacgc cgcttcccg aggaggaaga aggagggtgt 120
gaactg 126

```

```

SEQ ID NO: 121      moltype = DNA  length = 336
FEATURE            Location/Qualifiers
misc_feature        1..336
                    note = CD3 zeta effector domain no. 2
source              1..336
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 121
agggtgaaat ttctagaag cgccgatgct cccgcatac agcagggtca gaatcagctc 60
tacaatgaat tgaatctcgg caggcgagaa gactacgatg ttctggacaa gagacggggc 120
agggatcccg agatgggggg aaagcccccg agaaaaaac ctcaggaggg gttgtacaat 180
gagctgcaga aggacaagat ggctgaagcc tatagcgaga tcggaatgaa aggcgaaaga 240
cgacagagca aggggcatga cggtctgtac cagggtctct ctacagccac caaggacact 300
tatgatgcgt tgcatacgca agccttgcca ccccg 336

```

```

SEQ ID NO: 122      moltype = AA  length = 112
FEATURE            Location/Qualifiers
REGION             1..112
                    note = CD3 zeta effector domain no. 2
source              1..112
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 122
RVKFSRSADA PAYQQGQNQL YNELNLGRRE EYDVLDKRRG RDPEMGGKPR RKNPQEGLYN 60
ELQKDKMAEA YSEIGMKGER RRGKGHDGLY QGLSTATKDT YDALHMQALP PR 112

```

```

SEQ ID NO: 123      moltype = AA  length = 20
FEATURE            Location/Qualifiers
REGION             1..20
                    note = B2M signal peptide
source              1..20
                    mol_type = protein
                    organism = synthetic construct

```

```

SEQUENCE: 123
MSRSVALAVL ALLSLSGLEA

```

20

What is claimed is:

1. A polypeptide comprising an amino acid sequence at least 97% identical to the amino acid sequence set forth in SEQ ID NO: 90, wherein the polypeptide comprises an amino acid substitution at position 182.

2. The polypeptide of claim 1, further comprising an amino acid substitution at position 352.

3. The polypeptide of claim 2, wherein the amino acid substitution at position 352 is from threonine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, serine, tryptophan, tyrosine, and valine.

4. The polypeptide of claim 1, wherein the amino acid substitution at position 182 is from isoleucine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid,

glycine, histidine, leucine, lysine, methionine, phenylalanine, proline, serine, threonine, tryptophan, tyrosine, and valine.

5. The polypeptide of claim 4, further comprising an amino acid substitution at position 352.

6. The polypeptide of claim 5, wherein the amino acid substitution at position 352 is from threonine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, serine, tryptophan, tyrosine, and valine.

7. A nucleic acid encoding the polypeptide of claim 1.

8. A lentiviral vector comprising an envelope comprising the polypeptide of claim 1.

9. The lentiviral vector of claim 8, wherein the envelope further comprises a heterologous lymphocyte targeting protein comprising an extracellular domain comprising a lymphocyte targeting domain and a transmembrane domain.

10. The lentiviral vector of claim 9, further comprising an expression cassette that comprises a heterologous transgene encoding a chimeric antigen receptor.

11. A polypeptide comprising an amino acid sequence at least 98% identical to the amino acid sequence set forth in SEQ ID NO: 90, wherein the polypeptide comprises an amino acid substitution at position 182.

12. The polypeptide of claim 11, further comprising an amino acid substitution at position 352.

13. The polypeptide of claim 12, wherein the amino acid substitution at position 352 is from threonine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, serine, tryptophan, tyrosine, and valine.

14. The polypeptide of claim 11, wherein the amino acid substitution at position 182 is from isoleucine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, leucine, lysine, methionine, phenylalanine, proline, serine, threonine, tryptophan, tyrosine, and valine.

15. The polypeptide of claim 14, further comprising an amino acid substitution at position 352.

16. The polypeptide of claim 15, wherein the amino acid substitution at position 352 is from threonine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, serine, tryptophan, tyrosine, and valine.

17. A nucleic acid encoding the polypeptide of claim 11.

18. A lentiviral vector comprising an envelope comprising the polypeptide of claim 11.

19. The lentiviral vector of claim 18, wherein the envelope further comprises a heterologous lymphocyte targeting protein comprising an extracellular domain comprising a lymphocyte targeting domain and a transmembrane domain.

20. The lentiviral vector of claim 19, further comprising an expression cassette that comprises a heterologous transgene encoding a chimeric antigen receptor.

21. A polypeptide comprising an amino acid sequence at least 99% identical to the amino acid sequence set forth in SEQ ID NO: 90, wherein the polypeptide comprises an amino acid substitution at position 182.

22. The polypeptide of claim 21, further comprising an amino acid substitution at position 352.

23. The polypeptide of claim 22, wherein the amino acid substitution at position 352 is from threonine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, serine, tryptophan, tyrosine, and valine.

24. The polypeptide of claim 21, wherein the amino acid substitution at position 182 is from isoleucine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, leucine, lysine, methionine, phenylalanine, proline, serine, threonine, tryptophan, tyrosine, and valine.

25. The polypeptide of claim 24, further comprising an amino acid substitution at position 352.

26. The polypeptide of claim 25, wherein the amino acid substitution at position 352 is from threonine to an amino acid selected from the group consisting of: alanine, arginine, asparagine, aspartic acid, cysteine, glutamine, glutamic acid, glycine, histidine, isoleucine, leucine, lysine, methionine, phenylalanine, proline, serine, tryptophan, tyrosine, and valine.

27. A nucleic acid encoding the polypeptide of claim 21.

28. A lentiviral vector comprising an envelope comprising the polypeptide of claim 21.

29. The lentiviral vector of claim 28, wherein the envelope further comprises a heterologous lymphocyte targeting protein comprising an extracellular domain comprising a lymphocyte targeting domain and a transmembrane domain.

30. The lentiviral vector of claim 29, further comprising an expression cassette that comprises a heterologous transgene encoding a chimeric antigen receptor.

* * * * *